

Service Manual

LG-D290TR

Table Of Contents

1. INTRODUCTION.....	3	4. DOWNLOAD.....	155
1.1 Purpose	3	5. BLOCK DIAGRAM.....	172
1.2 Regulatory Information	3	6. CIRCUIT DIAGRAM	181
1.3 Specification.....	5	7. BGA PIN MAP	198
2. PERFORMANCE.....	8	8. PCB LAYOUT	204
2.1 Product Name.....	8	9. CALIBRATION.....	207
2.2 Supporting Standard.....	8	10. HIDDEN MENU.....	226
2.3 Main Parts : GSM Solution.....	8	11. DISASSEMBLE GUIDE.....	259
2.4 HW Features	9	11.1 DISASSEMBLE GUIDE.....	259
2.5 HW SPEC	11	11.2 Shield Can Rework Guide	264
3. TROUBLE SHOOTING	24	11.3 Rework Caution Points	269
3.1 WCDMA Rx Part.....	24	12. EXPLODED VIEW & REPLACEMENT PART LIST	270
3.2. WCDMA Tx Part Trouble(B1).....	47		
3.3 WCDMA Tx Part Trouble(B8).....	61		
3.4 GSM Rx Part Troubleshooting(850/900 Rx).....	75		
3.5 GSM Rx Part Troubleshooting(1800/1900 Rx)	84		
3.6GSM Tx Part Troubleshooting(850/900 Tx).....	95		
3.7 GSM Tx Part Troubleshooting(1800/1900_Tx)	103		
3.8 Power TS.....	111		
3.9 Charging TS.....	115		
3.10 USB TS.....	117		
3.11 Audio TS.....	118		
3.12 6 AXIS Sensor on/off TS	132		
3.13 Proximity Sensor on/off TS.....	135		
3.14 Touch Sensor Trouble Shooting.....	137		
3.15 Vibrator Trouble Shooting	139		
3.16 Camera TS	141		
3.17 Main LCD TS	145		
3.18 BT/Wifi TS	147		
3.19 GPS TS.....	150		
3.20 FM TS.....	153		

1. INTRODUCTION

1.1 Purpose

This manual provides the information necessary to repair, calibration, description and download the features of this model.

1.2 Regulatory Information

A. Security

Toll fraud, the unauthorized use of telecommunications system by an unauthorized part (for example, persons other than your company's employees, agents, subcontractors, or person working on your company's behalf) can result in substantial additional charges for your telecommunications services. System users are responsible for the security of own system. There are may be risks of toll fraud associated with your telecommunications system. System users are responsible for programming and configuring the equipment to prevent unauthorized use. The manufacturer does not warrant that this product is immune from the above case but will prevent unauthorized use of common carrier telecommunication service of facilities accessed through or connected to it. The manufacturer will not be responsible for any charges that result from such unauthorized use.

B. Incidence of Harm

If a telephone company determines that the equipment provided to customer is faulty and possibly causing harm or interruption in service to the telephone network, it should disconnect telephone service until repair can be done. A telephone company may temporarily disconnect service as long as repair is not done.

C. Changes in Service

A local telephone company may make changes in its communications facilities or procedure. If these changes could reasonably be expected to affect the use of the phones or compatibility with the net work, the telephone company is required to give advanced written notice to the user, allowing the user to take appropriate steps to maintain telephone service.

D. Maintenance Limitation

Maintenance limitations on the phones must be performed only by the manufacturer or its authorized agent. The user may not make any changes and/or repairs expect as specifically noted in this manual. Therefore, note that unauthorized alternations or repair may affect the regulatory status of the system and may void any remaining warranty.

E. Notice of Radiated Emission

This model complies with rules regarding radiation and radio frequency emission as defined by local regulatory agencies. In accordance with these agencies, you may be required to provide information such as the following to the end user.

F. Pictures

The pictures in this manual are for illustrative purposes only; your actual hardware may look slightly different.

G. Interference and Attenuation

A phone may interfere with sensitive laboratory equipment, medical equipment, etc. Interference from unsuppressed engines or electric motors may cause problems.

H. Electrostatic Sensitive Devices

ATTENTION

Boards, which contain Electrostatic Sensitive Device (ESD), are indicated by the  sign. Following information is ESD handling:

- Service personnel should ground themselves by using a wrist strap when exchange system boards.
- When repairs are made to a system board, they should spread the floor with anti-static mat which is also grounded.
- Use a suitable, grounded soldering iron.
- Keep sensitive parts in these protective packages until these are used.
- When returning system boards or parts like EEPROM to the factory, use the protective package as described.

1.3 Specification

KEY FEATURES

- Android 4.4-(KitKat)
- 4.5" Touch Screen WVGA (480 x 800)
- 8MP AF with Flash
- 3G Mobile Hotspot

ENHANCED USER EXPERIENCE

- Android 4.4
- Proximity Sensor – locks the touch screen and buttons while talking on the phone
- Motion Sensor – Accelerometer
- USB Charging via Computer
- Airplane Mode (RF Off)

DESIGN

- Sleek and Stylish Design
- 4.5" WVGA LCD Screen
- Capacitive Touch Screen – Plastic with fast and accurate response
- Micro USB/Charging Port

CAMERA/VIDEO

- 8MP AF
- Camera Resolutions: 3264(H) x 2448(V) Pixels
- Image Editor: zoom, rotate, crop images
- Brightness, ISO, White Balance,* Color Effects, Timer
- Cheese shutter – take photo with voice command.
- Video Resolutions: 640 x 480 320 x 240 176 x 144 Pixels

MULTIMEDIA

- Music Player for MP3, WMA, and Unprotected AAC
- Stereo Sound via Headset or Bluetooth
- Music Library – Organized by Artists, Albums, Songs, and Playlists
- Repeat and Shuffle Music Playback Modes

BLUETOOTH®

- Version: 4.0
- Supported Profiles: headset, hands-free, object push, advanced audio distribution (stereo), audio/video remote control, file transfer, phone book access, message access, serial port
- Listen to Music with Optional Stereo Bluetooth Headset**
- Send User-Generated Pictures (JPEG) and Videos via Bluetooth Wireless Technology

*Depends on available memory.

**Accessories sold separately.

VOICE/AUDIO

- One-Touch Speakerphone*
- Speaker-Independent Voice Commands
- Voice-Activated Dialing
- Text to Speech
- MP3 Music Ringer Support

TOOLS & DATA

- Android Apps: Alarm/Clock, Browser, Calculator, Calendar (Corporate or Google), Camera, Contacts, Email, Gallery, Gmail, Google+, Maps, Messaging, Messenger, Music Player, Phone, Places, Play Books, Play Movies, Play Music, Play Store, Talk, Video Player, Voice Recorder, Youtube
- Navigation – spoken turn-by-turn directions showing real-time traffic
- Wi-Fi Connectivity – IEEE 802.11 b/g/n
- A-GPS Support for Enhanced Location Accuracy

SPECIFICATIONS

- Technology: GSM/WCDMA
- Frequencies: GSM 850,900,1800,1900
WCDMA 2100, 900
- Data Transmission: WCDMA, GPRS
- Dimensions: 127.5X67.9X11.65(mm)
- Display: 4.5" , WVGA (480 x 800) pixels
- Standard Battery: 1,900 [mAh]

ACCESSORIES

- Standard Battery
- Travel Adapter and USB Cable*
- Stereo Headset

2. PERFORMANCE

2.1 Product Name

LG-D290TR : WCDMA900/1900+EGSM/GSM850/DCS/PCS(HSDPA 21Mbps HSUPA 5.76Mbps)

2.2 Supporting Standard

Item	Feature	Comment
Supporting Standard	WCDMA(FDD1,8)/EGSM/GSM850/DCS1800/PCS1900 with seamless handover Phase 2+(include AMR) SIM Toolkit : Class 1, 2, 3, C-E	
Frequency Range	WCDMA (FDD1) TX : 1850 – 1910 MHz WCDMA (FDD1) RX : 1930 – 1990 MHz WCDMA(FDD8) TX : 880 - 915 MHz WCDMA(FDD8) RX : 925 – 960 MHz EGSM TX : 880 – 915 MHz EGSM RX : 925 – 960 MHz GSM850 TX : 824 – 849 MHz GSM850 RX : 869 – 894 MHz GSM900 TX : 880 – 915 MHz GSM900 RX : 925 – 960 MHz DCS1800 TX : 1710 – 1785 MHz DCS1800 RX : 1805 – 1880 MHz PCS1900 TX : 1850 – 1910 MHz PCS1900 RX : 1930 – 1990 MHz	
Application Standard		

2.3 Main Parts : GSM Solution

Item	Part Name	Comment
AP Chip	MSM8212	
CP Chip	MSM8212	

2.4 HW Features

List	Type / Spec.	
1. Phone Type	DOP Type	
2. Size	127.5X67.9X11.9 (mm)	
3. Weight	142g (with battery)	
4. Battery	1,900mAh (typ.), 1,820mAh(min.) (Li-Ion)	
5. Chipset	MSM8212	
6. Memory	4GB(eMMC) +1GB(LPDDR2)	
7. LCD	Size	4.5 inch
	Dot	480(H) X 800(V)
	Color	16M
	Display Type	TFT, Incell Touch
	NTSC	50% (Typ)
	Contrast Ratio	1:1000 (Typ)
	Pixel Pitch	0. 0408mm X 0.1224mm
8. Main Camera (8M)	Type	BSI CIS Type
	Resolution	3264(H) X 2448(V) pixels.
	View Angle	71.2° ± 5% (Diagonal)
	Pixel x size	1.12um x1.12um
10. Audio	1) MIDI	MIDI
	2) Mono	18 X 10 X 3T Speaker
	3) Receiver	12 X 06 X 2.5T Receiver

2. PERFORMANCE

11. Bluetooth	Effective Distance	10M
	Distance	0 m ~ 10 m (depend on environment)
12. WLAN	standard	IEEE 802.11 b/g/n
	Throughput	Max 40Mbps (SDIO Driver performance)
	depend on environment	0 ~ 50m (depend on environment)
13. GPS	type	A-GPS

2.5 HW SPEC

1) WCDMA transmitter specification

Item	Spec.	Min	Typ.	Max	Unit
Transmit Frequency	B1(W1900): 1850 – 1910 MHz B8(W900) 880 - 915 MHz	-	-	-	MHz
Maximum Output Power	Power class III	21.0	23.0	25.0	dBm/3.84MHz
Frequency Accuracy	-	-200	-	+200	Hz
Minimum Output Power	-	-	-	-50	dBm/3.84MHz
Occupied Bandwidth	-	-	-	4.6	MHz
Adjacent Channel Leakage Power Ratio	$\Delta f = \pm 5\text{MHz}$	-	-	-33	dBm/3.84MHz
	$\Delta f = \pm 10\text{MHz}$	-	-	-43	dBm/3.84MHz
Error Vector Magnitude(EVM)	-	-	-	17	%
Peak Code Domain Error	-	-	-	15	dBm/3.84MHz
Open Loop Output Power	lor = -106.7dBm/3.84MHz	-46.7	-	-28.7	dBm/3.84MHz
	lor = -65.7dBm/3.84MHz	-23.0	-	-5.0	dBm/3.84MHz
	lor = -25.0dBm/3.84MHz	-0.0	-	14.0	dBm/3.84MHz
Transmit Off Power	-	-	-	-56	dBm/3.84MHz
Spectrum Emission Mask	$\Delta f = 2.5\sim 3.5\text{MHz}$	-35-15 * (Δf -2.5)			dBc/30KHz
	$\Delta f = 3.5\sim 7.5\text{MHz}$	-35-1 * (Δf -3.5)			dBc/1MHz

2. PERFORMANCE

	$\Delta f = 7.5 \sim 8.5 \text{MHz}$	$-39 \ 10 * (\Delta f - 7.5)$			dBc/1MHz
	$\Delta f = 8.5 \sim 12.5 \text{MHz}$	-49			dBc/1MHz
Transmit Spurious Emission	9kHz ~ 150kHz			-36	dBm
	150kHz ~ 30MHz			-36	dBm
	30MHz ~ 1GHz			-36	dBm
	1GHz ~ 12.5GHz			-30	dBm

2) WCDMA receiver specification

Item	Specification																			
Receive Frequency	B1(W1900): 1930 – 1990 MHz B8(W900): 925 – 960 MHz																			
Reference Sensitivity Level <REFÎ _{or} >	Band1 : BER < 0.001 when Î _{or} = -103.7 dBm / 3.84 MHz Band8 : BER < 0.001 when Î _{or} = -103.7 dBm / 3.84 MHz																			
Maximum Input Level	BER < 0.001 when Î _{or} = -25 dBm / 3.84 MHz																			
Adjacent Channel Selectivity (ACS)	ACS > 33 dB where BER < 0.001 when Î _{or} = <REFÎ _{or} > + 14 dB / 3.84 MHz & loac = -52 dBm / 3.84 MHz @ ±5 MHz																			
Blocking Characteristic	BER < 0.001 when Î _{or} = <REFÎ _{or} > + 3 dB / 3.84 MHz & lblocking = -56 dBm / 3.84 MHz @ Fuw(offset) = ±10 MHz or lblocking = -44 dBm / 3.84 MHz @ Fuw(offset) = ±15 MHz																			
Spurious Response	BER < 0.001 when Î _{or} = <REFÎ _{or} > + 3 dB / 3.84 MHz & lblocking = -44 dBm																			
Intermodulation	BER < 0.001 when Î _{or} = <REFÎ _{or} > + 3 dB / 3.84 MHz & louw1 = -46 dBm @ Fuw1(offset) = ±10 MHz & louw2 = -46 dBm / 3.84 MHz @ Fuw2(offset) = ±20 MHz																			
Spurious Emissions	< -57 dBm / 100 kHz BW @ 9 kHz ≤ f < 1 GHz < -47 dBm / 1 MHz BW @ 1 GHz ≤ f ≤ 12.75 GHz																			
Inner Loop Power Control In Uplink	Adjust output(TPC command) <table><tr><td>cmd</td><td>1dB</td><td>2dB</td><td>3dB</td></tr><tr><td>+1</td><td>+0.5/1.5</td><td>+1/3</td><td>+1.5/4</td></tr><tr><td>0</td><td>-0.5/+0.5</td><td>-0.5/+0.5</td><td>-0.5/+0.5</td></tr><tr><td>-1</td><td>-0.5/-1.5</td><td>-1/-3</td><td>-1.5/-4</td></tr></table> group(10equal command group) <table><tr><td>+1</td><td>+8/+12</td><td>+16/+24</td></tr></table>	cmd	1dB	2dB	3dB	+1	+0.5/1.5	+1/3	+1.5/4	0	-0.5/+0.5	-0.5/+0.5	-0.5/+0.5	-1	-0.5/-1.5	-1/-3	-1.5/-4	+1	+8/+12	+16/+24
cmd	1dB	2dB	3dB																	
+1	+0.5/1.5	+1/3	+1.5/4																	
0	-0.5/+0.5	-0.5/+0.5	-0.5/+0.5																	
-1	-0.5/-1.5	-1/-3	-1.5/-4																	
+1	+8/+12	+16/+24																		

3) HSDPA transmitter specification

	Item	Spec.		Min	Typ.	Max	Unit
Transmitter	5.2.A Maximum Output Power with HS-DPCCH	$1/15 \leq \beta_c/\beta_d \leq 12/15$ $13/15 \leq \beta_c/\beta_d \leq 15/8$ $15/7 \leq \beta_c/\beta_d \leq 15/0$		21.0 20 19	23.0 23.0 22.5	25.0 25 25	dBm/3.84M Hz
	5.7A HS-DPCCH	MAX / 0dB m	Step 0	0 dB +/- 1 dB			dB
			Step 1 – Step 0	6 dB +/- 2.3 dB			dB
			Step 2 – Step 3	1 dB +/- 0.6 dB			dB
			Step 5 – Step 4	0 dB +/- 0.6 dB			dB
			Step 6 – Step 7	5 dB +/- 2.3 dB			dB
	5.9A Spectrum Emission Mask with HS-DPCCH	$\Delta f = 2.5 \sim 3.5\text{MHz}$		-35-15 * (Δf -2.5)			dBc/30KHz
		$\Delta f = 3.5 \sim 7.5\text{MHz}$		-35-1 * (Δf -3.5)			dBc/1MHz
		$\Delta f = 7.5 \sim 8.5\text{MHz}$		-39 10 * (Δf -7.5)			dBc/1MHz
		$\Delta f = 8.5 \sim 12.5\text{MHz}$		-49			dBc/1MHz
	5.10A Adjacent Channel Leakage Power Ratio with HS-DPCCH	$\Delta f = \pm 5\text{MHz}$		-	-	-33	dBm/3.84M Hz
		$\Delta f = \pm 10\text{MHz}$		-	-	-43	dBm/3.84M Hz
	5.13.1A Error Vector Magnitude (EVM) with HS-DPCCH	MAX Power -20 dBm		-	-	17.5	%
Receiver	6.3A Maximum Input Level for HS-PDSCH Reception	Pin = -25dBm		-	-	10	BLER(%)

4) GSM receiver specification

Item	Spec.
Phase Error	Rms : 5° Peak : 20 °
Frequency Error	GSM : 0.1 ppm DCS/PCS : 0.1 ppm
EMC(Radiated Spurious Emission Disturbance)	GSM/DCS : < -28dBm
Transmitter Output power and Burst Timing	GSM : 5dBm – 33dBm ± 3dB DCS/PCS : 0dBm – 30dBm ± 3dB
Burst Timing	<3.69us
Spectrum due to modulation out to less than 1800kHz offset	200kHz : -36dBm 600kHz : -51dBm/-56dBm
Spectrum due to modulation out to larger than 1800kHz offset to the edge of the transmit band	GSM : 1800-3000kHz : < -63dBc(-46dBm) 3000kHz-6000kHz : < -65dBc(-46dBm) 6000kHz < : < -71dBc(-46dBm) DCS : 1800-3000kHz : < -65dBc(-51dBm) 6000kHz < : < -73dBc(-51dBm)
Spectrum due to switching transient	400kHz : -19dBm/-22dBm(5/0), -23dBm 600kHz : -21dBm/-24dBm(5/0), -26dBm
Reference Sensitivity – TCH/FS	Class II(RBER) : -105dBm(2.439%)
Usable receiver input level range	0.012(-15 - -40dBm)
Intermodulation rejection – Speech channels	± 800kHz, ± 1600kHz : -98dBm/-96dBm (2.439%)
AM Suppression - GSM : -31dBm - DCS : -29dBm	-98dBm/-96dBm (2.439%)
Timing Advance	± 0.5T

5) Bluetooth transmitter specification

Item	Spec.		Min	Typ.	Max	Unit
Output Power	DH5 PRBS9	Power (Class I)	0	-	20	dBm
Power Density	DH5 PRBS9	@100kHz RBW		-	+20	dBm
Power Control	DH1 PRBS9	Step size	2	-	8	dB
		Power(min)		-	4	dBm
Frequency Range	DH5 PRBS9	-30dBc @100kHz RBW	2400	-	2483.5	MHz
Tx Output Spectrum (20dB BW)	DH5 PRBS9	$\Delta f = f_H - f_L$		-	1	MHz
Tx Output Spectrum (Adjacent Output Power)	DH5 PRBS9	@M-N = 2		-	-20	dBm
		@M-N ≥ 3		-	-40	dBm
Modulation Index	DH5 11110000	$f1_{AVG}$	140	-	175	kHz
	DH5 10101010	$f2_{MAX}$	115	-		kHz
		$f2_{AVG}/f1_{AVG}$	0.8	-		
Initial Carrier Freq. Tolerance	DH5 PRBS9		-75	-	+75	kHz
Carrier Frequency Drift	DH1 10101010	Freq. Drift	-25	-	+25	kHz
		Drift Rate	-20	-	+20	kHz/50us
	DH3 10101010	Freq. Drift	-40	-	+40	kHz
		Drift Rate	-20	-	+20	kHz/50us
	DH5	Freq. Drift	-40	-	+40	kHz
	10101010	Drift Rate	-20	-	+20	kHz/50us

6) Bluetooth receiver specification

Item	Spec. BER=0.1%	Min	Typ.	Max	Unit
Sensitivity (Single Slot Packet)	DH1 PRBS9 Input power @BT_w	-	-	-70	dBm
Sensitivity (Multi Slot Packet)	DH5 PRBS9 Input power @BT_w	-	-	-70	dBm
C/I Performance*	DH1 PRBS9 Input power @BT_i	C/I@Co-Channel -60dBm @ BT	-	11	dB
		C/I @1MHz -60dBm@ BT	-	0	
		C/I @2MHz -60dBm@ BT	-	-30	
		C/I @≥3MHz -67dBm@ BT	-40	-37	
		C/I @Image -67dBm@ BT	-	-9	
		C/I @ Image±1MHz -67dBm@ BT	-	-20	
Blocking Performance	DH1 PRBS9 Input power @CW	30M ~2 GHz -67dBm@ BT	-10	-	dBm
		2 ~ 2.4 GHz -67dBm@ BT	-27	-	
		2.5 ~ 3 GHz -67dBm@ BT	-27	-	
		3 ~ 12.75 GHz -67dBm@ BT	-10	-	
Intermodulation Performance	DH1 PRBS9 Input power @CW_i	$\Delta f = +5\text{MHz}$ (CW_i) $\Delta f = +10\text{MHz}$ (BT_i) -64dBm@ BT_w	-39	-	dBm
		$\Delta f = -5\text{MHz}$ (CW_i) $\Delta f = -10\text{MHz}$ (BT_i) -64dBm@ BT_w	-39	-	dBm
Maximum Input Level	DH1 PRBS9 Input power @BT_w	-	-20	-	dBm

7) WLAN 802.11b transceiver specification

Item	Specification
Transmit Frequency	2400 MHz ~ 2483.5 MHz (CH1~CH13)
Tx Power Level	$\leq 20\text{dBm}$ under (Europe), $\leq 30\text{dBm}$ under (USA)
Frequency Tolerance	within ± 25 PPM
Chip clock Frequency Tolerance	within ± 25 PPM
Spectrum Mask	≤ -30 @ $f_c - 22\text{MHz} < f < f_c - 11\text{MHz}$ and $f_c + 11\text{MHz} < f < f_c + 22\text{MHz}$ ≤ -50 @ $f < f_c - 22\text{MHz}$ and $f > f_c + 22\text{MHz}$
Power ramp on/off time	$\leq 2\mu\text{s}$
Carrier Suppression	$\leq -15\text{dB}$
Modulation Accuracy (Peak EVM)	$\leq 35\%$
Spurious Emissions	$< -36\text{ dBm}$ @ 30MHz ~ 1GHz $< -30\text{ dBm}$ above @ 1GHz ~ 12.75GHz $< -47\text{ dBm}$ @ 1.8GHz ~ 1.9GHz $< -47\text{ dBm}$ @ 5.15GHz ~ 5.3GHz
Rx Min input Sensitivity	$\leq -76\text{dBm}$ (1Mbps,2Mbps,5.5Mbps,11Mbps) @ FER $\leq 8\%$
Rx Max input Sensitivity	$\geq -10\text{dBm}$ (1Mbps,2Mbps,5.5Mbps,11Mbps) @ FER $\leq 8\%$
Rx Adjacent Channel Rejection	$\geq 35\text{dB}$ @ FER $\leq 8\%$, interference input signal -70dBm @ $f_c \pm 25\text{MHz}$ (11Mbps)

8) WLAN 802.11g transceiver specification

Item	Specification
Transmit Frequency	2400 MHz ~ 2483.5 MHz (CH1~CH13)
Tx Power Level	≤ 20dBm under (Europe), ≤ 30dBm under (USA)
Frequency Tolerance	within ±25 PPM
Chip clock Frequency Tolerance	within ±25 PPM
Spectrum Mask	≤ -20 @ ±11MHz offset (9Mhz ~ 11MHz) ≤ -28 @ ±20MHz offset (11MHz ~ 20Mhz) ≤ -40 @ ±30MHz offset (20MHz ~ 30Mhz)
Transmitter constellation error (rms EVM)	≤ -5dB
Spurious Emissions	< -36 dBm @ 30MHz ~ 1GHz < -30 dBm above @ 1GHz ~ 12.75GHz < -47 dBm @ 1.8GHz ~ 1.9GHz < -47 dBm @ 5.15GHz ~ 5.3GHz
Rx Min input Sensitivity	PER ≤ 10% -82dBm@6Mbps, -81dBm@9Mbps, -79dBm@12Mbps -77dBm@18Mbps, -74dBm@24Mbps, -70dBm@36Mbps -66dBm@48Mbps, -65dBm@54Mbps
Rx Max input Sensitivity	≥ -20dBm(6,9,12,18,24,36,48,54Mbps) @ PER ≤ 10%
Rx Adjacent Channel Rejection	PER ≤ 10%, ACR ≥ 16dB@6Mbps, ACR ≥ 15dB@9Mbps, ACR ≥ 13dB@12Mbps, ACR ≥ 11dB@18Mbps, ACR ≥ 8dB@24Mbps, ACR ≥ 4dB@36Mbps ACR ≥ 0dB@48Mbps, ACR ≥ -1dB@54Mbps ※ ACR shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in min input sensitivity

9) WLAN 802.11n transceiver specification

Item	Specification
Transmit Frequency	2400 MHz ~ 2483.5 MHz (CH1~CH13)
Tx Power Level	$\leq 20\text{dBm}$ under (Europe), $\leq 30\text{dBm}$ under (USA)
Frequency Tolerance	within ± 25 PPM
Chip clock Frequency Tolerance	within ± 25 PPM
Spectrum Mask	≤ -20 @ $\pm 11\text{MHz}$ offset (9Mhz ~ 11MHz) ≤ -28 @ $\pm 20\text{MHz}$ offset (11MHz ~ 20Mhz) ≤ -45 @ $\pm 30\text{MHz}$ offset (20MHz ~ 30Mhz)
Transmitter constellation error (rms EVM)	$\leq -5\text{dB}@6.5\text{Mbps}$, $\leq -10\text{dB}@13\text{Mbps}$, $\leq -13\text{dB}@19.5\text{Mbps}$, $\leq -16\text{dB}@26\text{Mbps}$, $\leq -19\text{dB}@39\text{Mbps}$, $\leq -22\text{dB}@52\text{Mbps}$, $\leq -25\text{dB}@58.5\text{Mbps}$, $\leq -28\text{dB}@65\text{Mbps}$
Spurious Emissions	< -36 dBm @ 30MHz ~ 1GHz < -30 dBm above @ 1GHz ~ 12.75GHz < -47 dBm @ 1.8GHz ~ 1.9GHz < -47 dBm @ 5.15GHz ~ 5.3GHz
Rx Min input Sensitivity	PER $\leq 10\%$ $-82\text{dBm}@6.5\text{Mbps}$, $-79\text{dBm}@13\text{Mbps}$, $-77\text{dBm}@19.5\text{Mbps}$ $-74\text{dBm}@26\text{Mbps}$, $-70\text{dBm}@39\text{Mbps}$, $-66\text{dBm}@52\text{Mbps}$ $-65\text{dBm}@58.5\text{Mbps}$, $-64\text{dBm}@65\text{Mbps}$
Rx Max input Sensitivity	$\geq -20\text{dBm}(6.5, 13, 19.5, 26, 39, 52, 58.5, 65\text{Mbps})$ @ PER $\leq 10\%$
Rx Adjacent Channel Rejection	PER $\leq 10\%$, ACR $\geq 16\text{dB}@6.5\text{Mbps}$, ACR $\geq 13\text{dB}@13\text{Mbps}$, ACR $\geq 11\text{dB}@19.5\text{Mbps}$, ACR $\geq 8\text{dB}@26\text{Mbps}$, ACR $\geq 4\text{dB}@39\text{Mbps}$, ACR $\geq 0\text{dB}@52\text{Mbps}$ ACR $\geq -1\text{dB}@58.5\text{Mbps}$, ACR $\geq -2\text{dB}@65\text{Mbps}$ ※ACR shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in min input sensitivity

10) GPS receiver specification

Item	Spec.
Sensitivity Test	Success rate $\geq$ 60% @ -154 dBm

11) Current consumption

** Test condition for Standby current consumption should be as below.

: Measurement time 1hr, with Agilent & CMT JIG

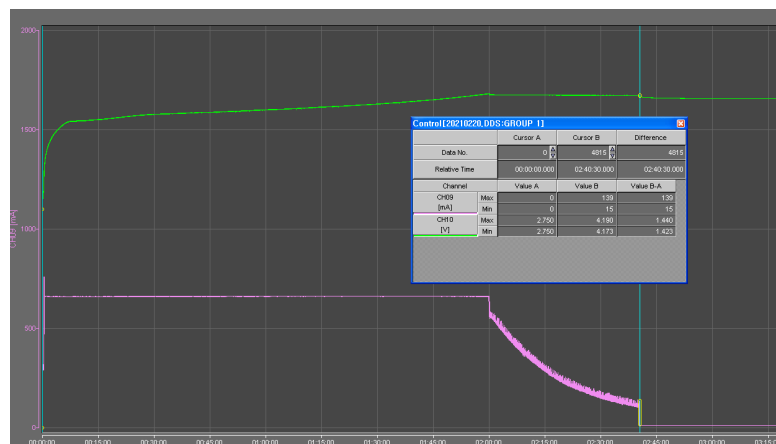
Item	Specification	
	WCDMA	GSM
1. Sleep Mode	Under 5mA	Under 5mA
2. Sleep : connector Ear jack	Under 6mA	Under 6mA
3. Current(Sleep & Idle AVG)	Under 10mA @ DRX 7	Under 10mA @ P.P 5
4. BT Connected idle	Under 11.5mA	Under 11.5 mA
5. Talk Mode	- Max Pwr Under 600mA @ Avg - Tx 10dBm Under 300 mA @ Avg	GSM LVL 5 under 300 mA @ Avg
6. Power Off Mode	under 300uA	Under 300 uA
7. Backup Battery	under 5uA	under 5uA

12) RSSI indicator (Based on Cell power)

BAR	WCDMA	GSM/DCS/PCS
5	Over -86 ±3dBm	Over -90 ±2dBm
5→4	-86± 3dBm	-90± 2dBm
4→3	-92± 3dBm	-96± 2dBm
3→2	-99± 3dBm	-98± 2dBm
2→1	-102± 3dBm	-102± 2dBm
1→0	-109± 3dBm	-104± 2dBm

13) Charging Time

Under 3 hour 30 min (1900mAh battery, 700mA TA)



14) Cut-off Voltage : 3.5V ± 0.1V

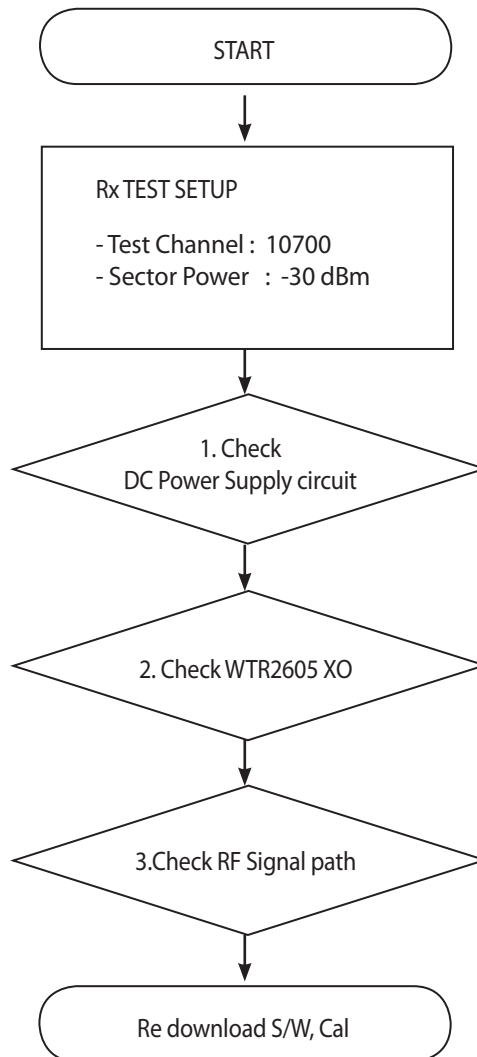
15) Battery indicator

Battery Bar	Specification	
BAR 20 (Full)	98% over	remain%
BAR 20 --> 19	98% ◇ 97%	
BAR 19 --> 18	93% ◇ 92%	
BAR 18 --> 17	88% ◇ 87%	
BAR 17 --> 16	83% ◇ 82%	
BAR 16 --> 15	78% ◇ 77%	
BAR 15 --> 14	73% ◇ 72%	
BAR 14 --> 13	68% ◇ 67%	
BAR 13 --> 12	63% ◇ 62%	
BAR 12 --> 11	58% ◇ 57%	
BAR 11 --> 10	53% ◇ 52%	
BAR 10 --> 9	48% ◇ 47%	
BAR 9 --> 8	43% ◇ 42%	
BAR 8 --> 7	38% ◇ 37%	
BAR 7 --> 6	33% ◇ 32%	
BAR 6 --> 5	28% ◇ 27%	
BAR 5 --> 4	23% ◇ 22%	
BAR 4 --> 3	16% ◇ 15%	
BAR 3 --> 2	13% ◇ 12%	
BAR 2 --> 1	8% ◇ 7%	
BAR 1 --> 0	3% ◇ 2%	
Low Battery Pop-up	16% → 15% : One Time popup (No call) There is no pop-up until 5% if you delete pop-up	
Critical Low Battery Pop-up	6% → 5% : One Time popup (No call)	
Power Off	1%	

3. TROUBLE SHOOTING

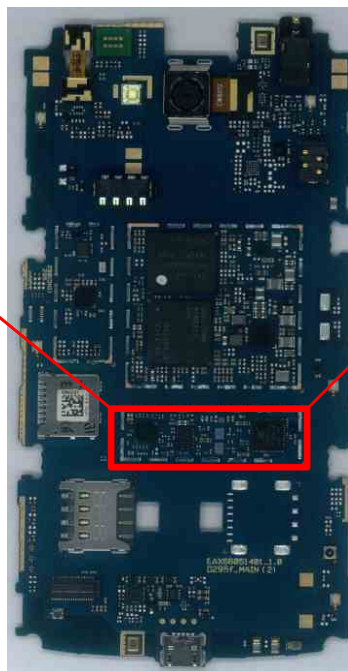
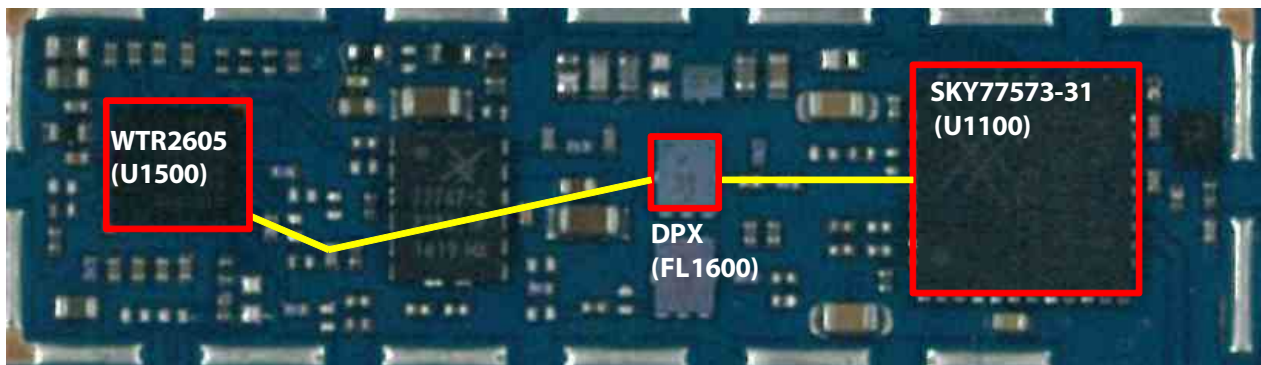
3.1 WCDMA Rx Part

3.1.1 WCDMA_B1 Rx

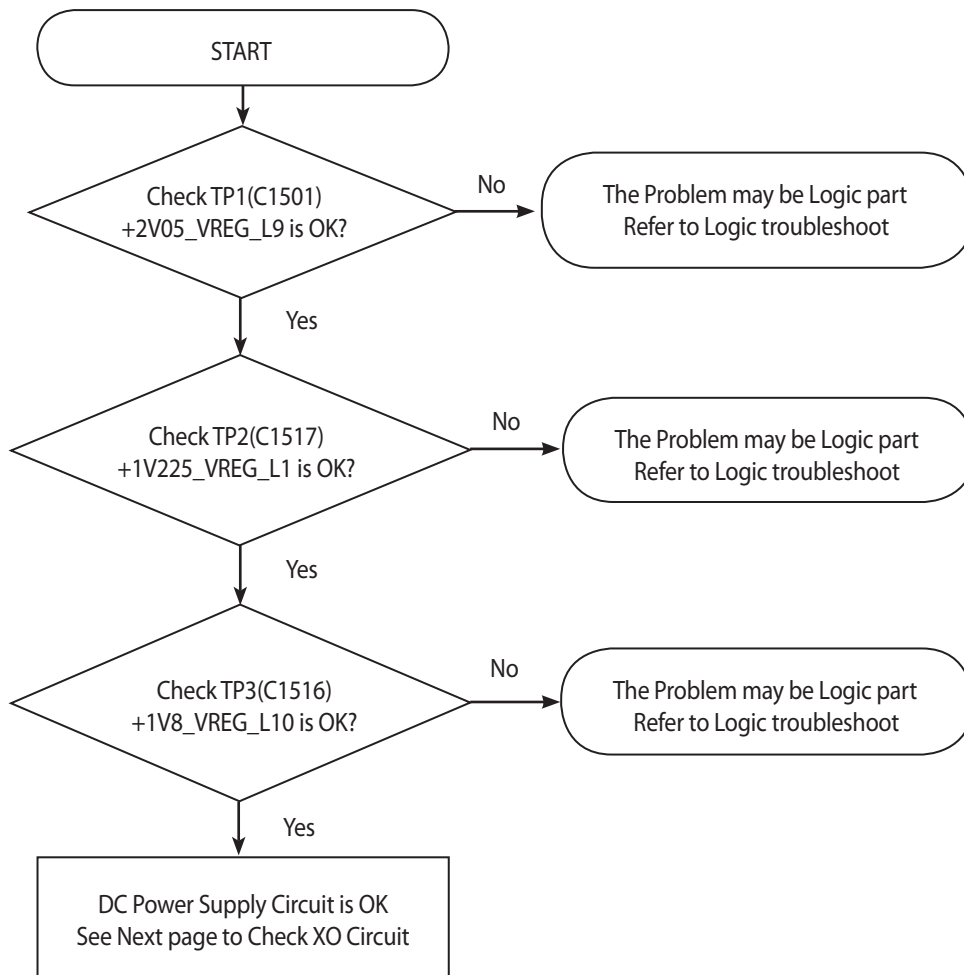


* there are no Test Points on the IQ Data Line. So It is impossible to check the IQ Data Wave form

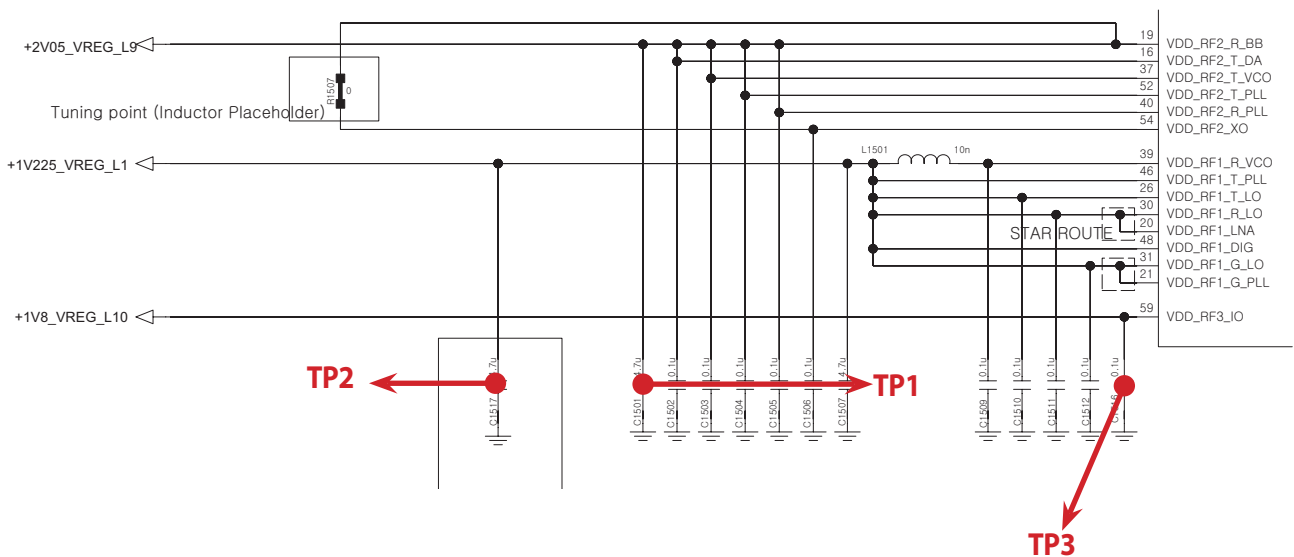
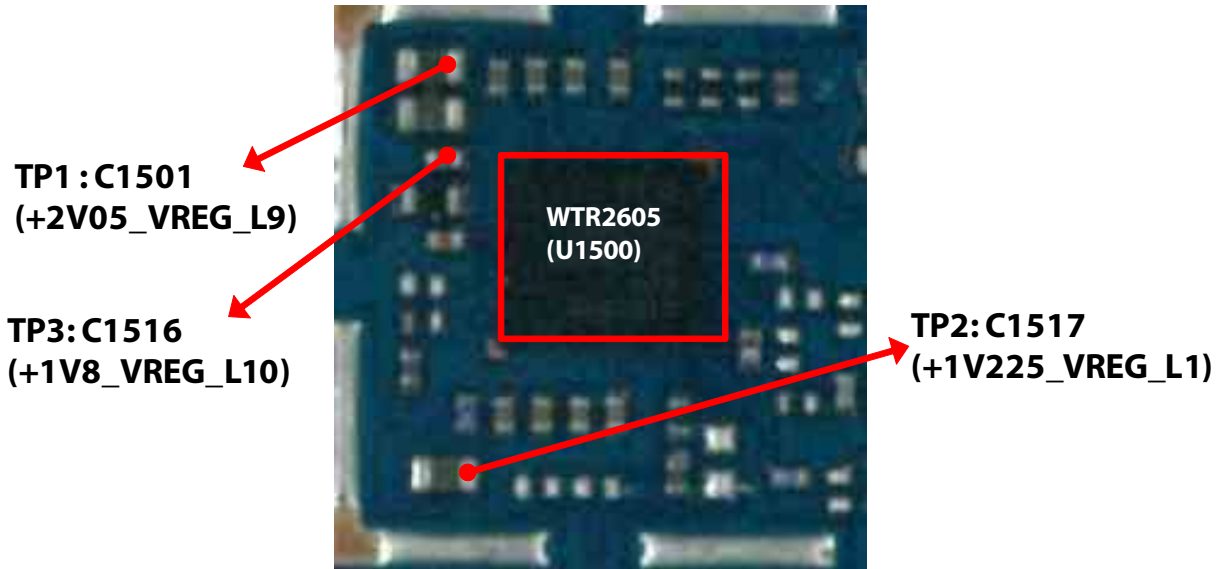
3. TROUBLE SHOOTING



3.1.1.1 Checking DC Power supply circuit



3. TROUBLE SHOOTING



3.1.1.2 Checking XO circuit

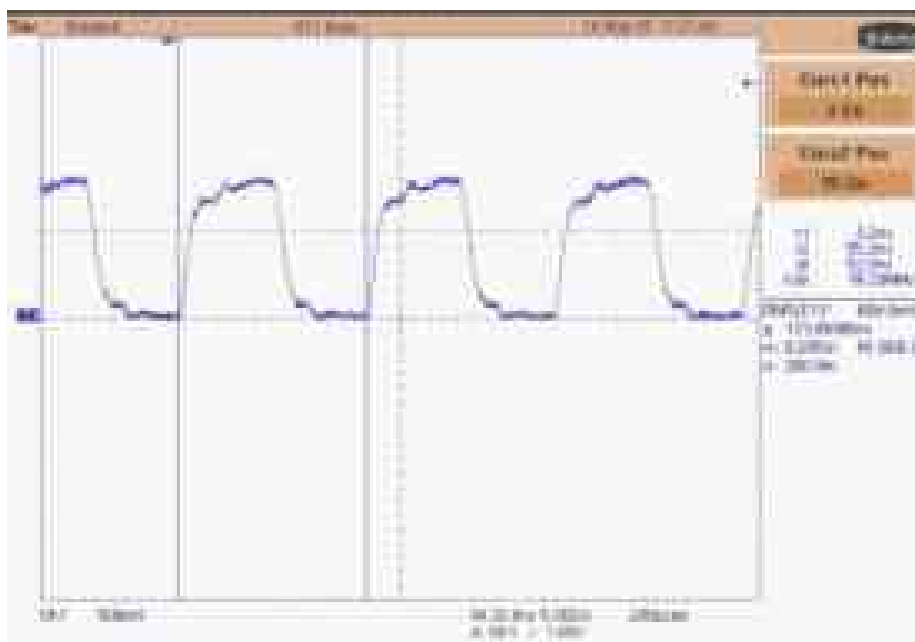
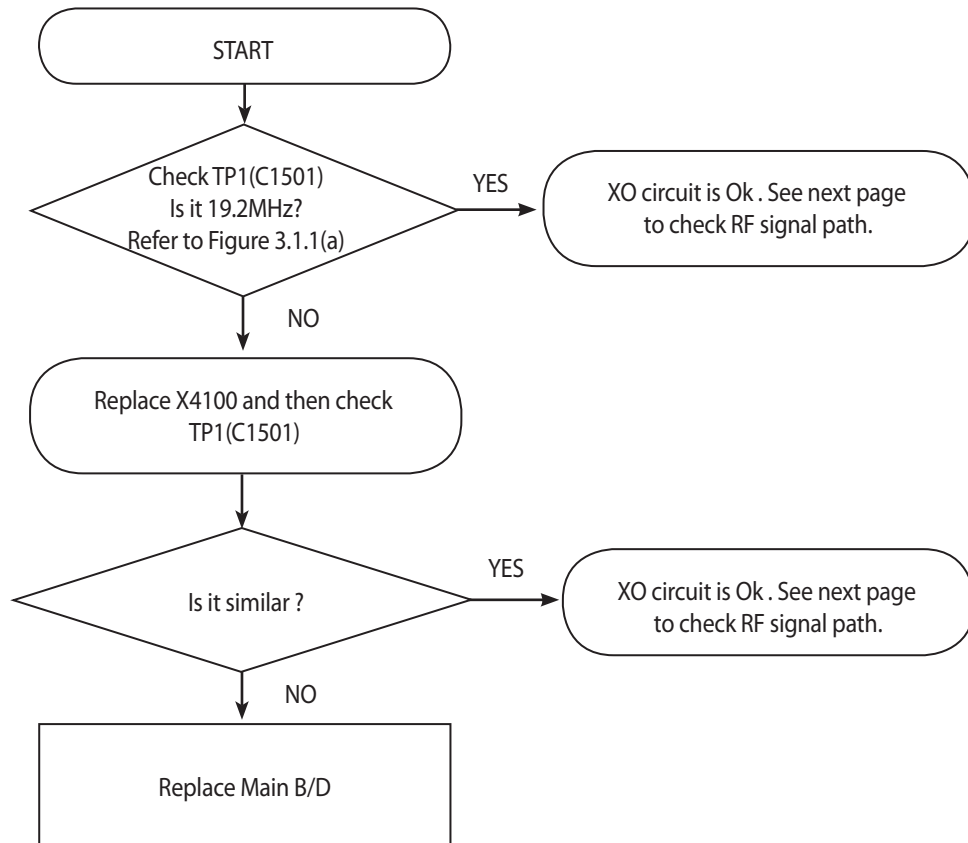
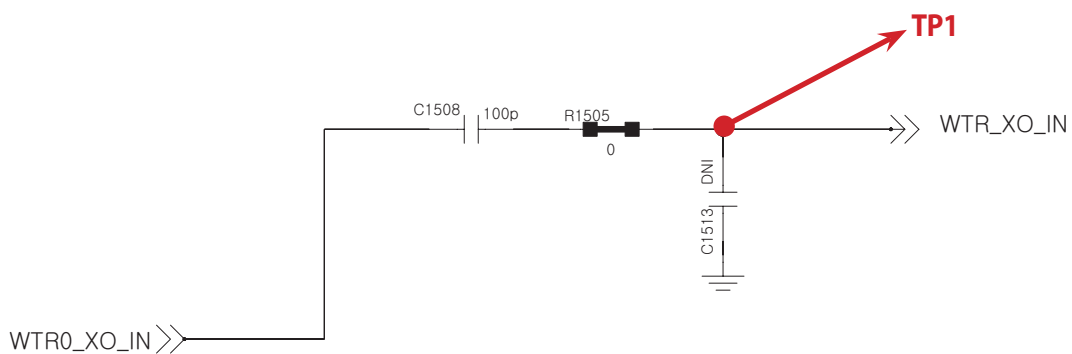
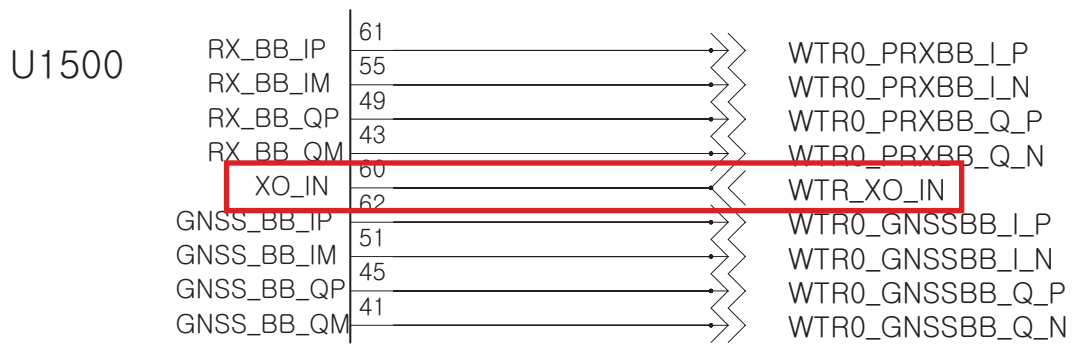
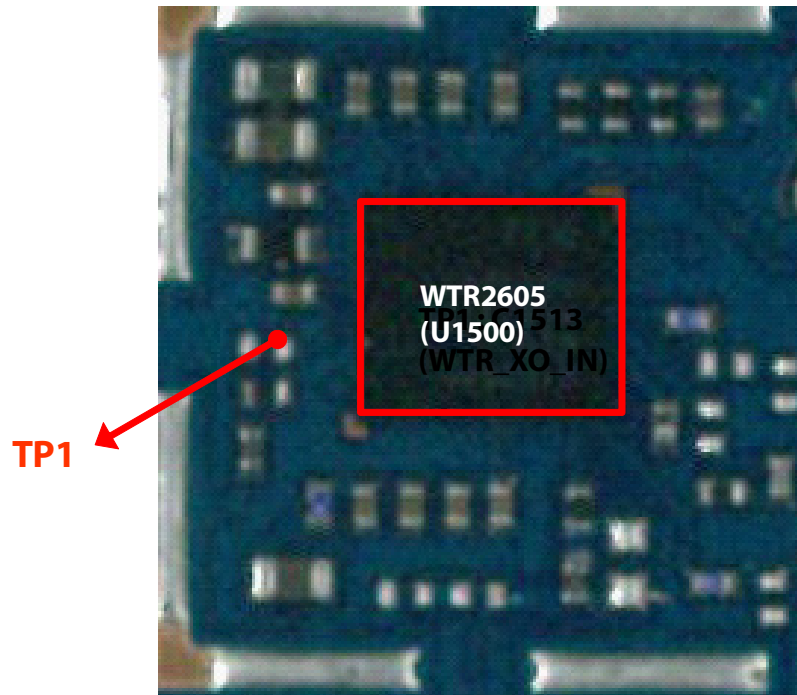
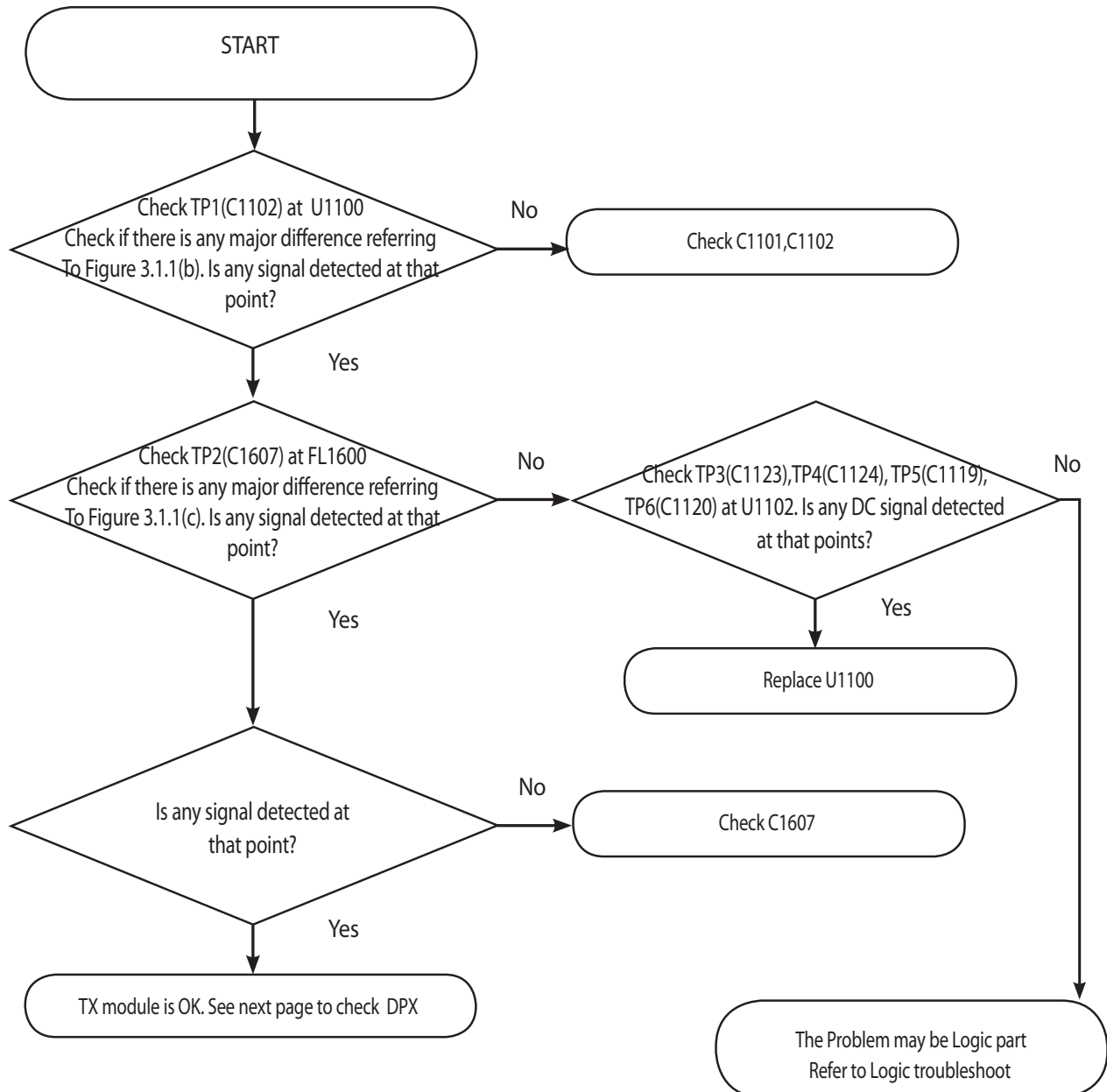


Figure 3.1.1 (a)

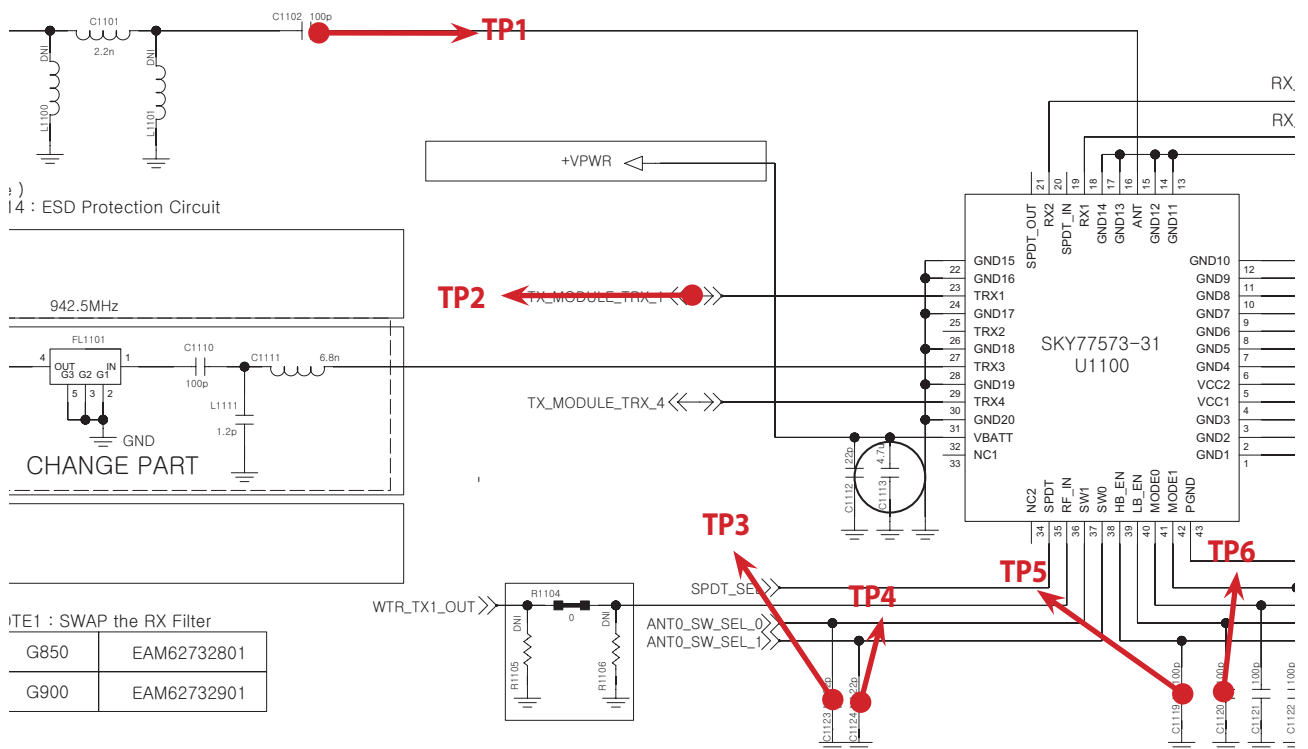
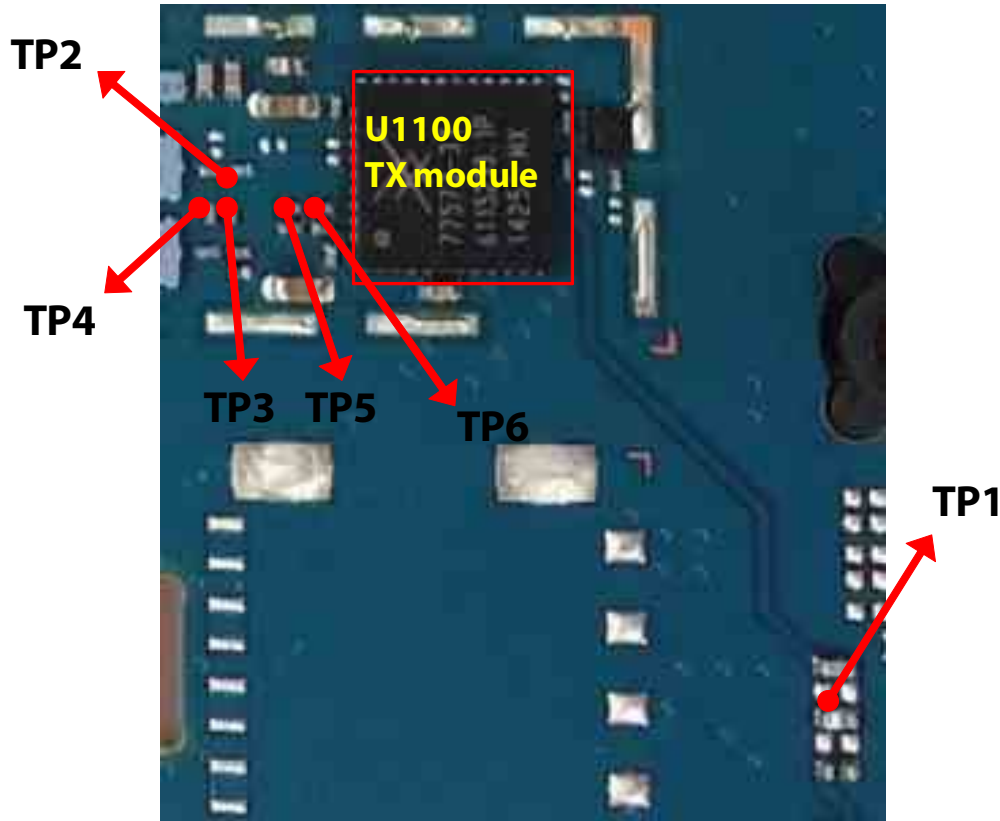
3. TROUBLE SHOOTING



3.1.1.3 Checking RF signal path (TX Module)



3. TROUBLE SHOOTING



TP1

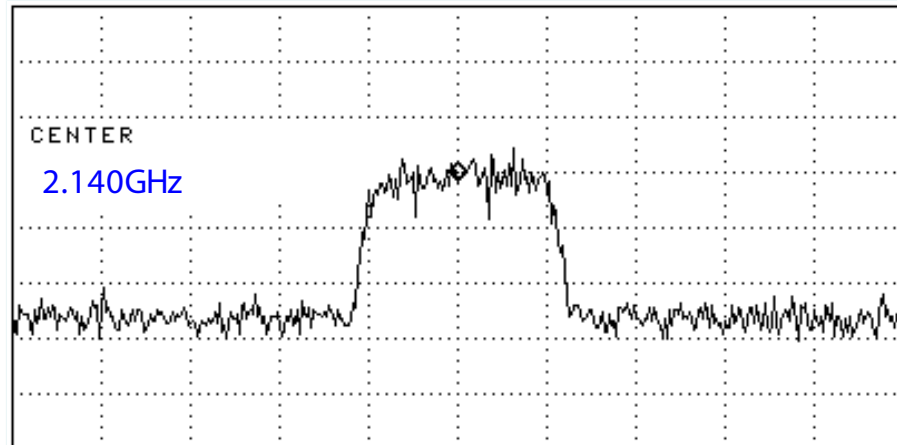


Figure 3.1.1 (b)

TP2

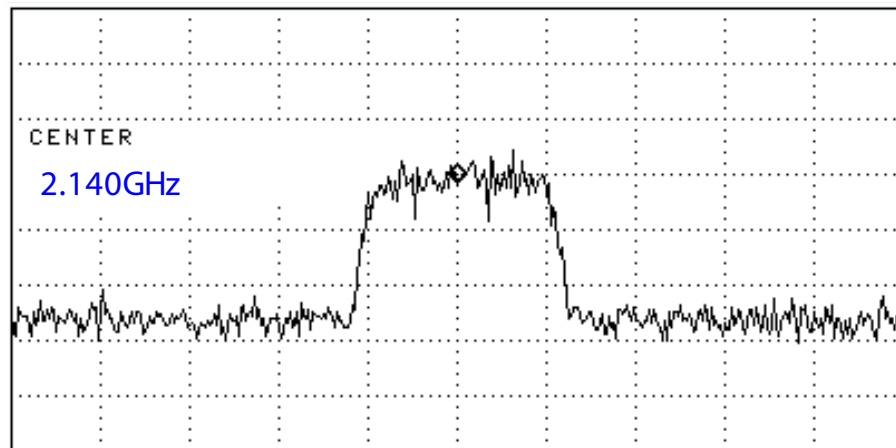
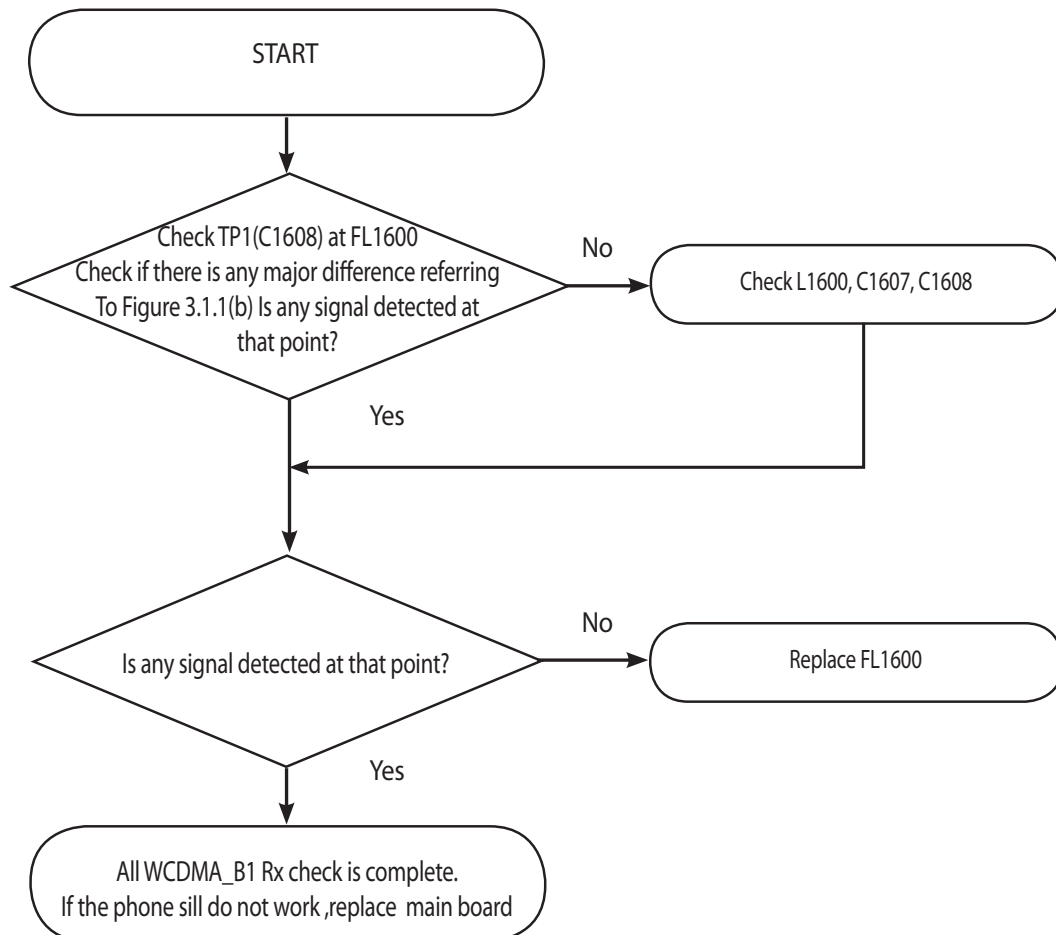
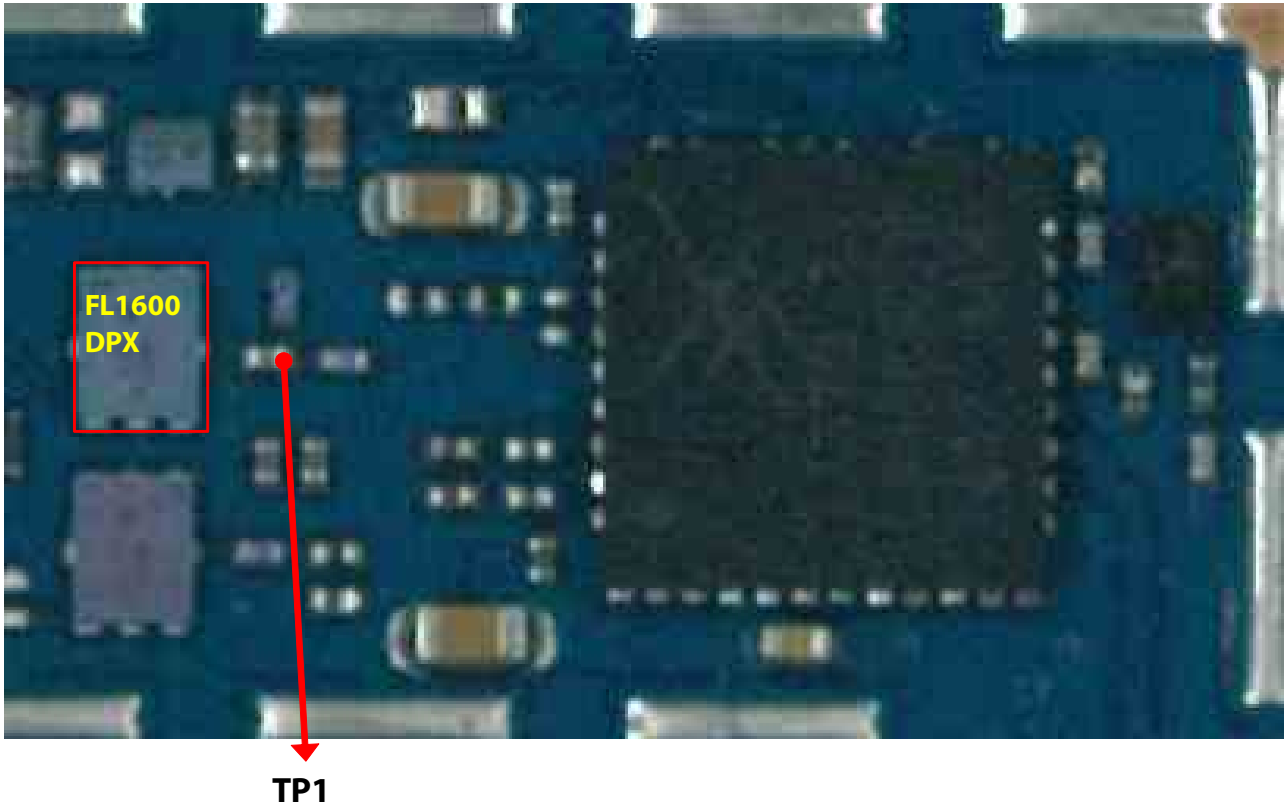


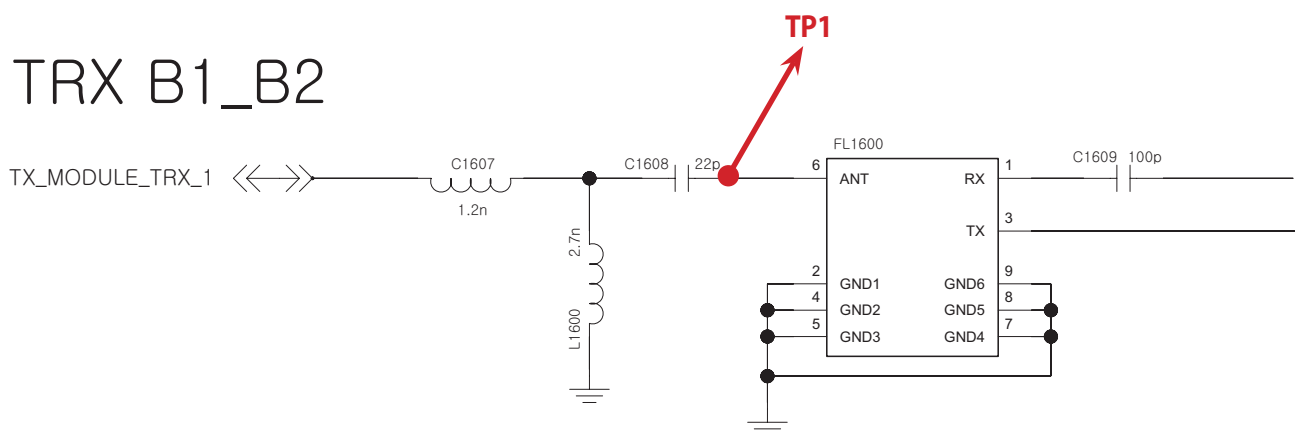
Figure 3.1.1 (c)

3.1.1.4 Checking RF signal path (Duplexer)





TRX B1_B2



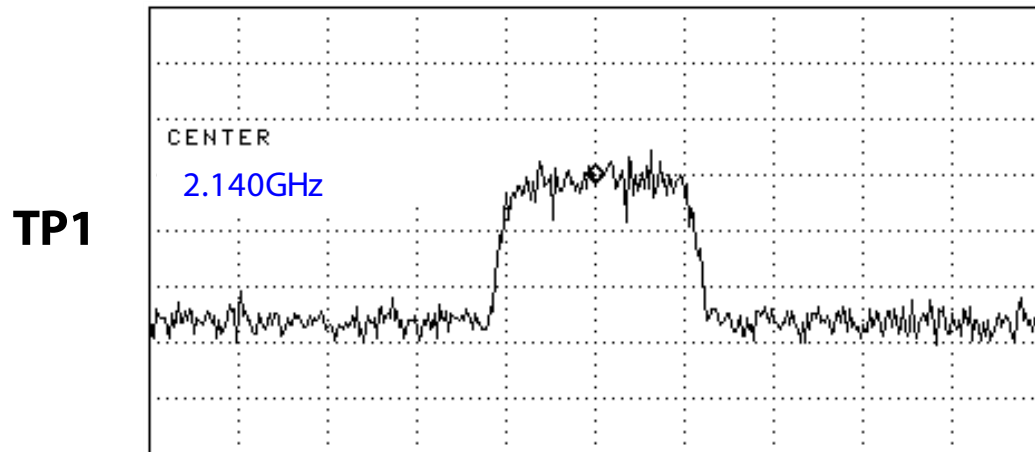
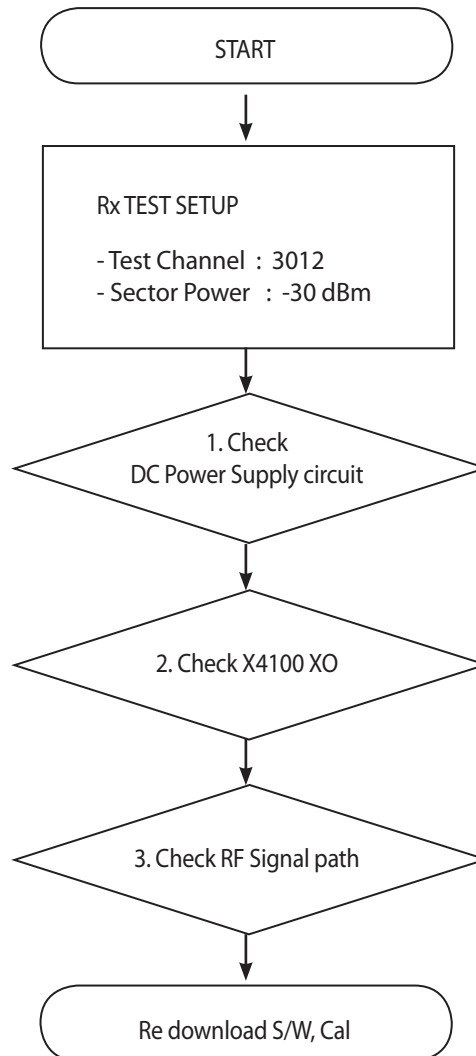


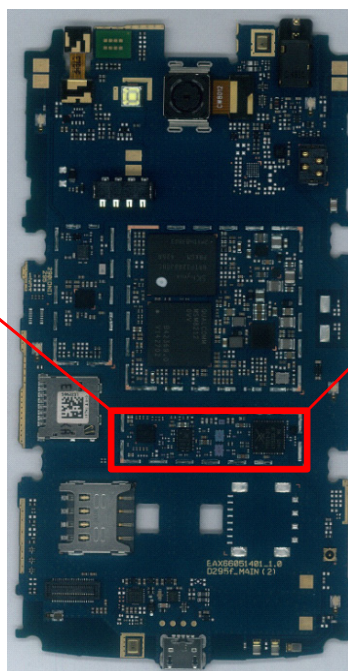
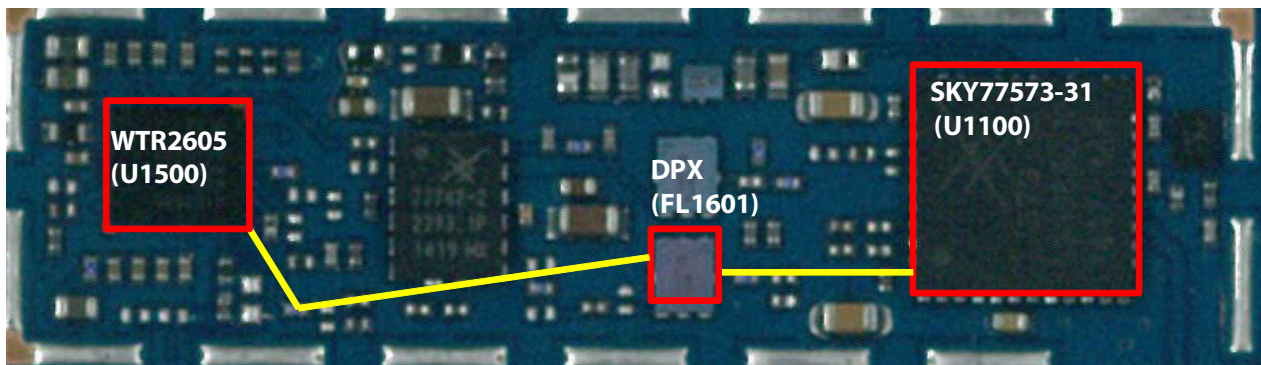
Figure 3.1.1 (b)

3.1.2 WCDMA_B8 Rx

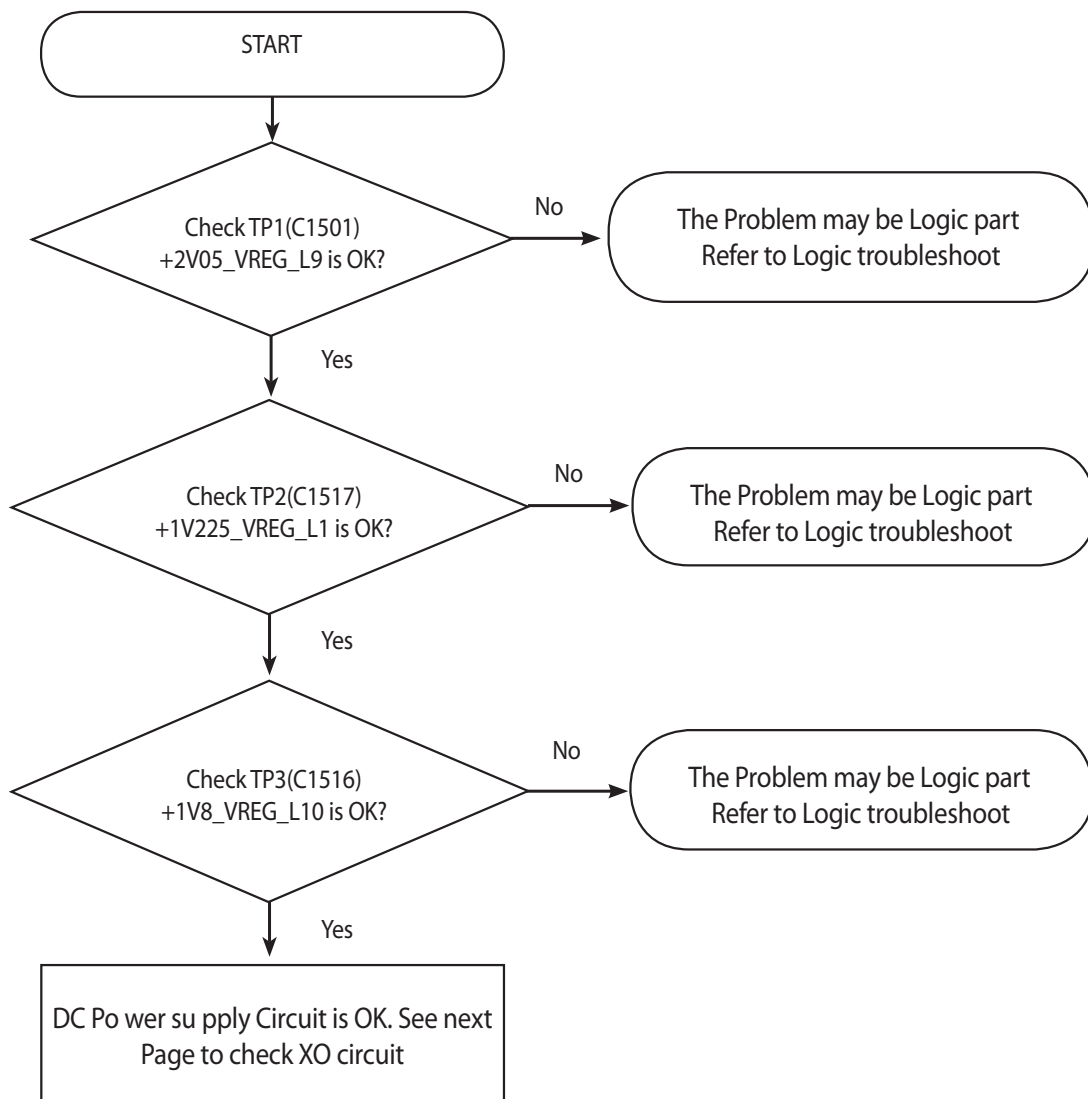


* there are no Test Points on the IQ Data Line. So It is impossible to check the IQ Data Wave form

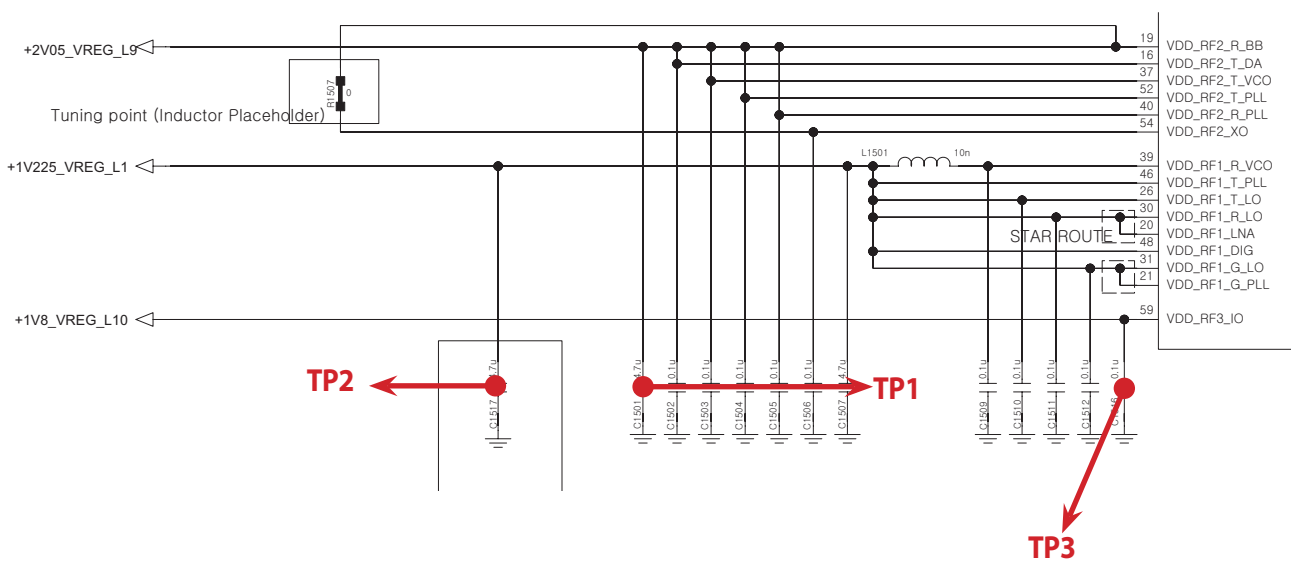
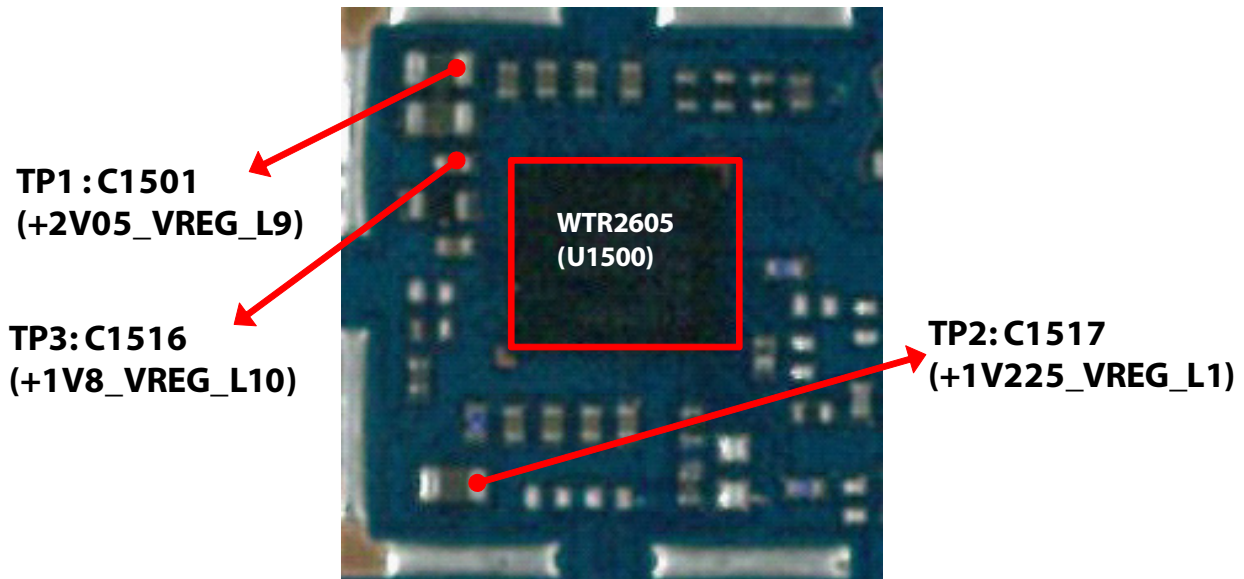
3. TROUBLE SHOOTING



3.1.2.1 Checking DC Power supply circuit



3. TROUBLE SHOOTING



3.1.2.2 Checking XO circuit

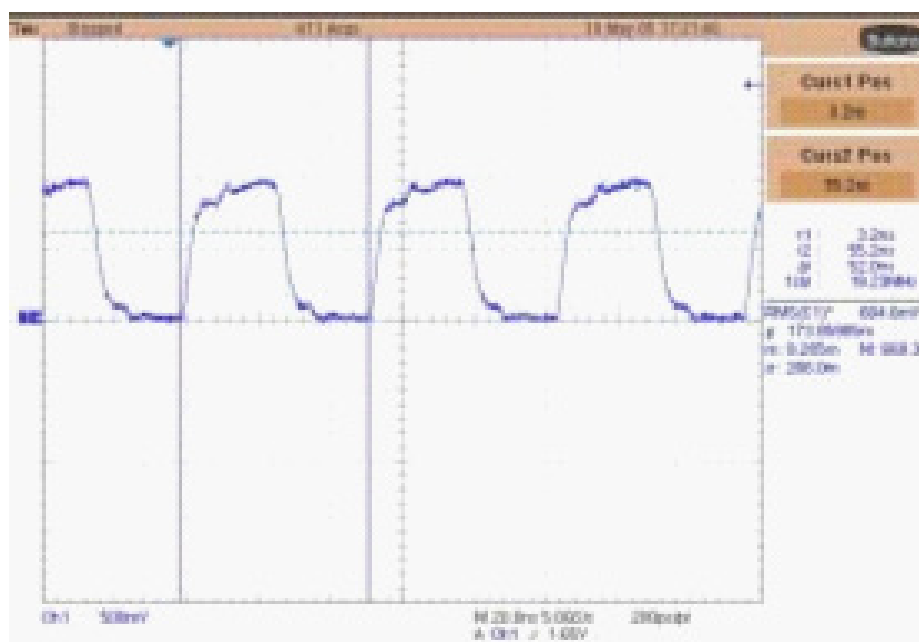
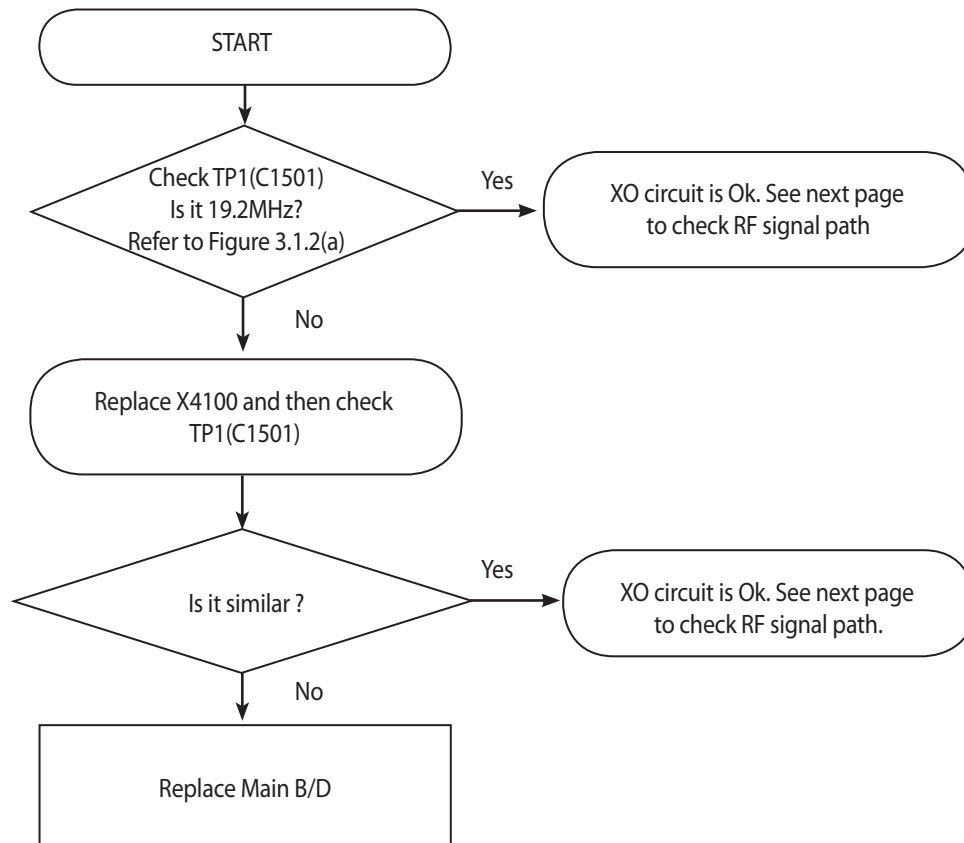
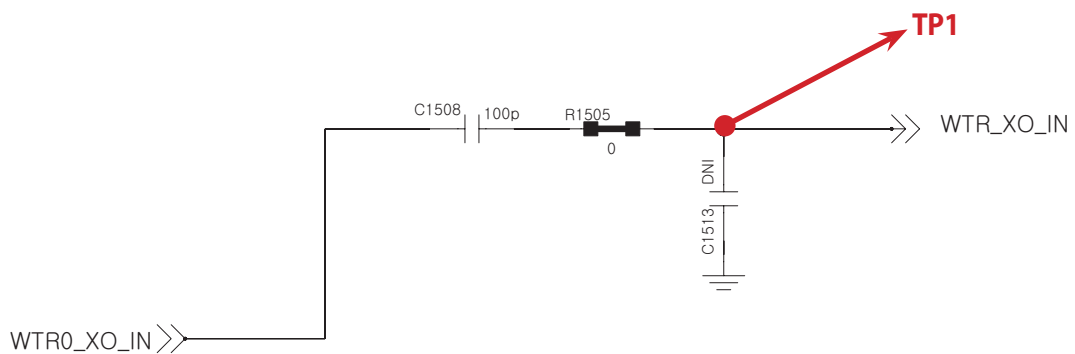
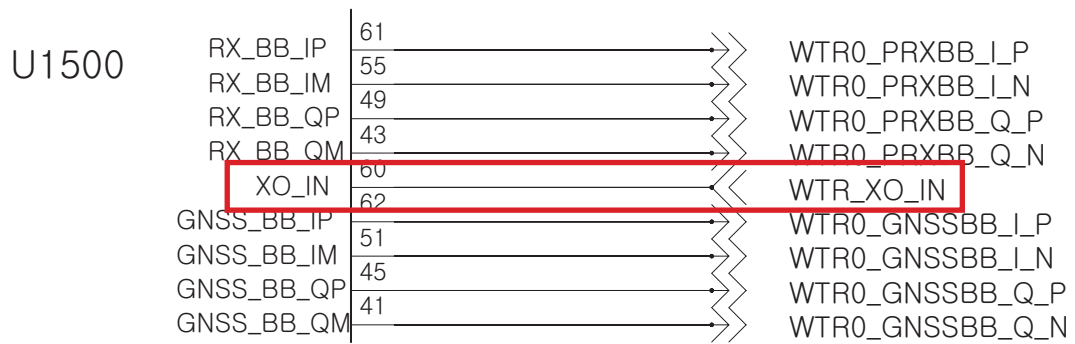
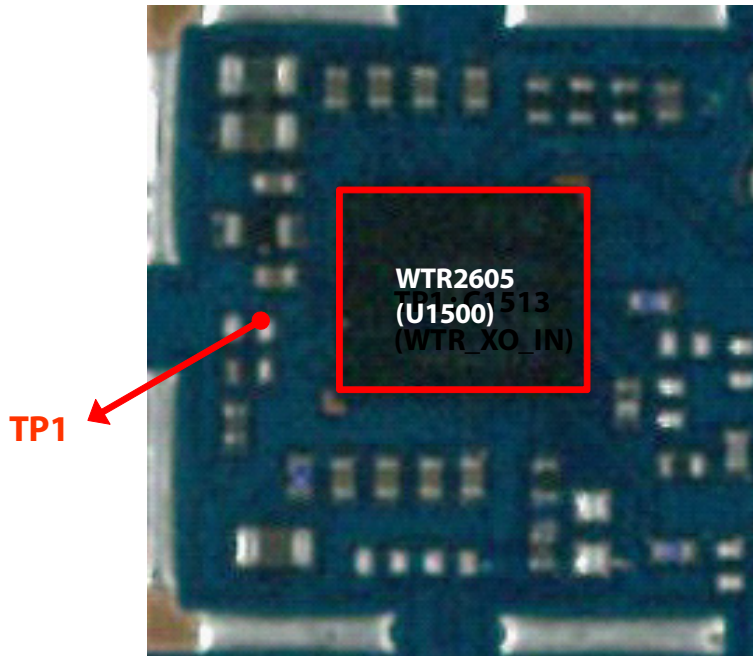
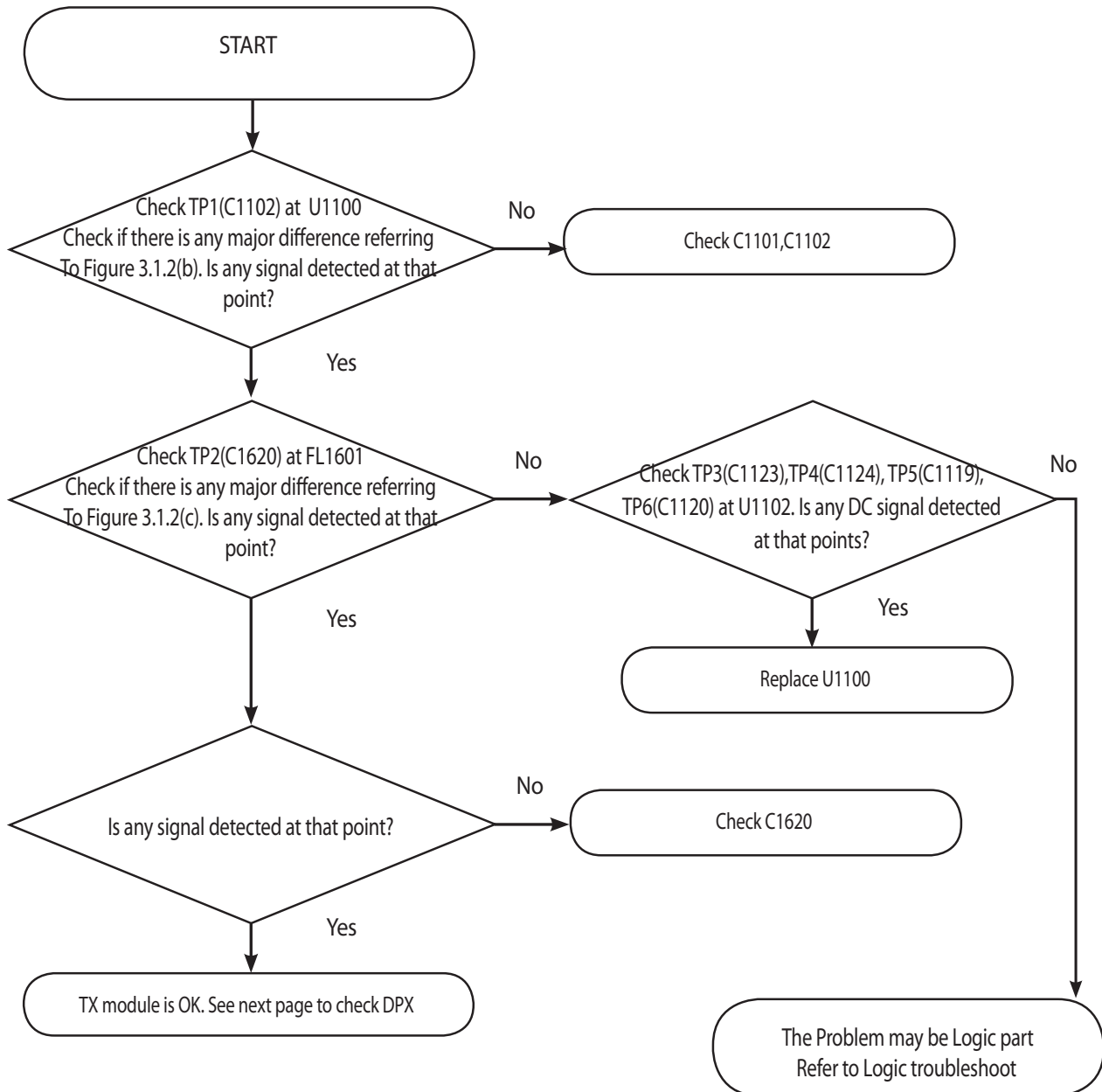


Figure 3.1.2 (a)

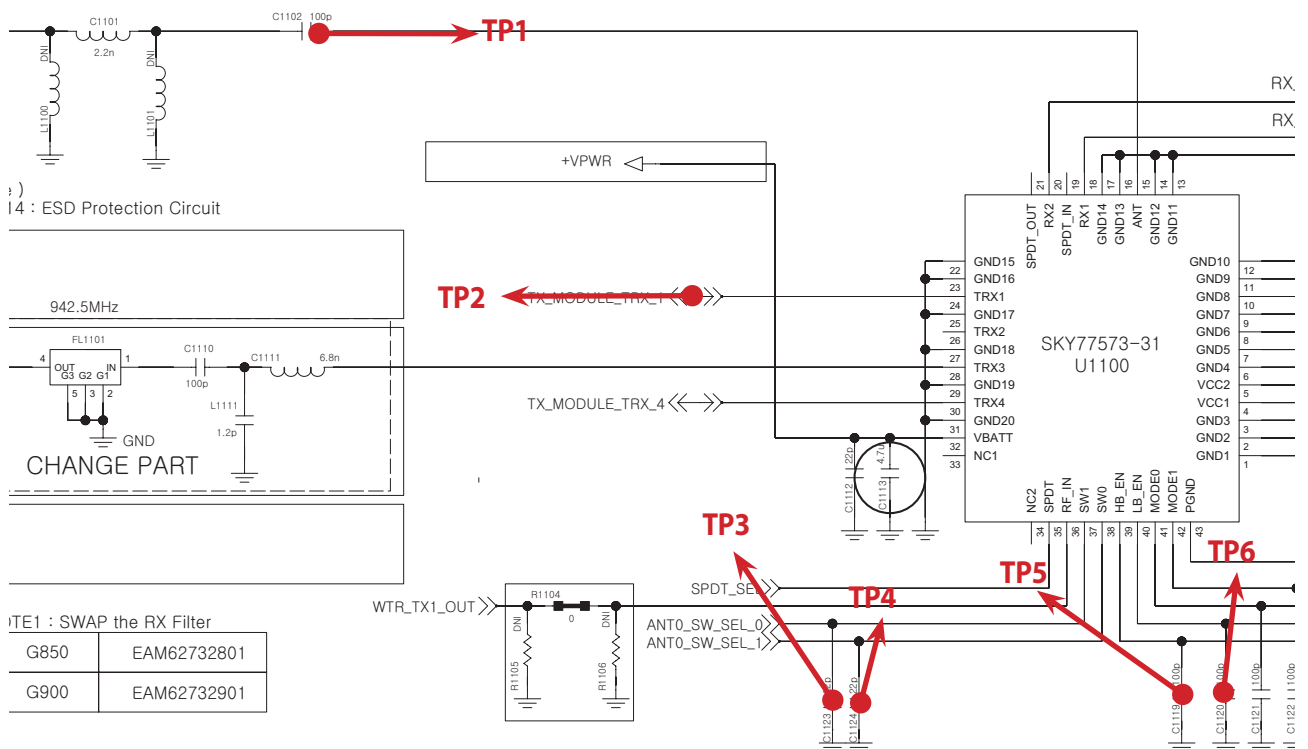
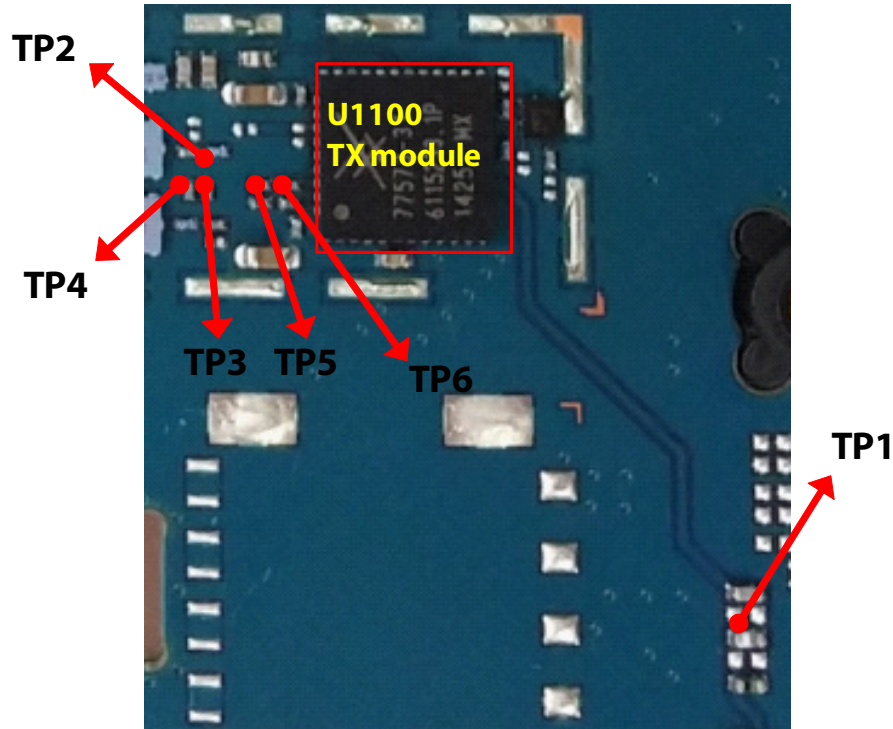
3. TROUBLE SHOOTING



3.1.2.3 Checking RF signal path (TX Module)



3. TROUBLE SHOOTING



TP1

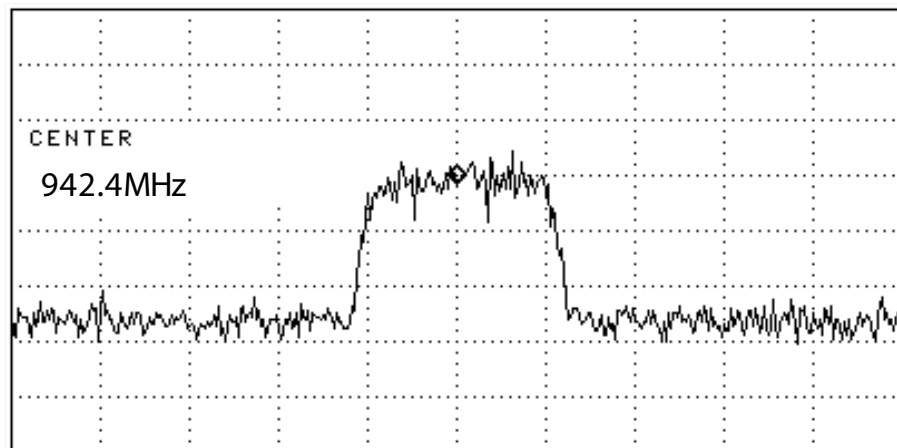


Figure 3.1.2 (b)

TP2

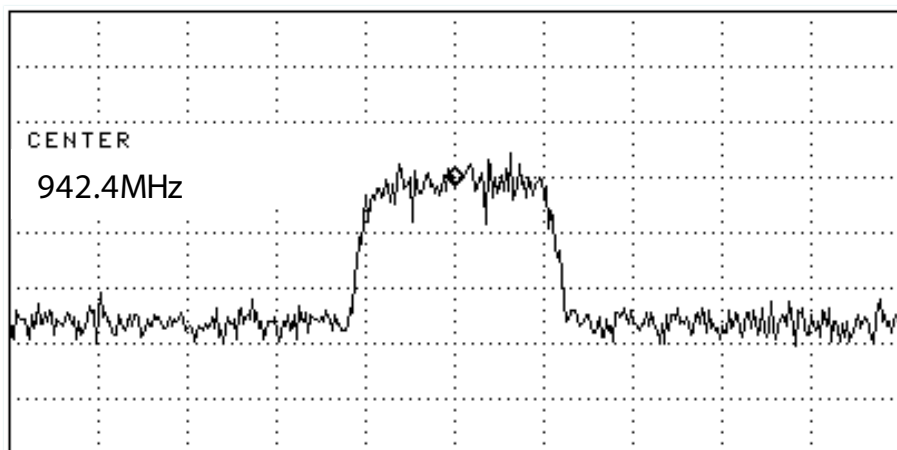


Figure 3.1.2 (c)

3.1.2.4 Checking RF signal path (Duplexer)

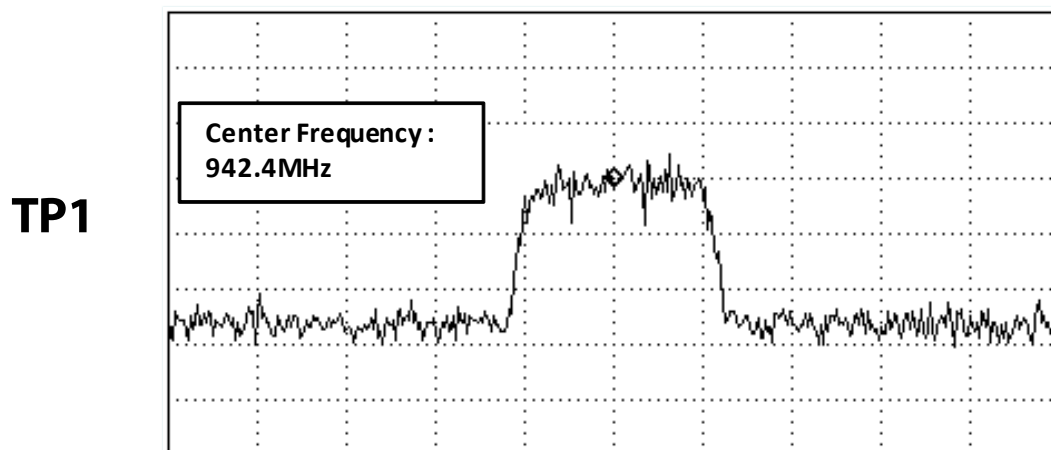
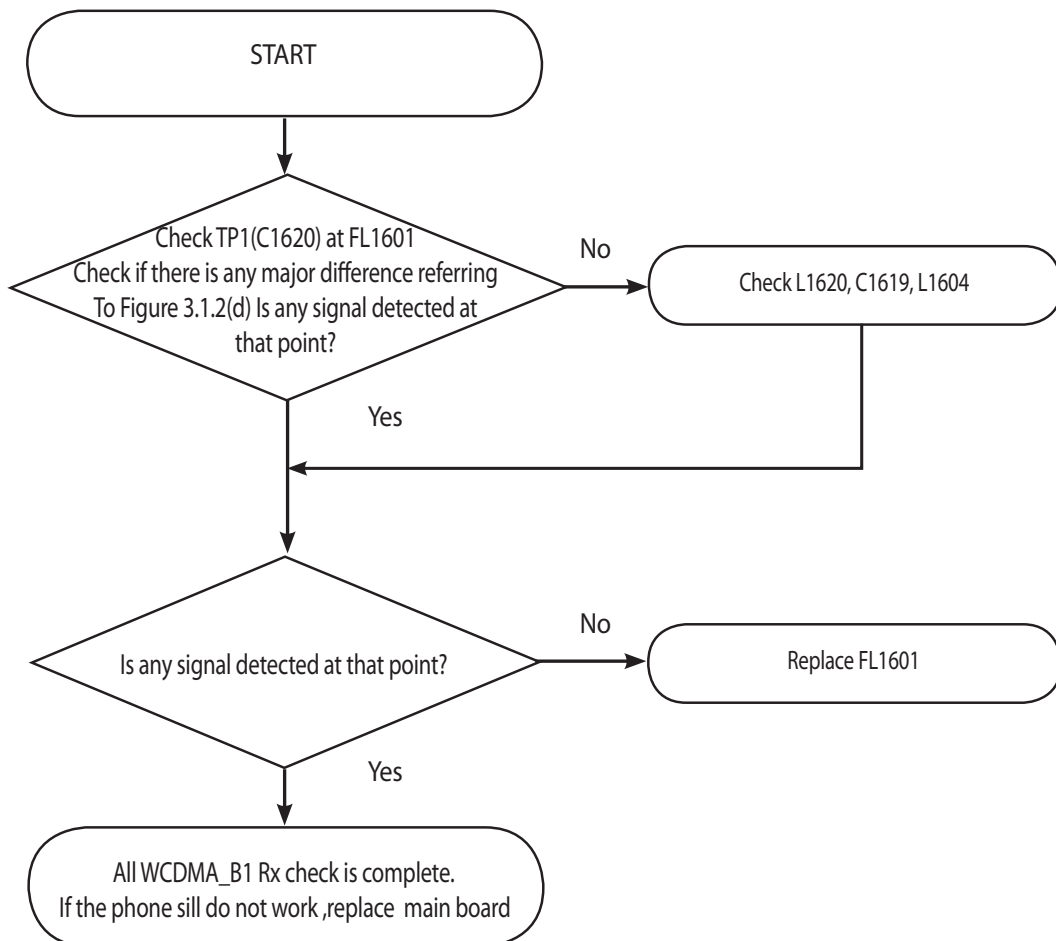
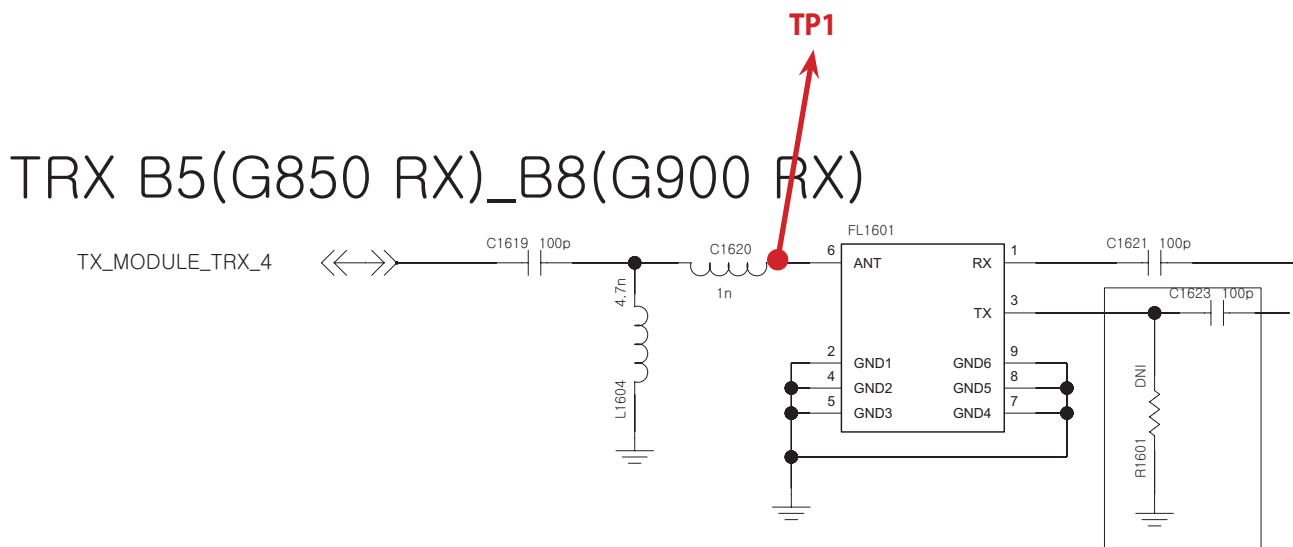
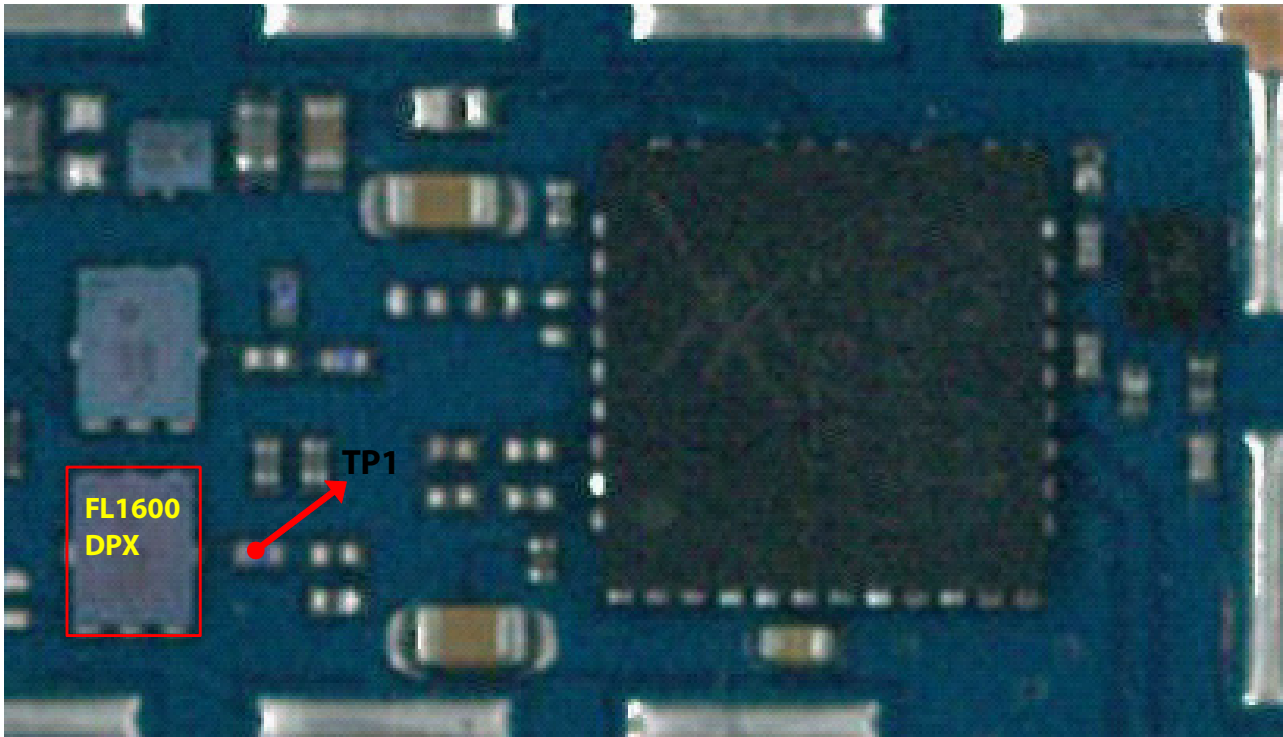
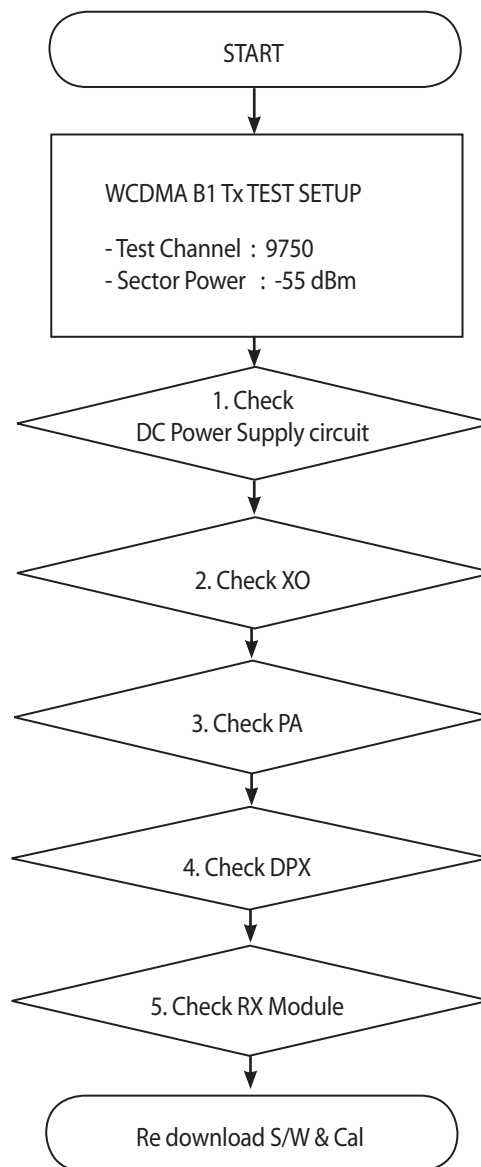


Figure 3.1.2 (d)



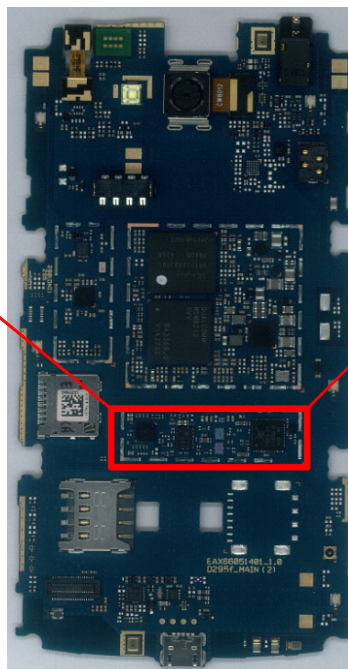
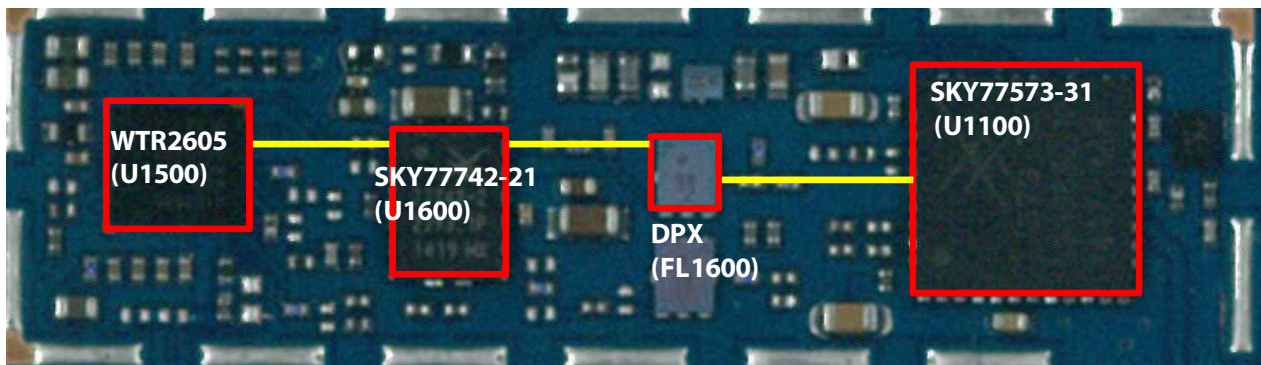
3.2 WCDMA Tx Part Trouble

3.2.1 WCDMA_B1 Tx

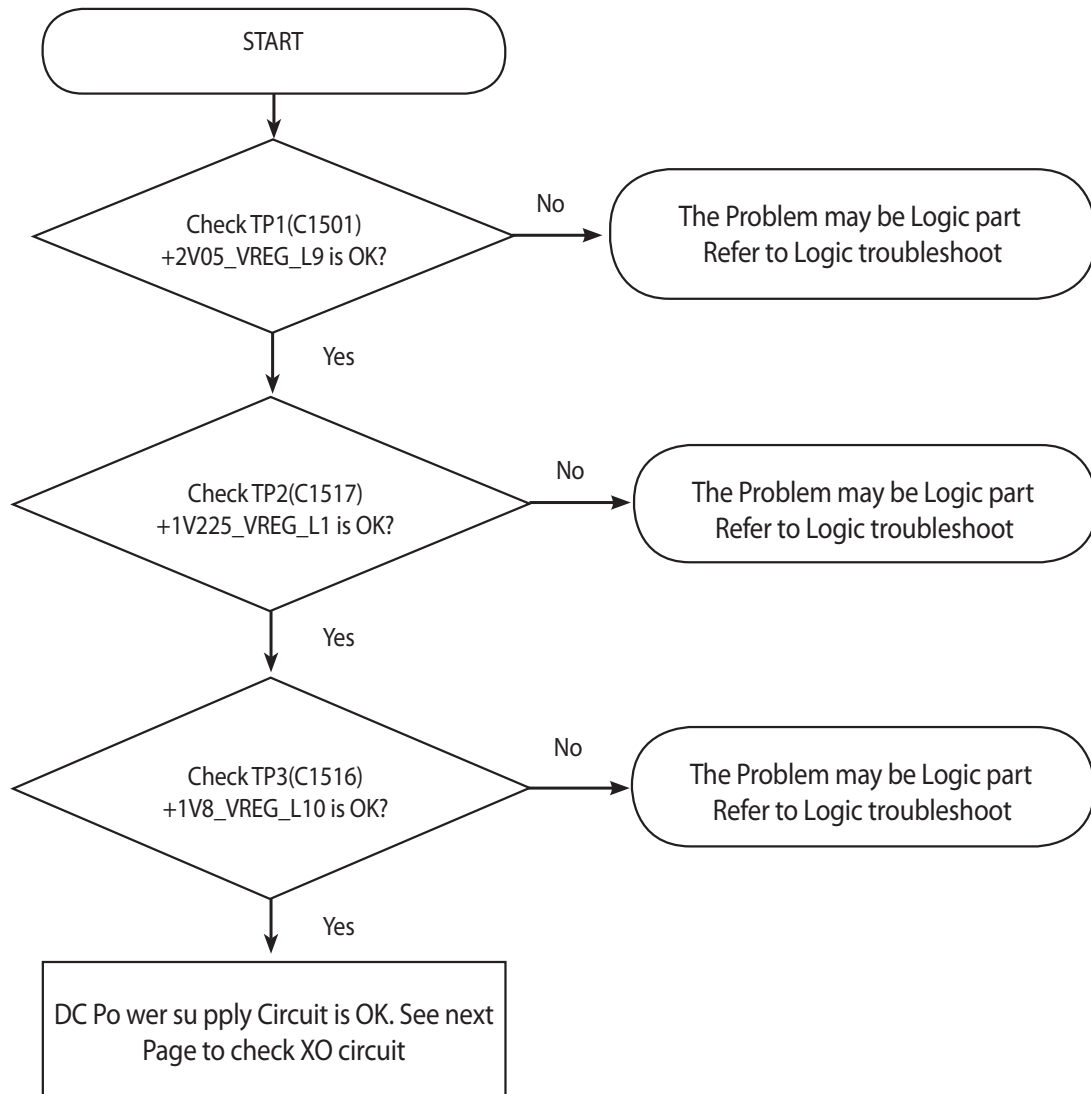


* there are no Test Points on the IQ Data Line. So It is impossible to check the IQ Data Wave form

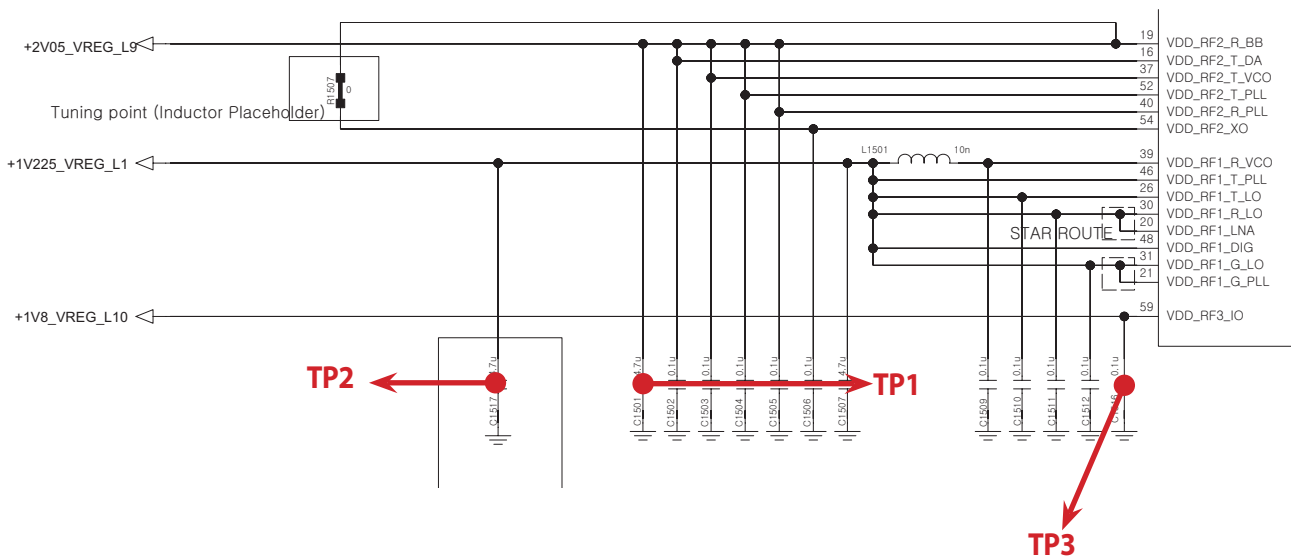
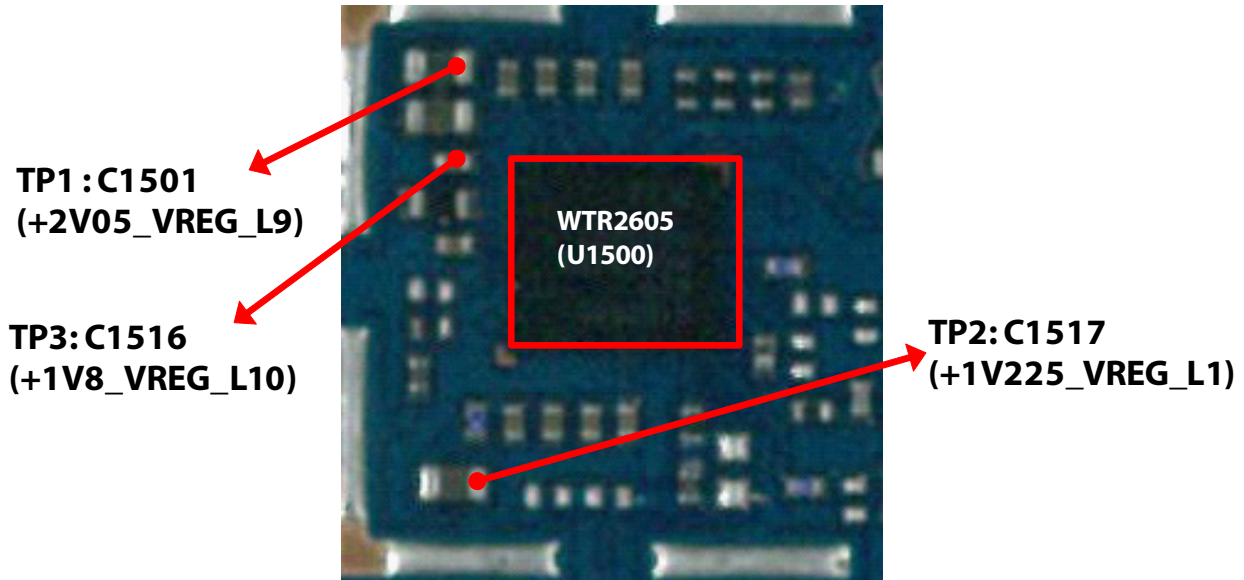
3. TROUBLE SHOOTING



3.2.1.1 Checking DC Power supply circuit



3. TROUBLE SHOOTING



3.2.1.2 Checking XO circuit

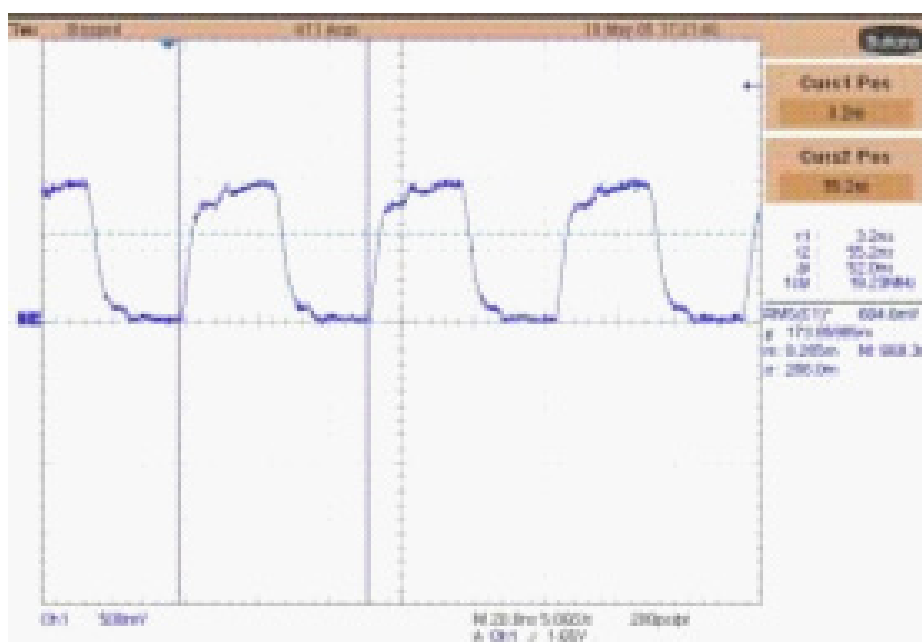
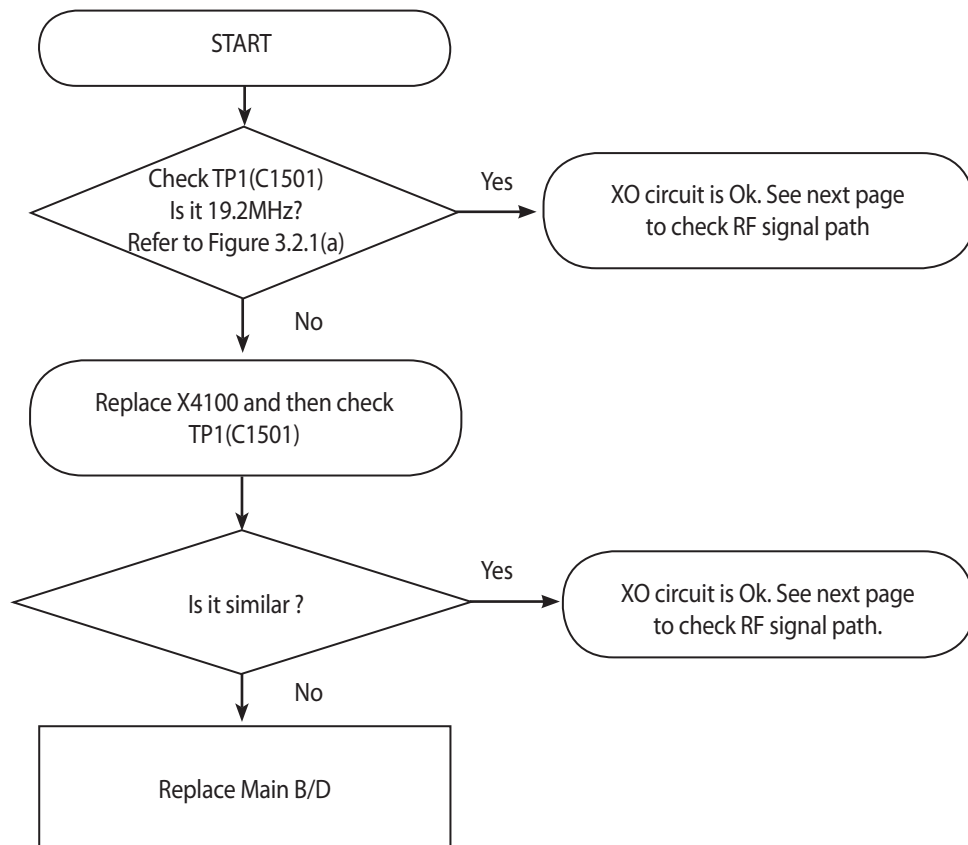
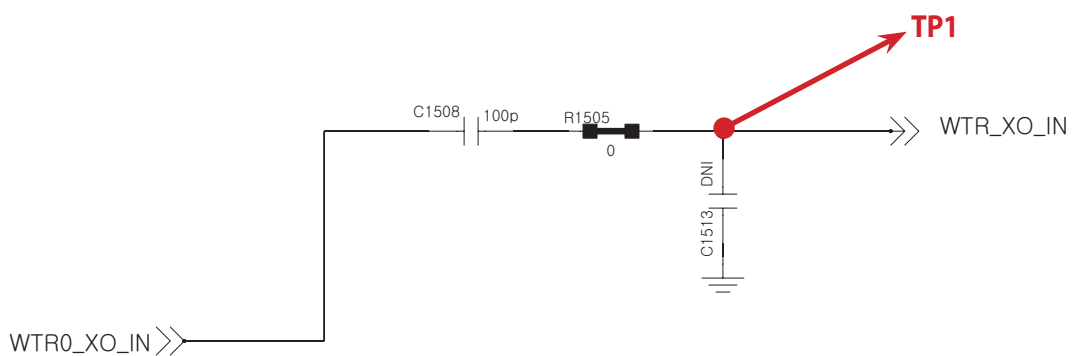
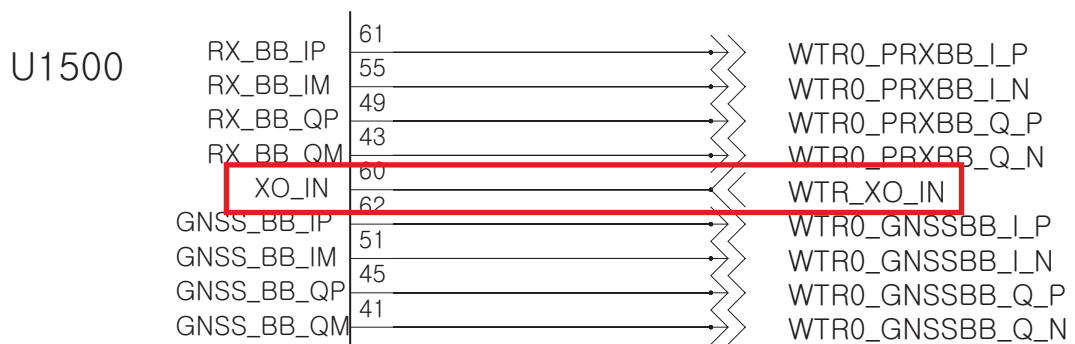
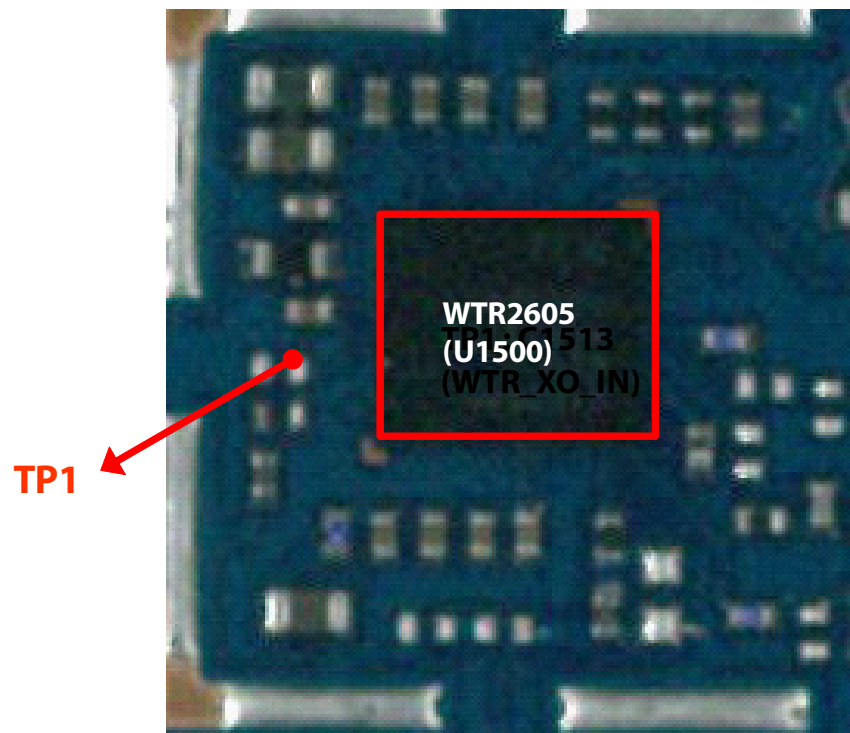
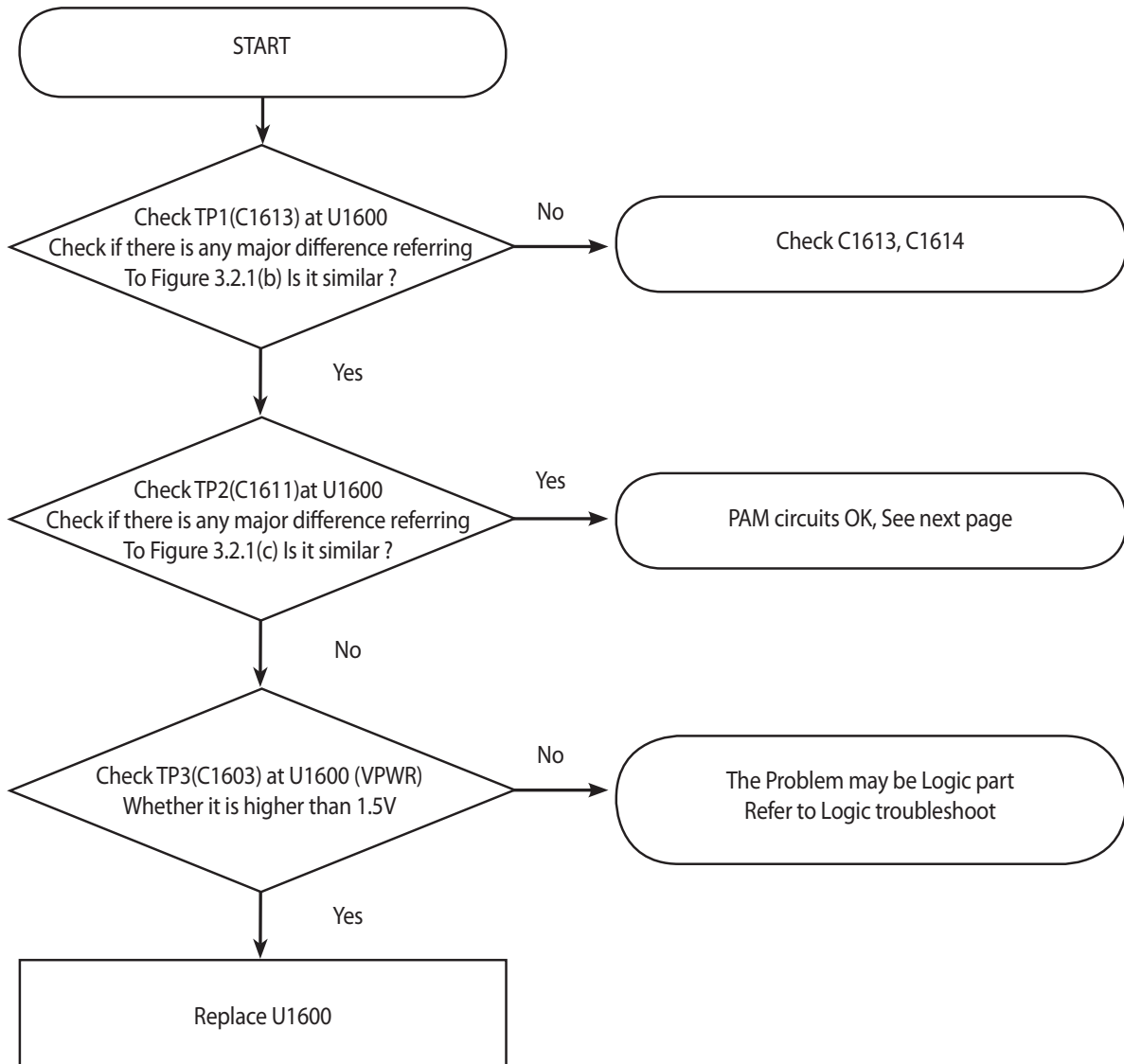


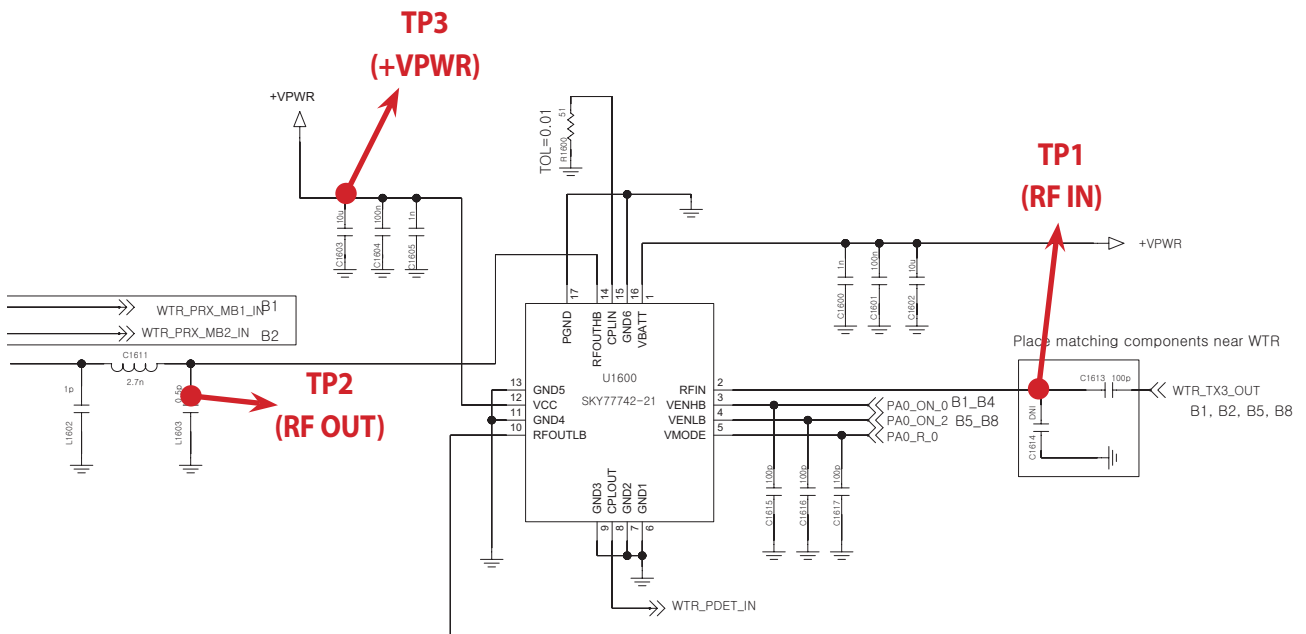
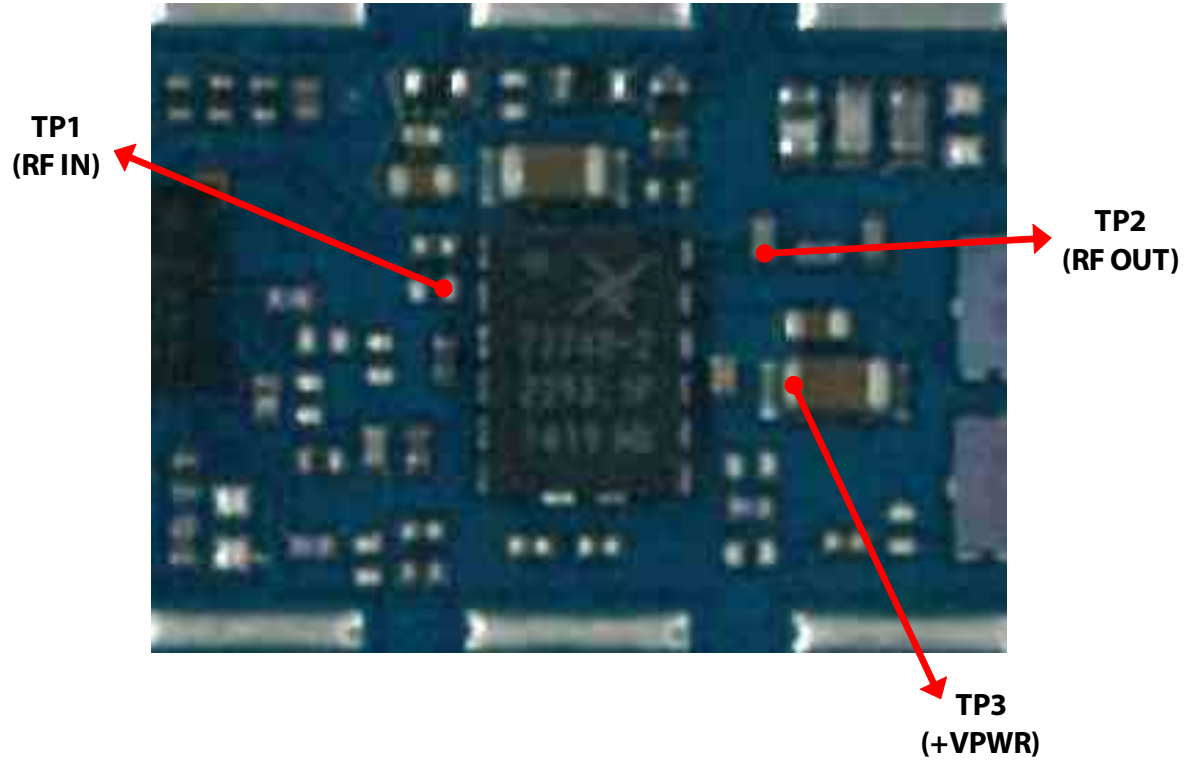
Figure 3.2.1 (a)



3.2.1.3 Checking B1 TX PA



3. TROUBLE SHOOTING



TP1

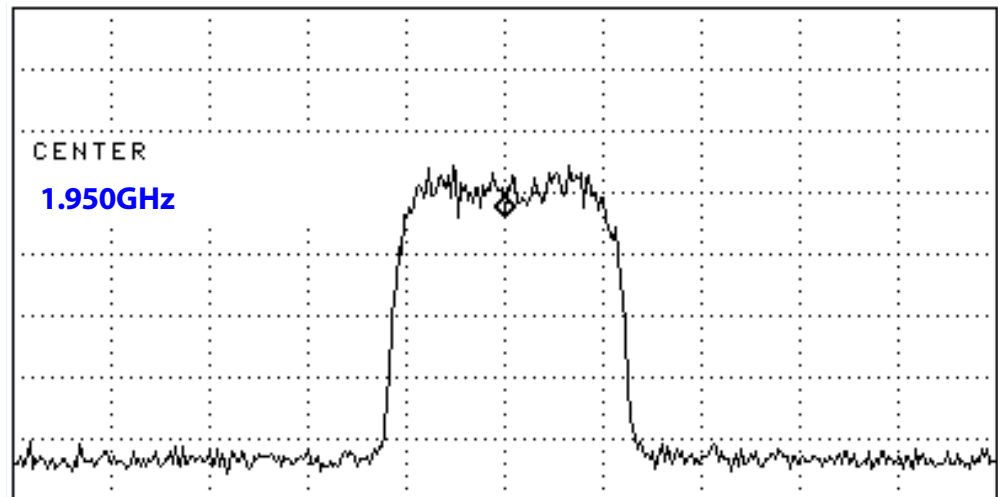


Figure 3.2.1 (b)

TP2

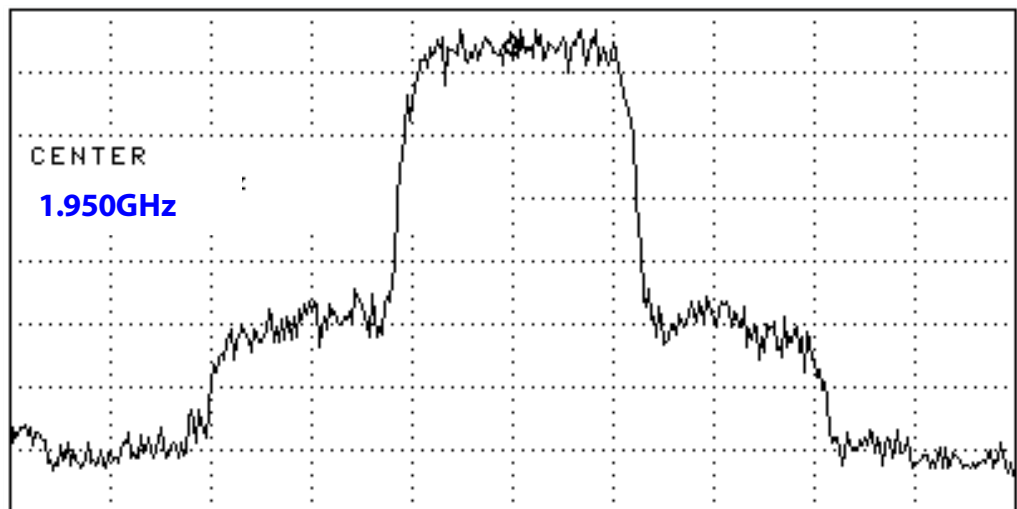
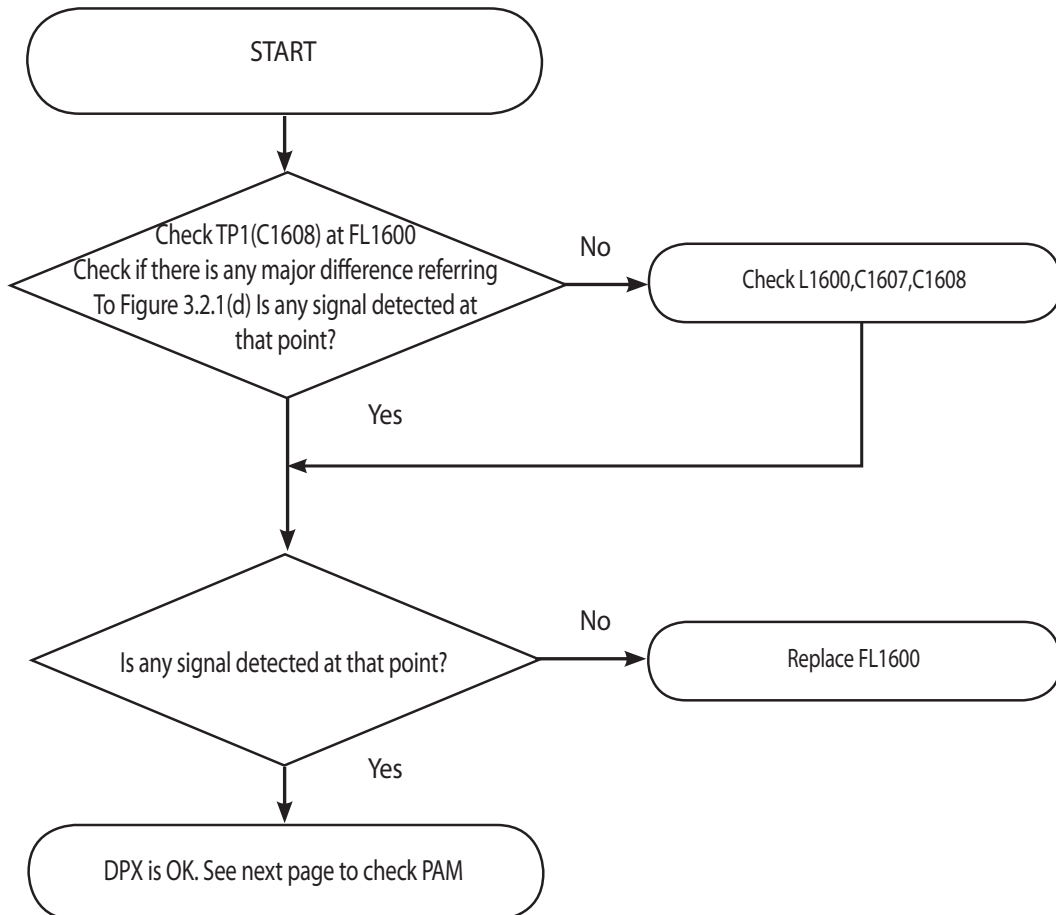


Figure 3.2.1 (c)

3.2.1.4 Checking RF signal path (DPX)



TP1

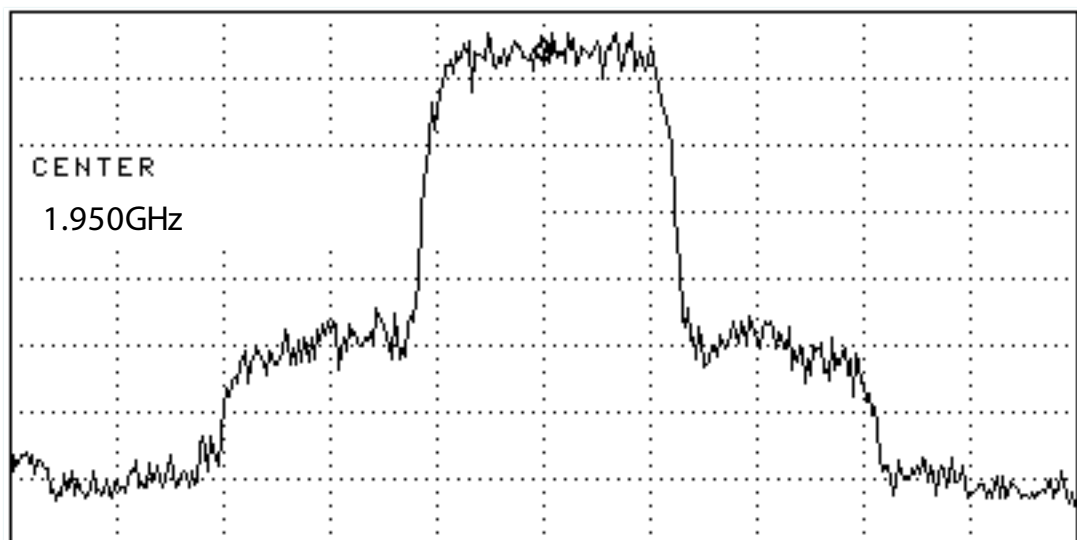
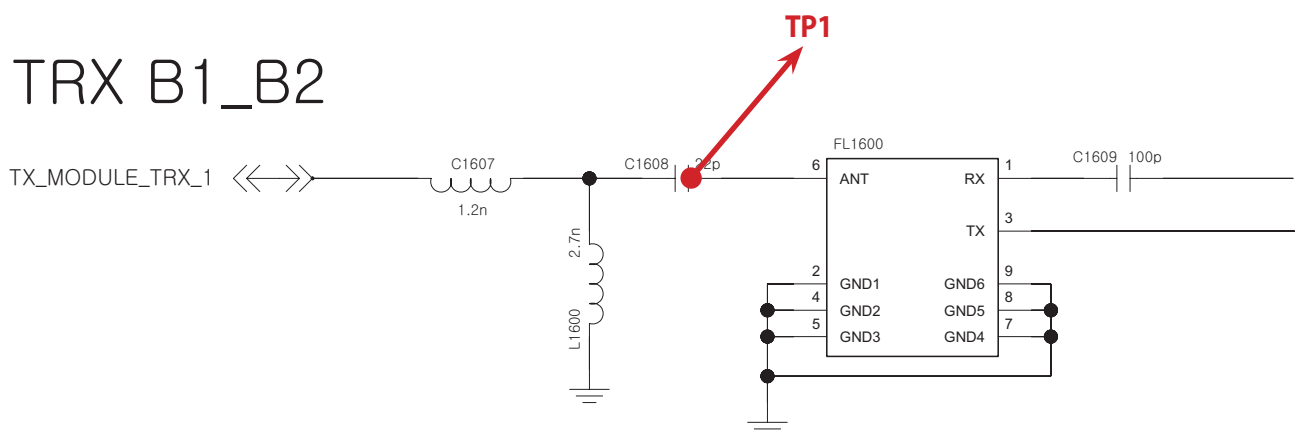
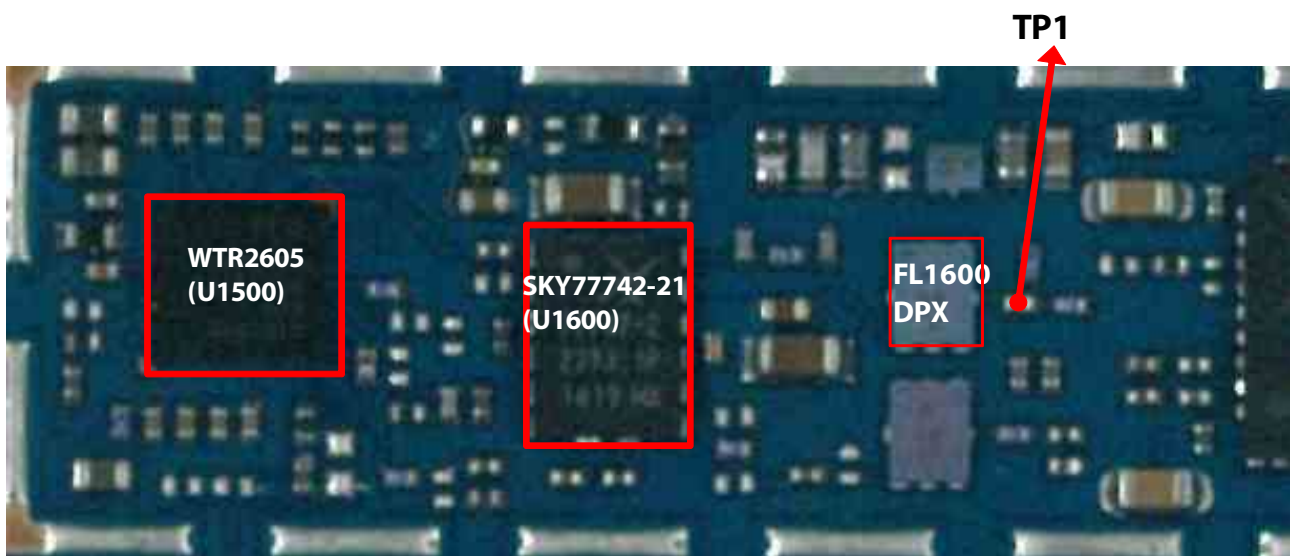
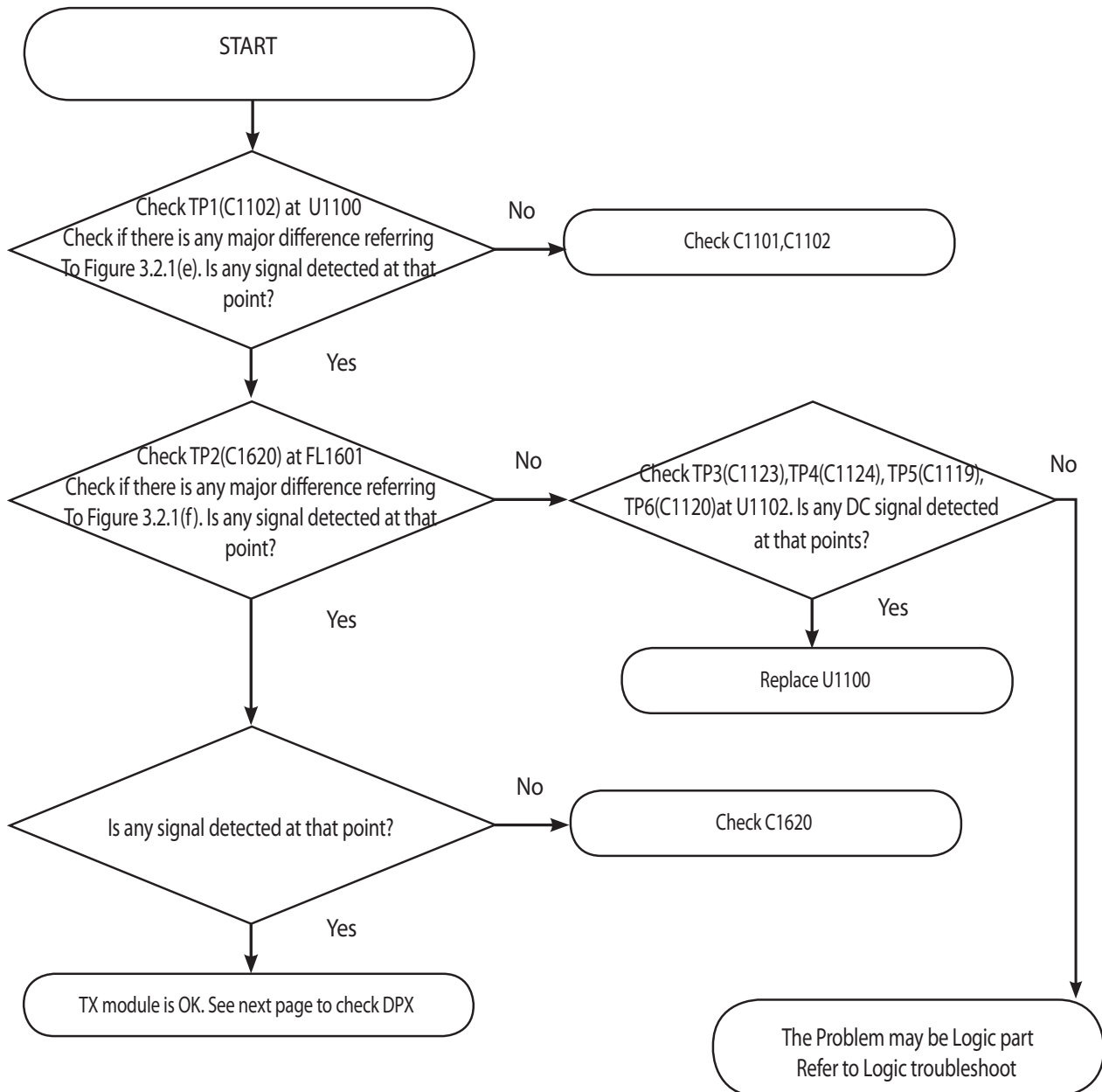


Figure 3.2.1 (d)

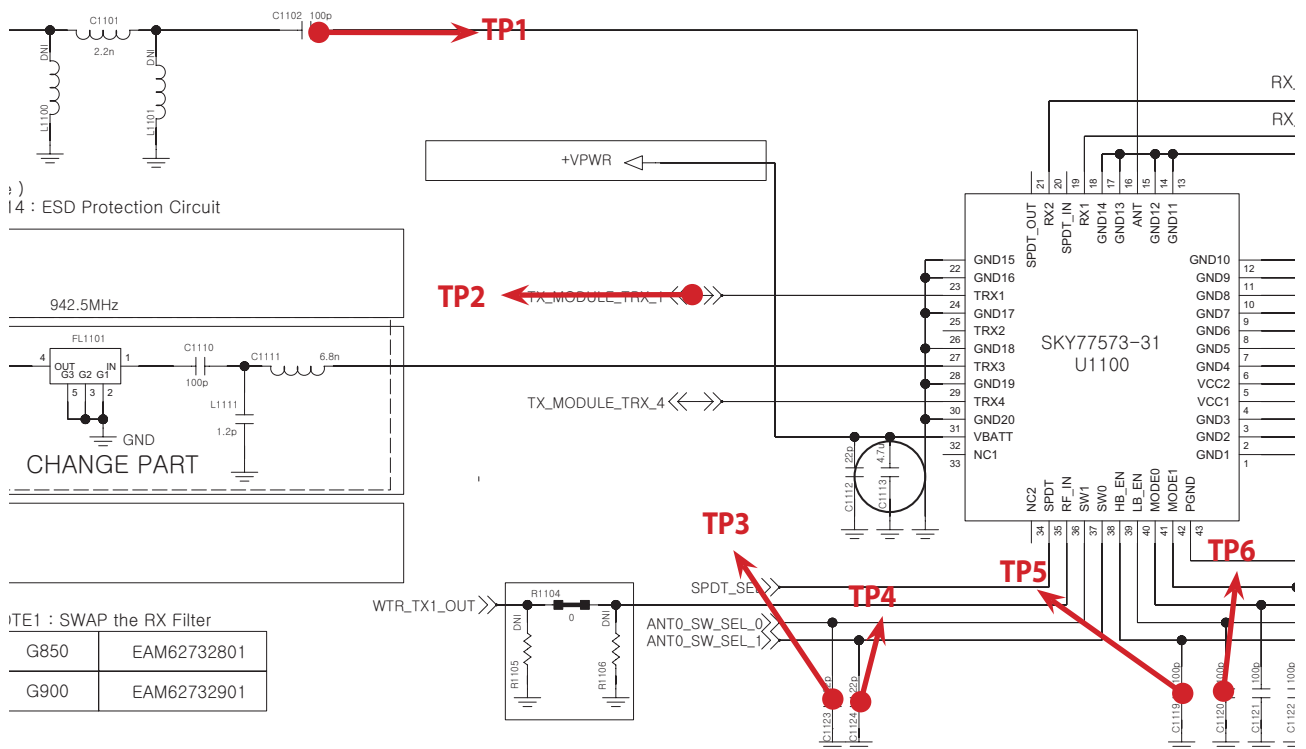
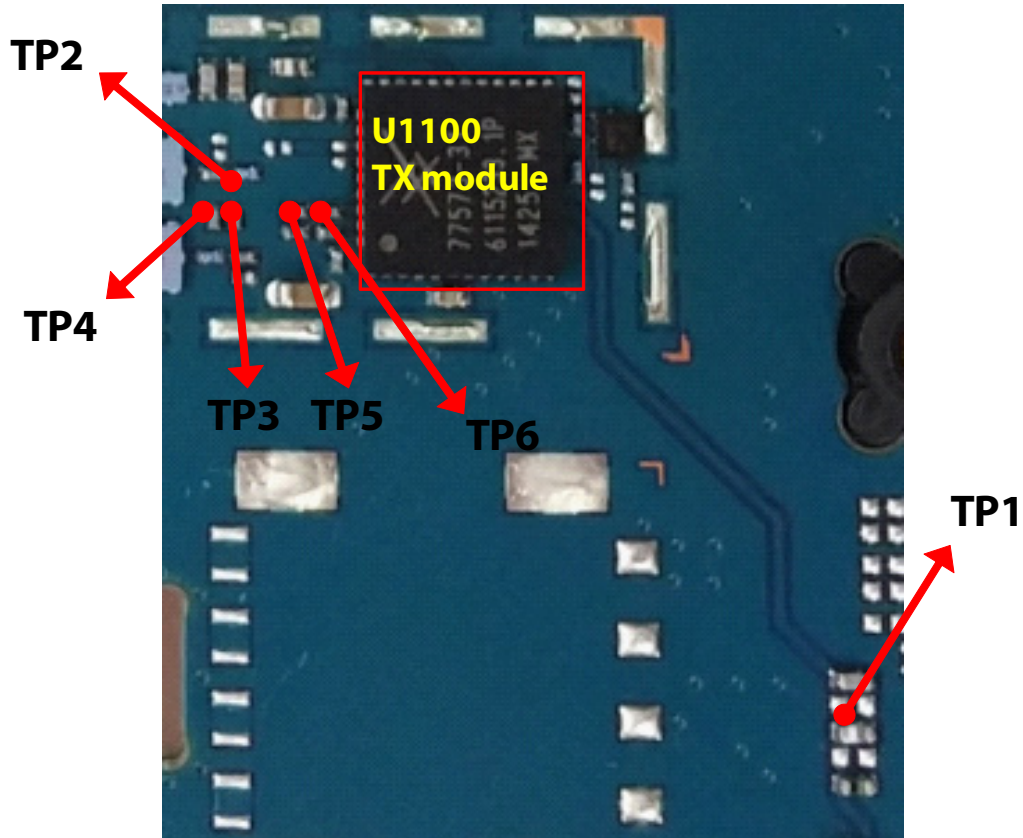
3. TROUBLE SHOOTING



3.2.1.5 Checking RF signal path (TX Module)



3. TROUBLE SHOOTING



TP1

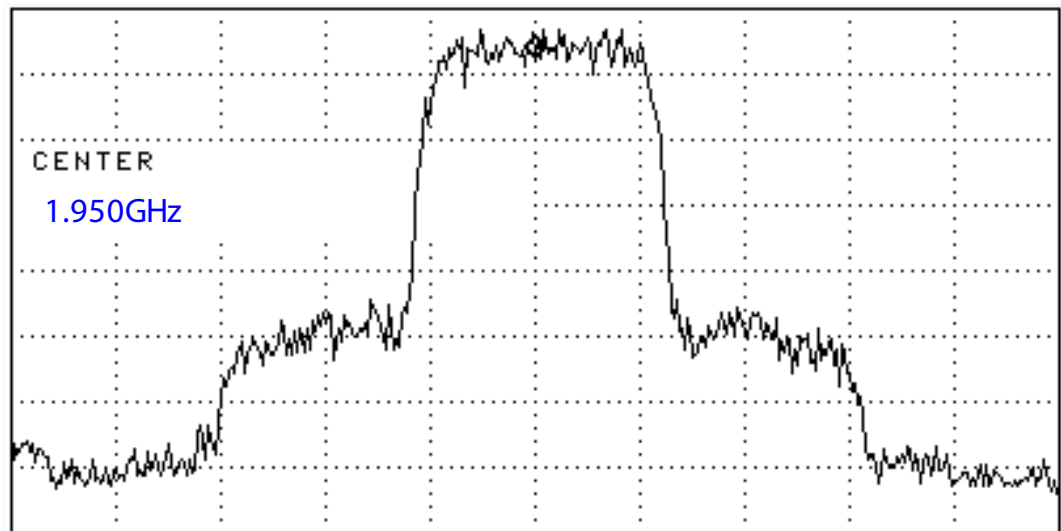


Figure 3.2.1 (e)

TP2

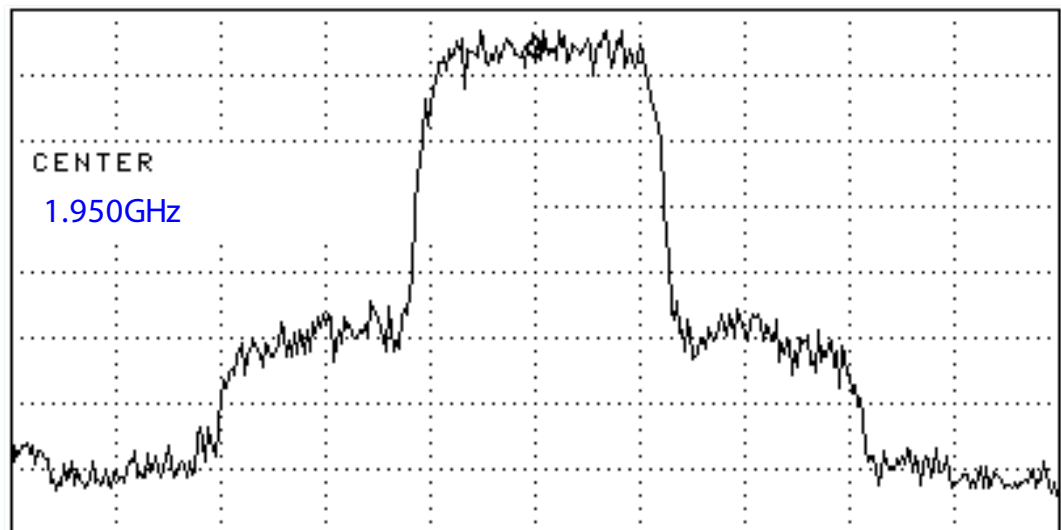
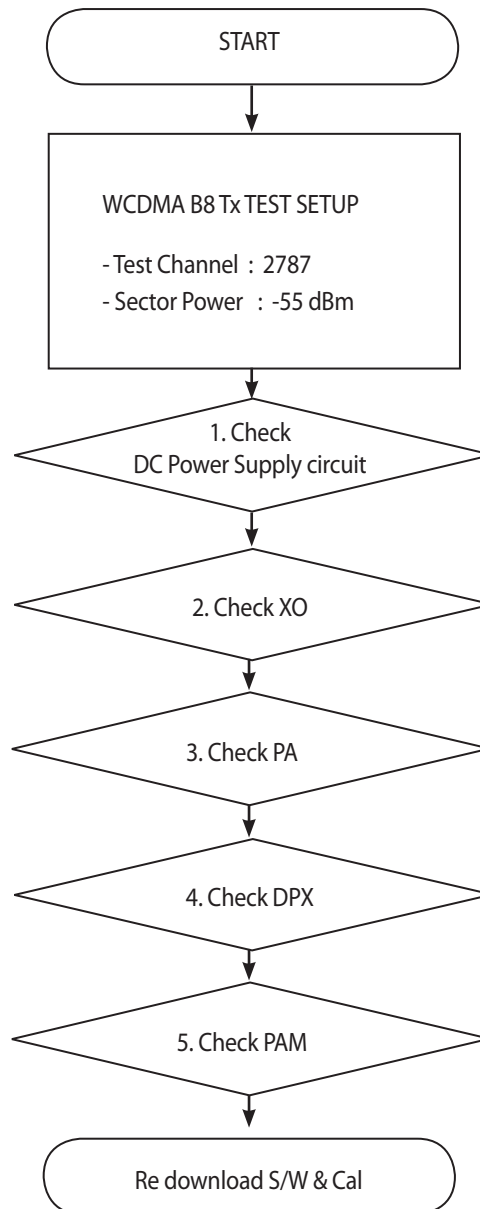


Figure 3.2.1 (f)

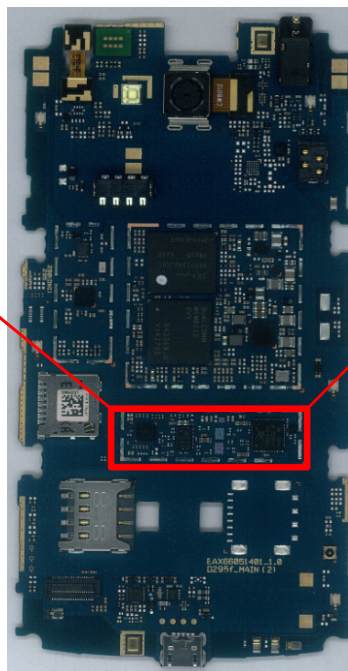
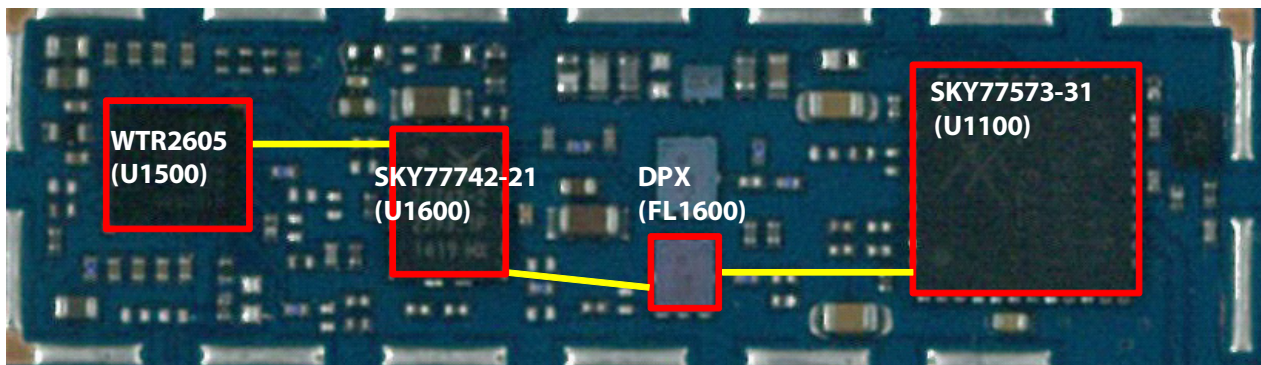
3.3 WCDMA Tx Part Trouble

3.3.1 WCDMA_B8 Tx

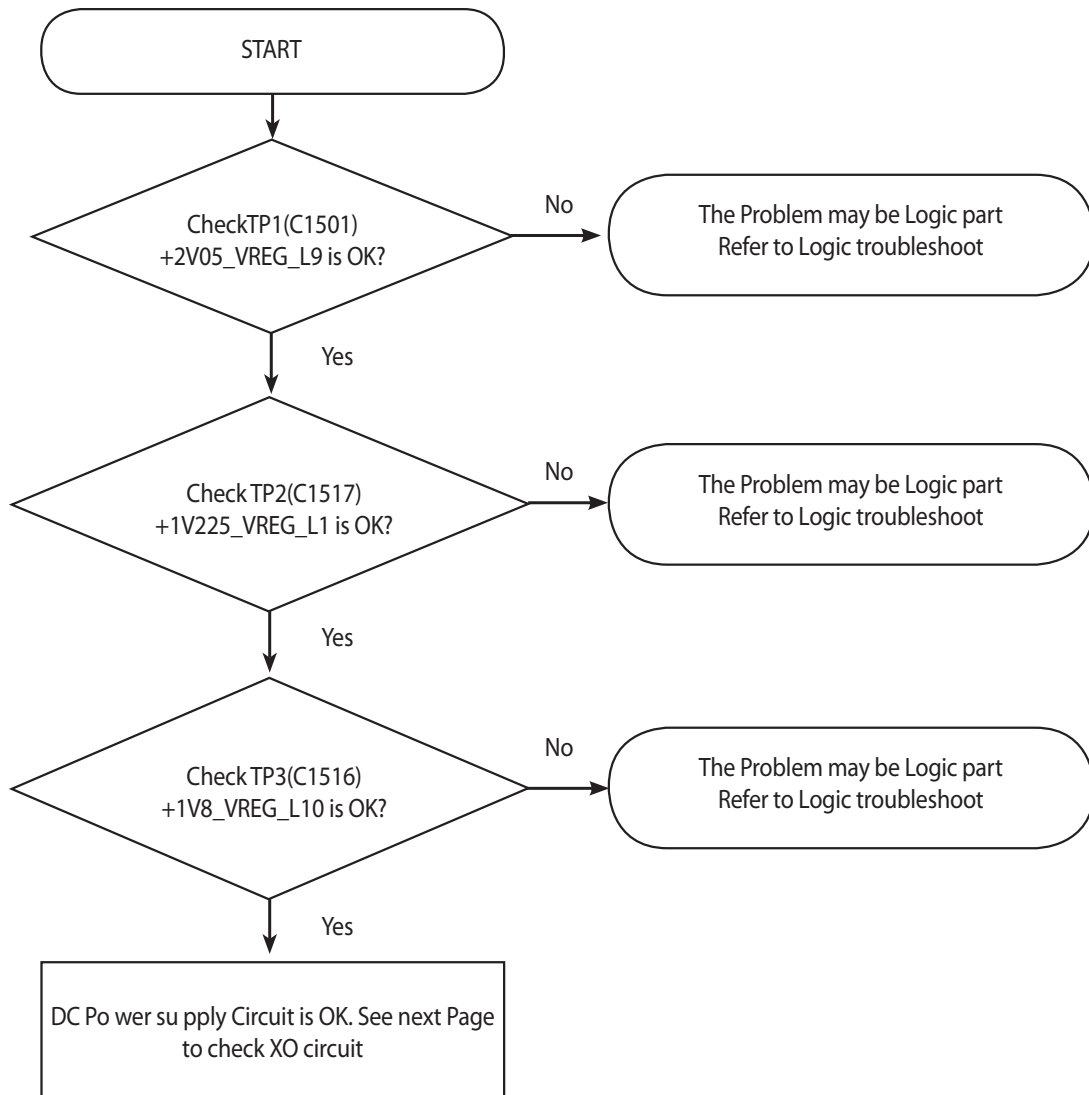


* there are no Test Points on the IQ Data Line. So It is impossible to check the IQ Data Wave form

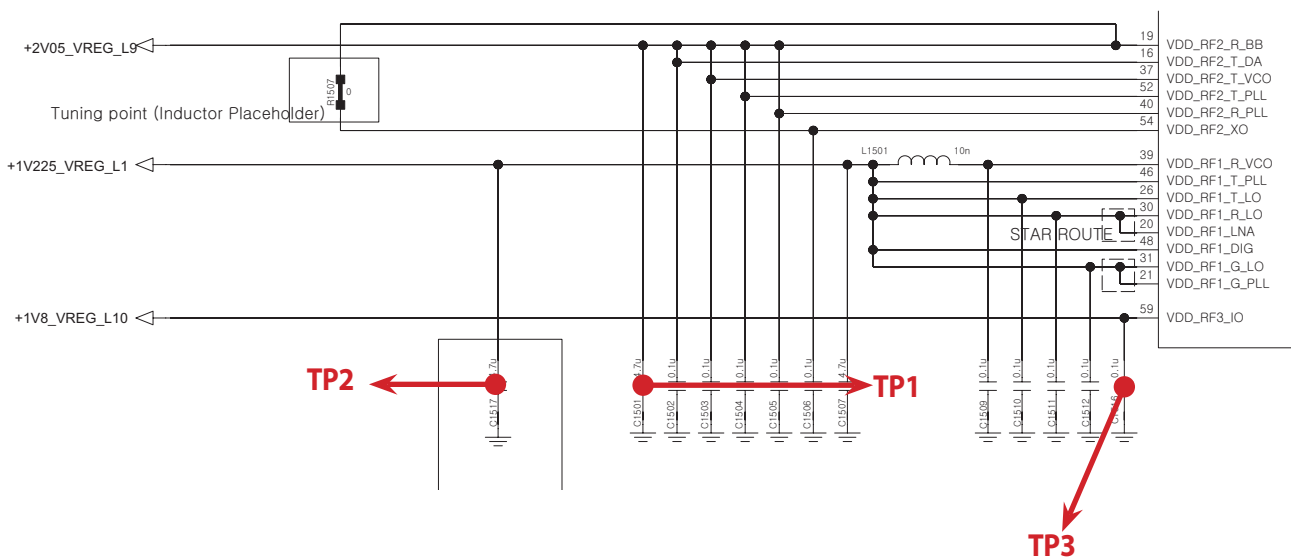
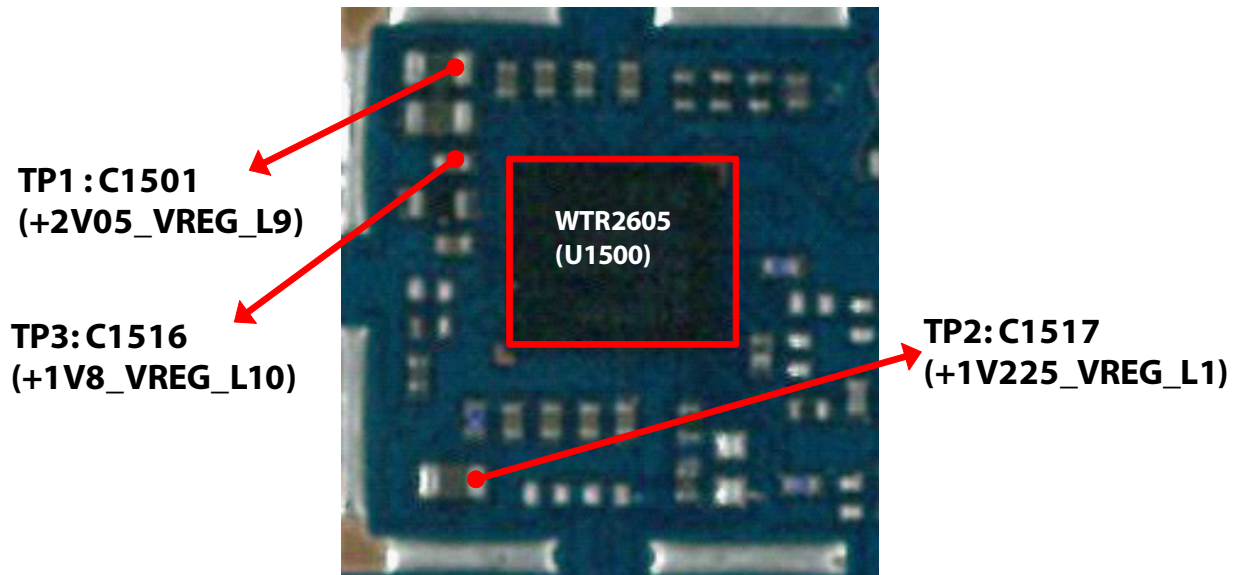
3. TROUBLE SHOOTING



3.3.1.1 Checking DC Power supply circuit



3. TROUBLE SHOOTING



3.3.1.2 Checking XO circuit

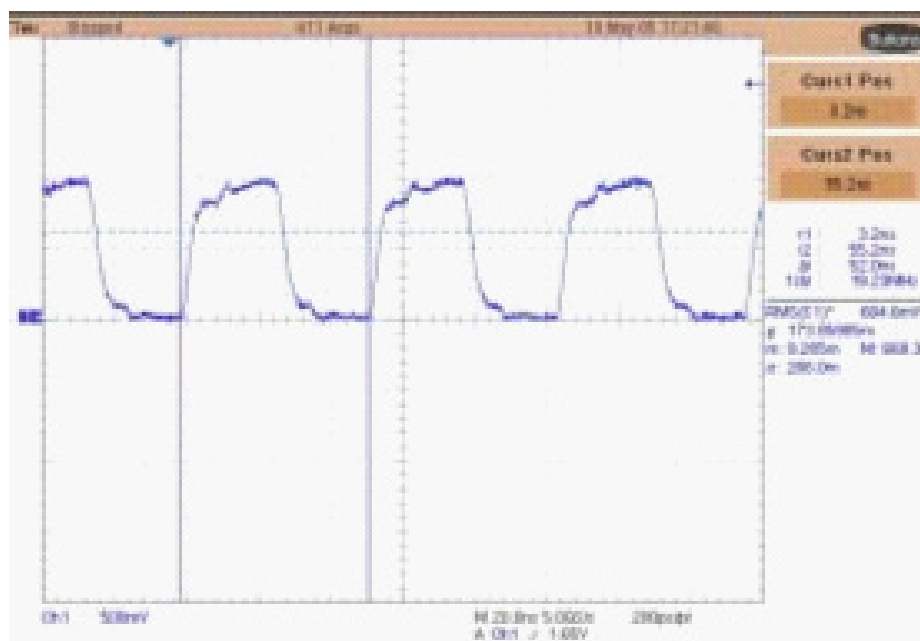
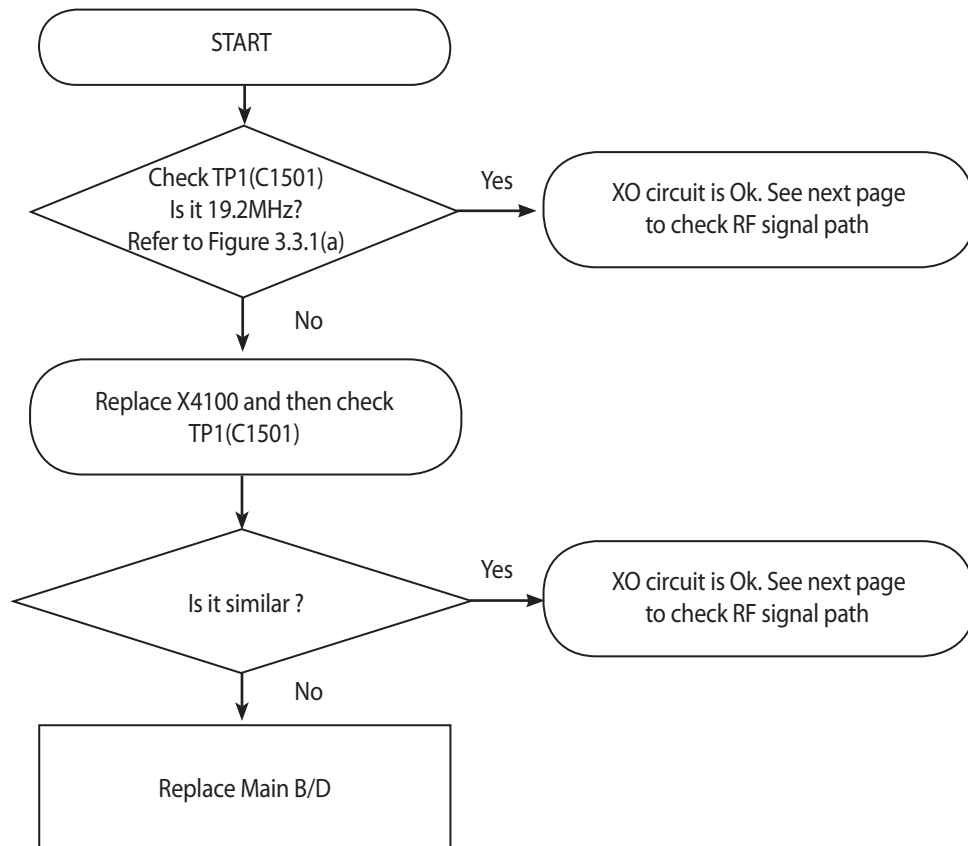
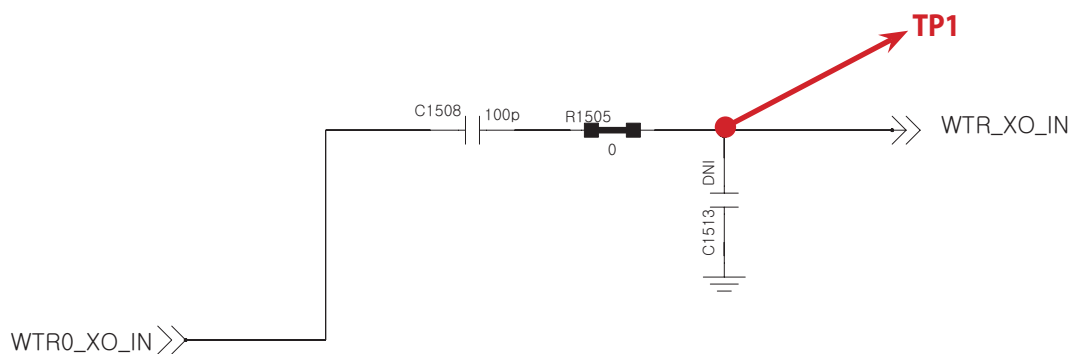
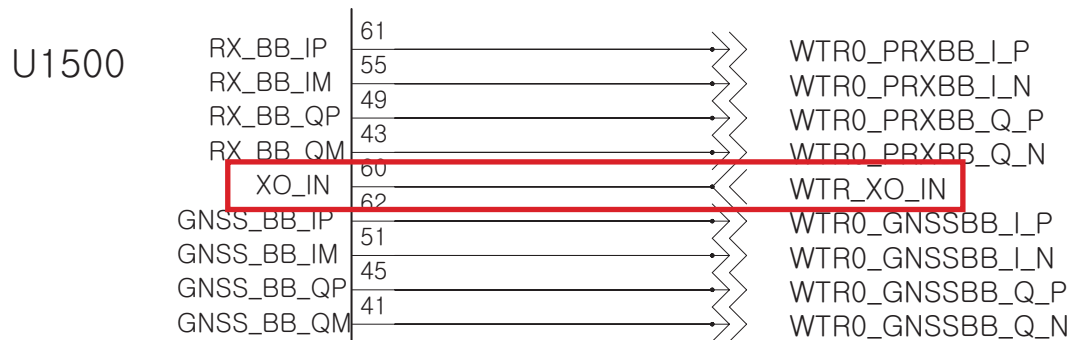
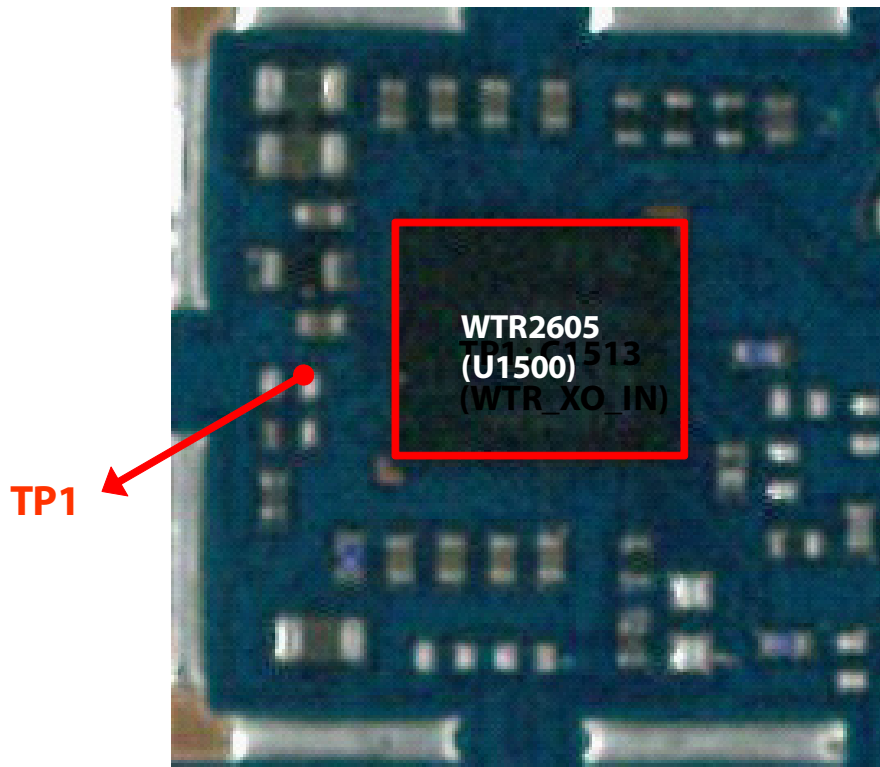
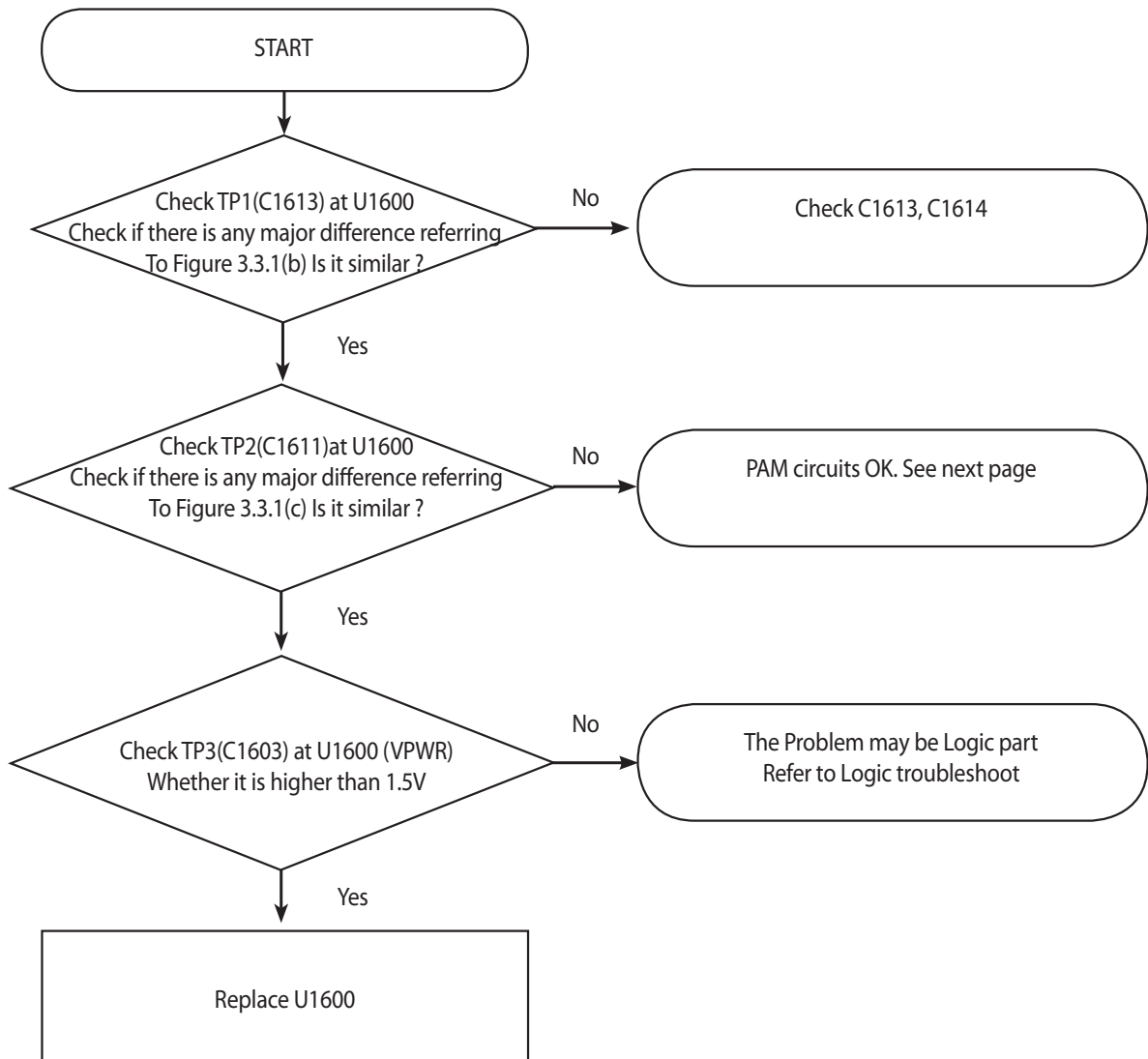


Figure 3.3.1 (a)

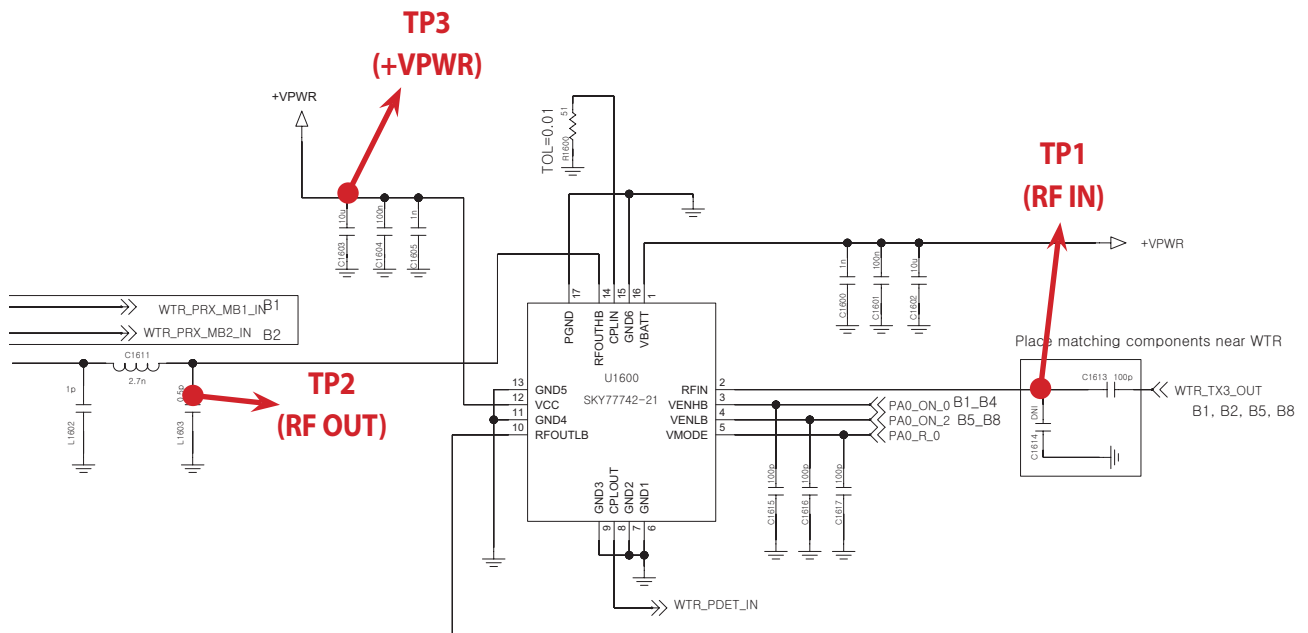
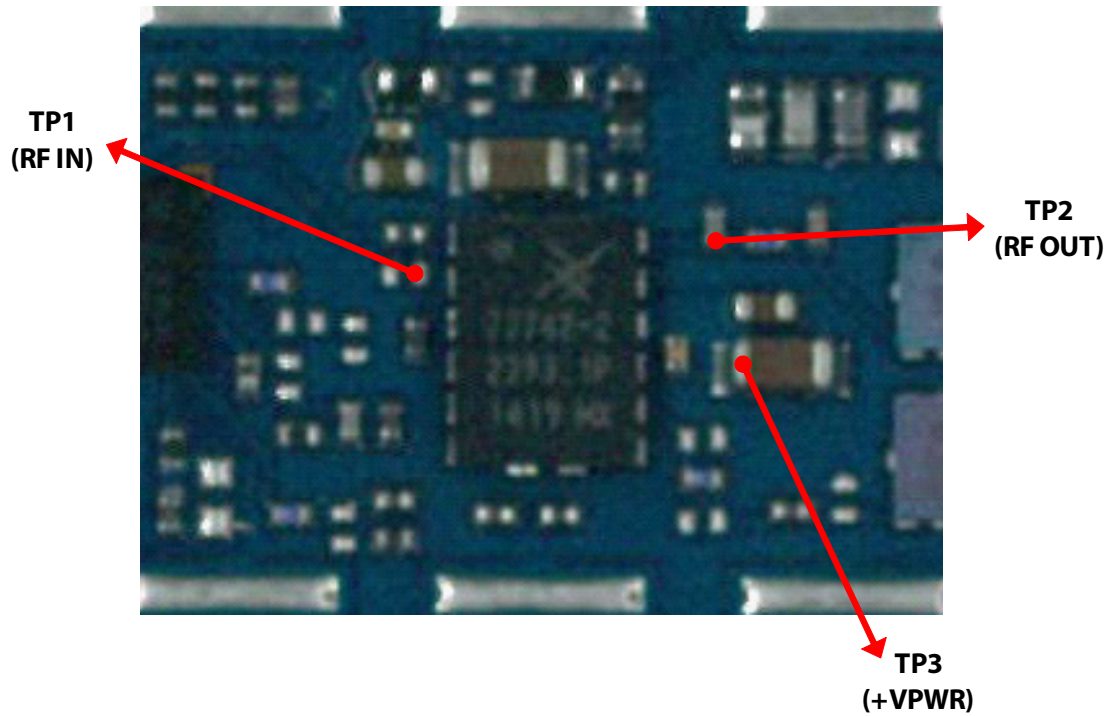
3. TROUBLE SHOOTING



3.3.1.3 Checking B8 TX PA



3. TROUBLE SHOOTING



TP1

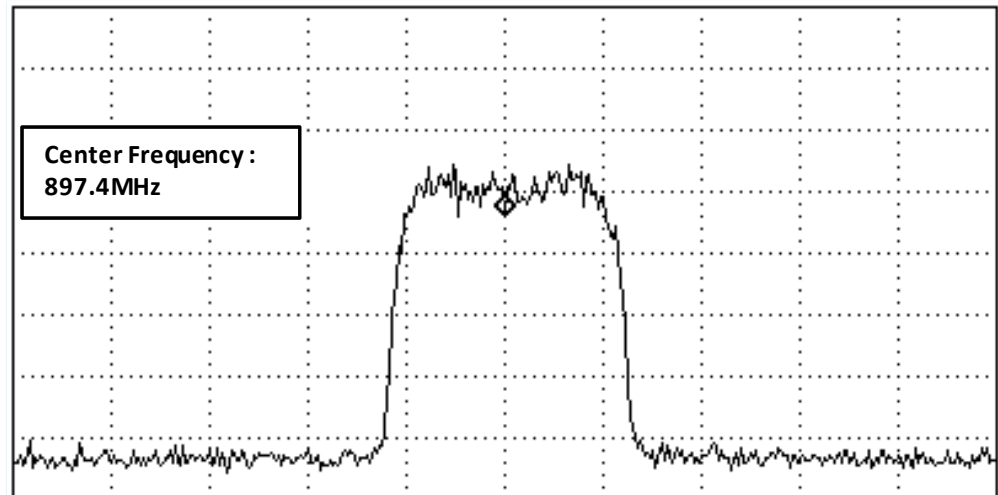


Figure 3.3.1 (b)

TP2

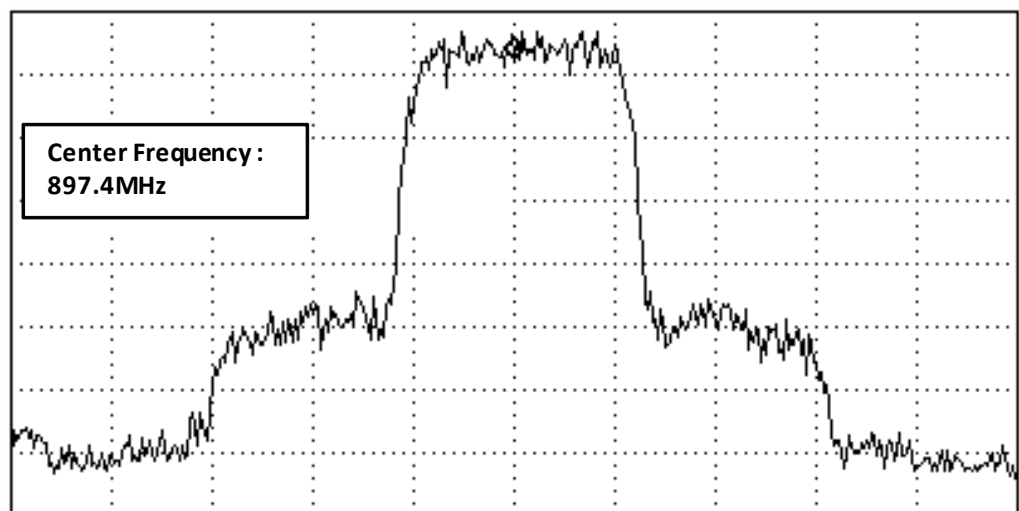
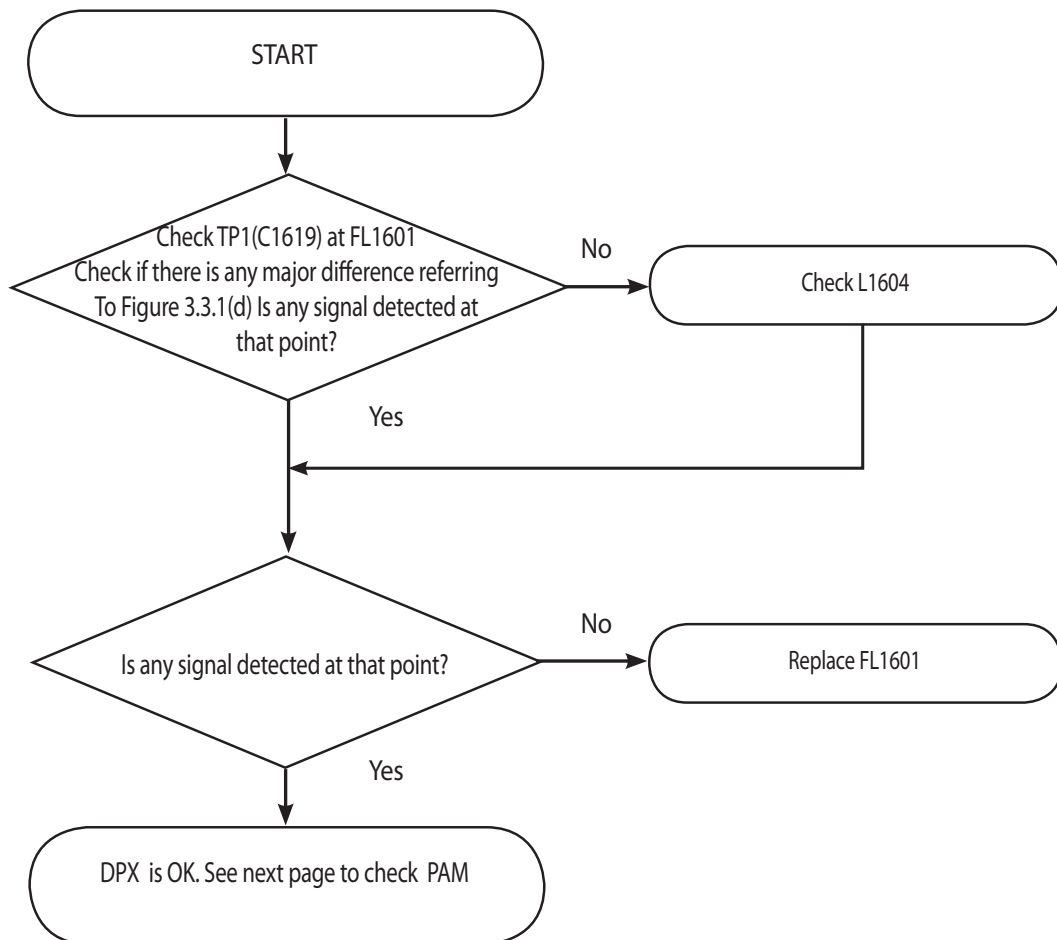


Figure 3.3.1 (C)

3.3.1.4 Checking RF signal path (DPX)



TP1

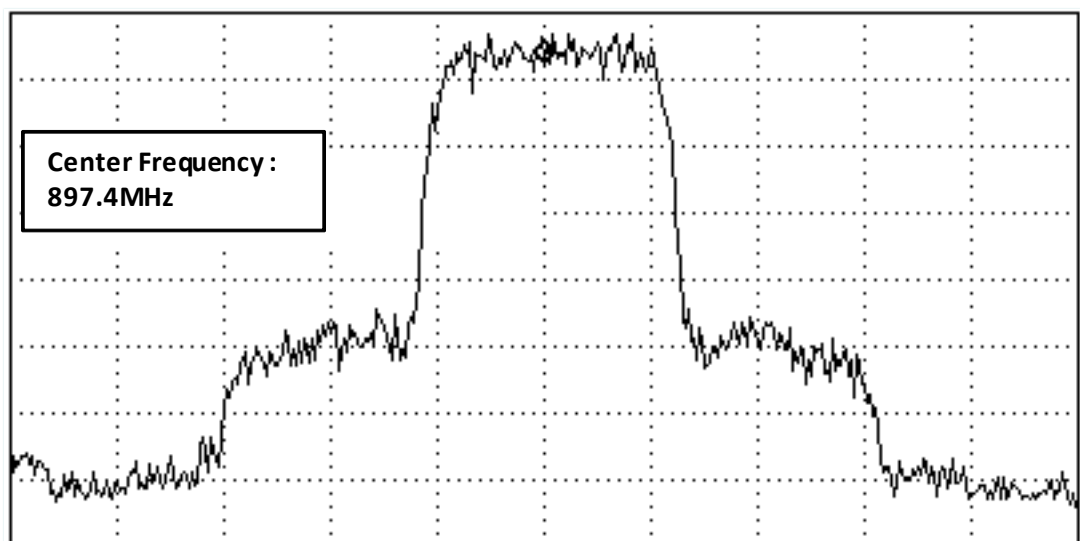
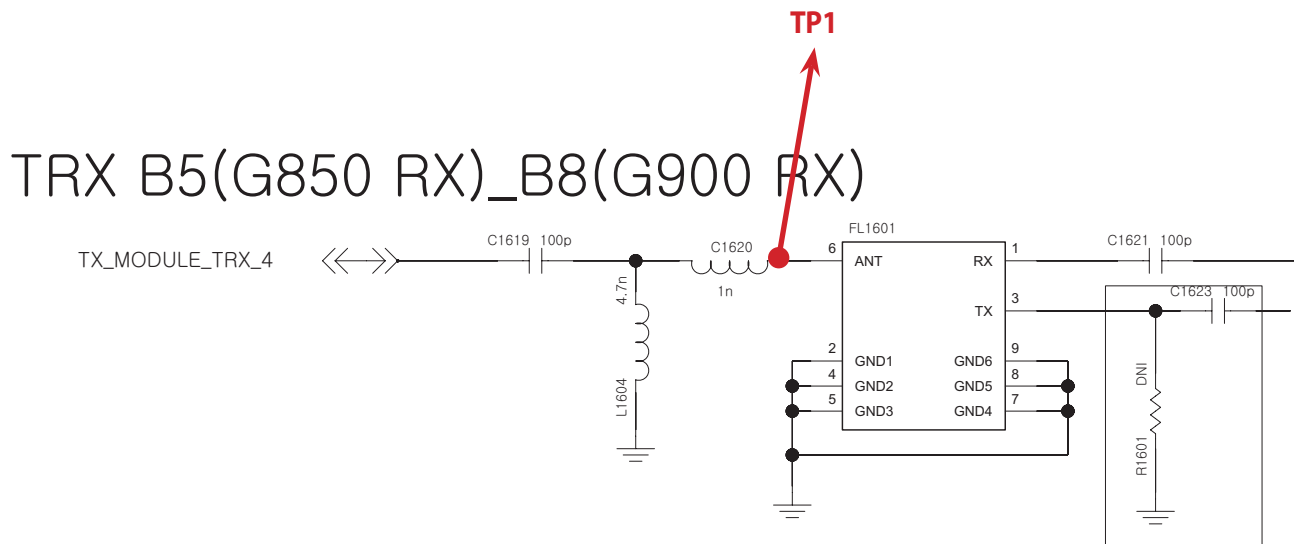
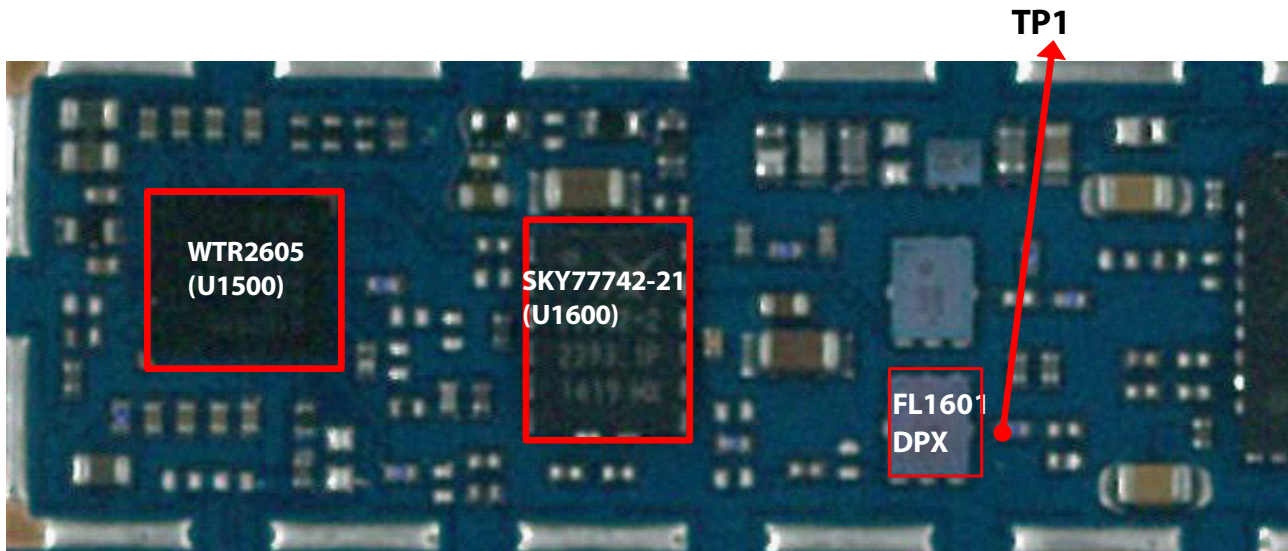
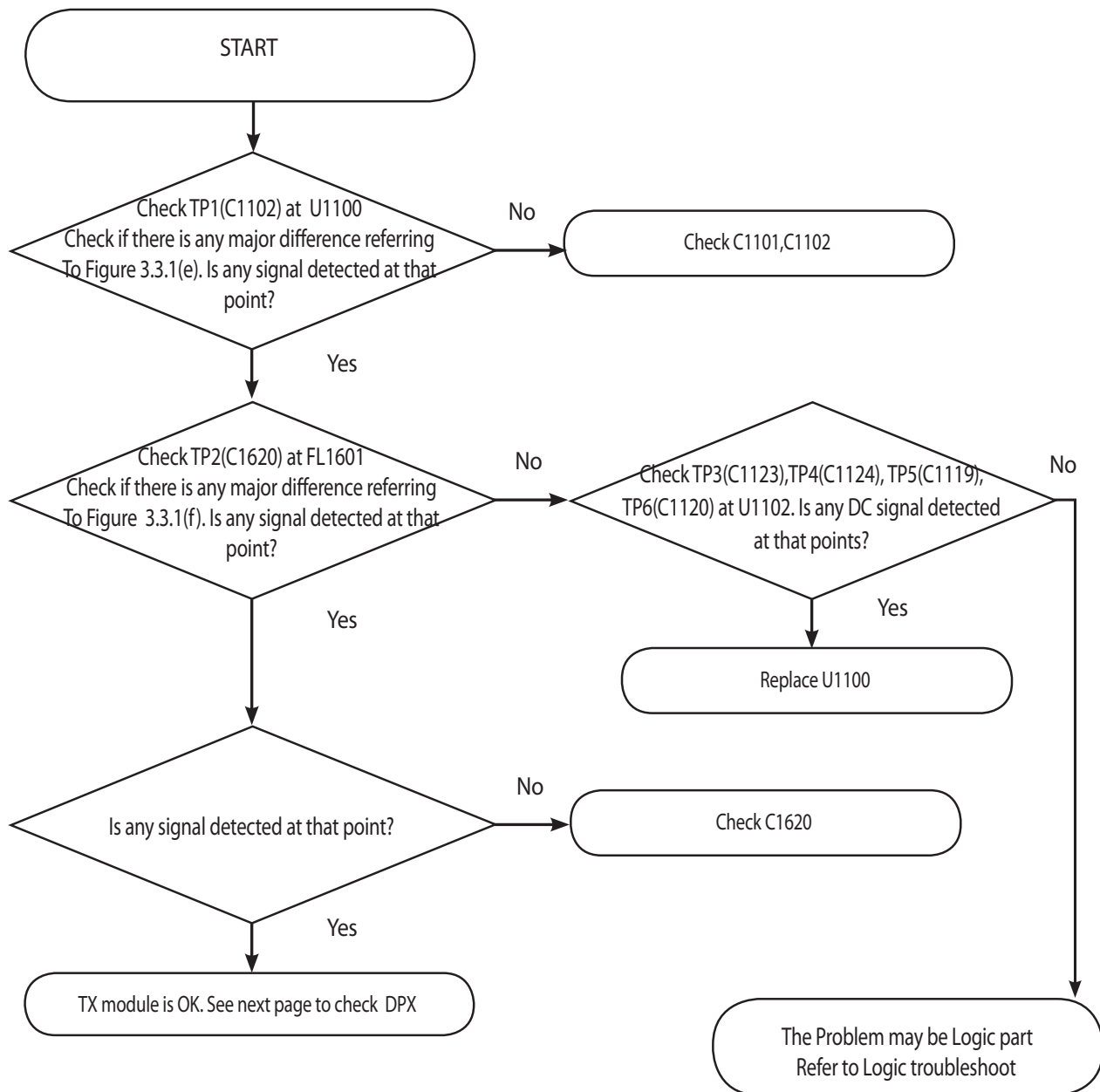


Figure 3.3.1 (d)

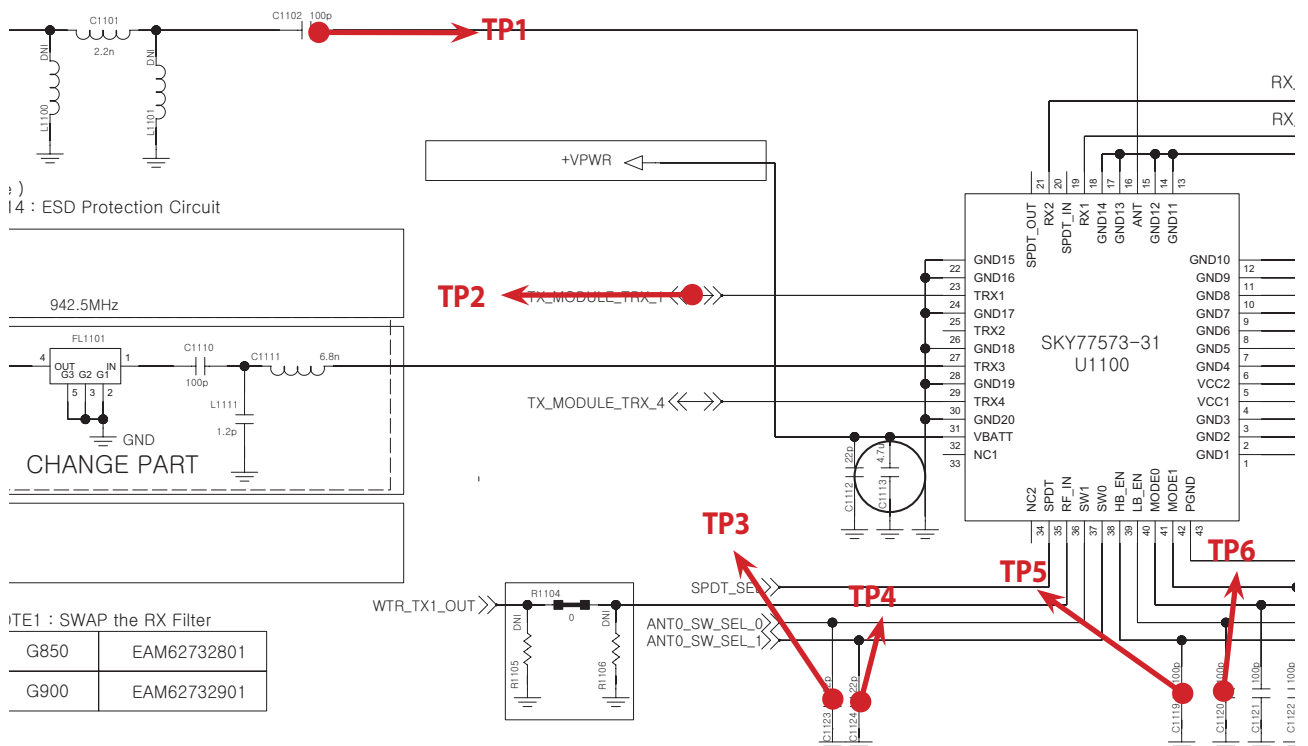
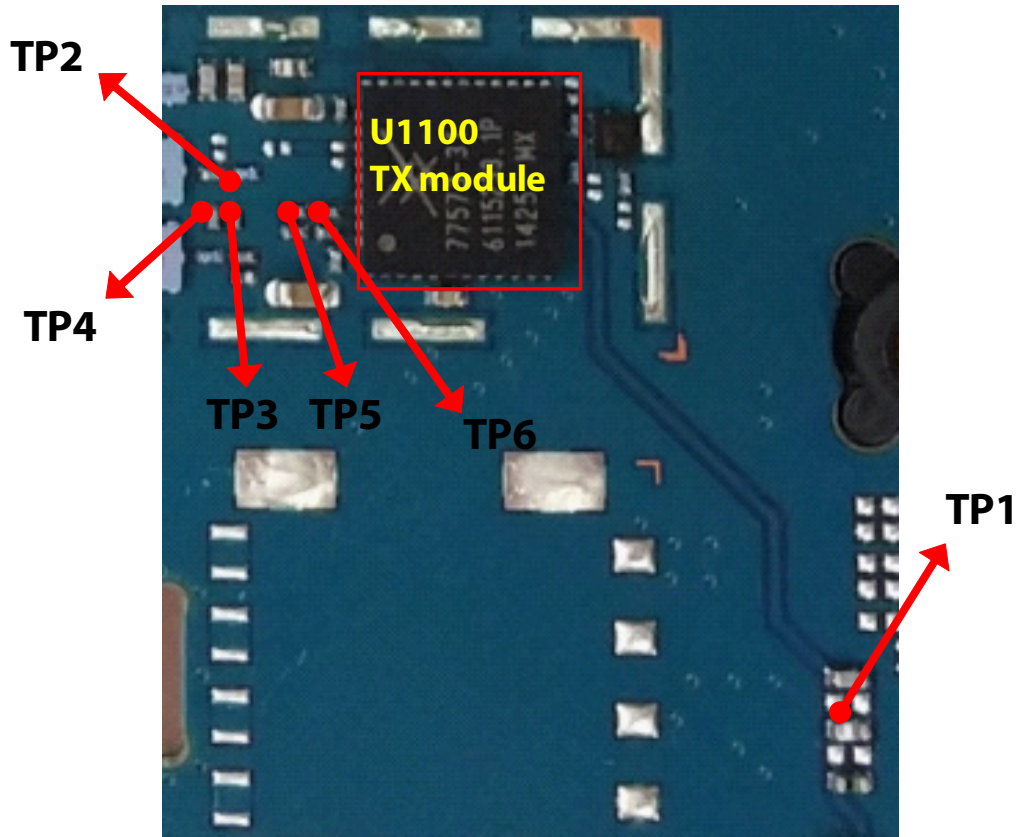
3. TROUBLE SHOOTING



3.3.1.5 Checking RF signal path (TX Module)



3. TROUBLE SHOOTING



TP1

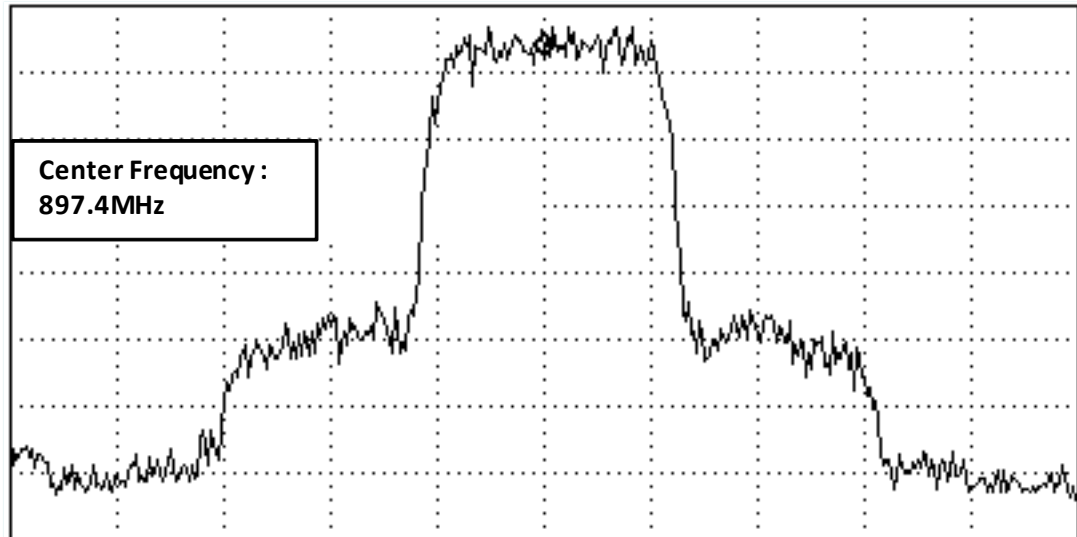


Figure 3.3.1 (e)

TP2

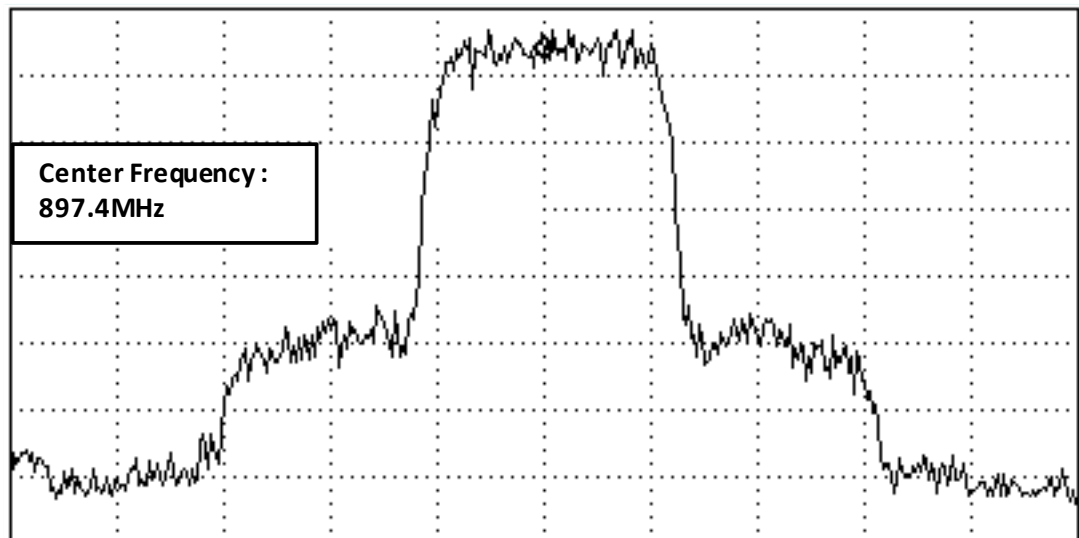
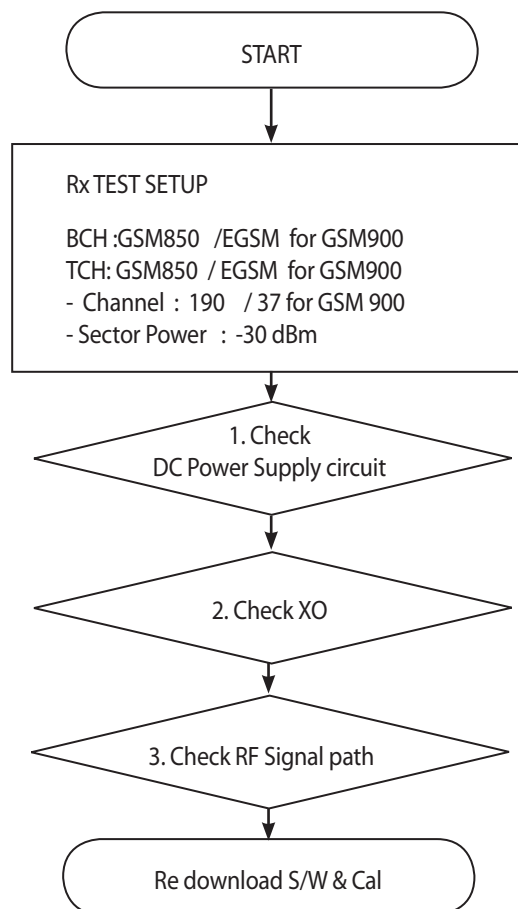


Figure 3.3.1 (f)

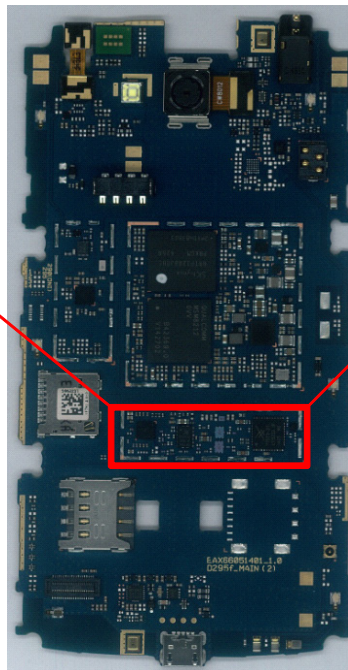
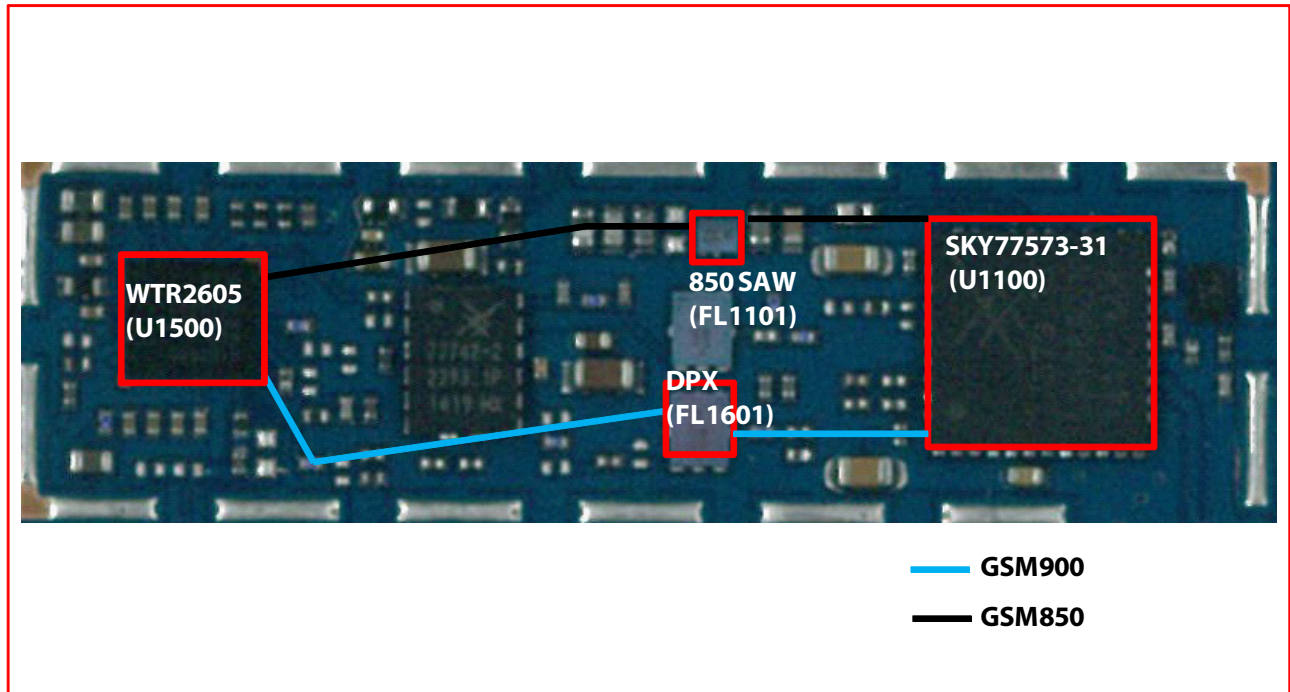
3.4 GSM Rx Part Troubleshooting

3.4.1 GSM850/900 Rx

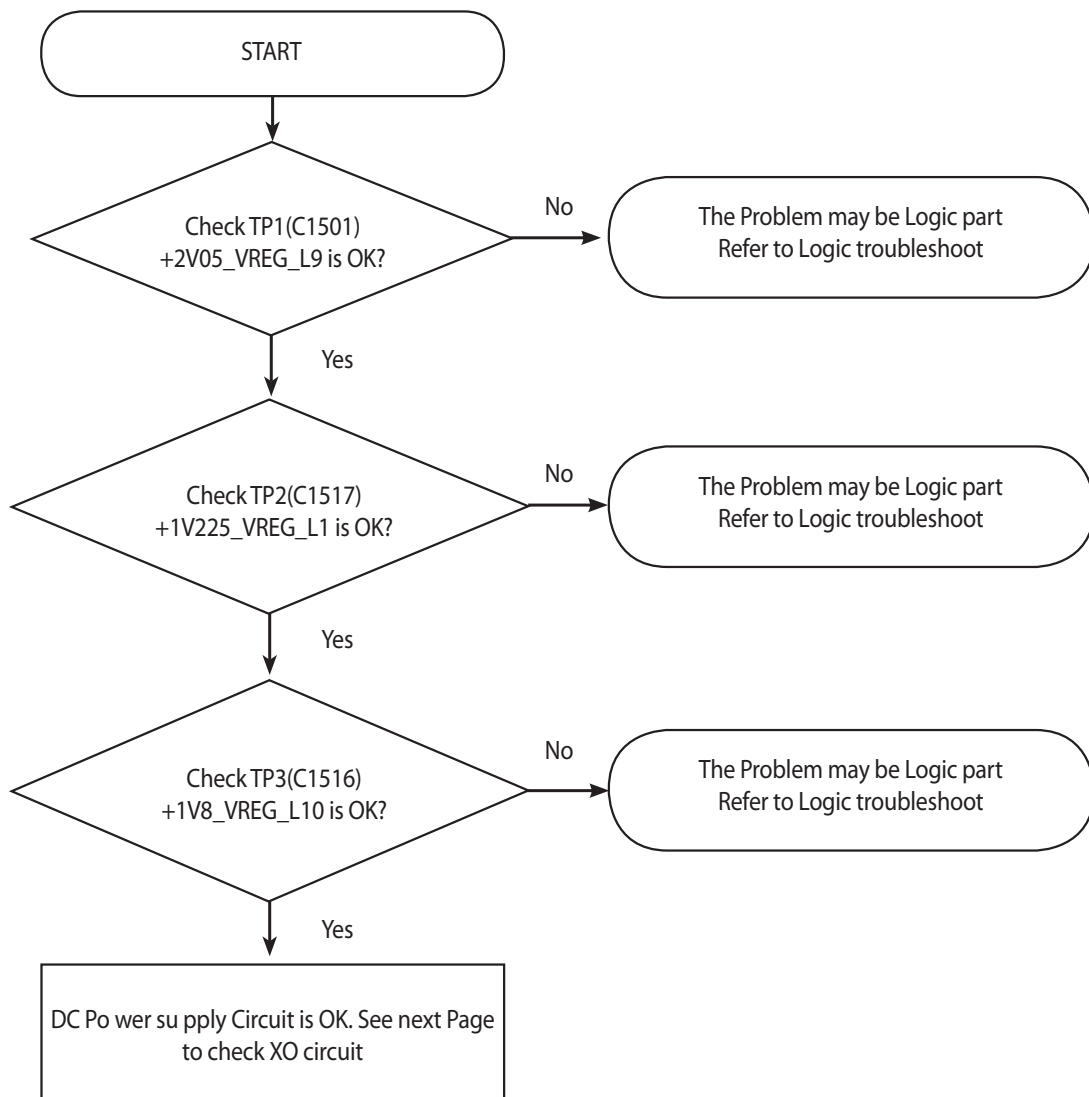


* there are no Test Points on the IQ Data Line. So It is impossible to check the IQ Data Wave form

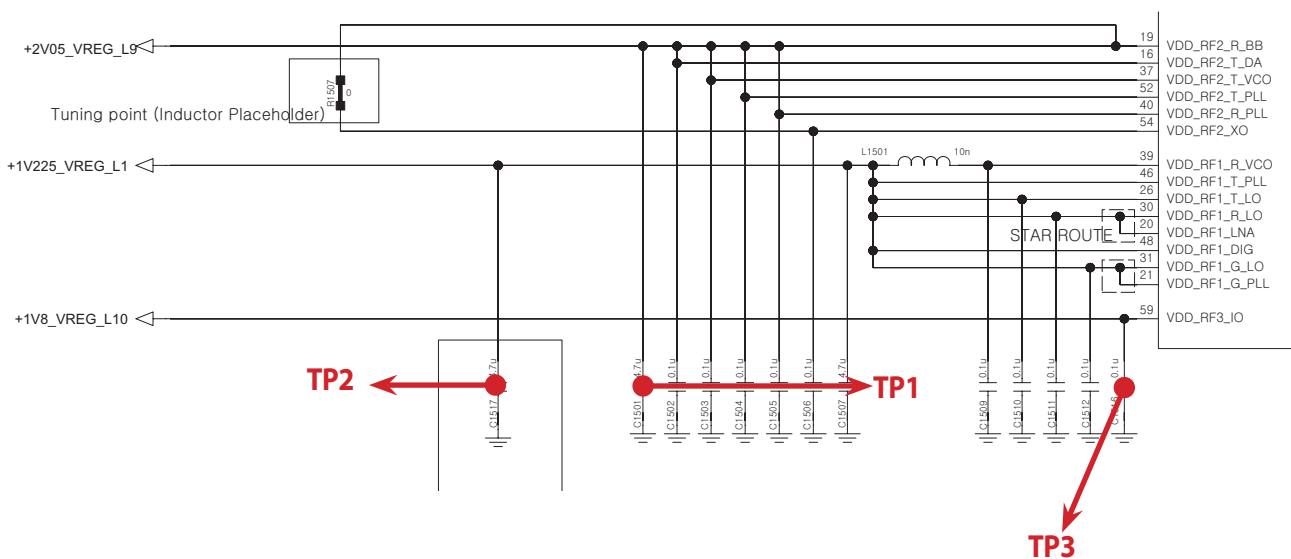
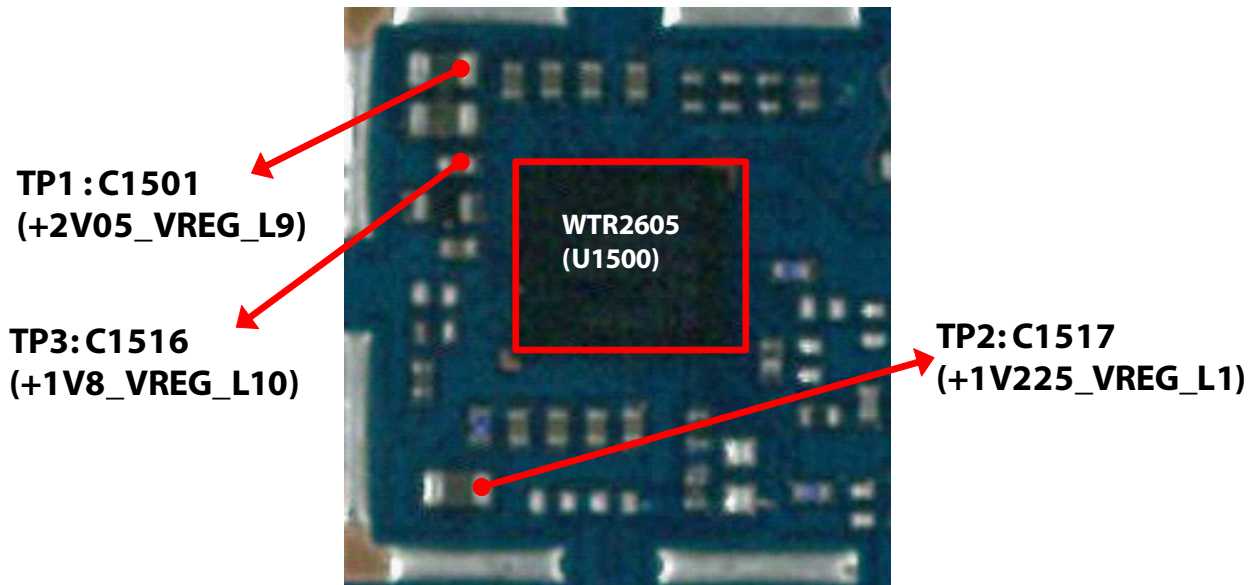
3. TROUBLE SHOOTING



3.4.1.1 Checking DC Power supply circuit



3. TROUBLE SHOOTING



3.4.1.2 Checking XO circuit

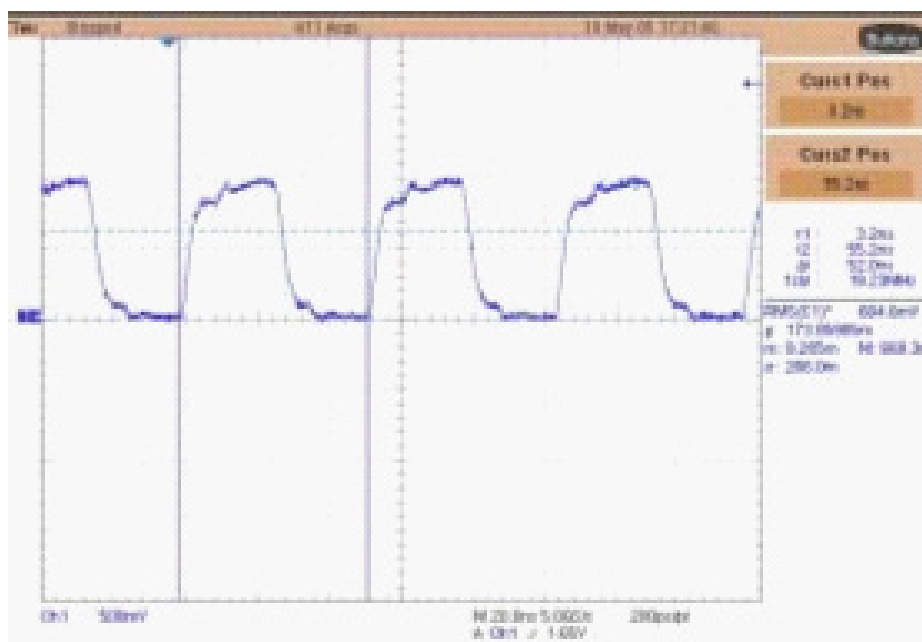
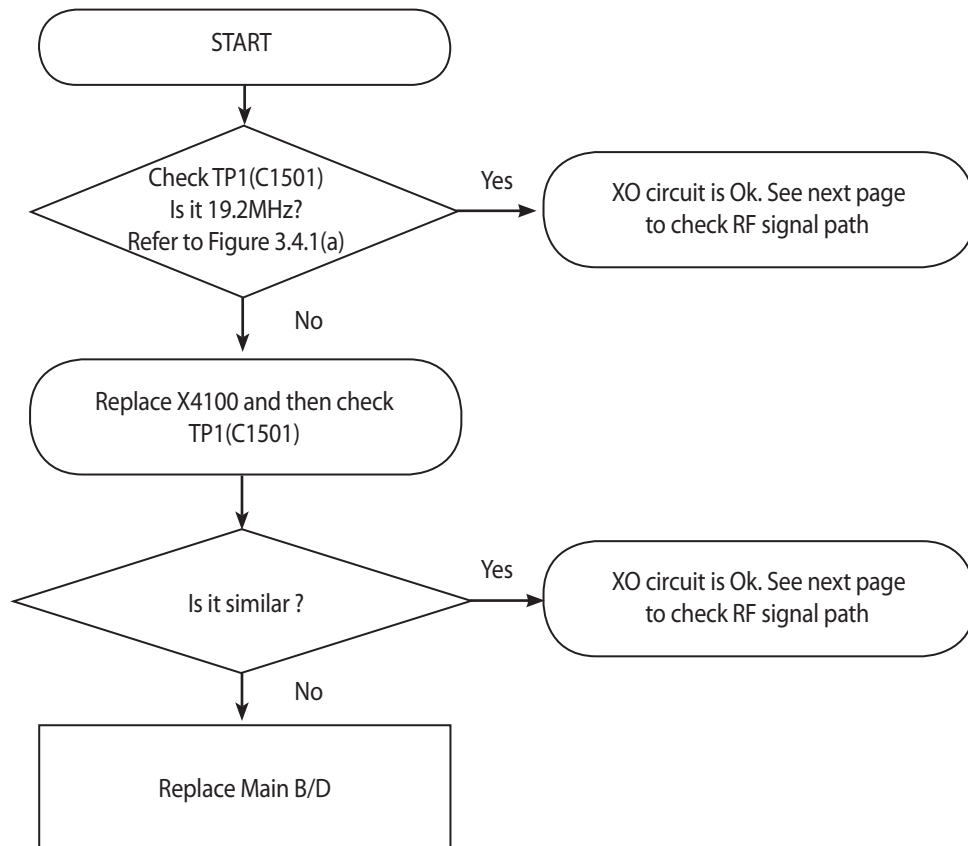
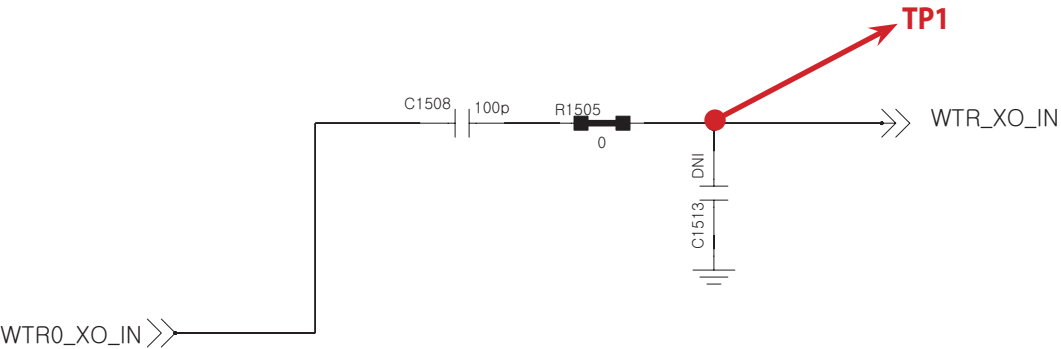
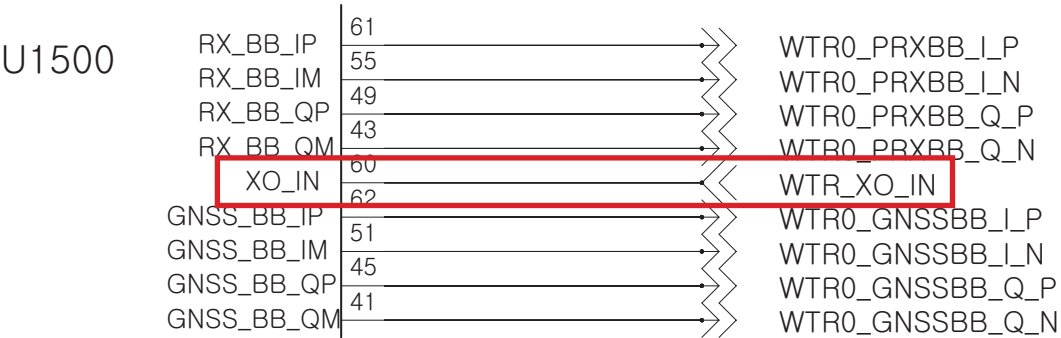
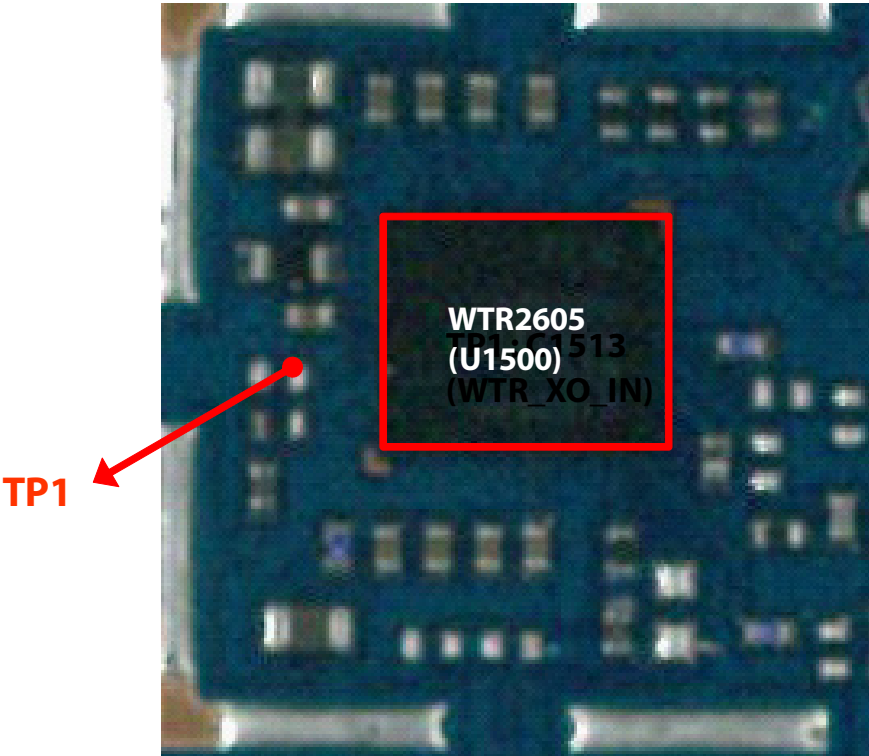
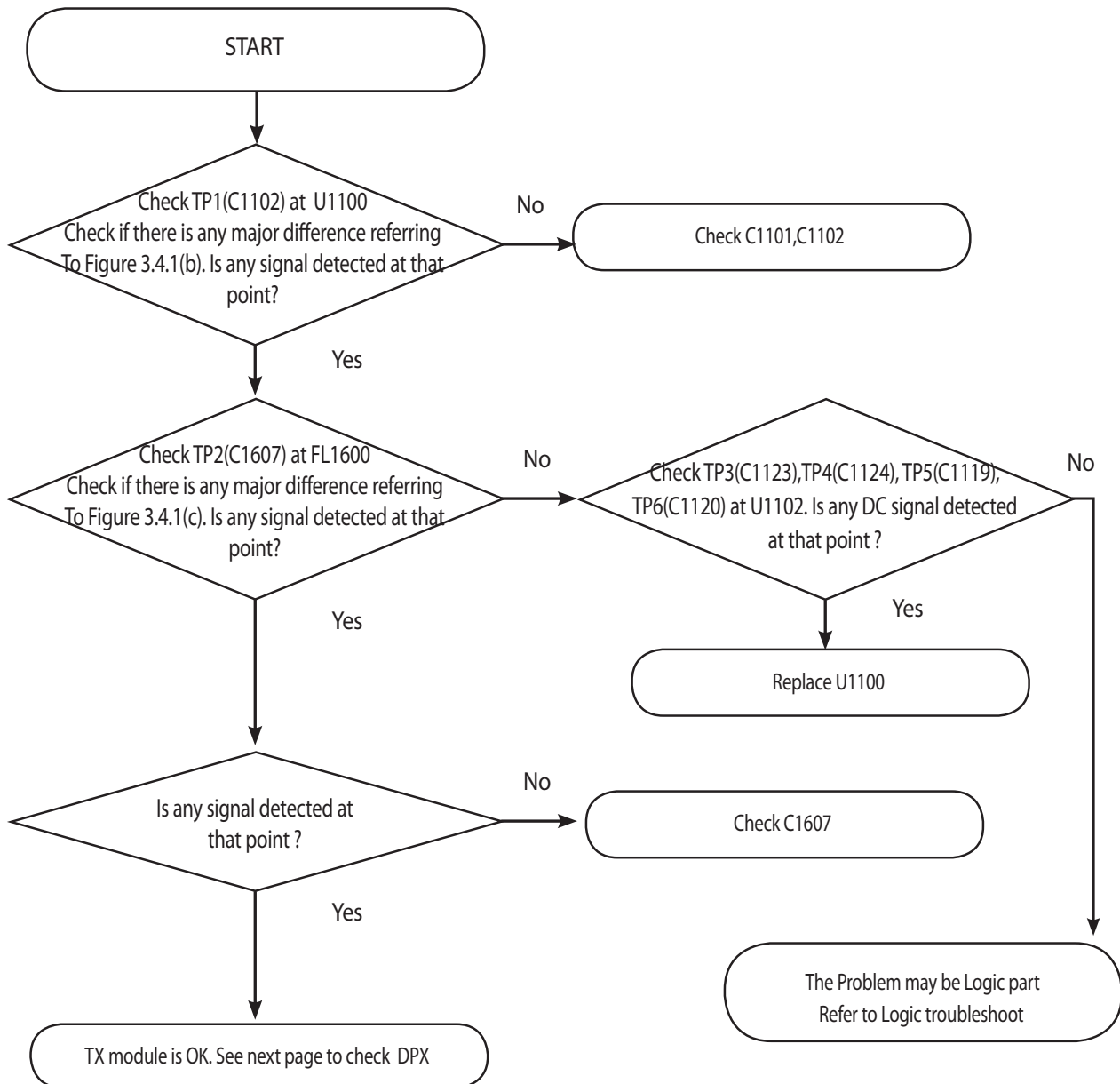


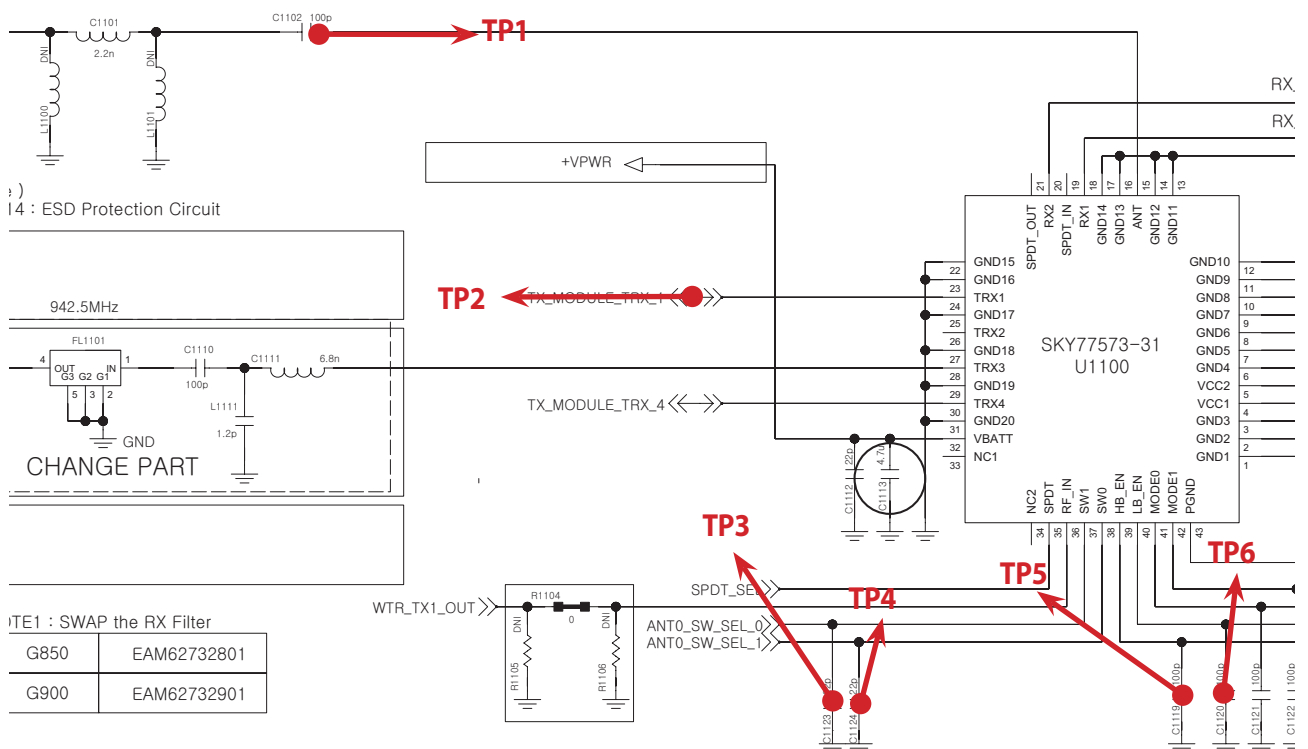
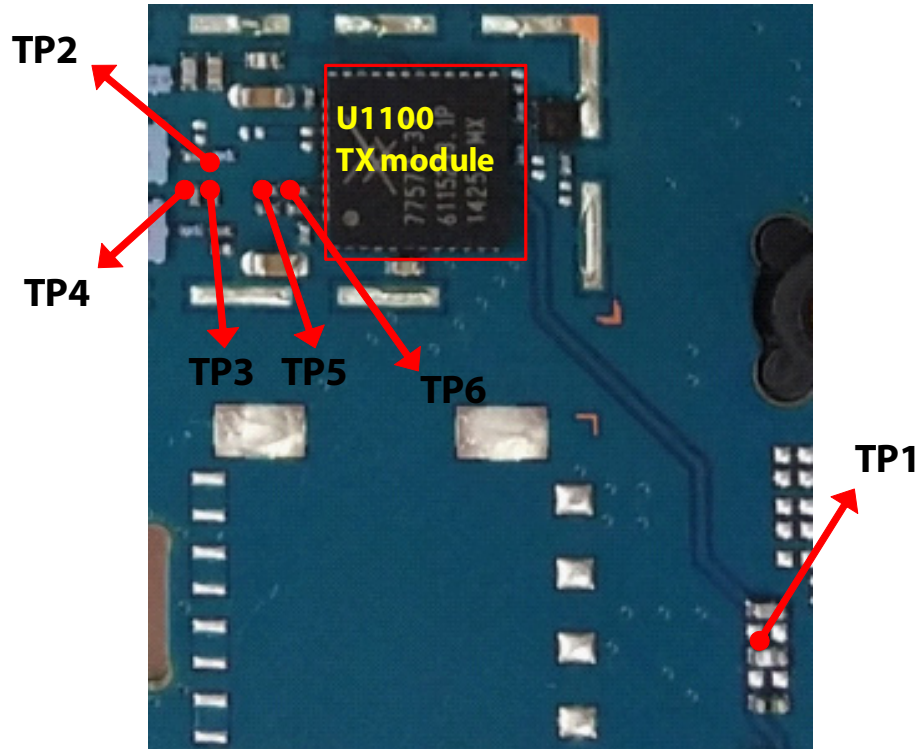
Figure 3.4.1 (a)



3.4.1.3 Checking RF signal path (TX Module)



3. TROUBLE SHOOTING



TP1

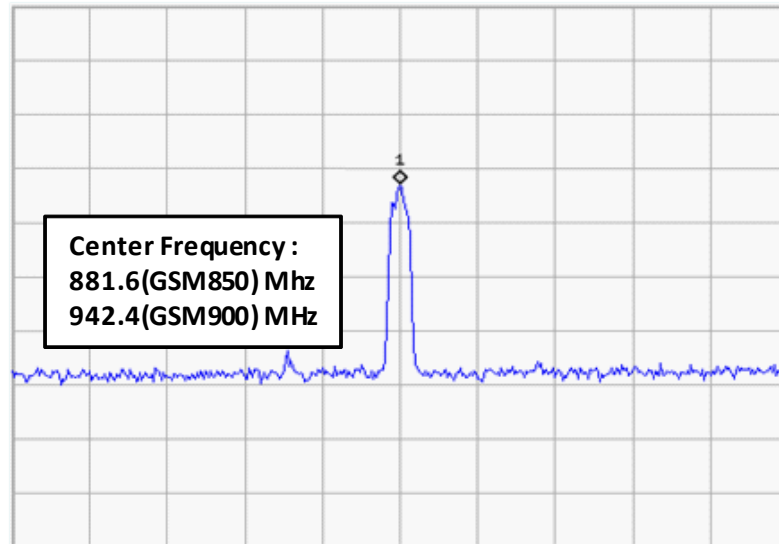


Figure 3.4.1 (b)

TP2

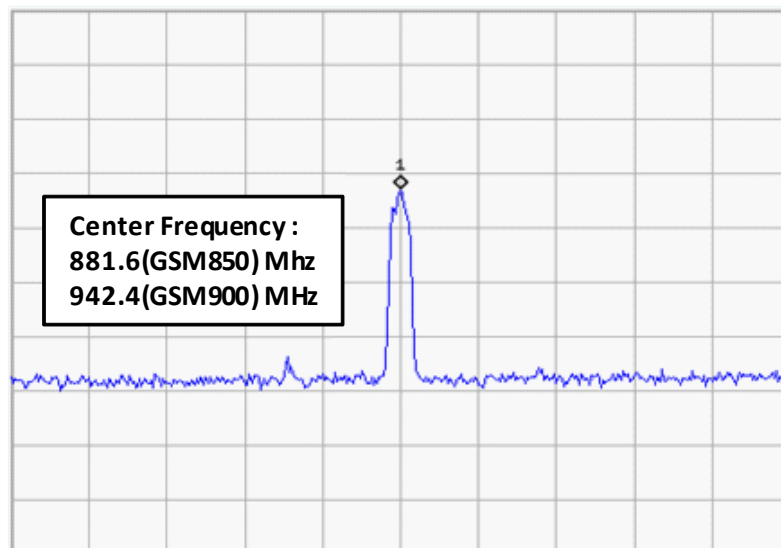
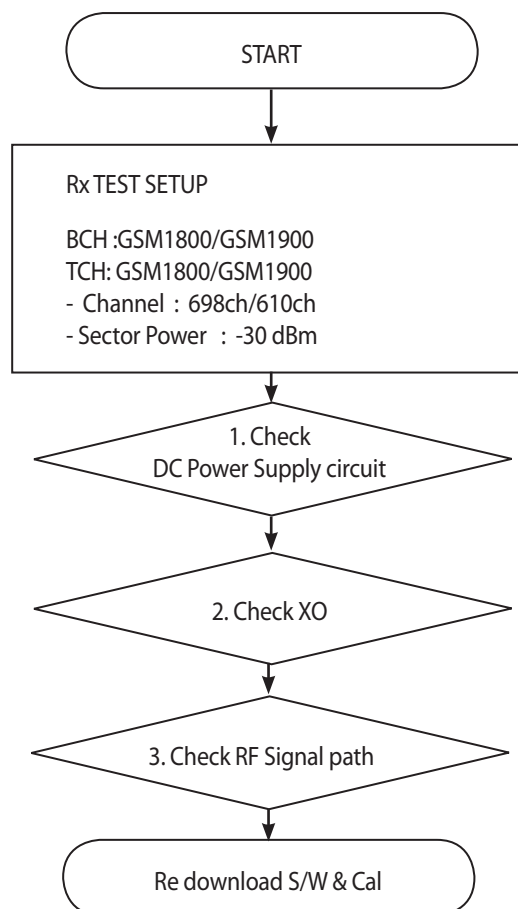


Figure 3.4.1 (C)

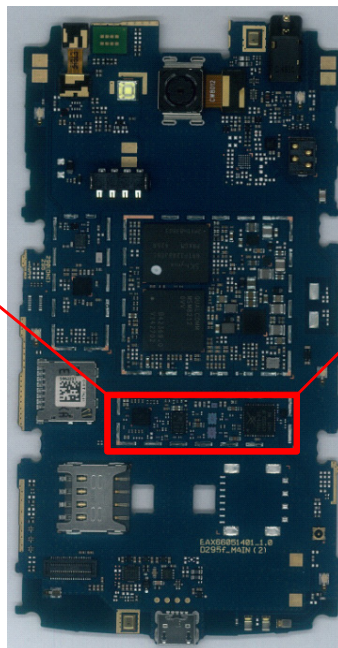
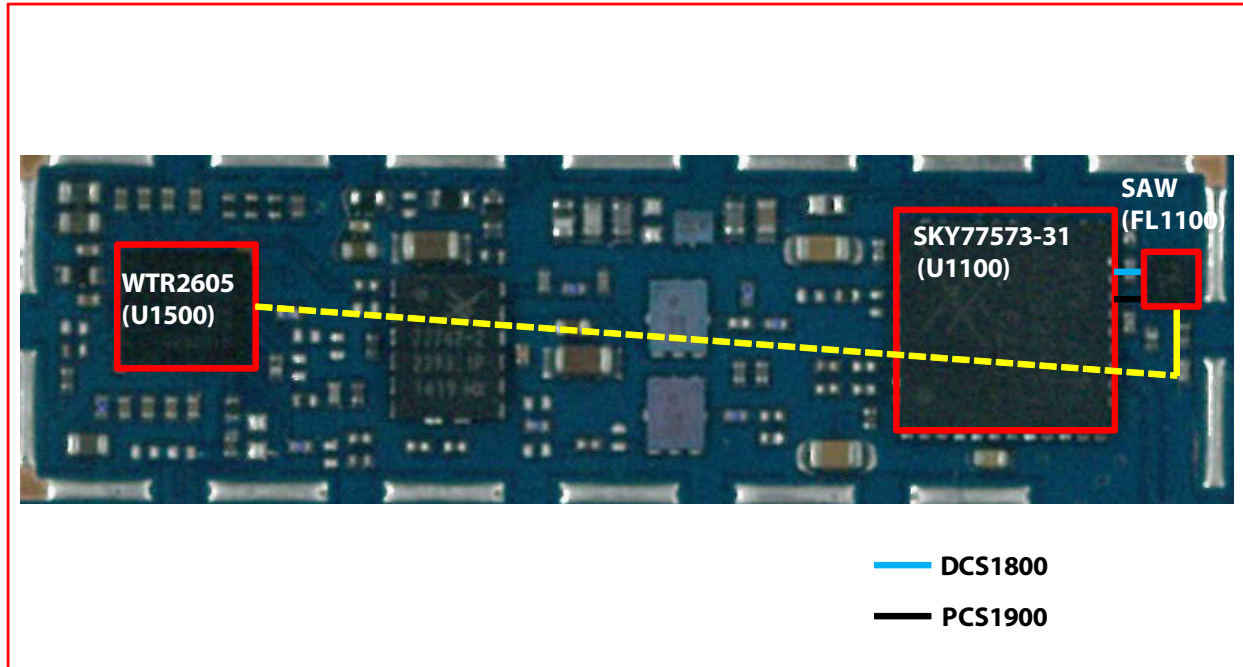
3.5 GSM Rx Part Troubleshooting

3.5.1 GSM1800/1900 Rx

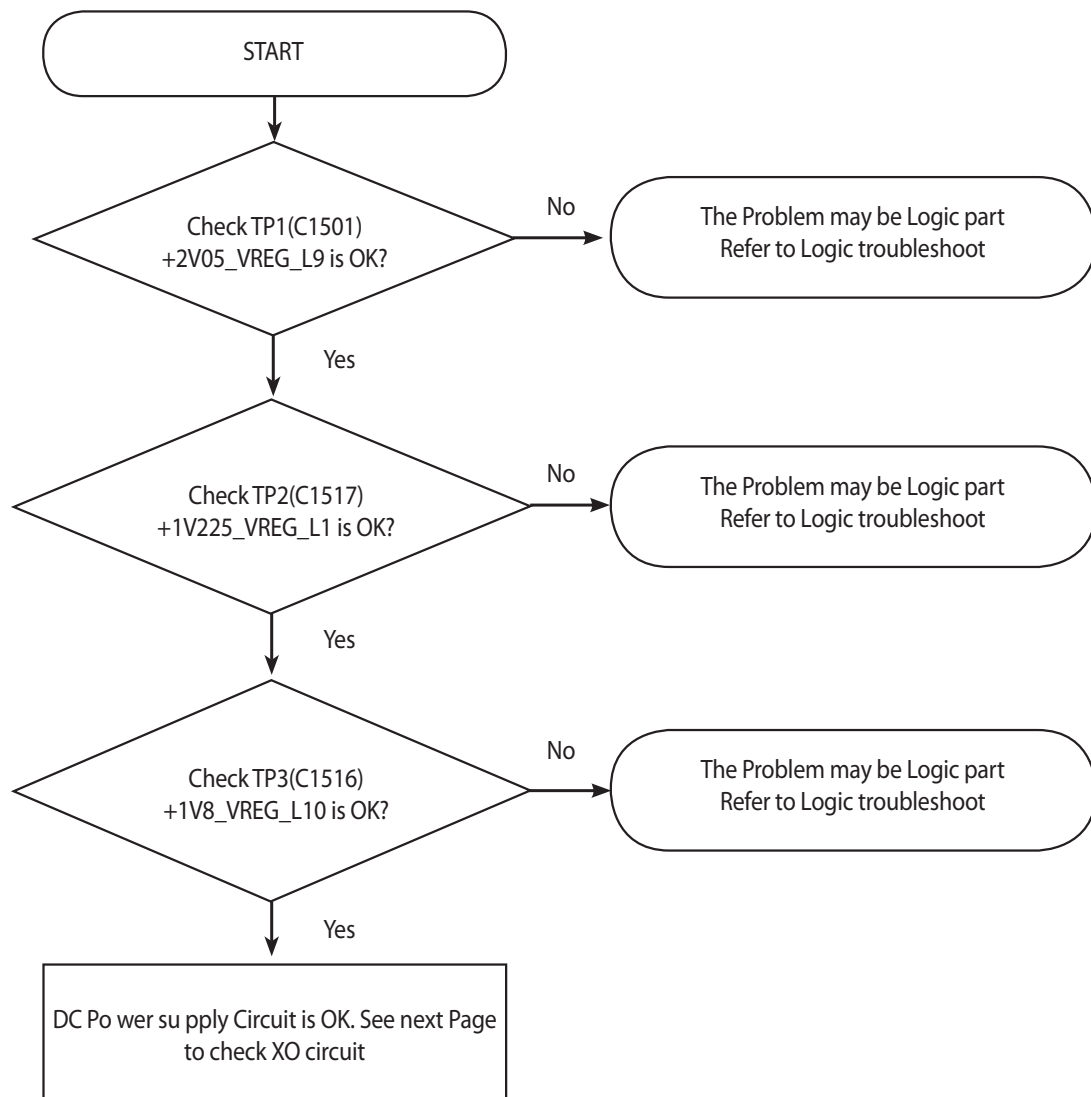


* there are no Test Points on the IQ Data Line. So It is impossible to check the IQ Data Wave form

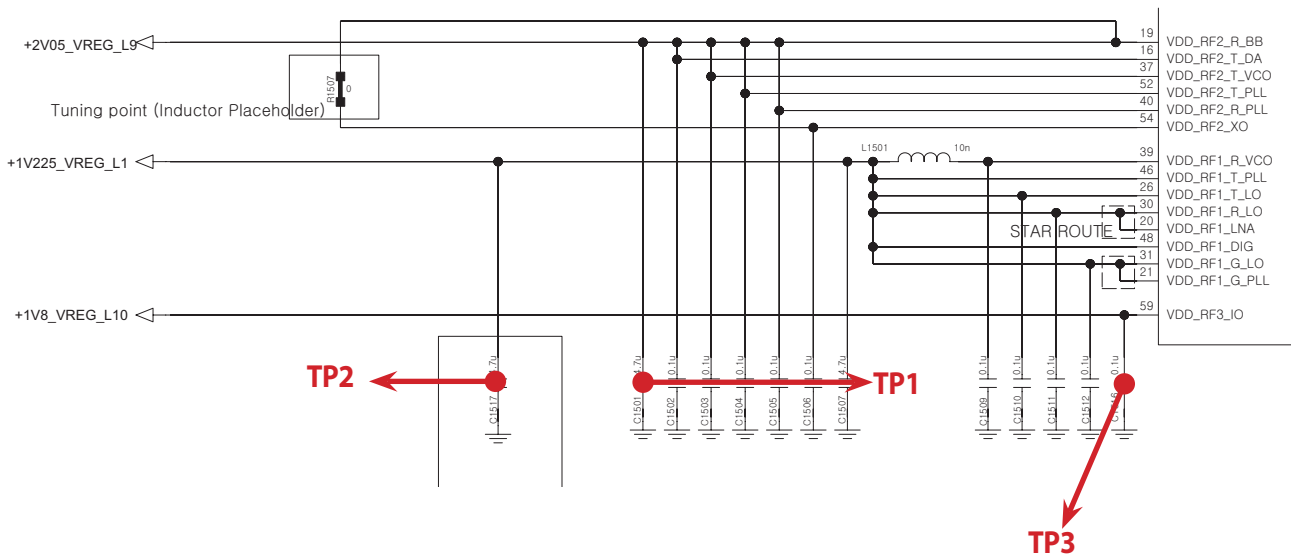
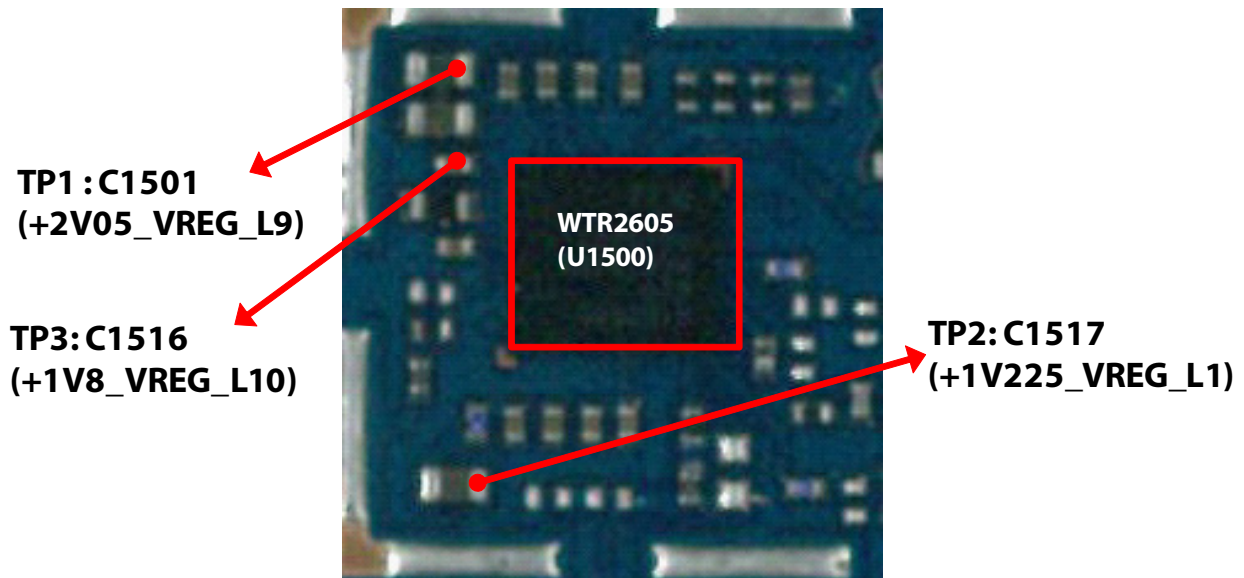
3. TROUBLE SHOOTING



3.5.1.1 Checking DC Power supply circuit



3. TROUBLE SHOOTING



3.5.1.2 Checking XO circuit

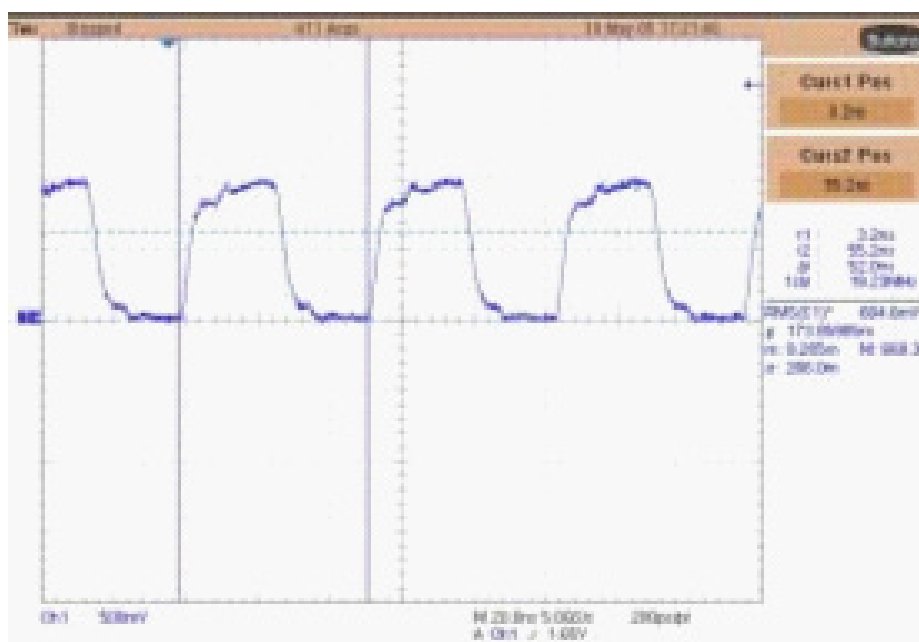
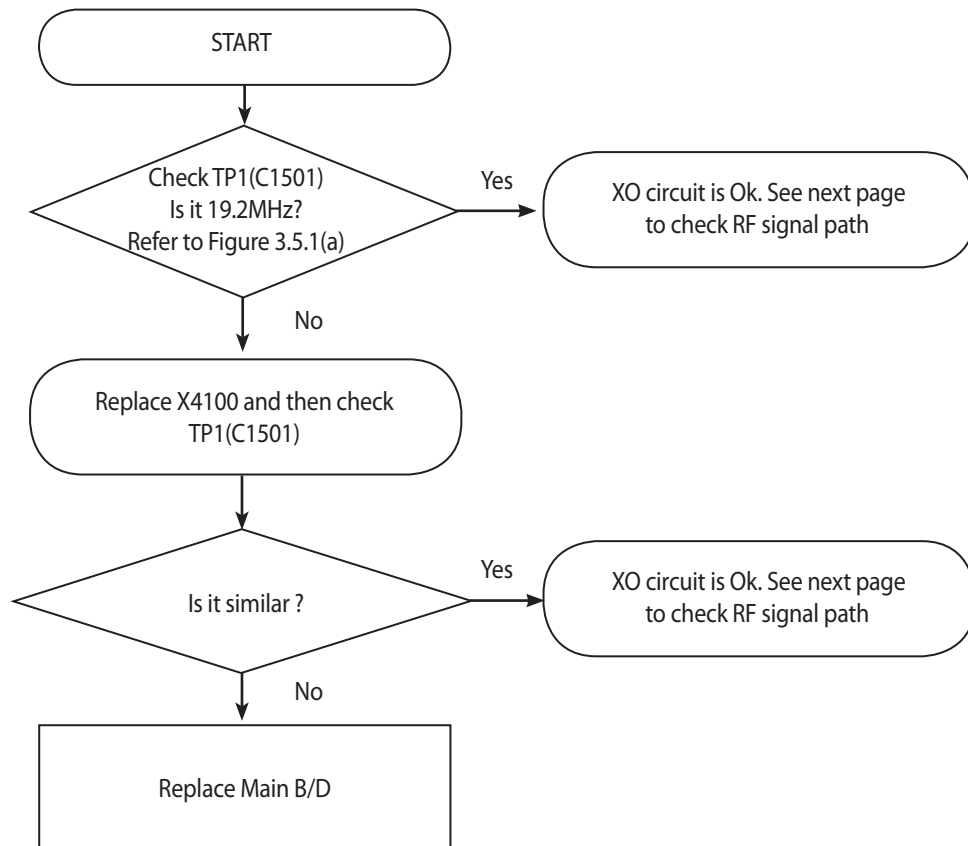
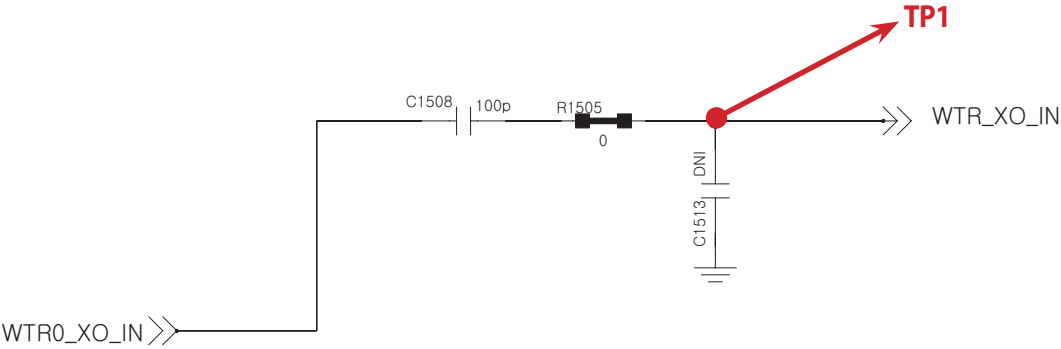
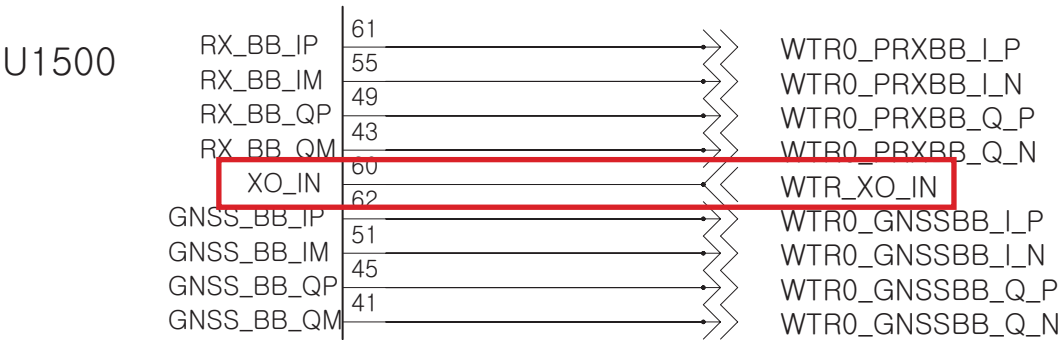
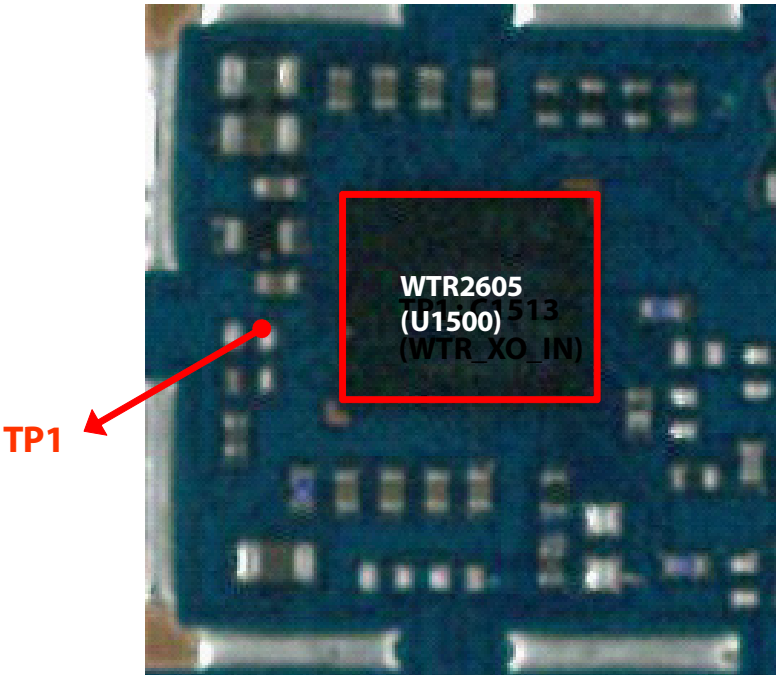
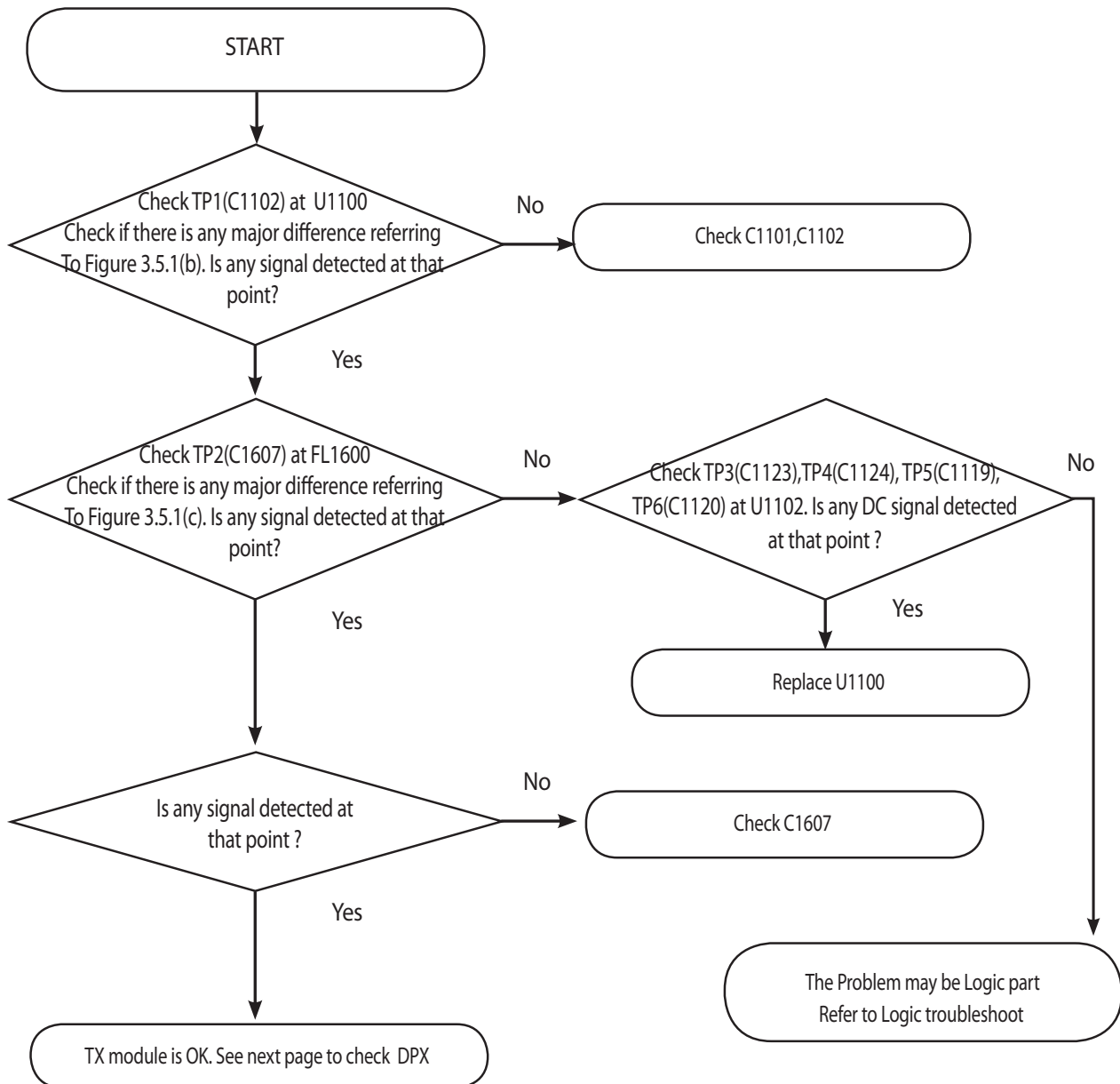


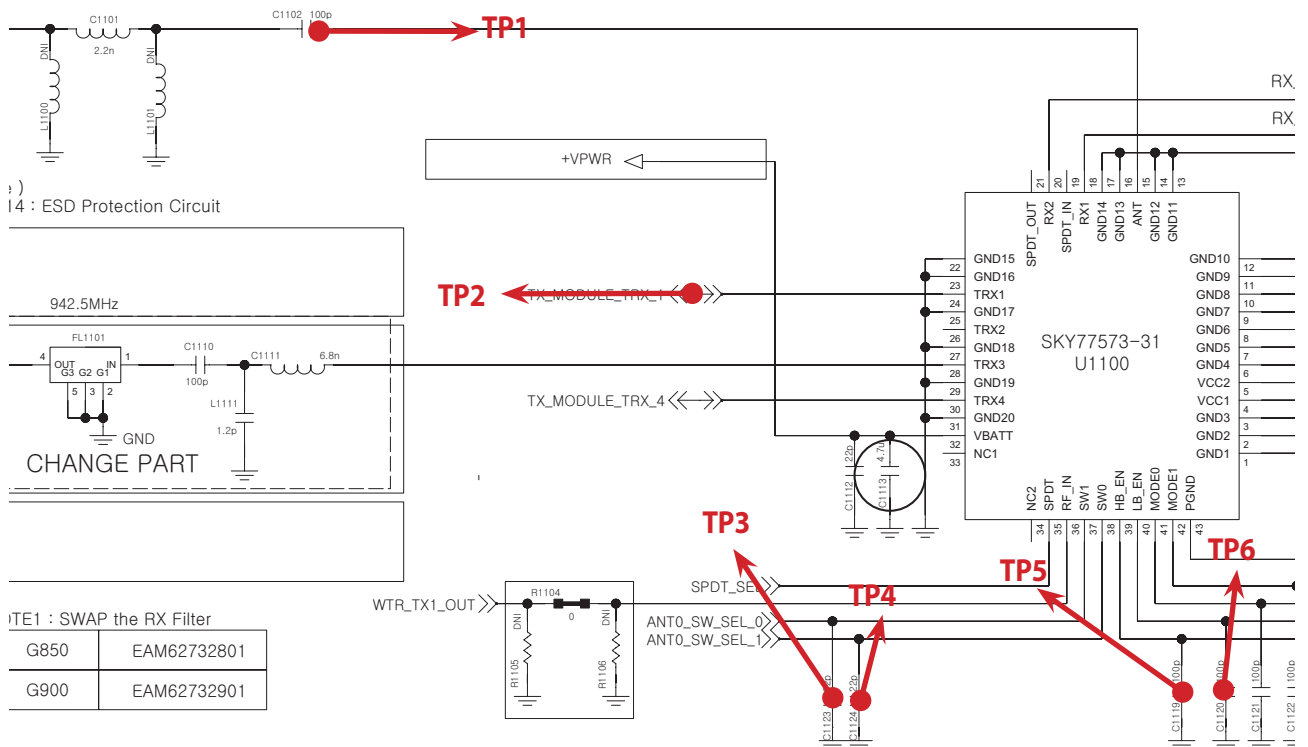
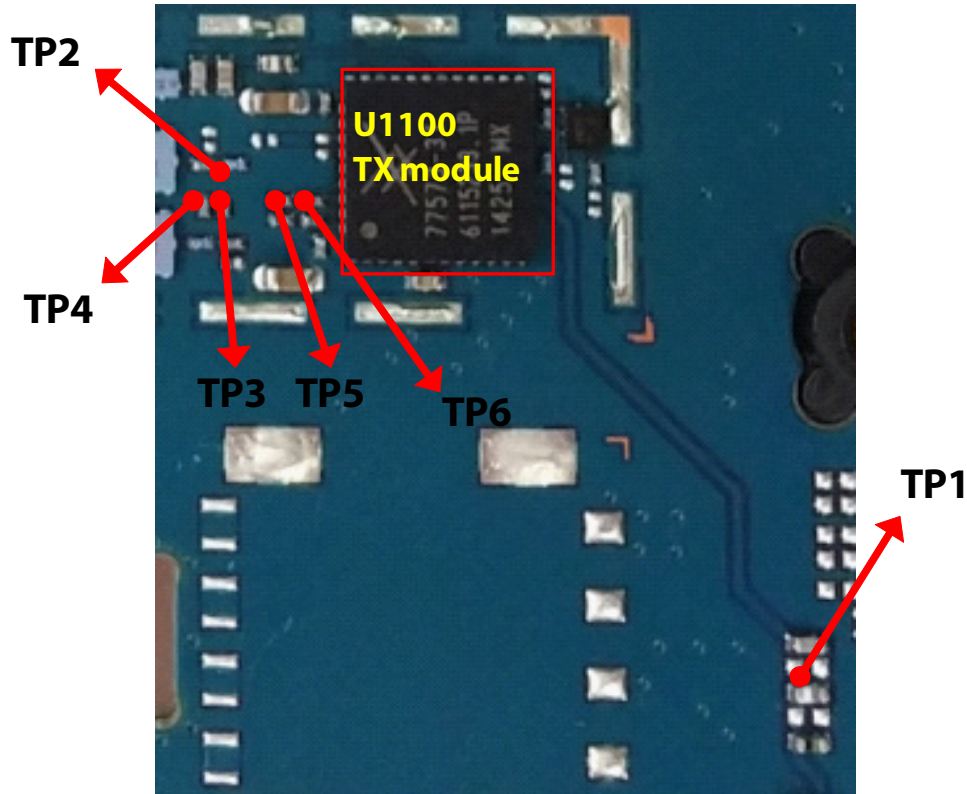
Figure 3.5.1 (a)



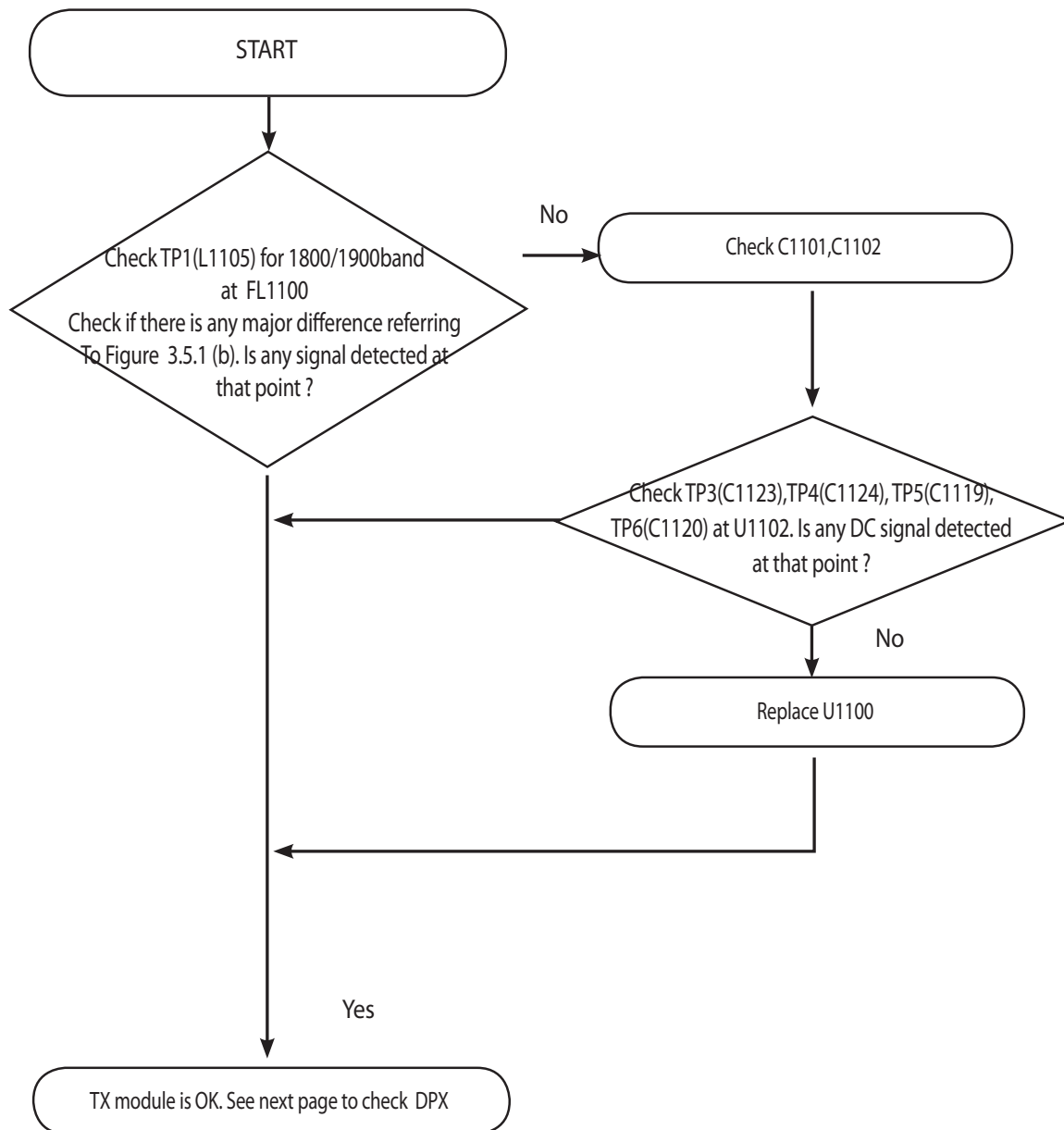
3.5.1.3 Checking RF signal path (TX Module)

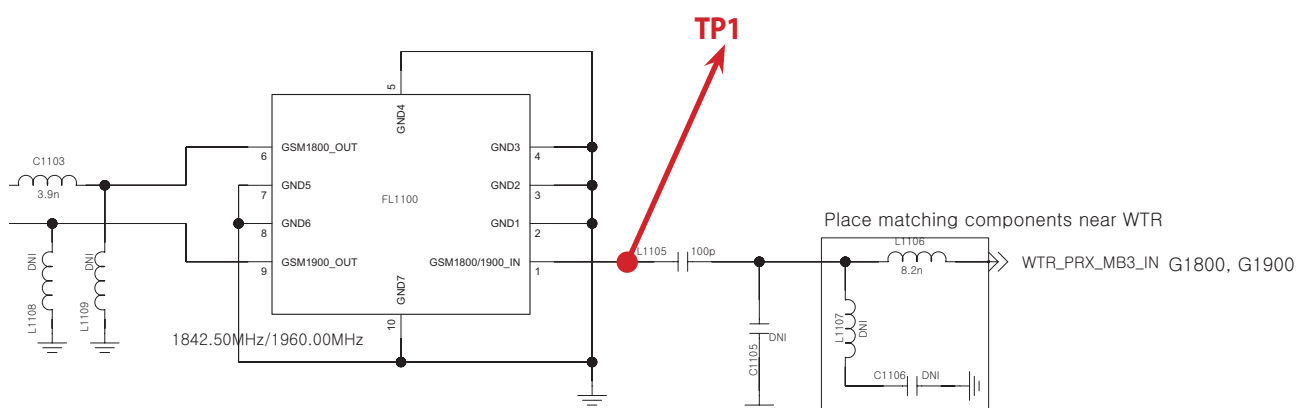
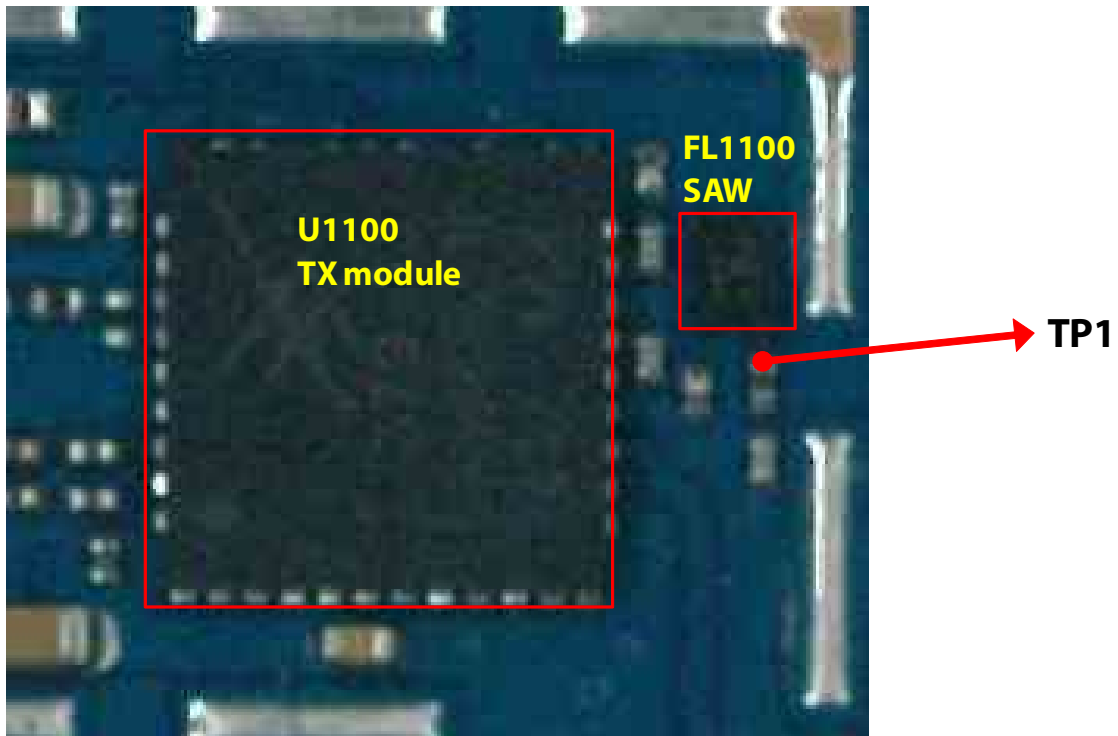


3. TROUBLE SHOOTING



3.5.1.4 Checking RF Flow(SAW filter)





TP1

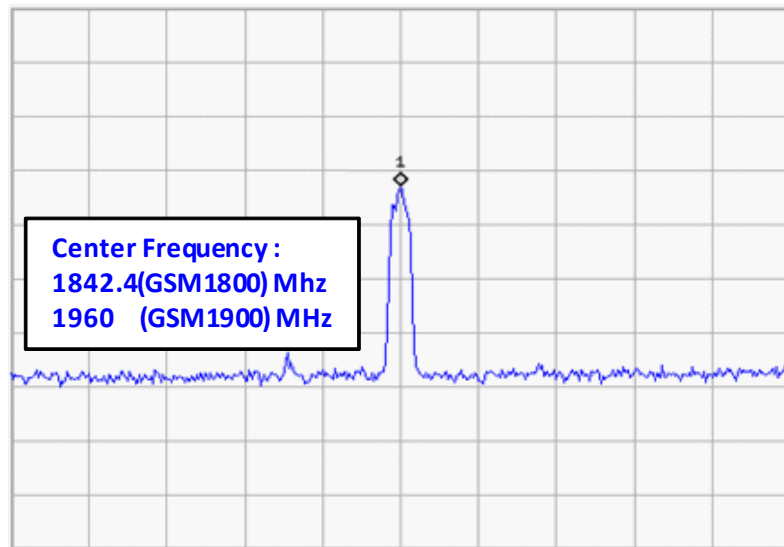


Figure 3.5.1 (b)

TP2

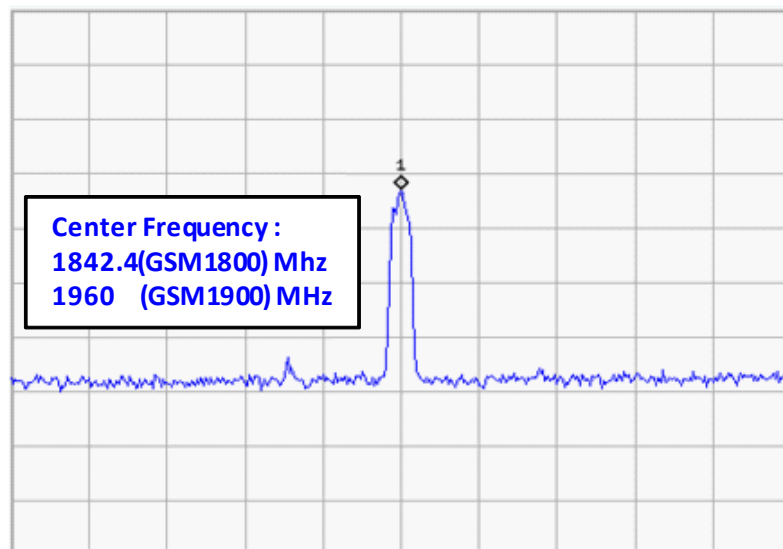
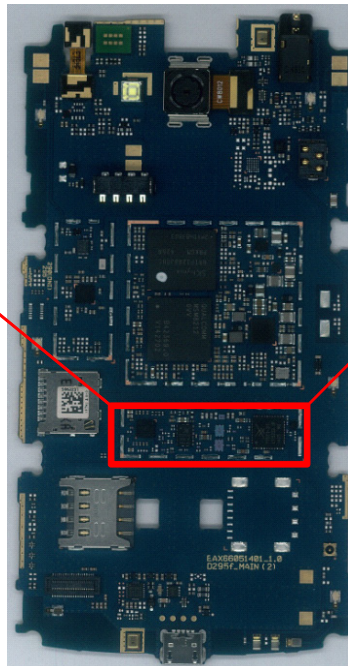
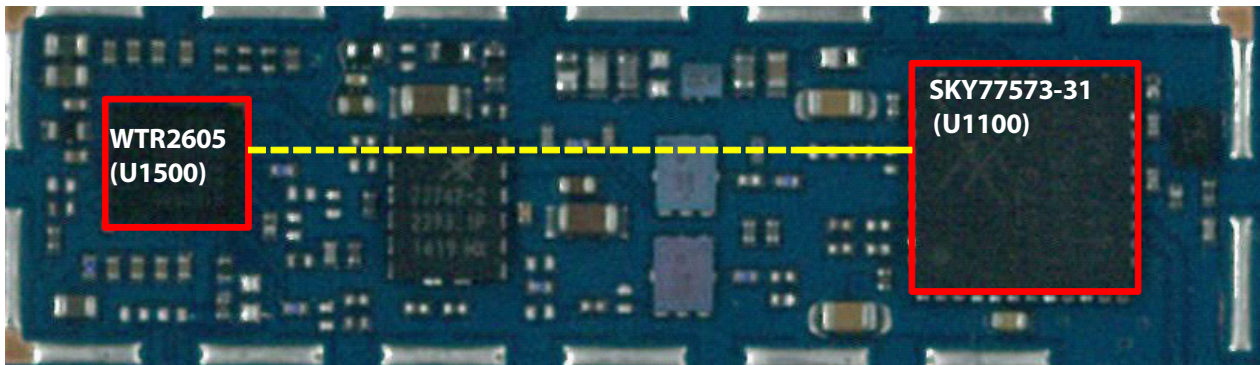


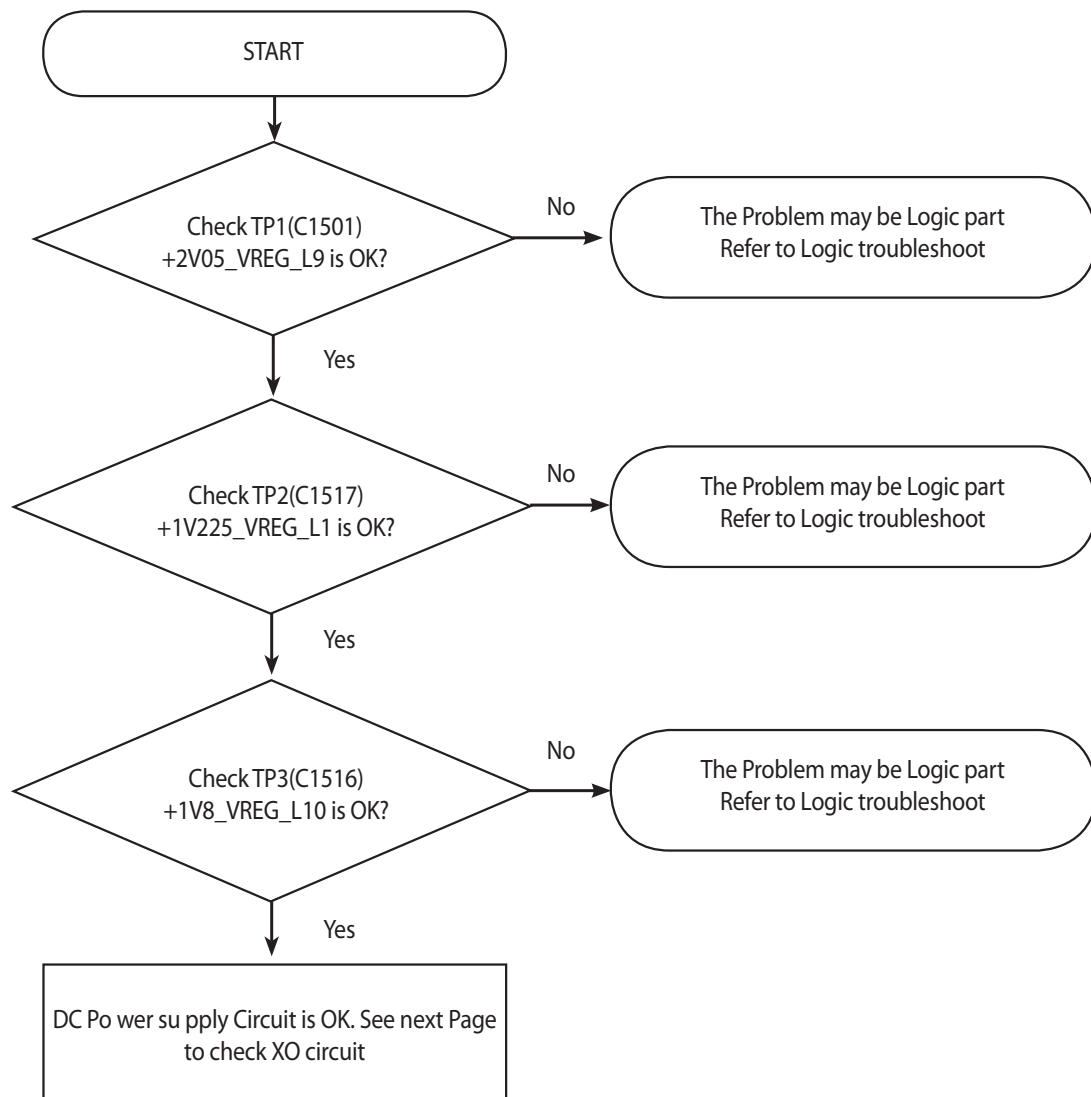
Figure 3.5.1 (c)

3.6 GSM Tx Part Troubleshooting

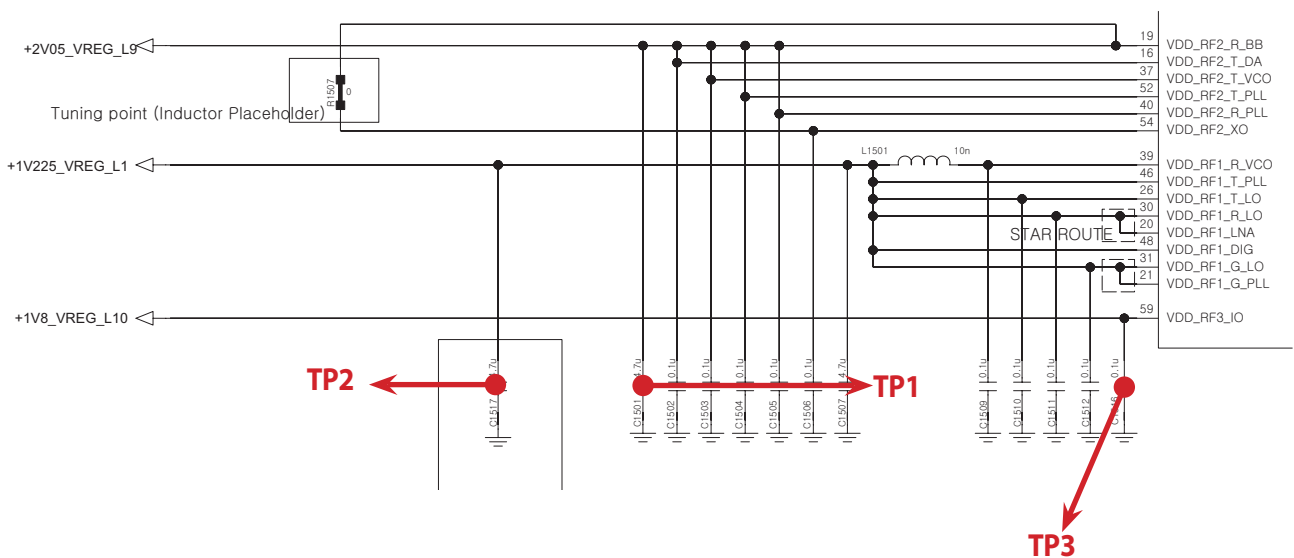
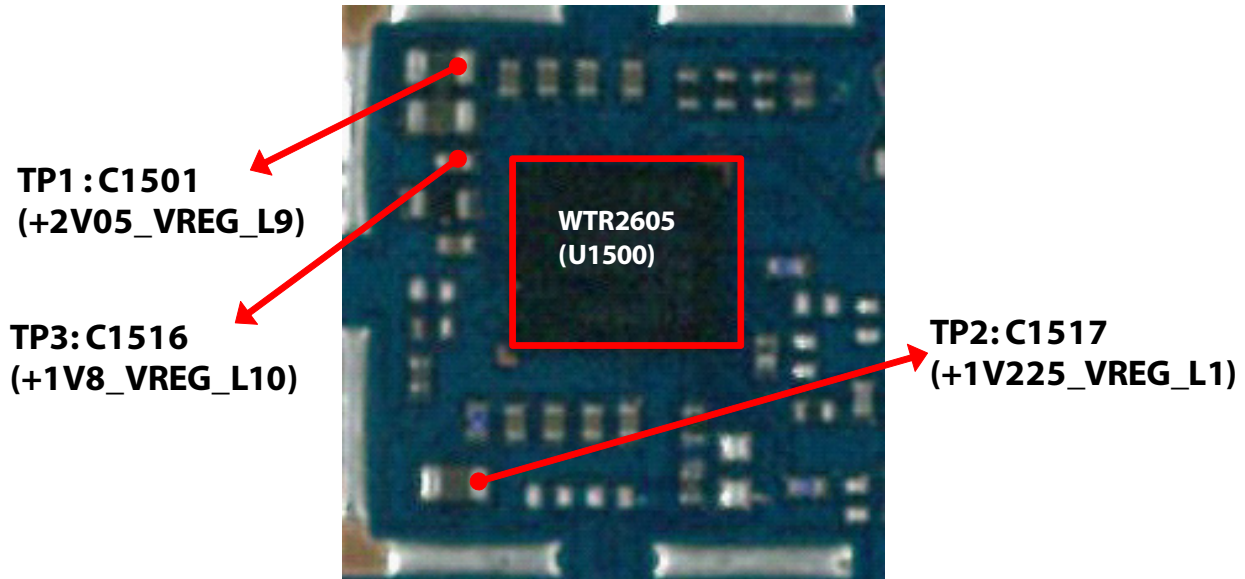
3.6.1 GSM850/900 Tx



3.6.1.1 Checking DC Power supply circuit



3. TROUBLE SHOOTING



3.6.1.2 Checking XO circuit

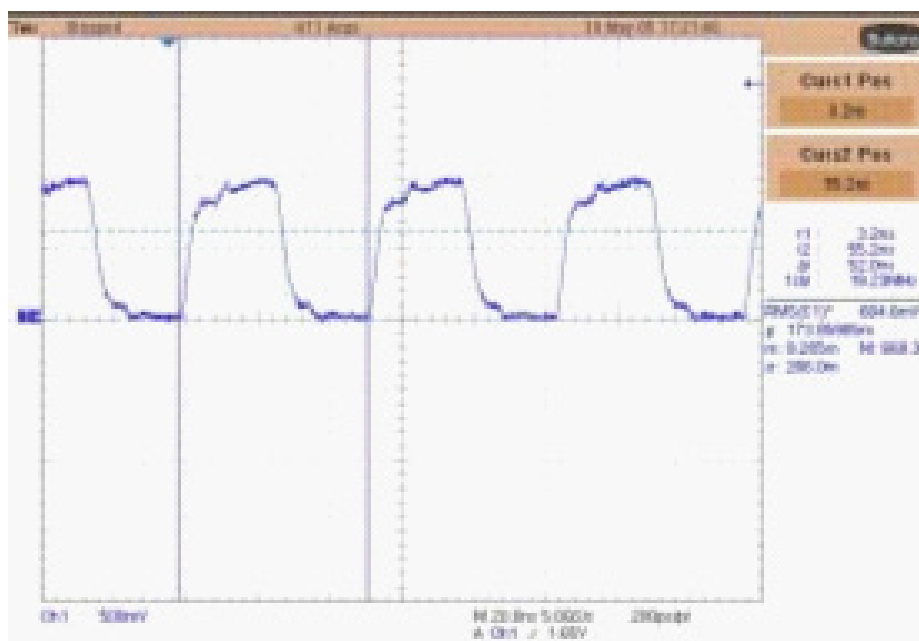
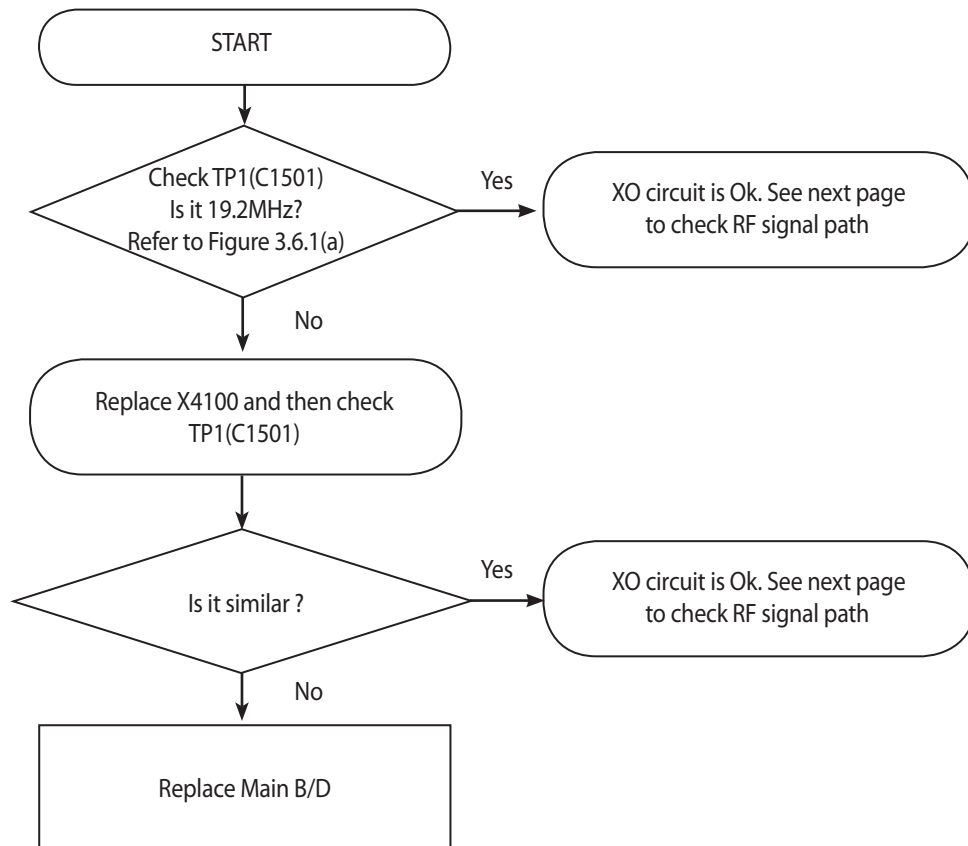
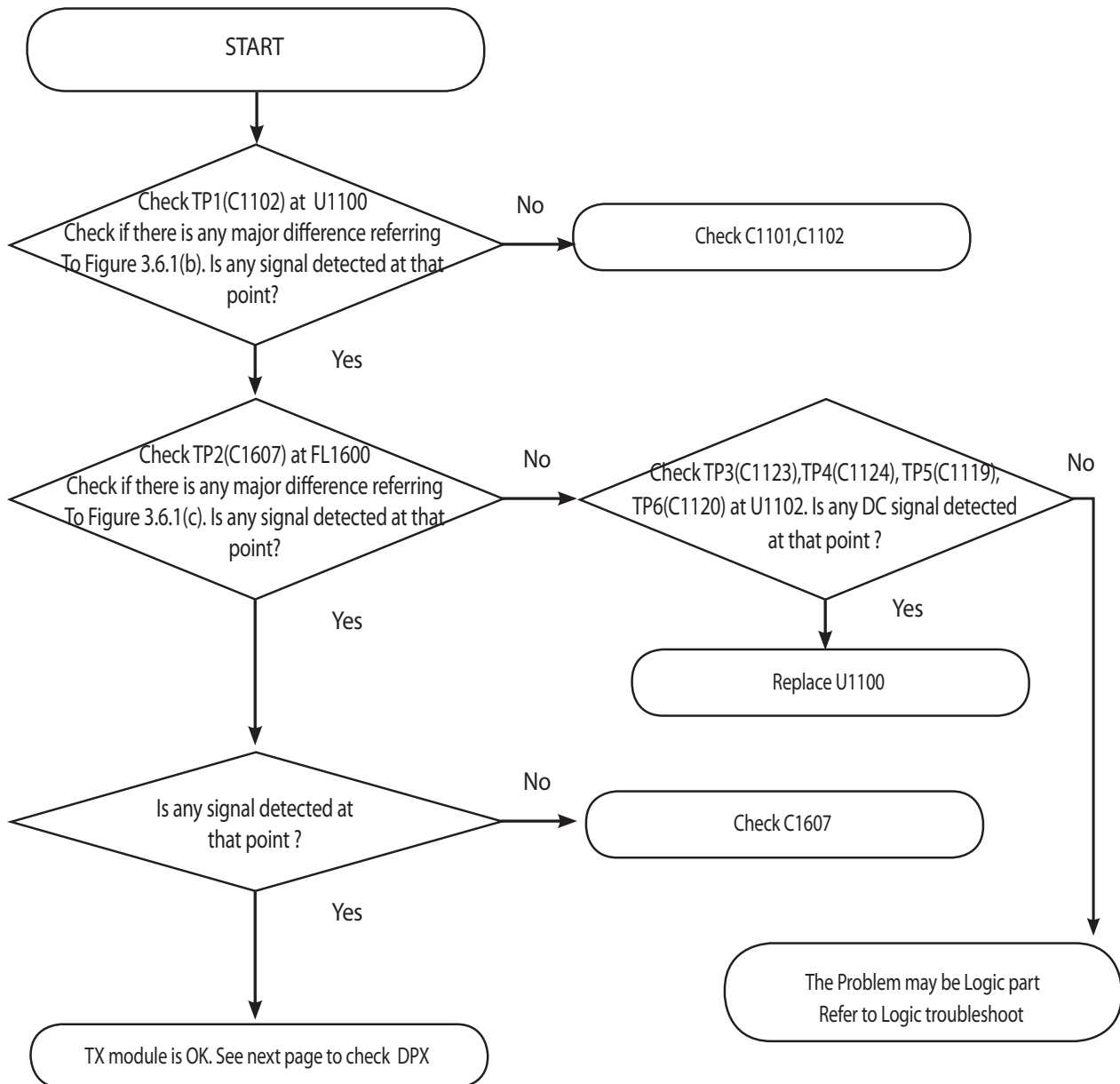


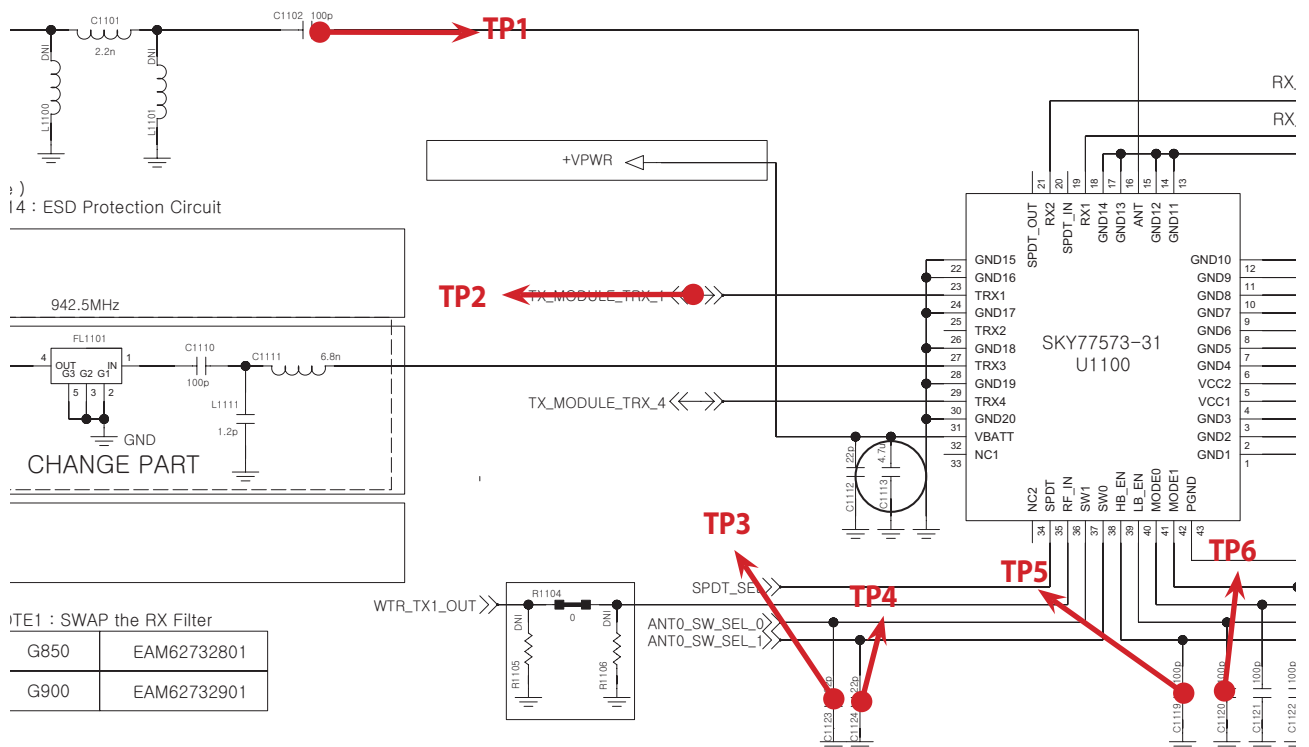
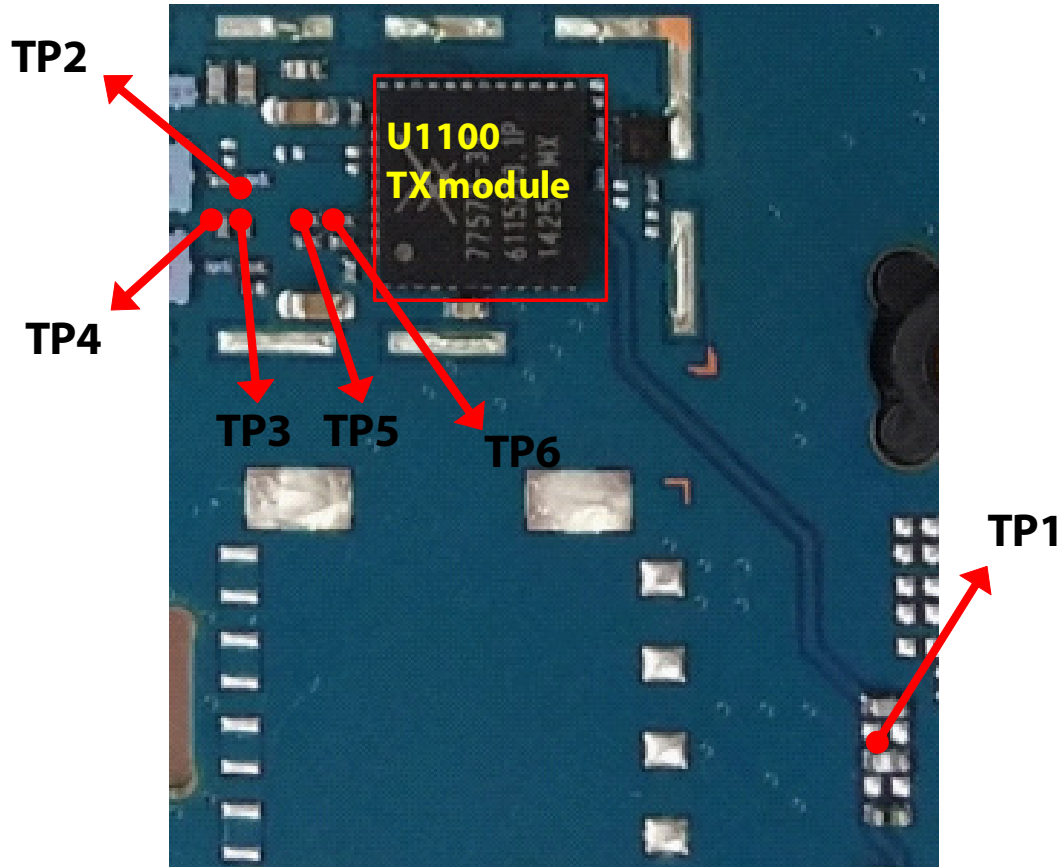
Figure 3.6.1 (a)



3.6.1.3 Checking RF signal path (TX Module)



3. TROUBLE SHOOTING



TP1

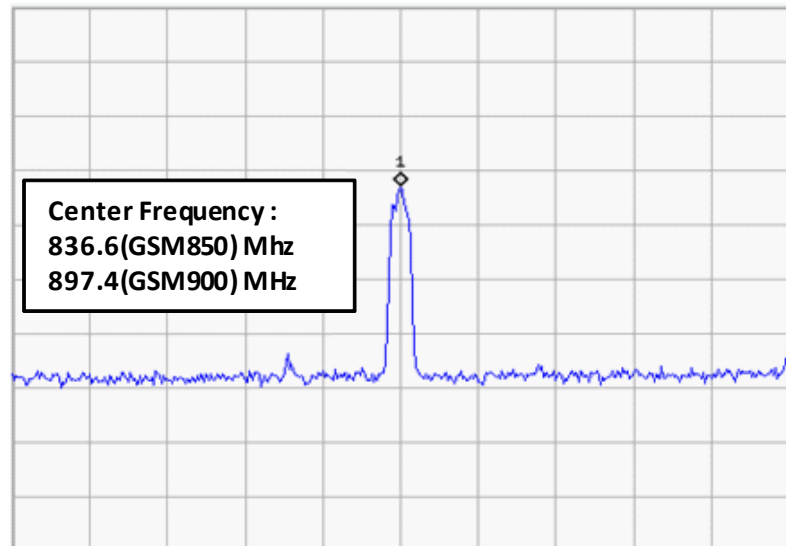


Figure 3.6.1 (b)

TP2

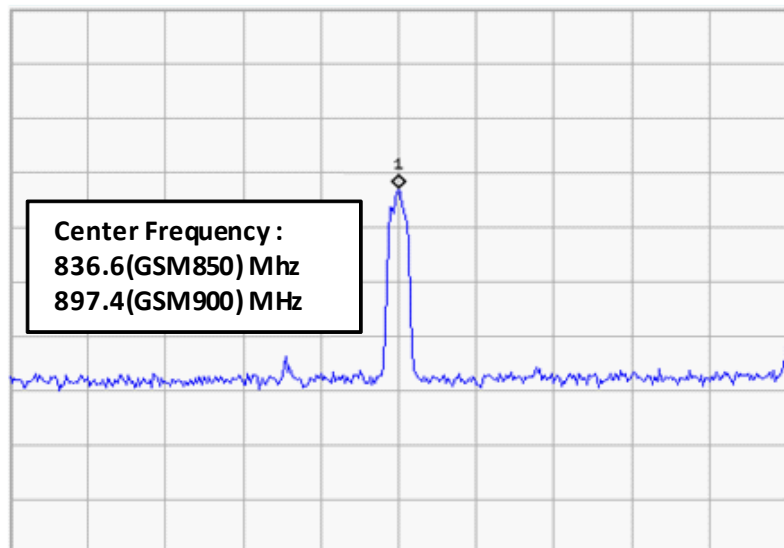
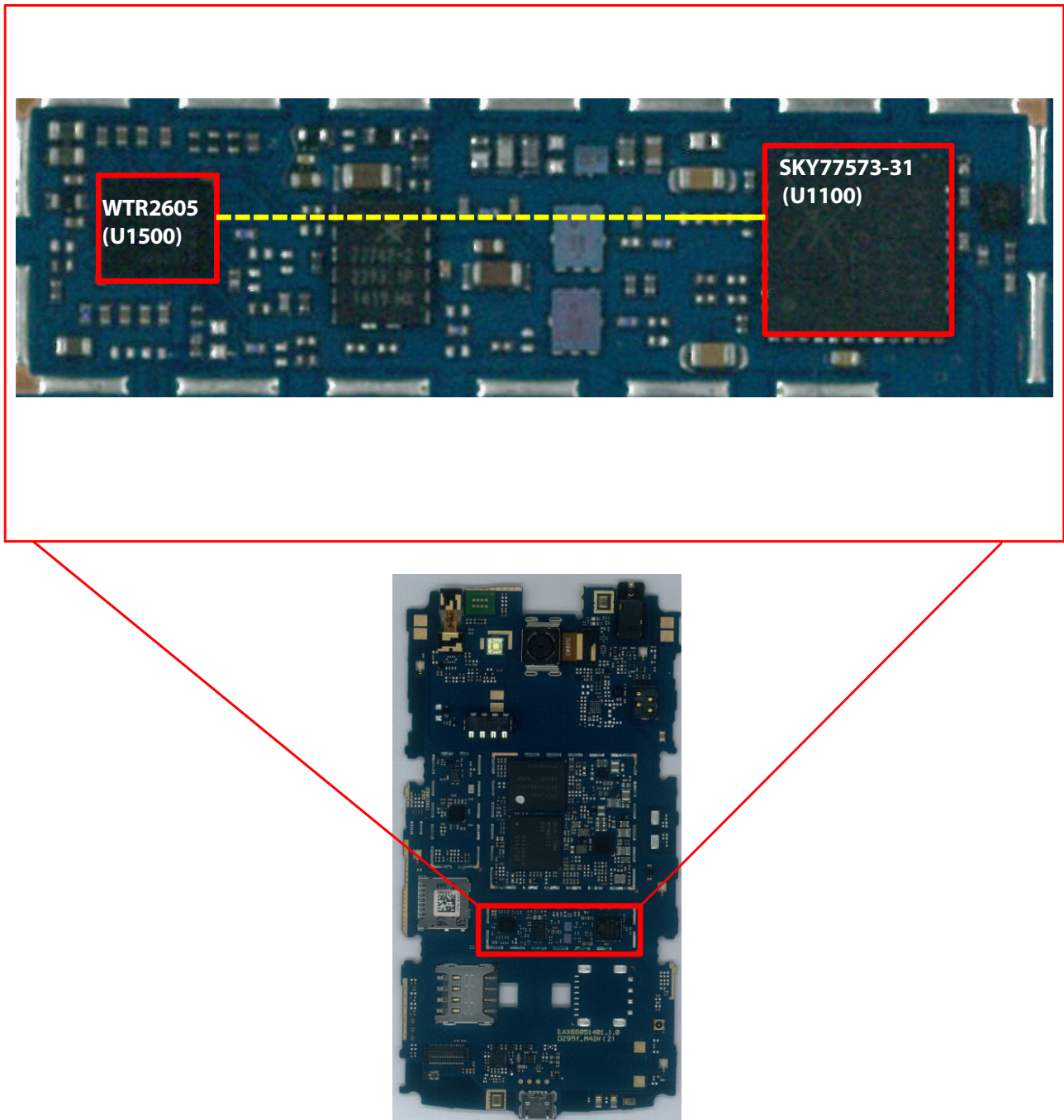


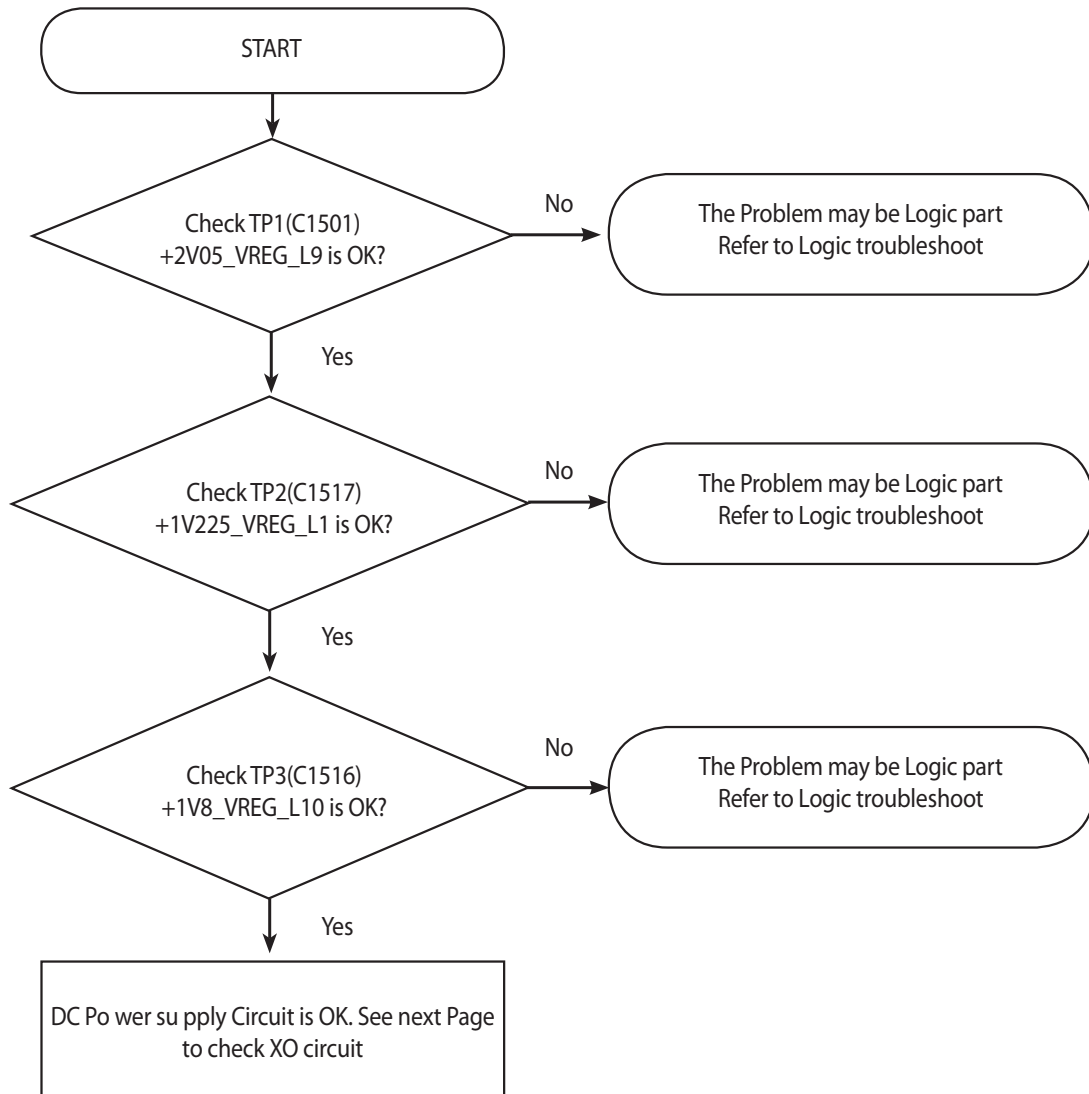
Figure 3.6.1 (c)

3.7 GSM Tx Part Troubleshooting

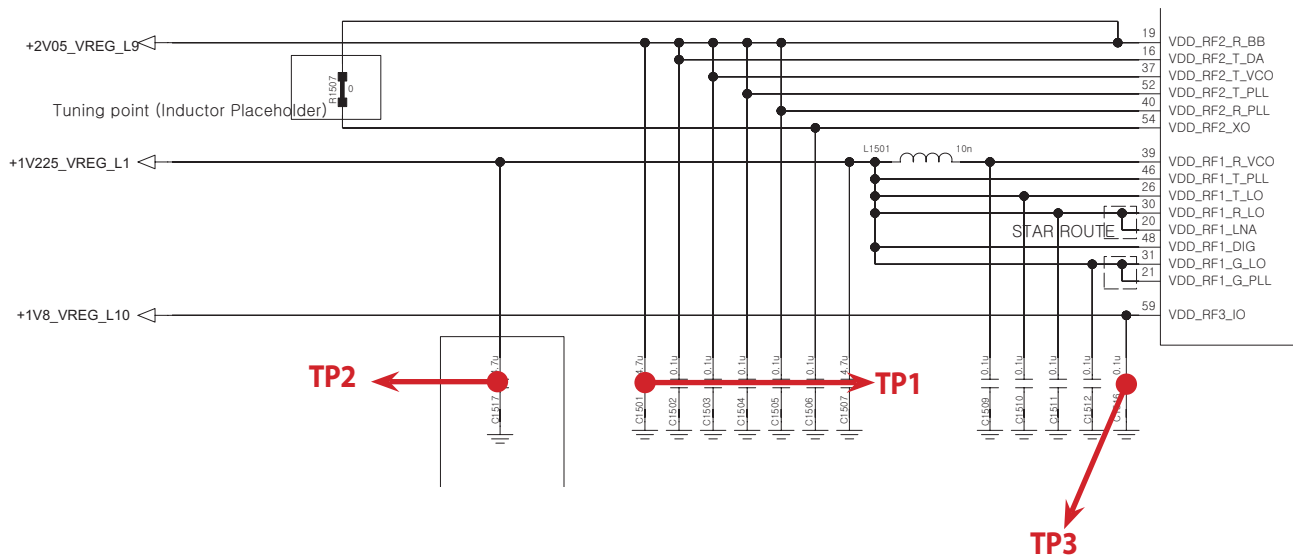
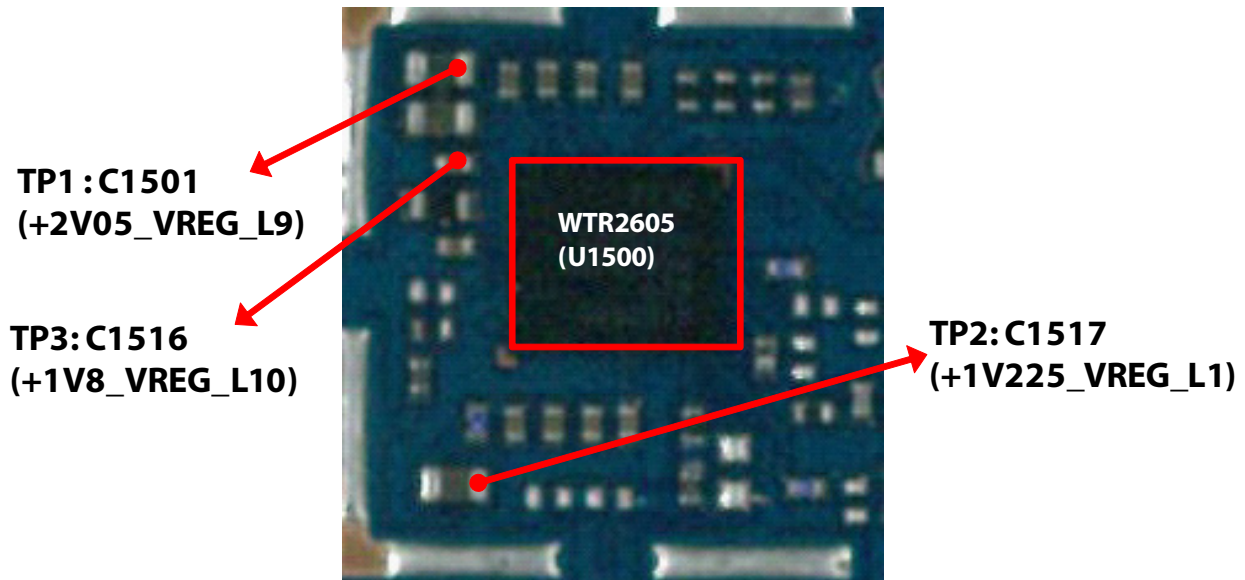
3.7.1 GSM1800/1900_Tx



3.7.1.1 Checking DC Power supply circuit



3. TROUBLE SHOOTING



3.7.1.2 Checking XO circuit

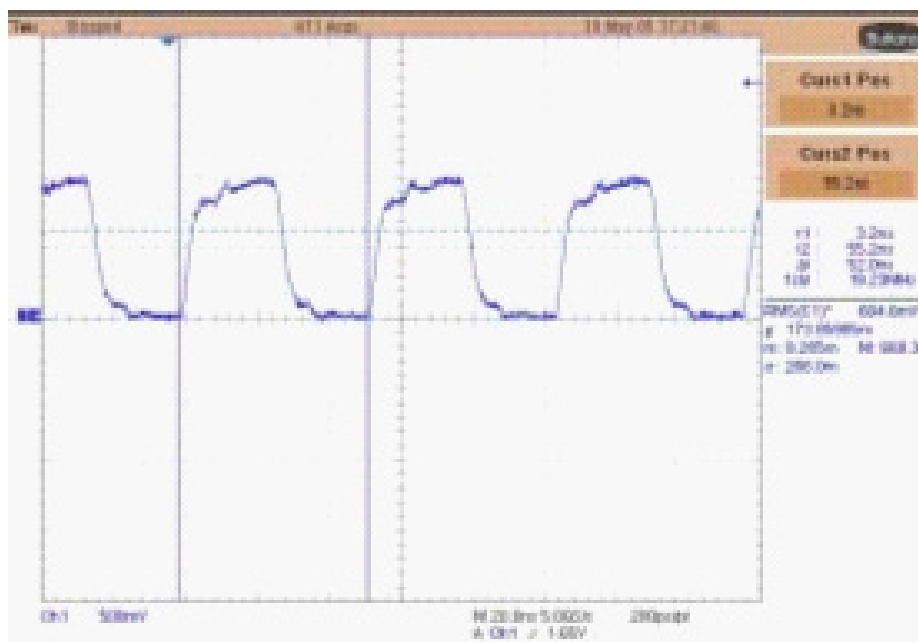
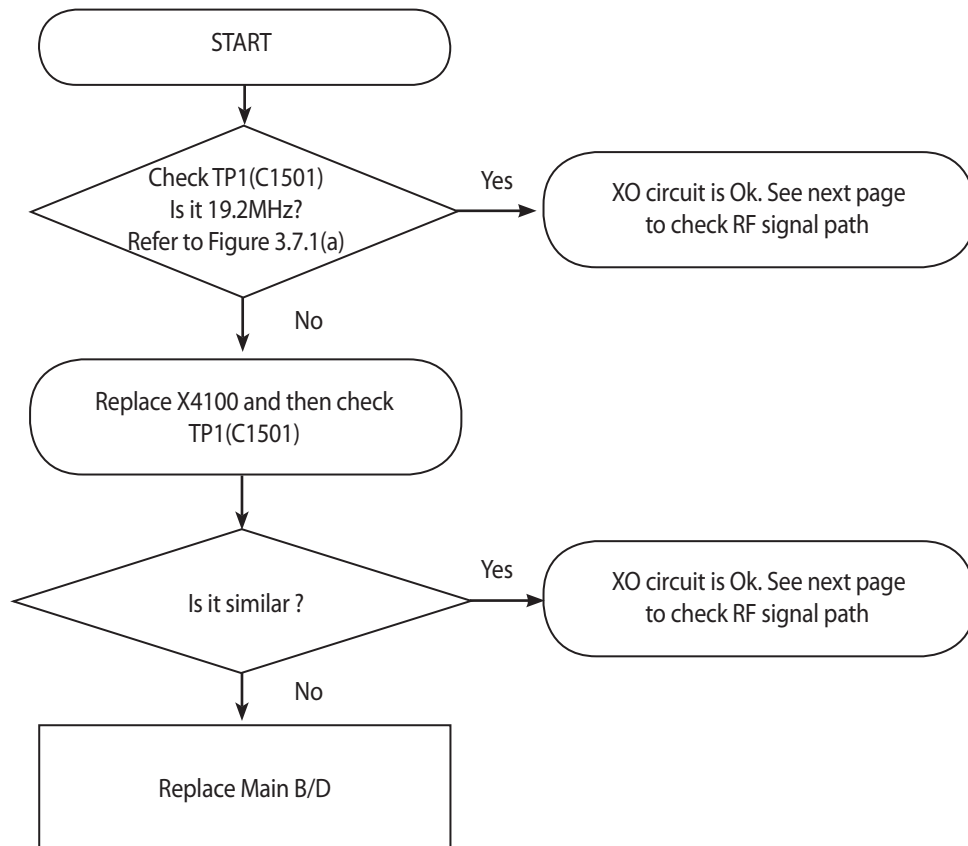
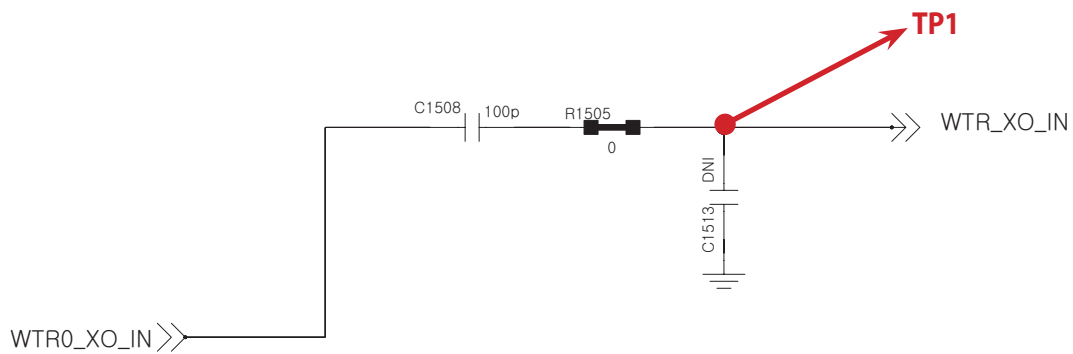
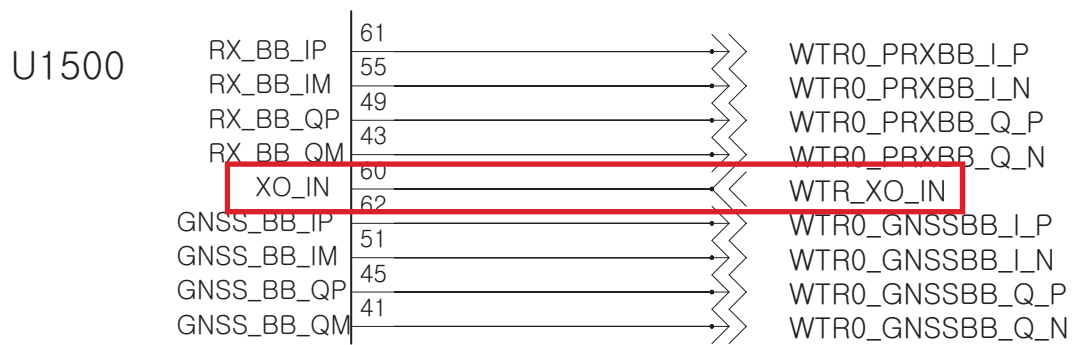
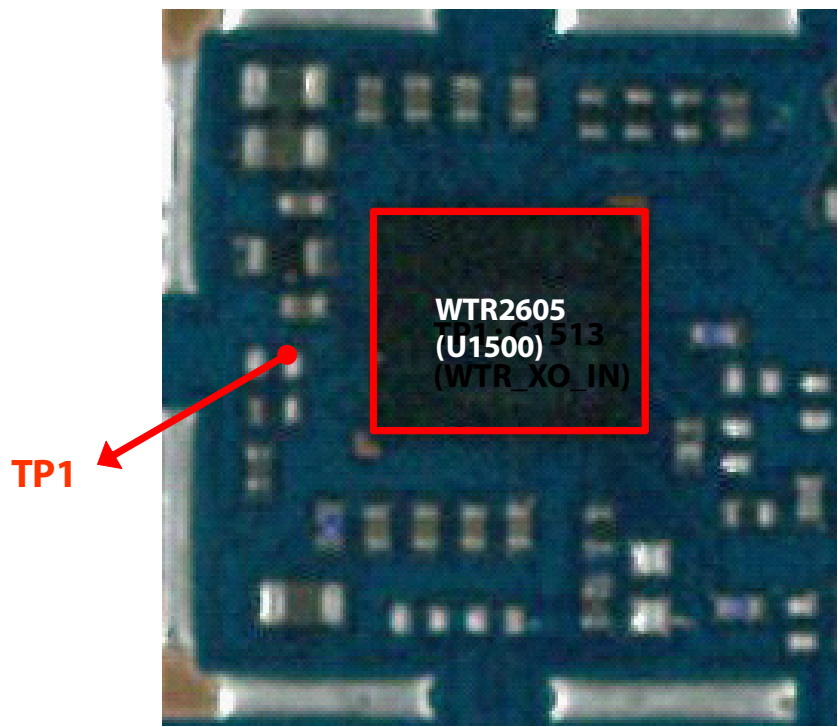
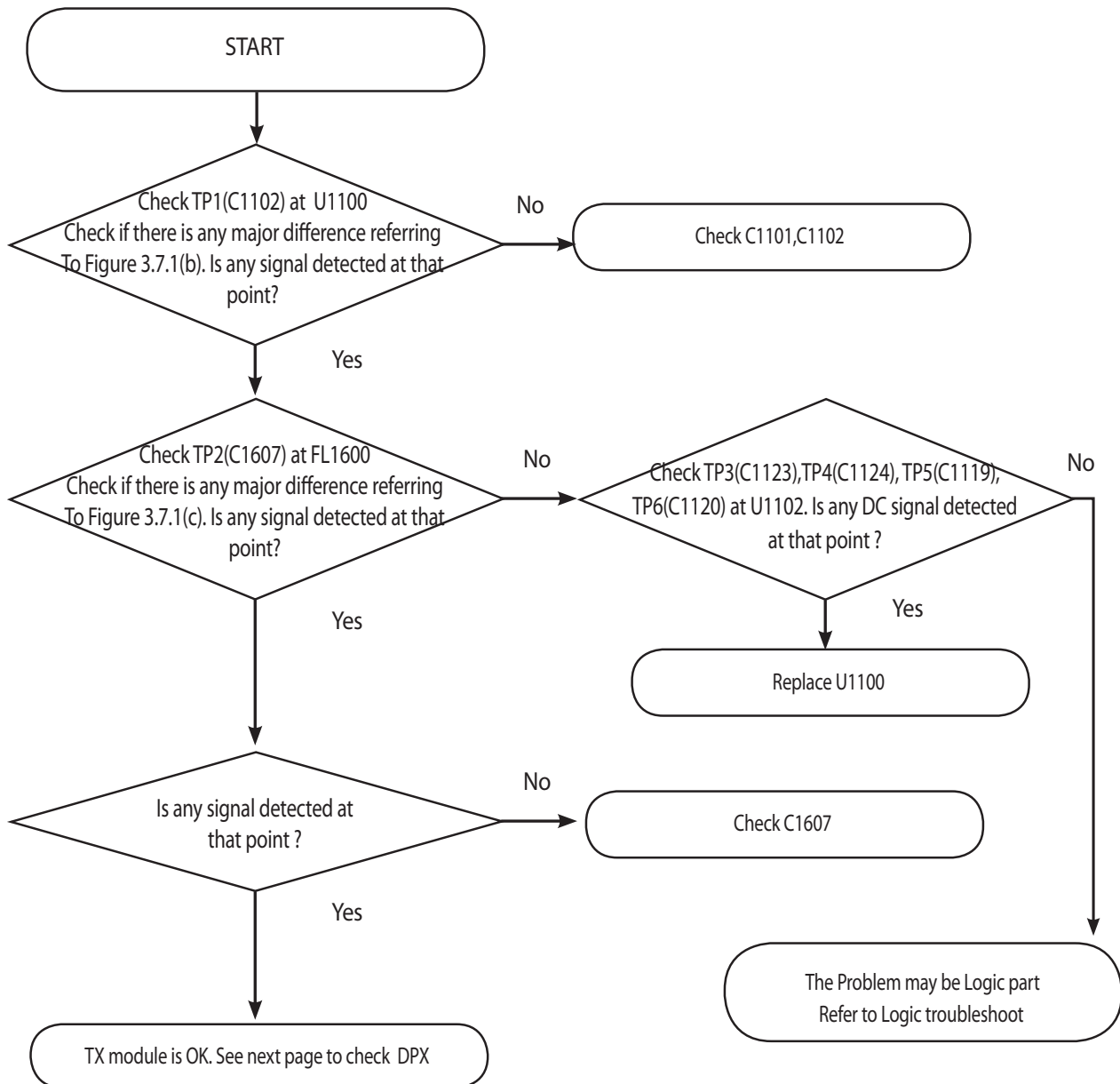


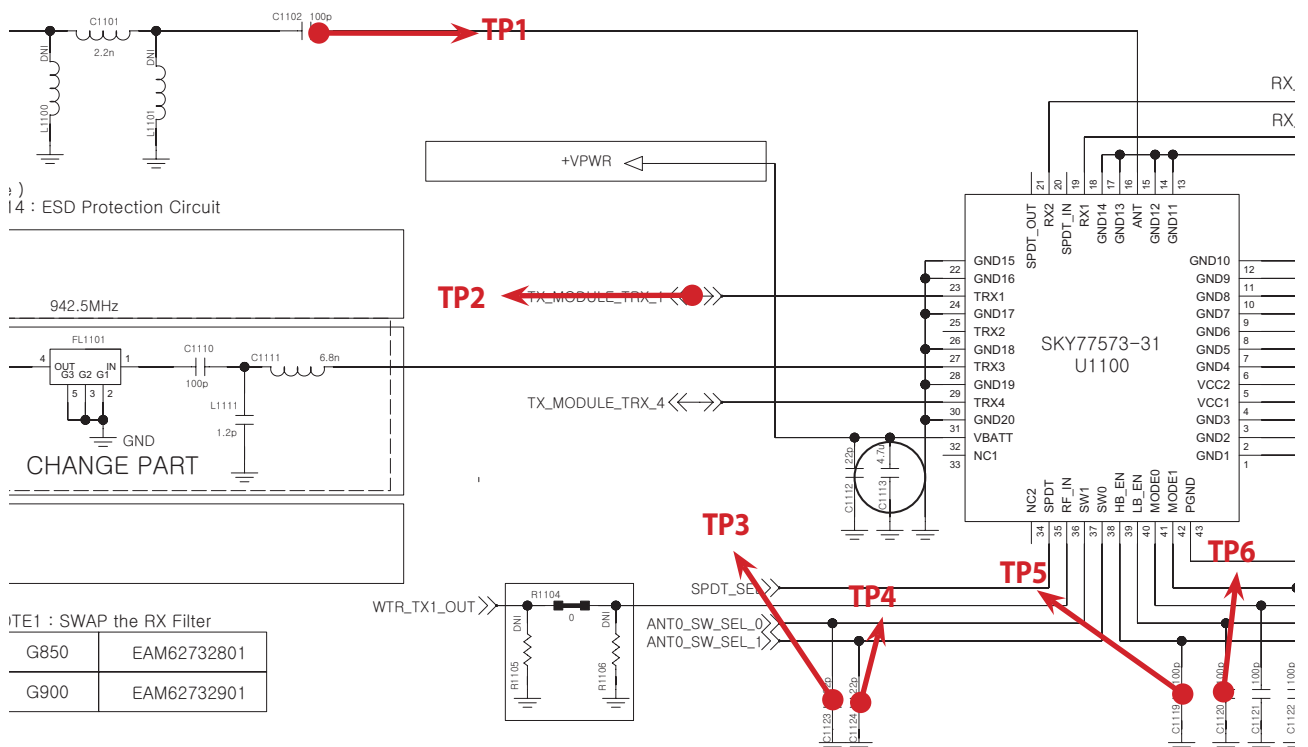
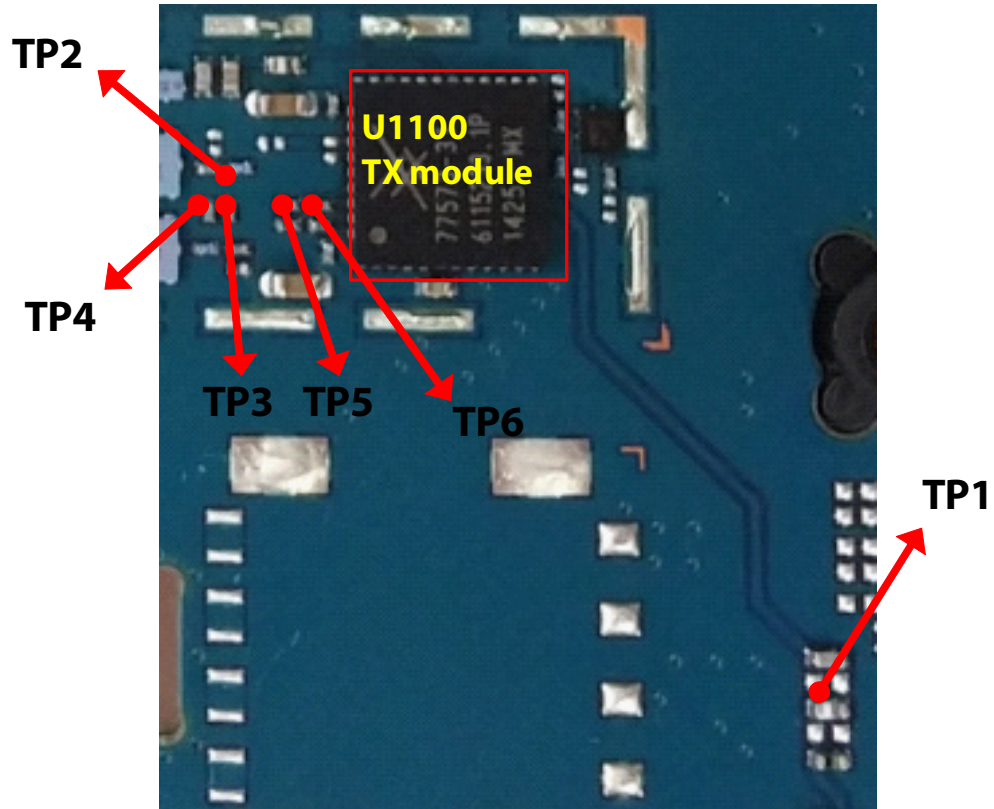
Figure 3.7.1 (a)



3.7.1.3 Checking RF signal path (TX Module)



3. TROUBLE SHOOTING



TP1

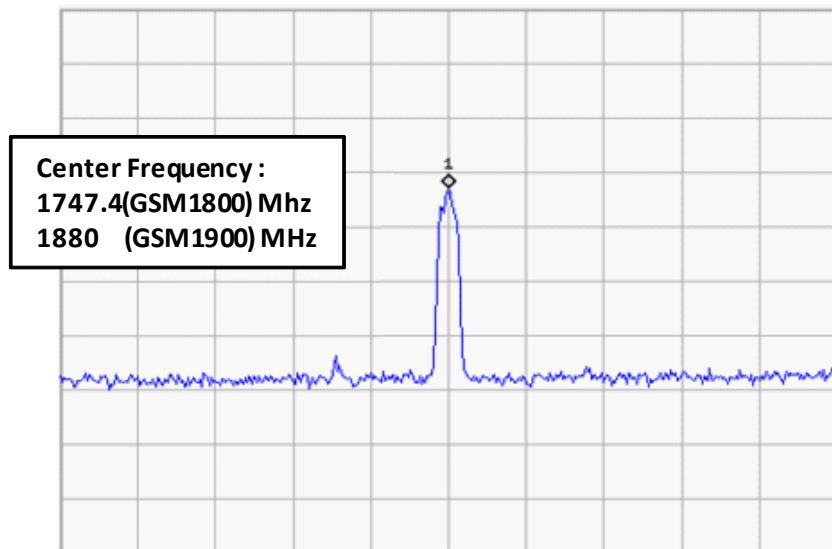


Figure 3.7.1 (b)

TP2

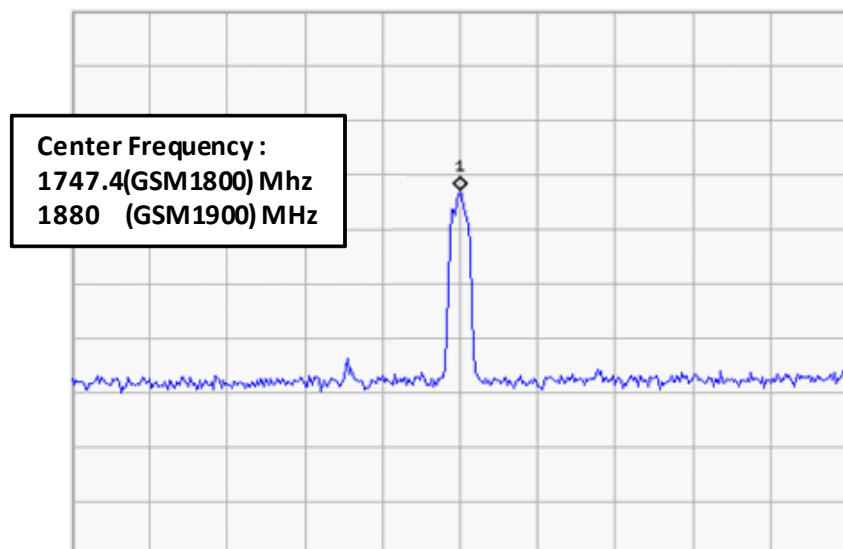
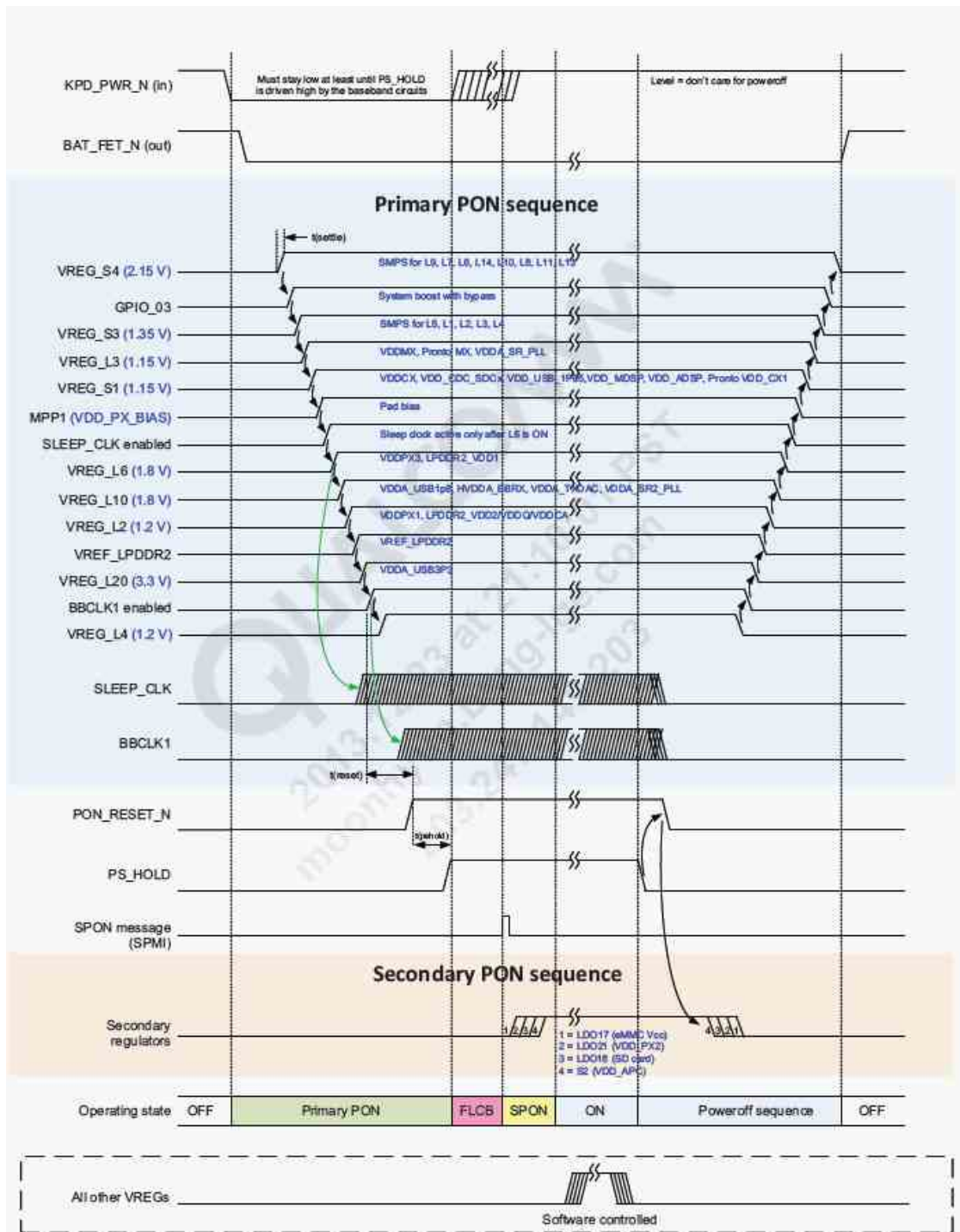


Figure 3.7.1 (c)

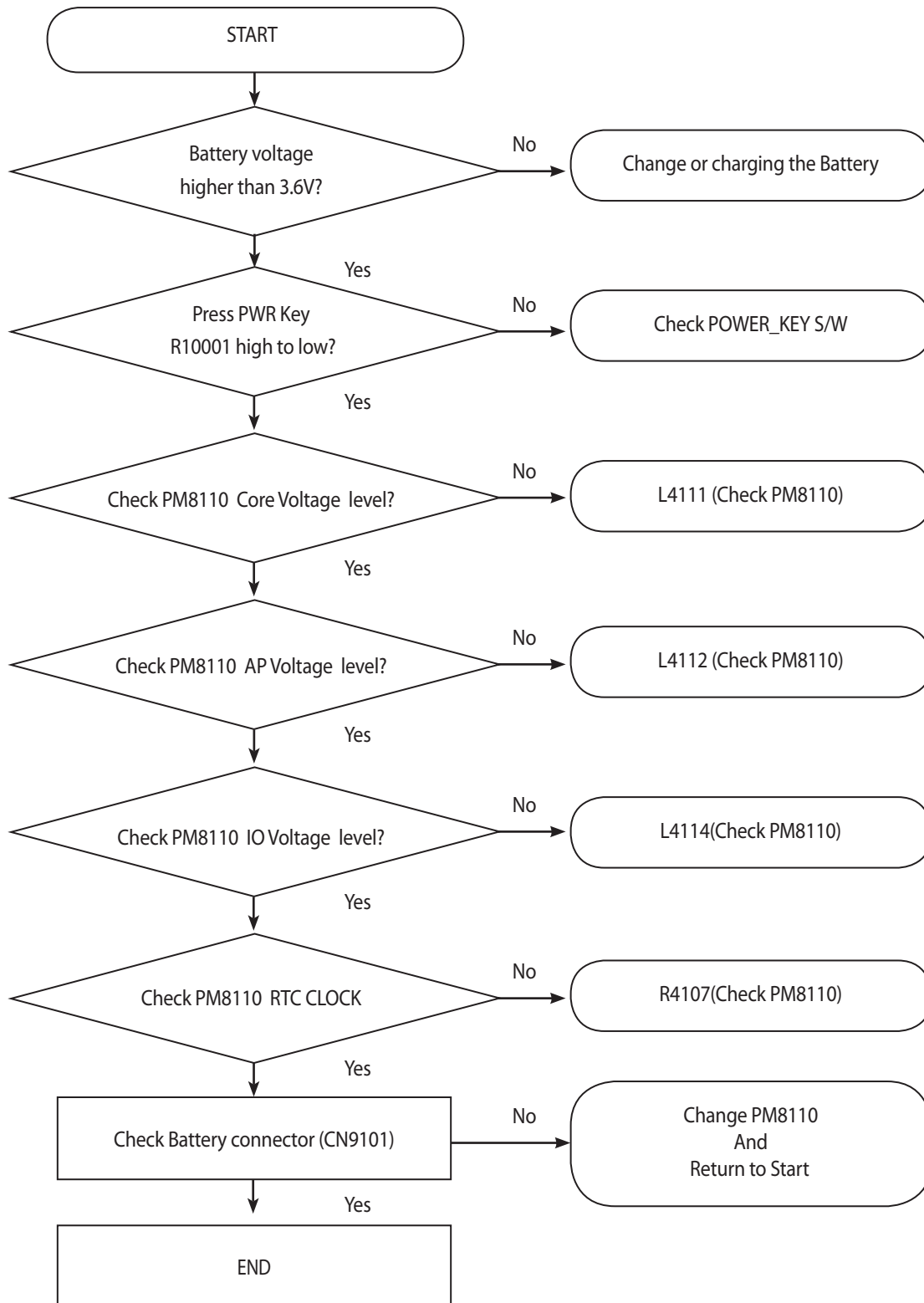
3.8. Power TS

The main power source of D290TR is provided by 1 chip which is PM8110

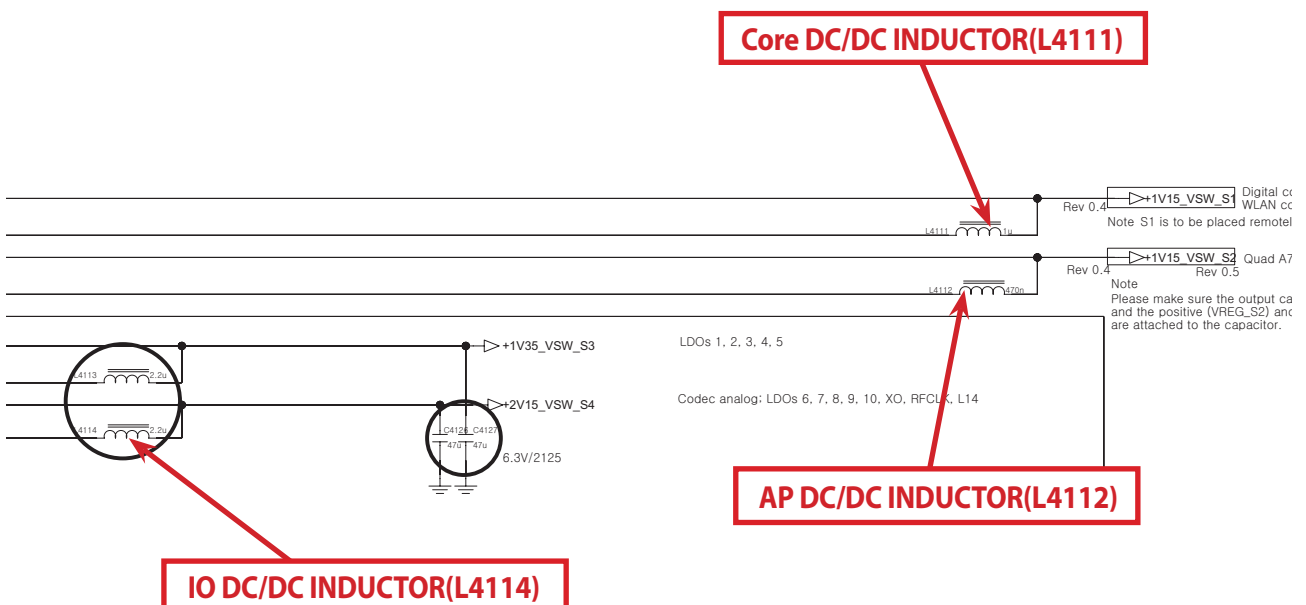
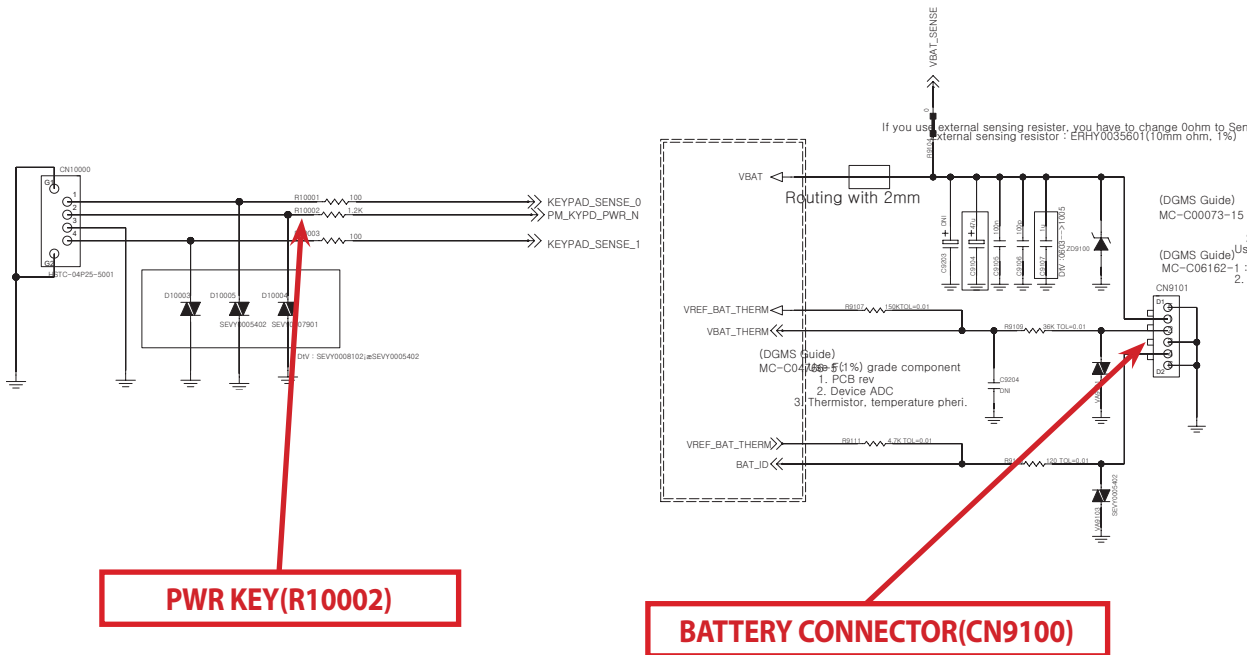
Power ON sequence of D290TR is



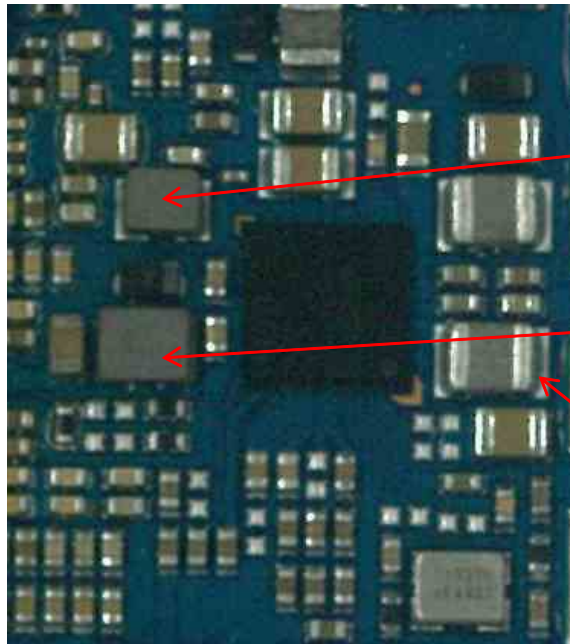
3. TROUBLE SHOOTING



3. TROUBLE SHOOTING



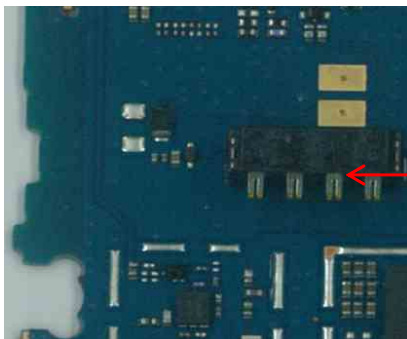
3. TROUBLE SHOOTING



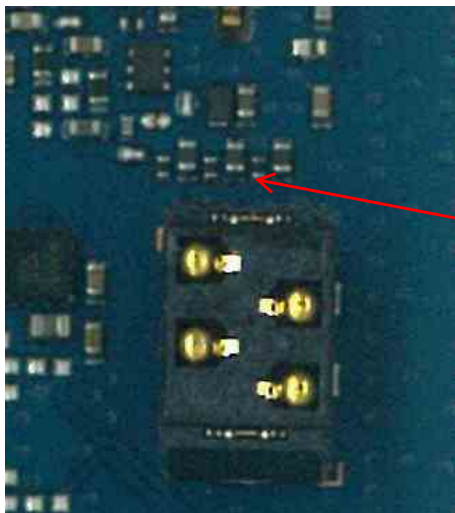
Core DC/DC INDUCTOR(L4111)

AP DC/DC INDUCTOR(L4112)

IO DC/DC INDUCTOR(L4114)



BATTERY CONNECTOR(CN9100)



PWR KEY(R10002)

3.9 Charging TS

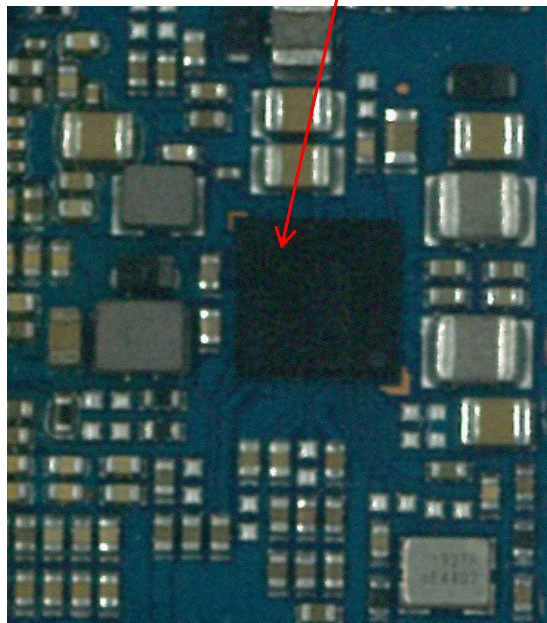
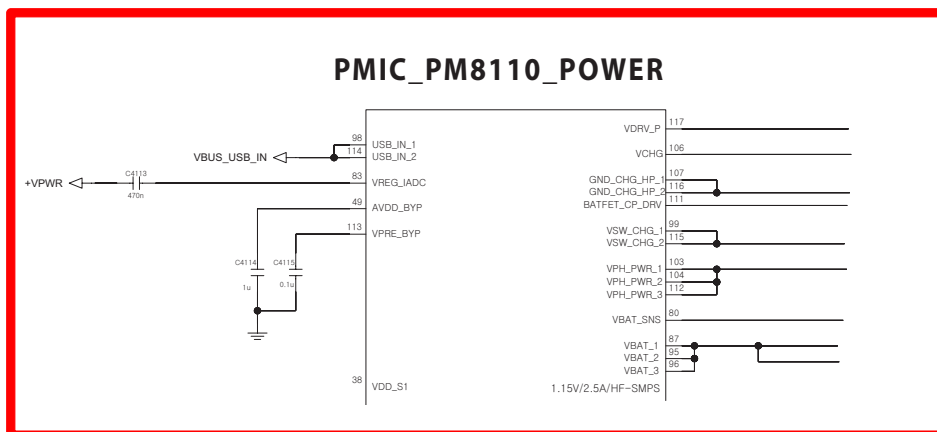
D290TR uses PMIC(liner Charger) for charging control. The PMIC circuit includes the charging pass transistor detects the VDD and VBAT voltages, and its control driver. The PMIC maximum charging current is set around 700mA.

Charging Procedure

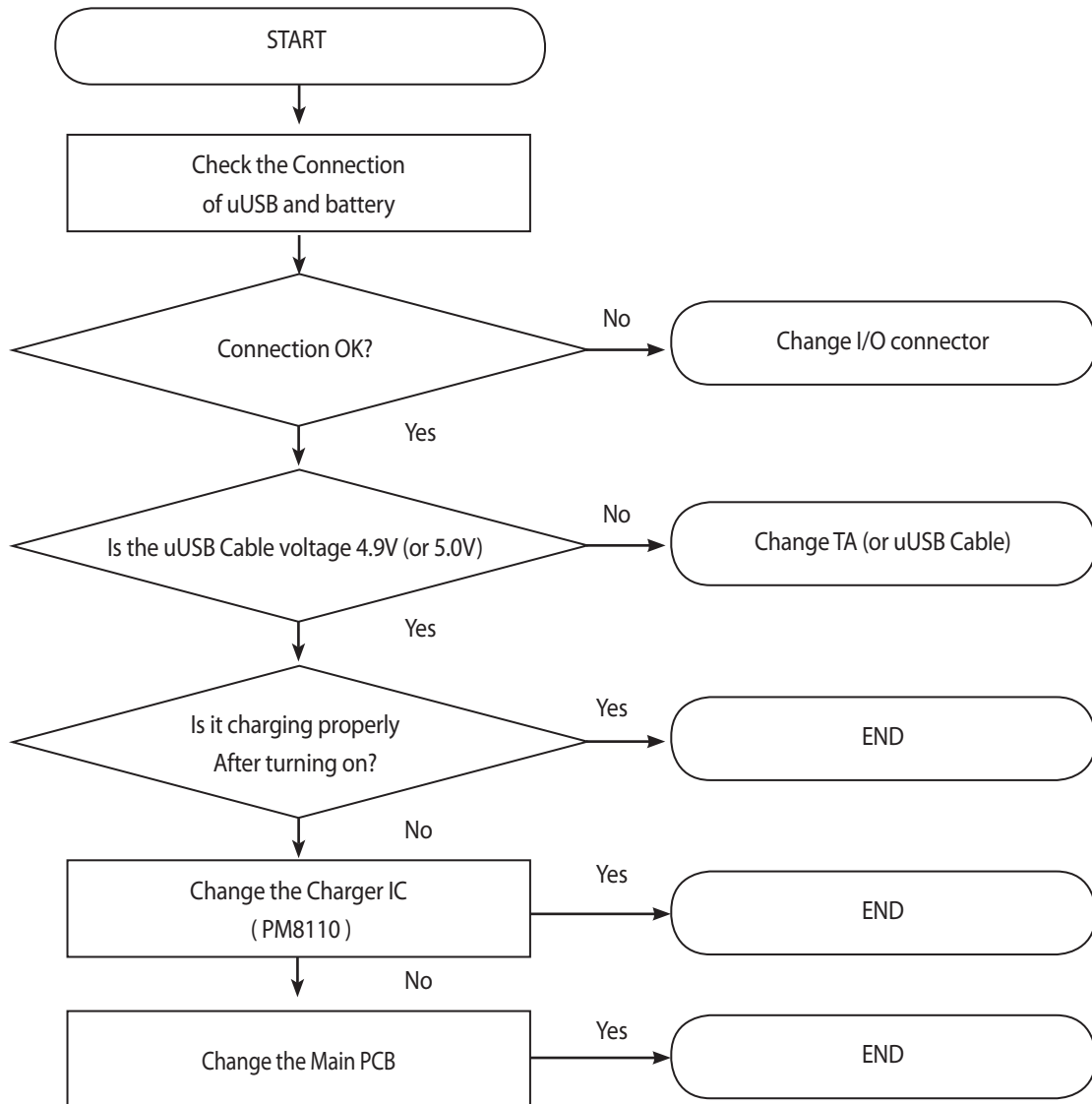
- Supply the VCHG use a TA or u-USB cable
- The PMIC control the charging current
- Charging current flows into the main battery

Troubleshooting Check Point

- Connection of the TA or u-USB cable
- Main battery

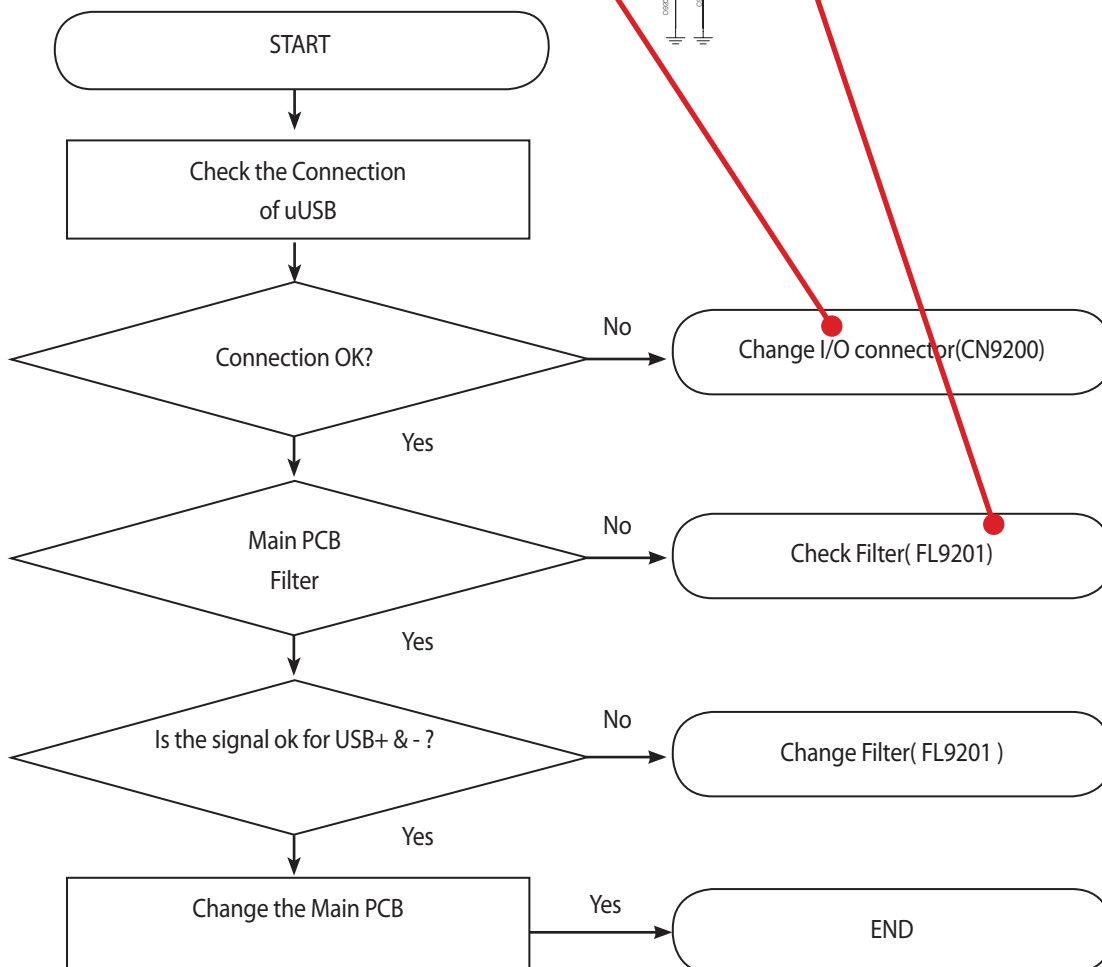
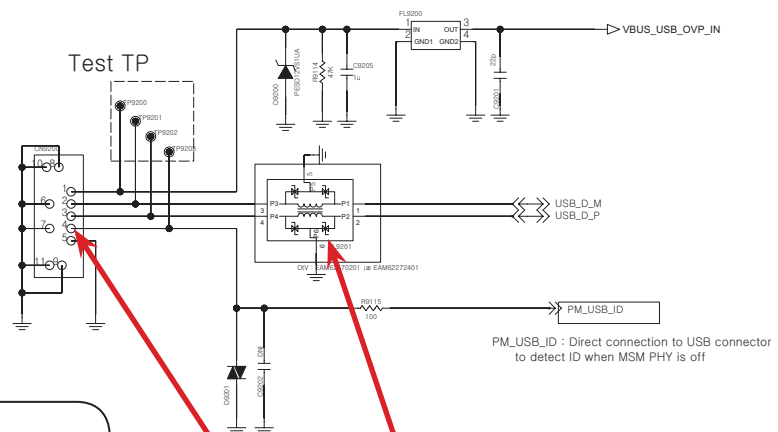


3. TROUBLE SHOOTING



3.10. USB TS

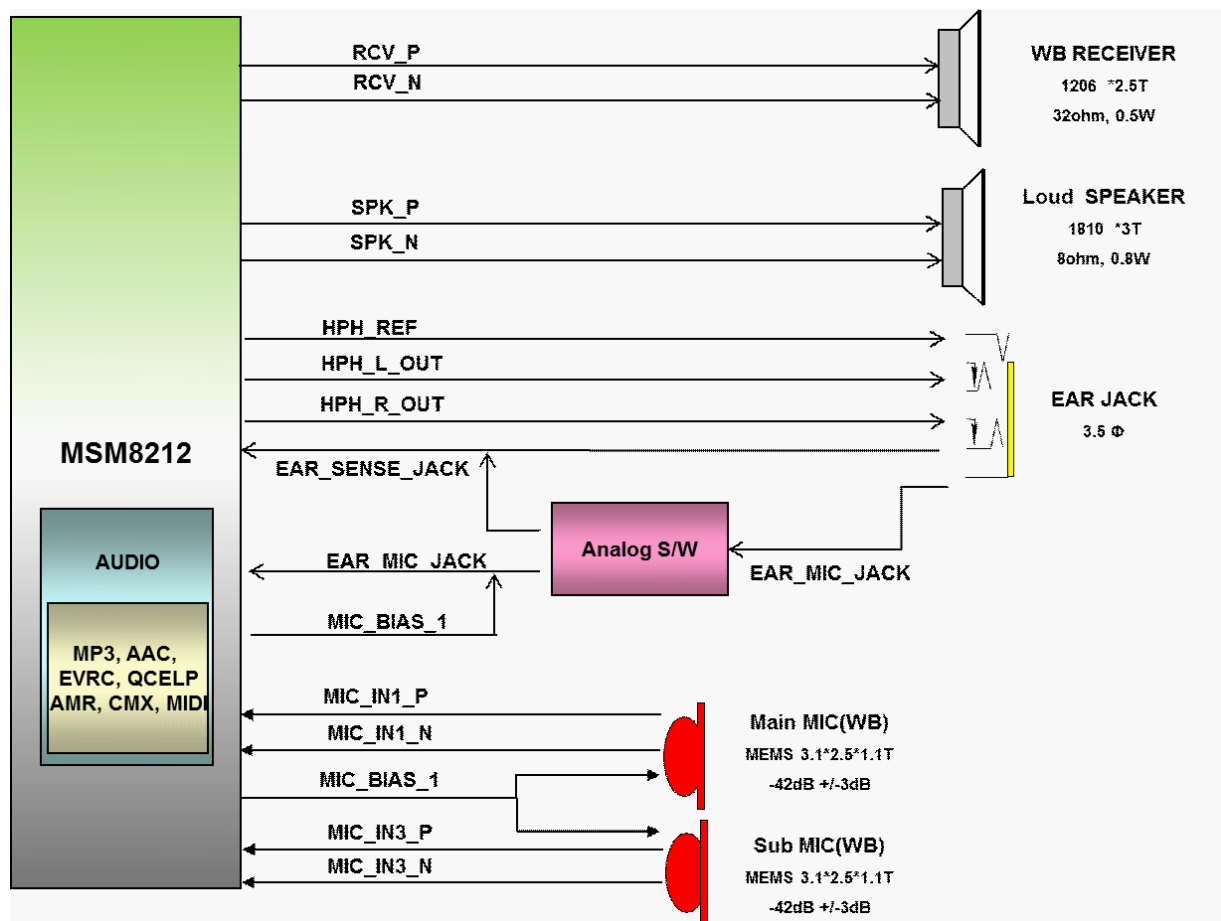
< 9-2-1_IO_USB2.0 > Rev_0.5



3.11. Audio TS

3.11.1 Overview

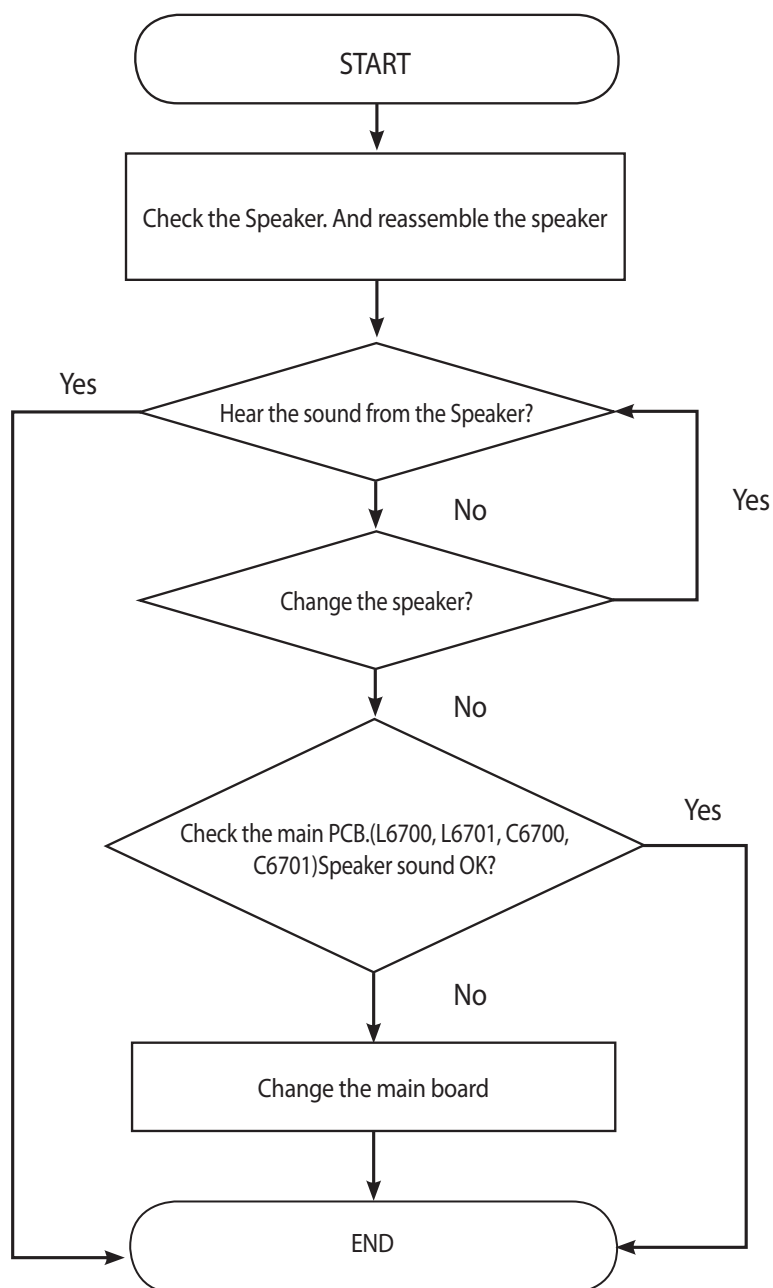
Audio Block Diagram



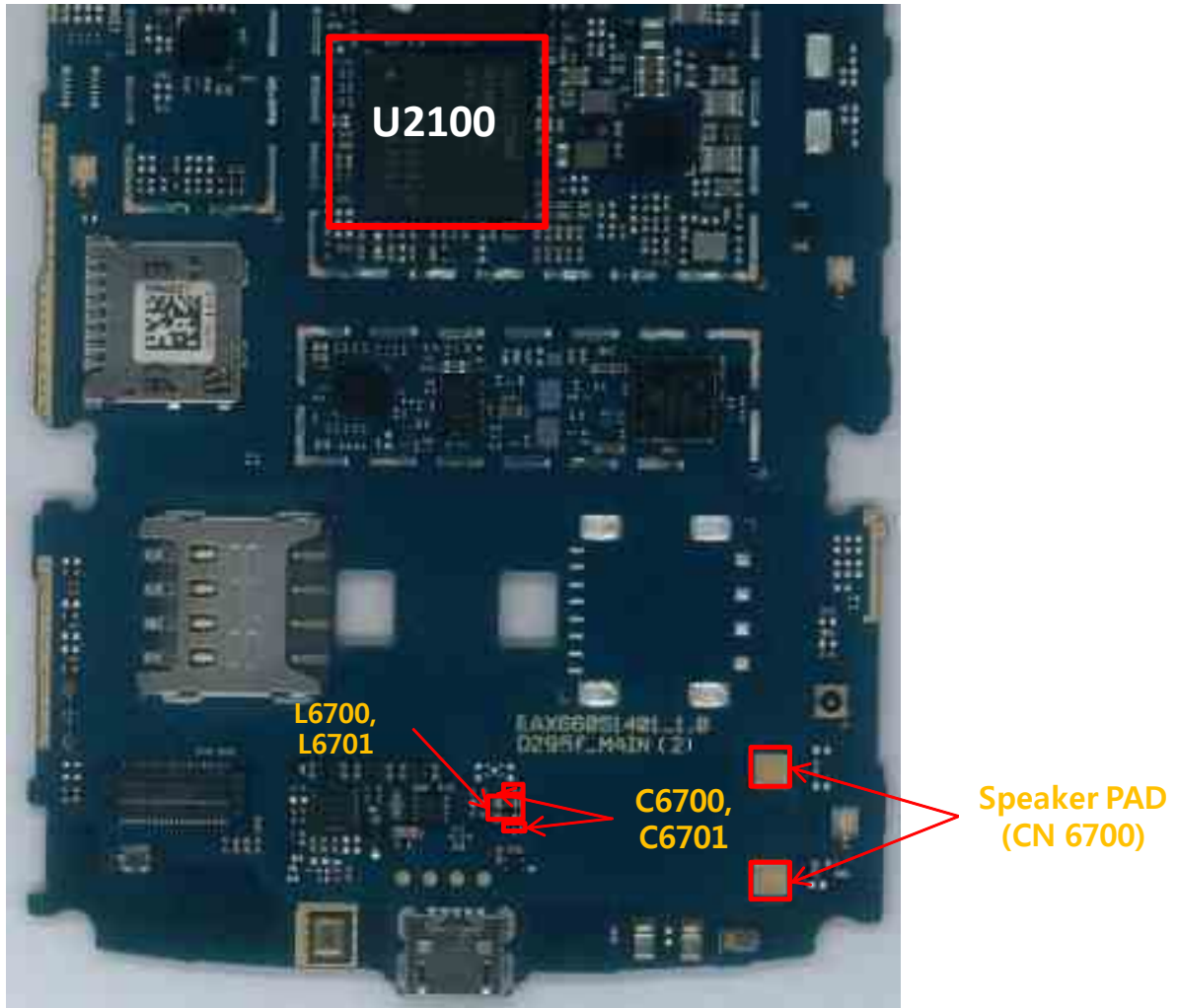
3.11.2 Speaker Trouble Shooting

It's trouble shooting guide for no sound case.

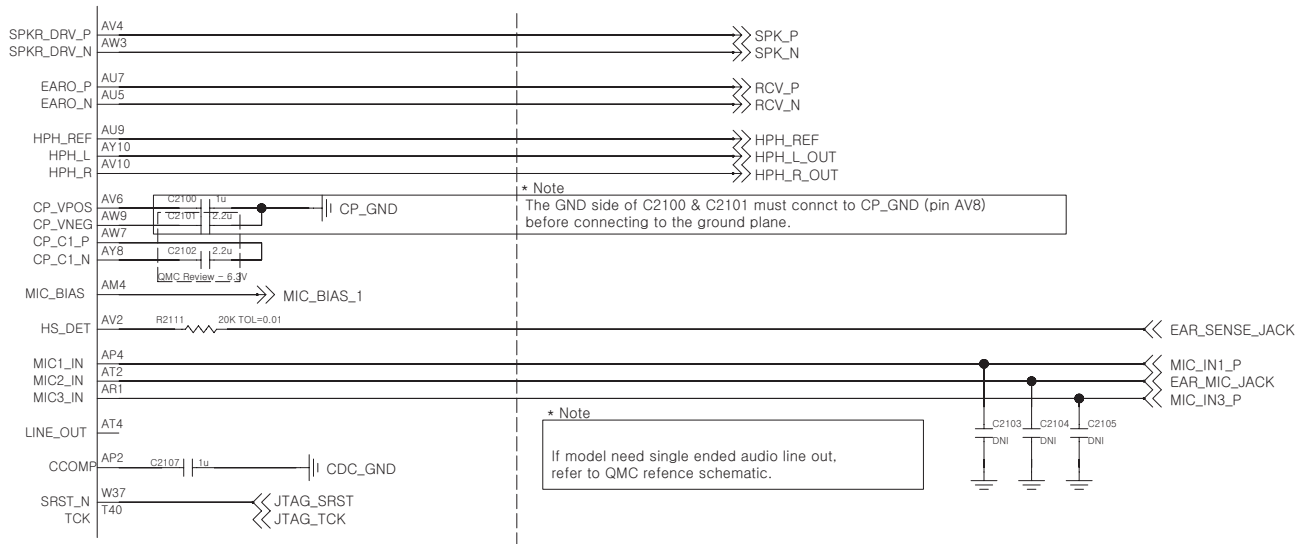
Speaker & Receiver control signals are generate by MSM8212 (U2100)If low level volume or breaking the sound, change the speaker



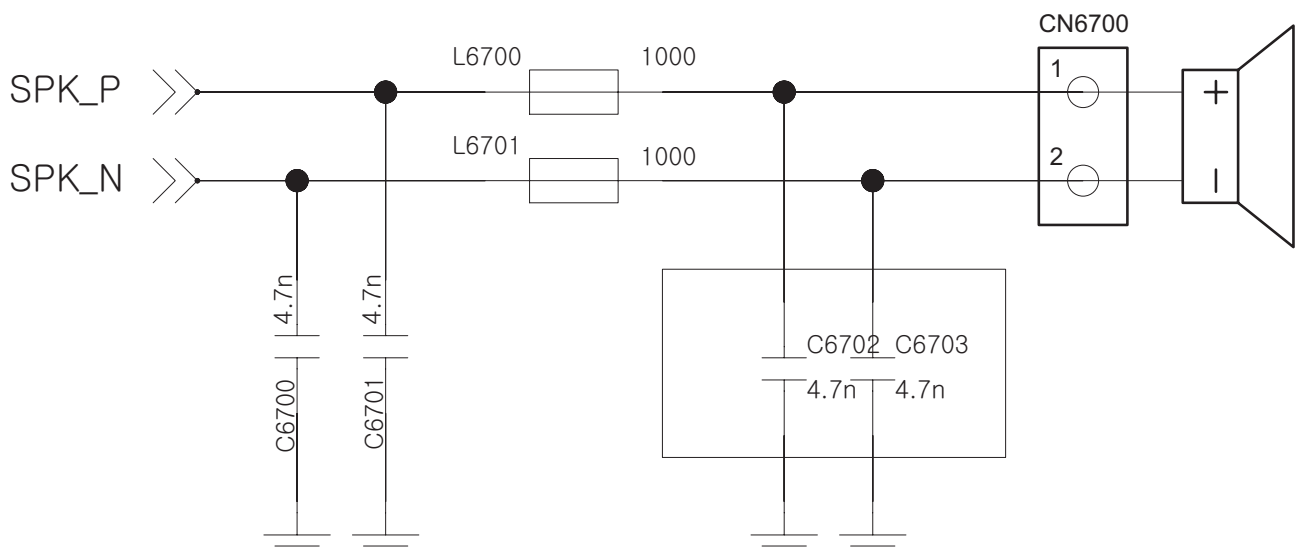
3. TROUBLE SHOOTING



Main PCB



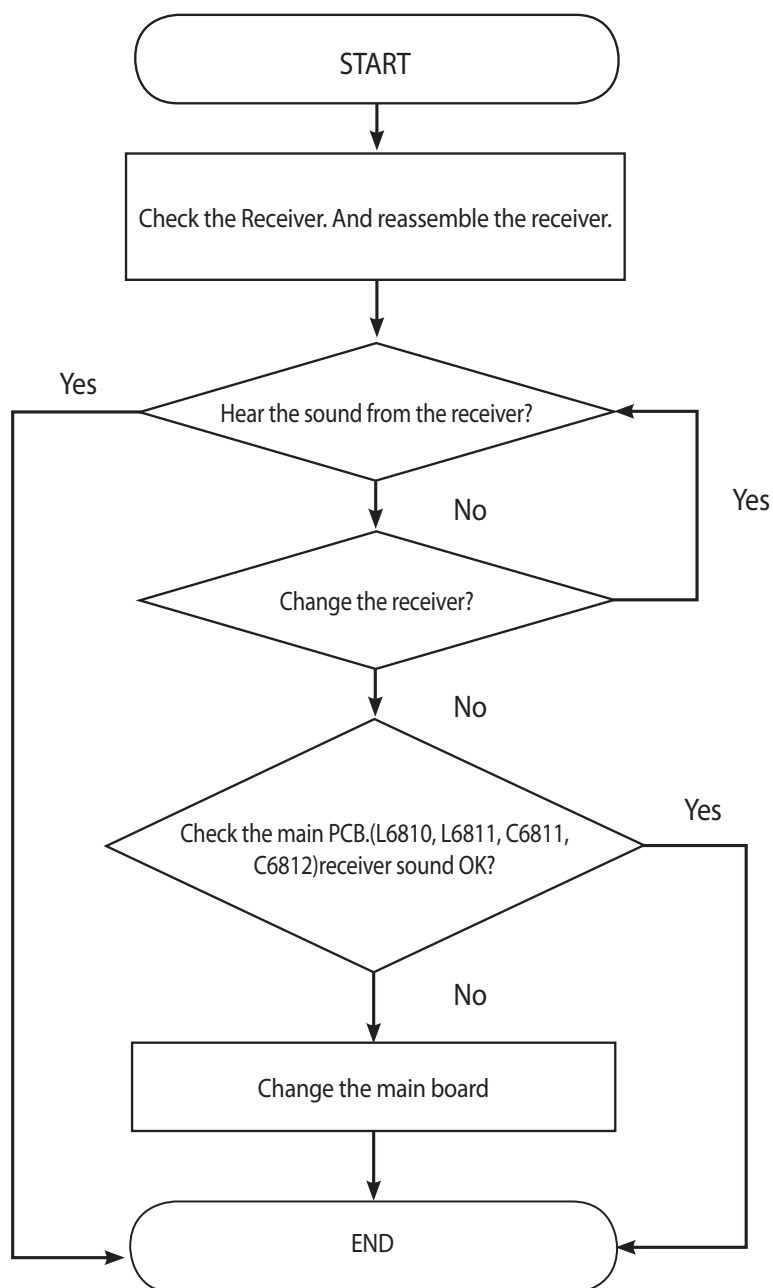
Speaker



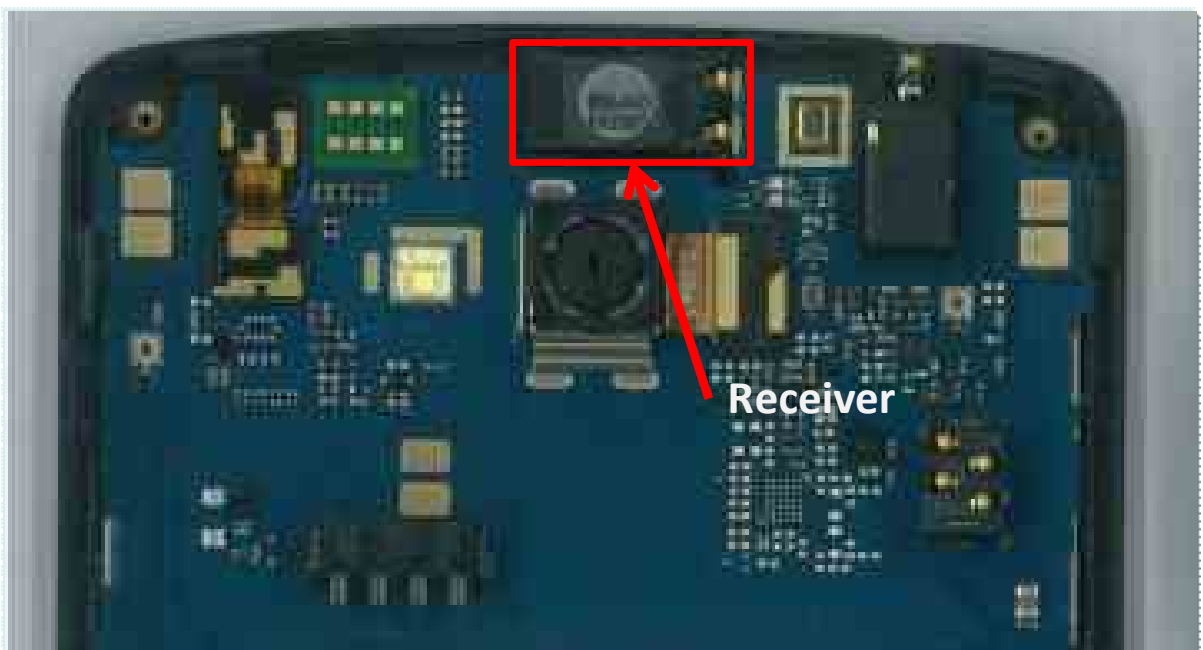
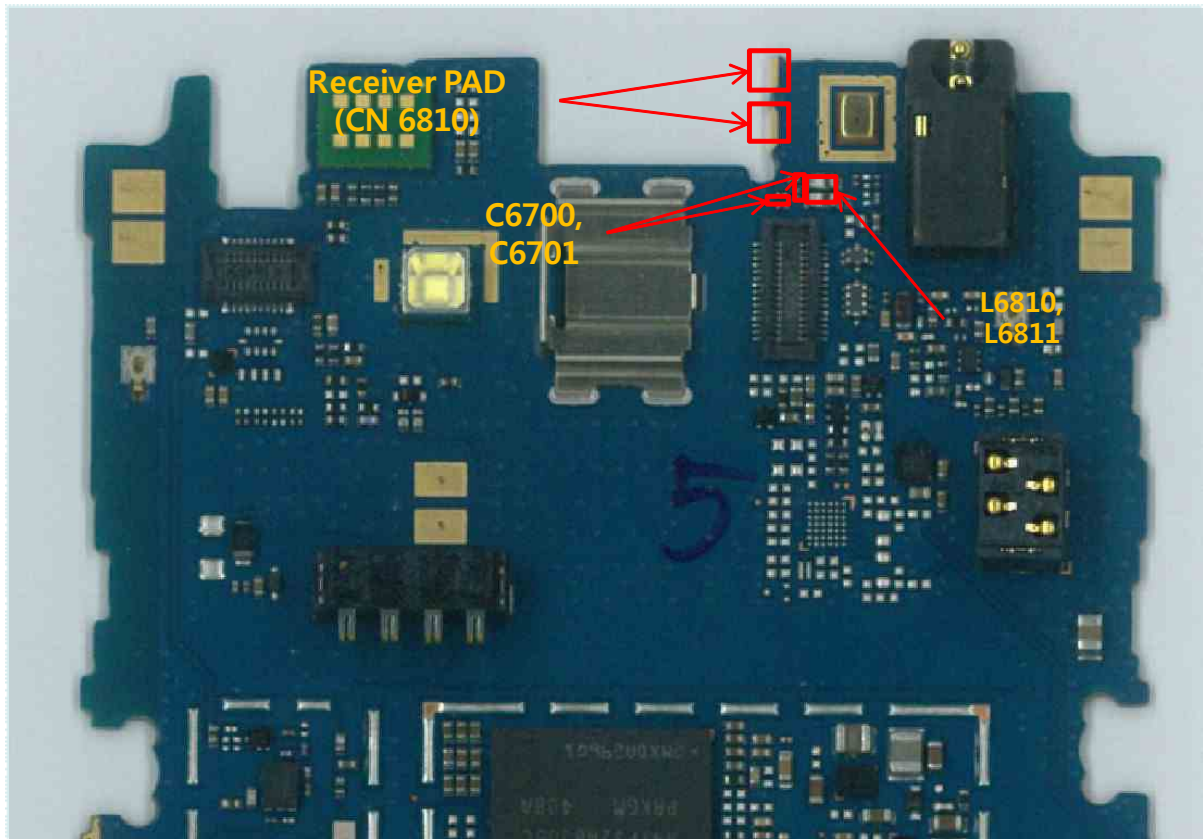
3.11.3 Receiver Trouble Shooting

It's trouble shooting guide for no sound case.

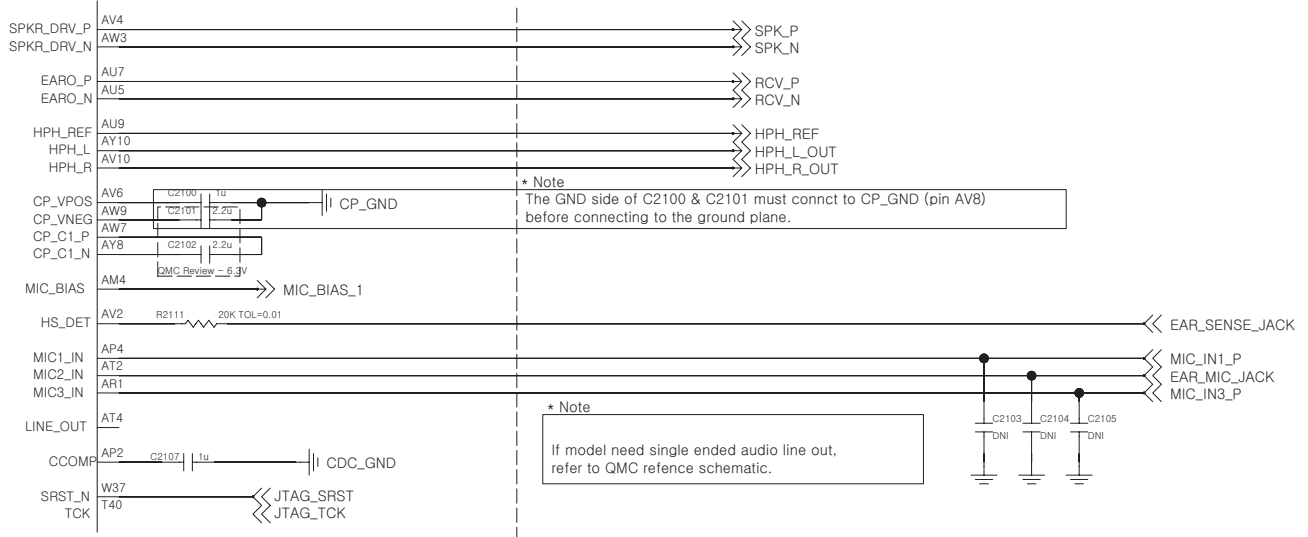
Speaker & Receiver control signals are generate by MSM8212 (U2100)If low level volume or breaking the sound, change the Receiver



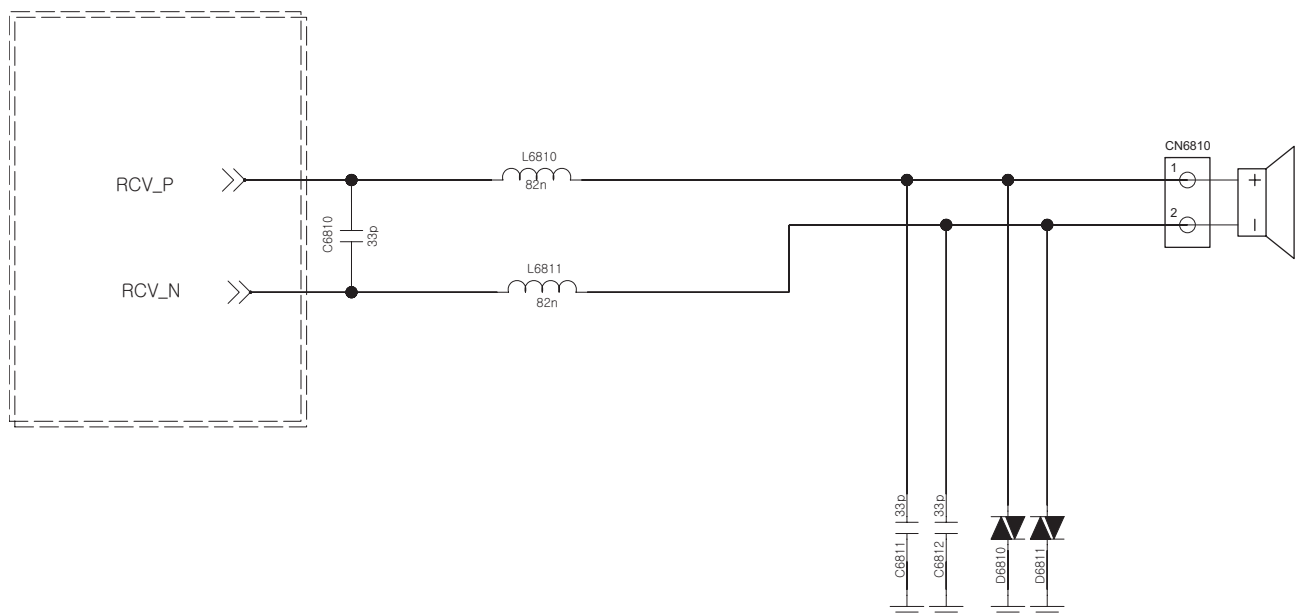
3. TROUBLE SHOOTING



Main PCB

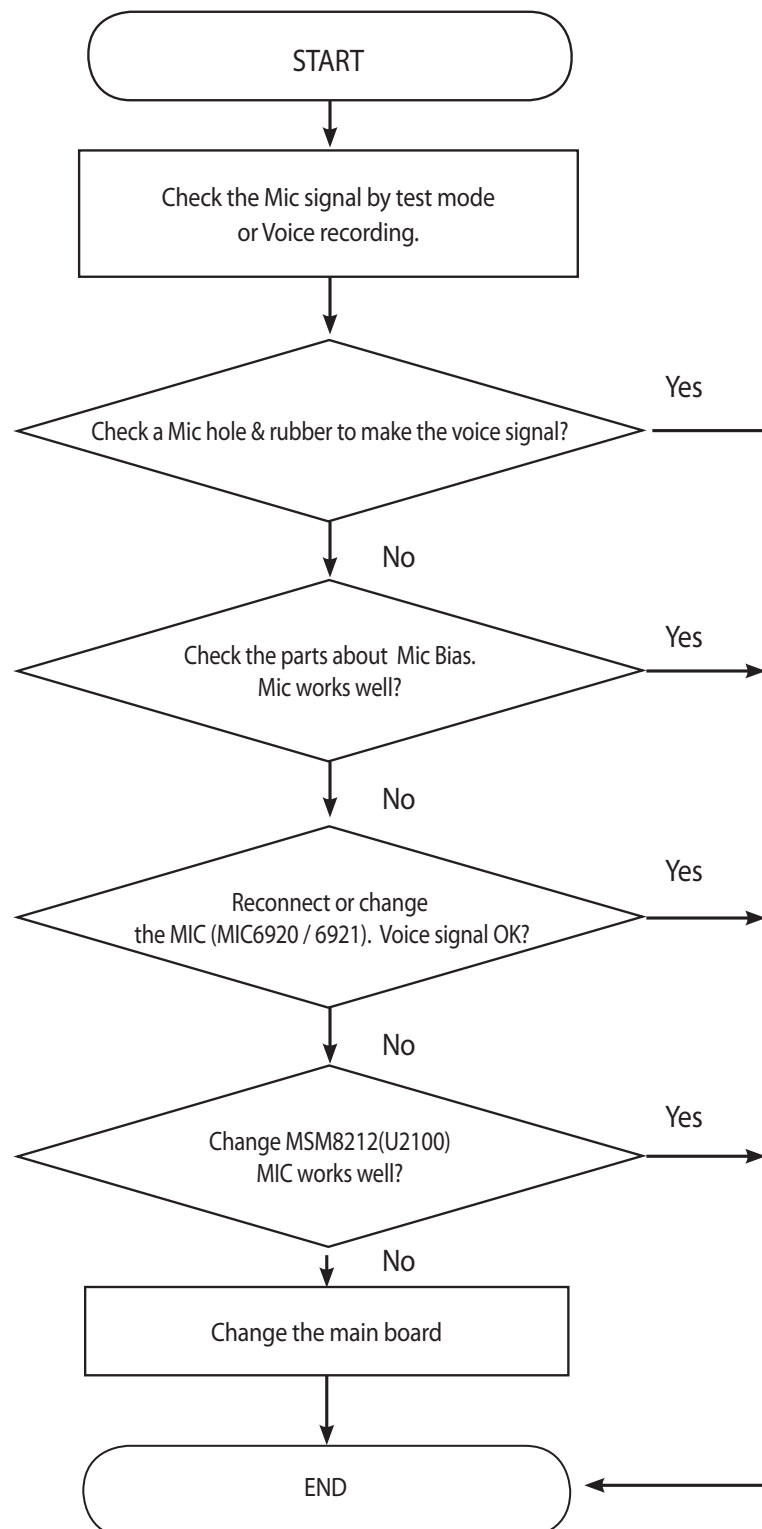


Receiver

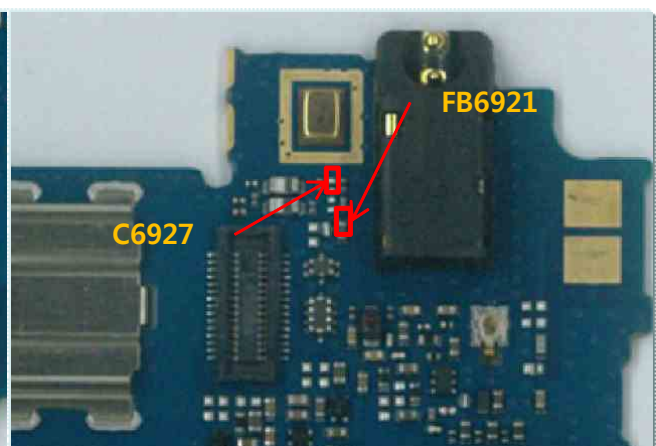
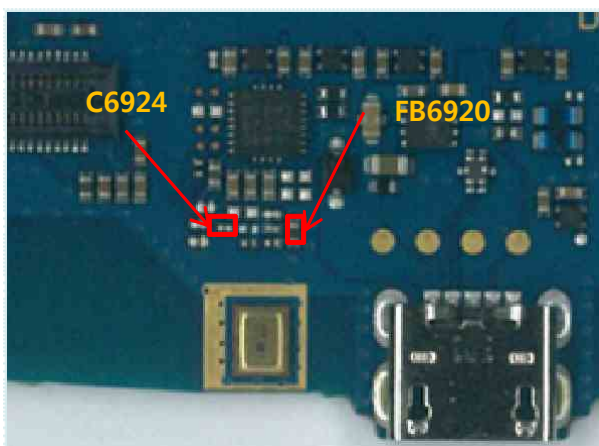


3.11.4 Main MIC / Sub MIC Trouble Shooting

It's operating voice call(including speaker phone), voice recording and camcorder recording.

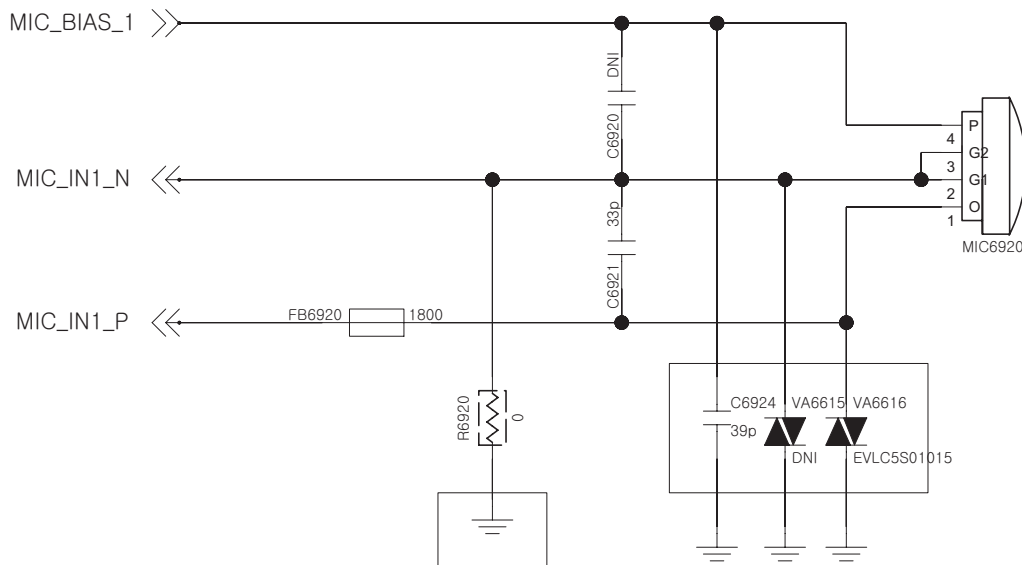


Check a Main Mic hole.

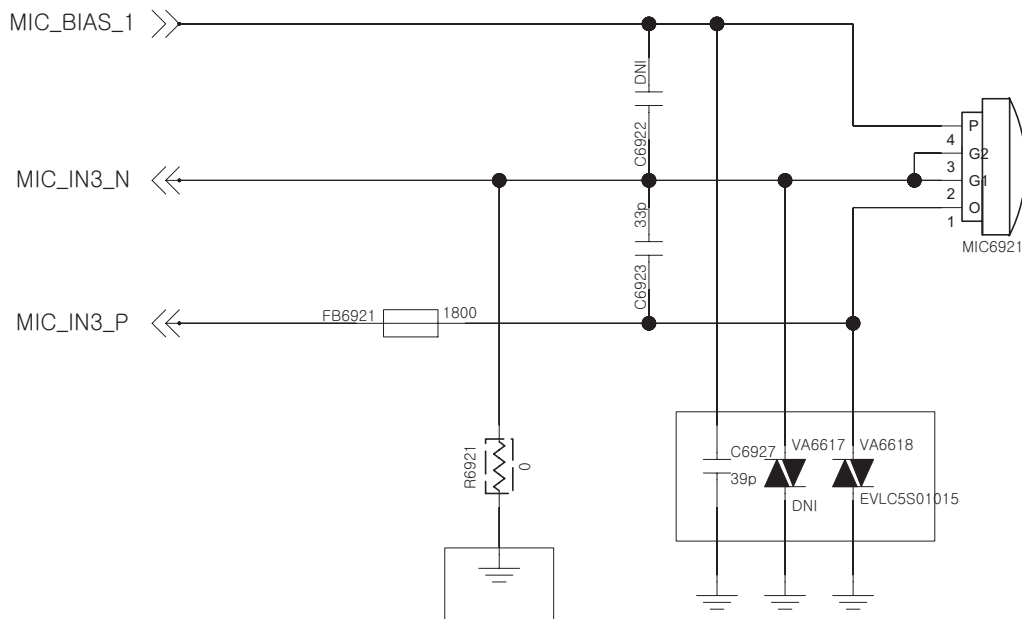


Check the MIC bias/output level

Main MIC



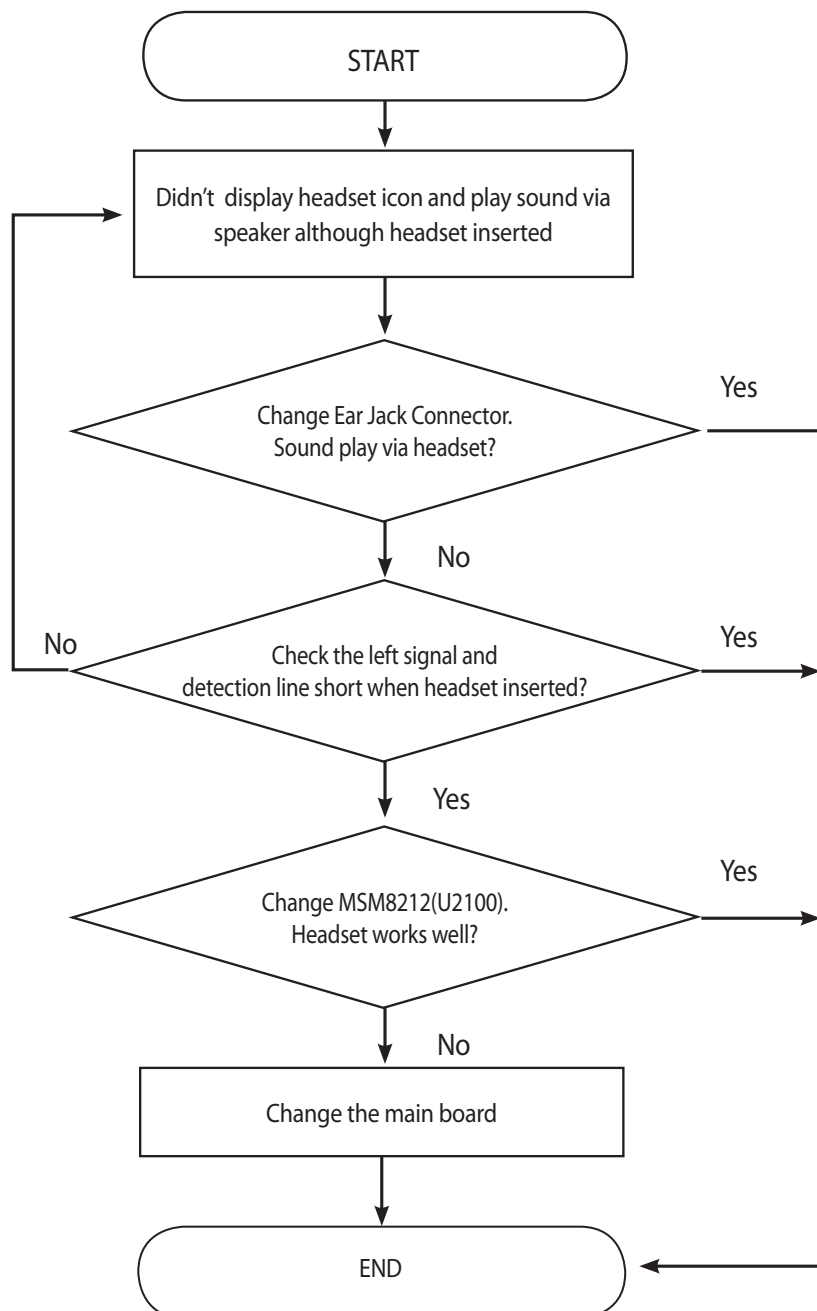
Sub MIC

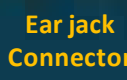


3.11.5. Ear-Mic Trouble Shooting

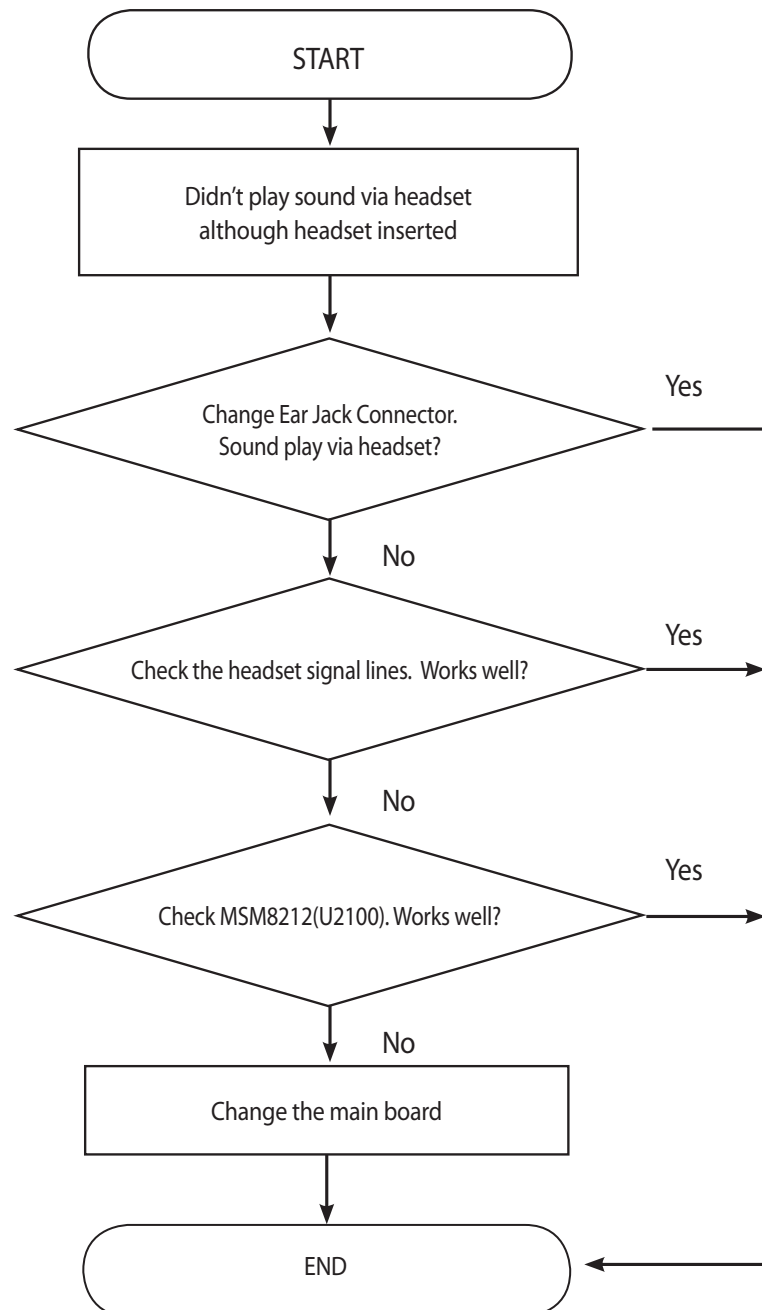
Ear MIC control signals are generate by MSM8212(U2100)

Case A. Disable detected headset insert.

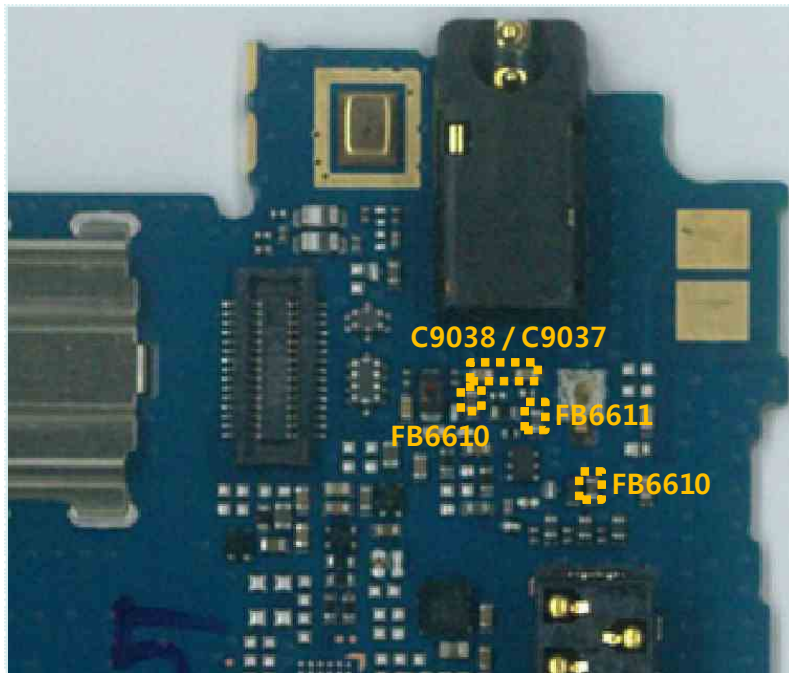
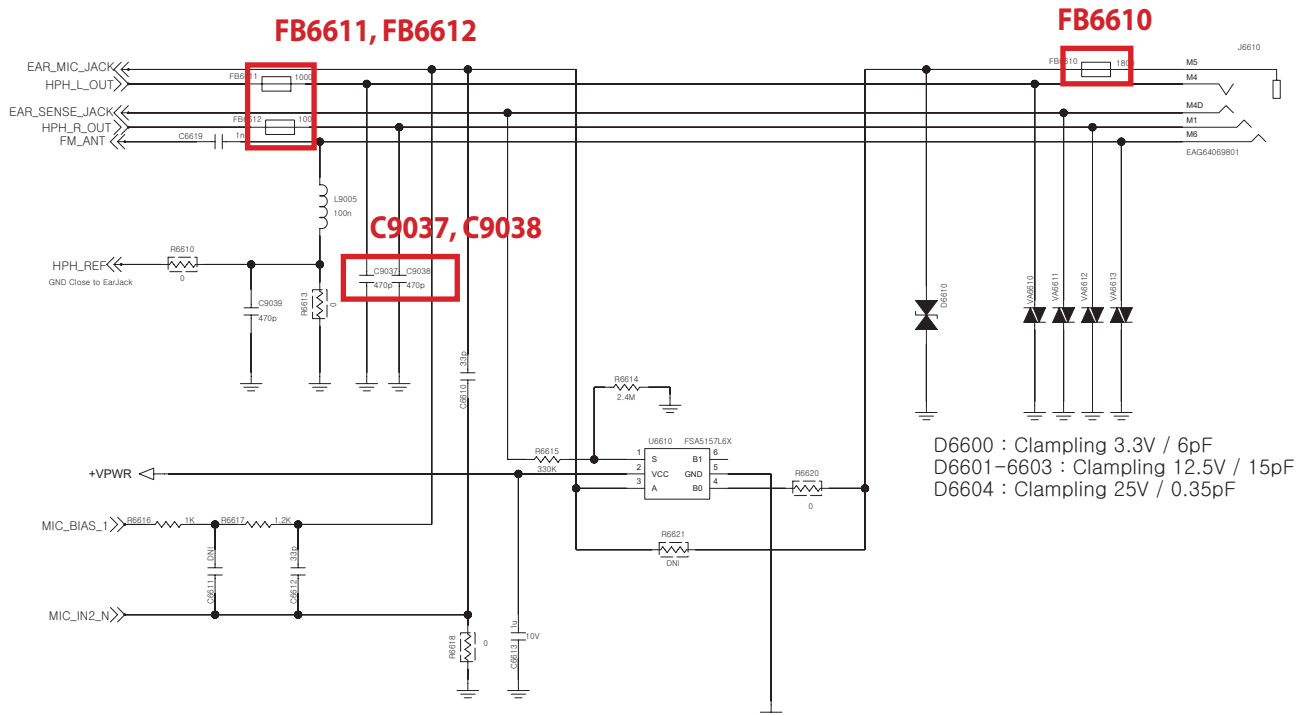




Case B. No Sound from Headset



3. TROUBLE SHOOTING



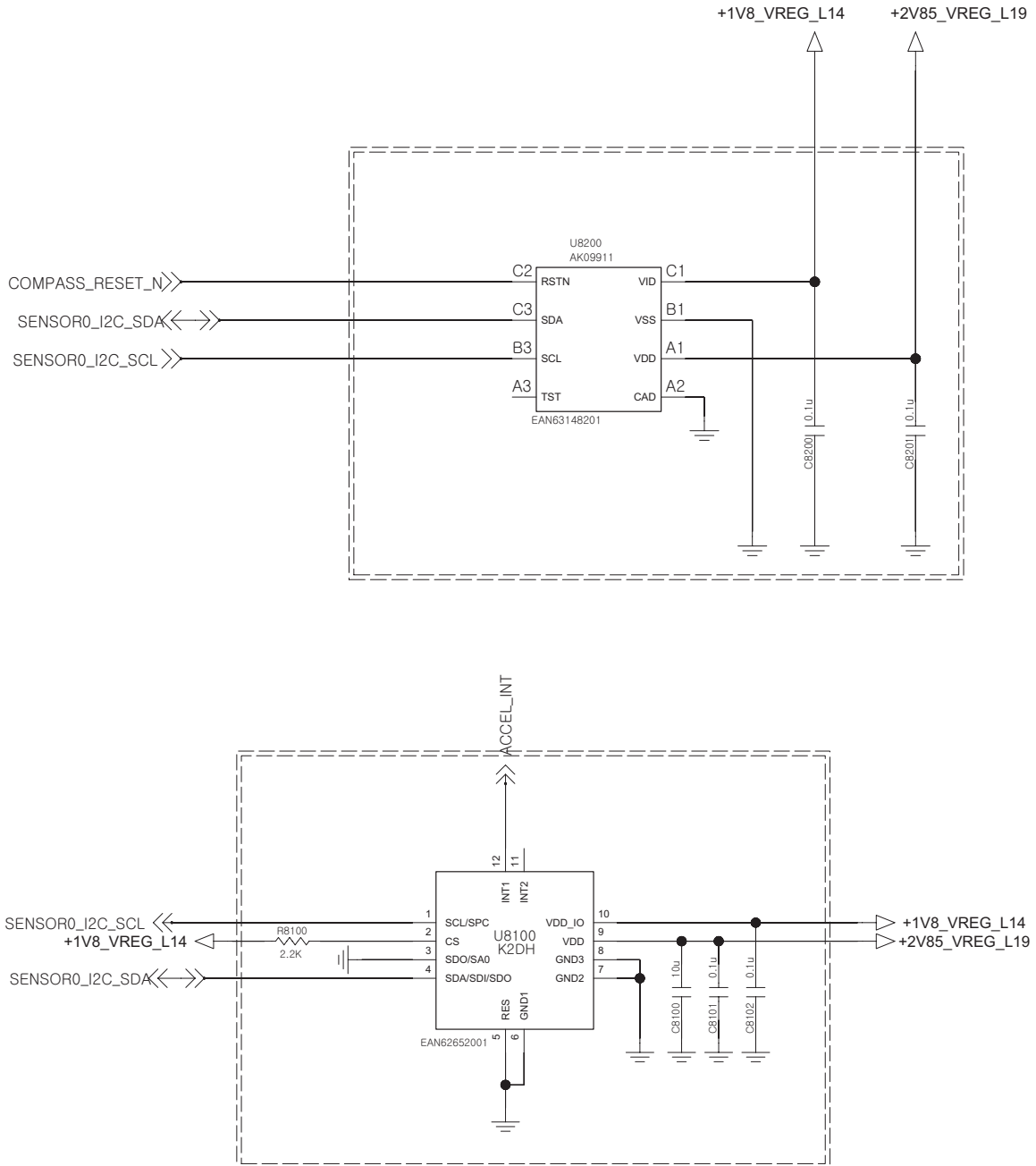
Check the headset MIC bias. (FB6610)

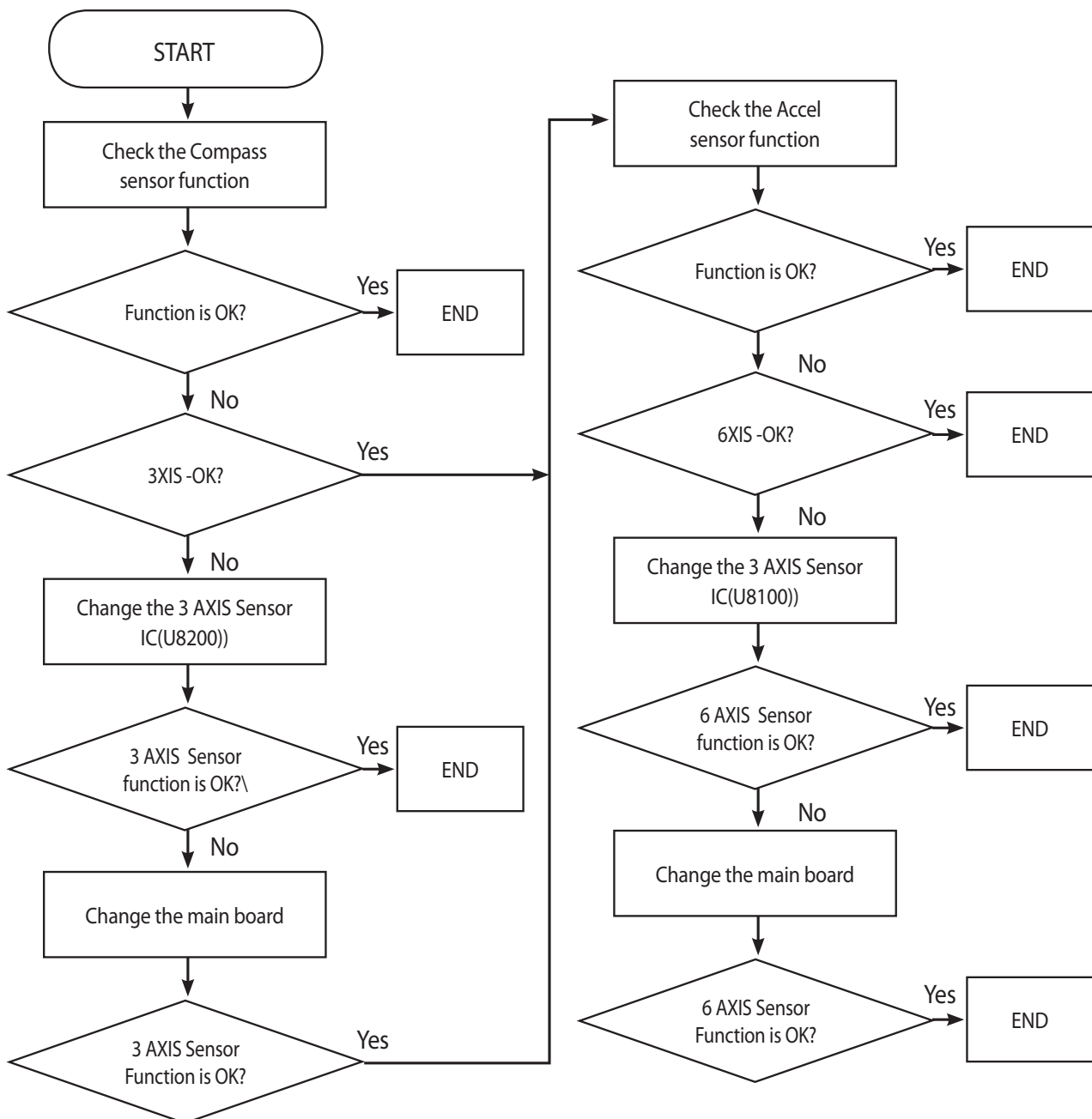
In case of you can't check the headset signal lines (FB6611, FB6612)

Can heard sound but sound is abnormal, check C6922, C6923

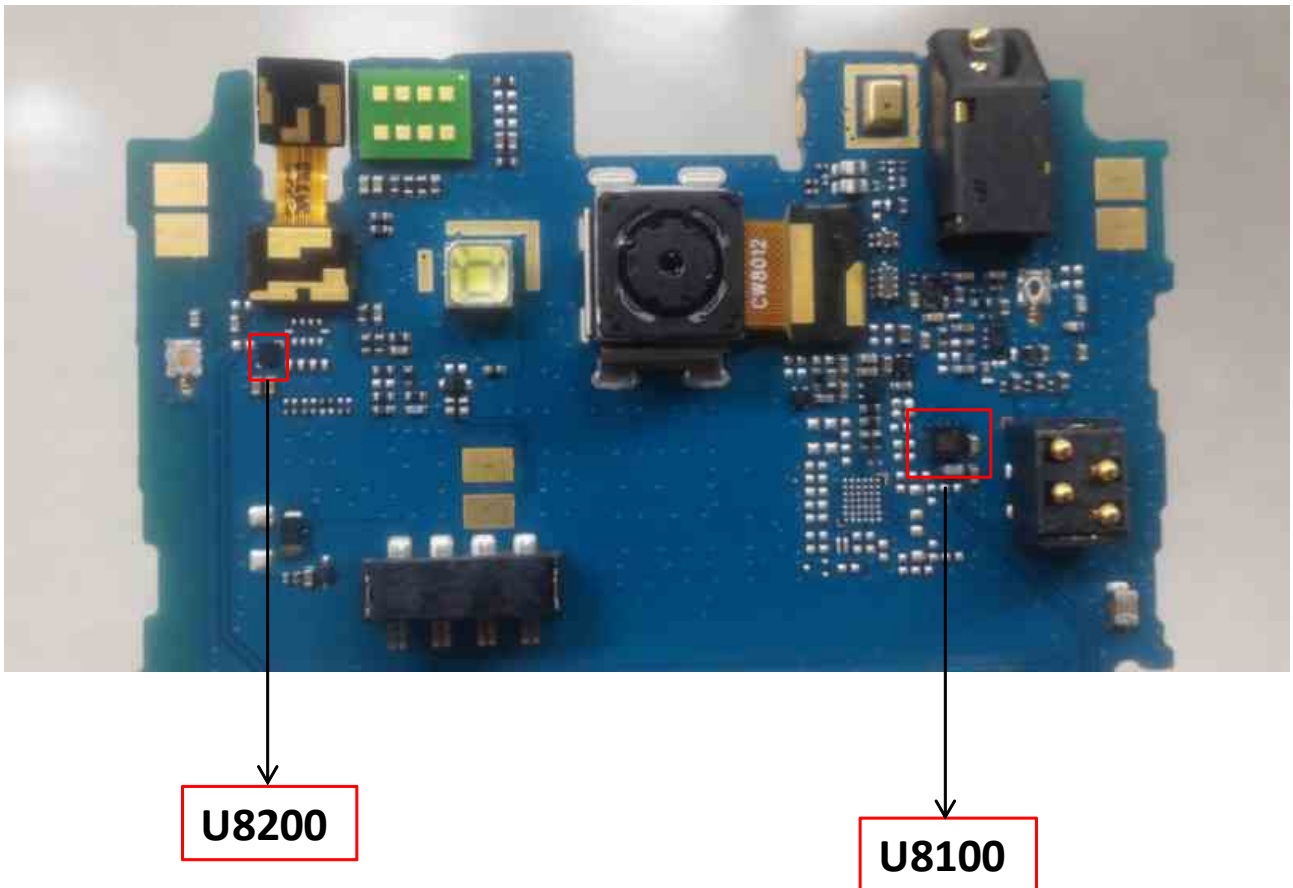
3.12. AXIS Sensor on/off TS

The **6 AXIS** sensor is calibrated by the sensor data using SW algorithm.





3. TROUBLE SHOOTING

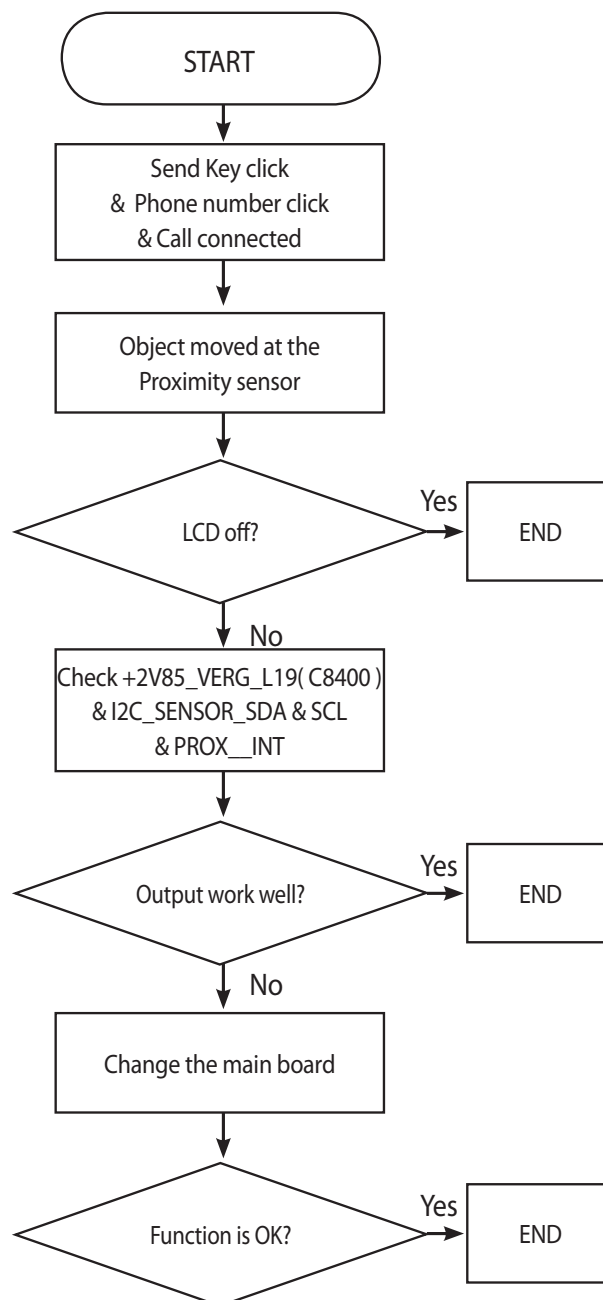


3.13 Proximity Sensor on/off TS

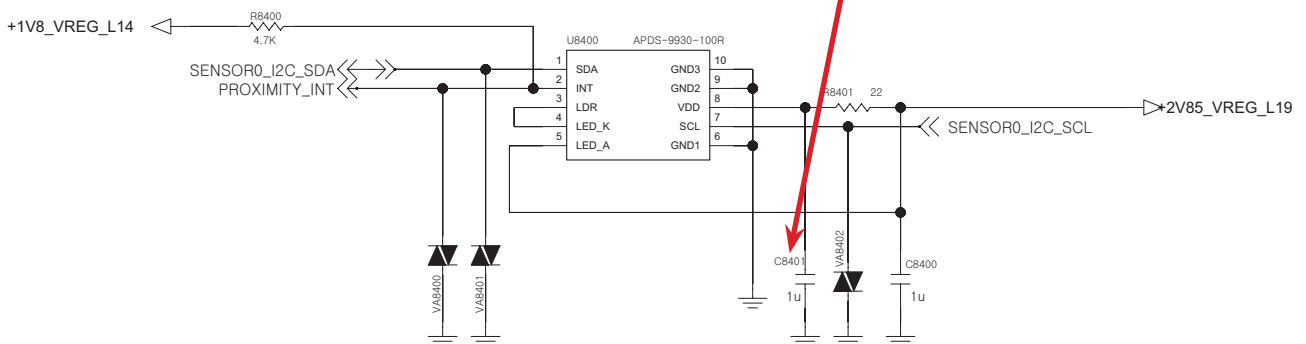
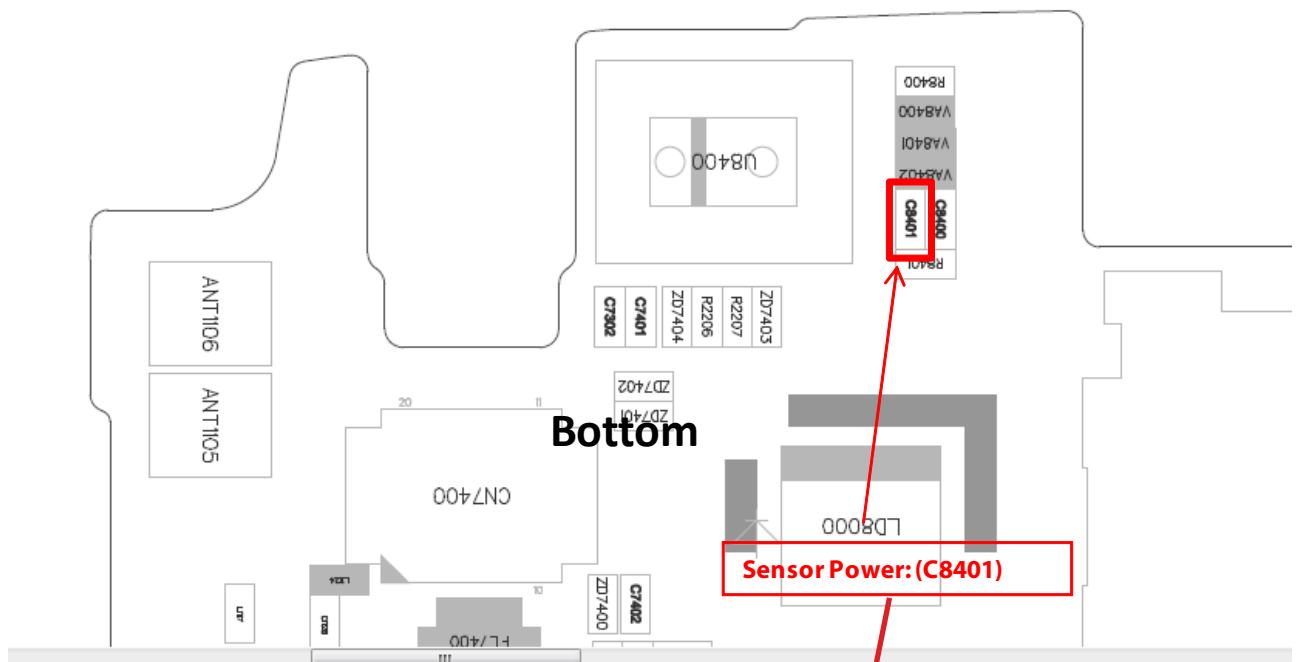
Proximity Sensor is worked as below:

Send Key click -> Phone number click -> Call connected -> Object moved at the sensor

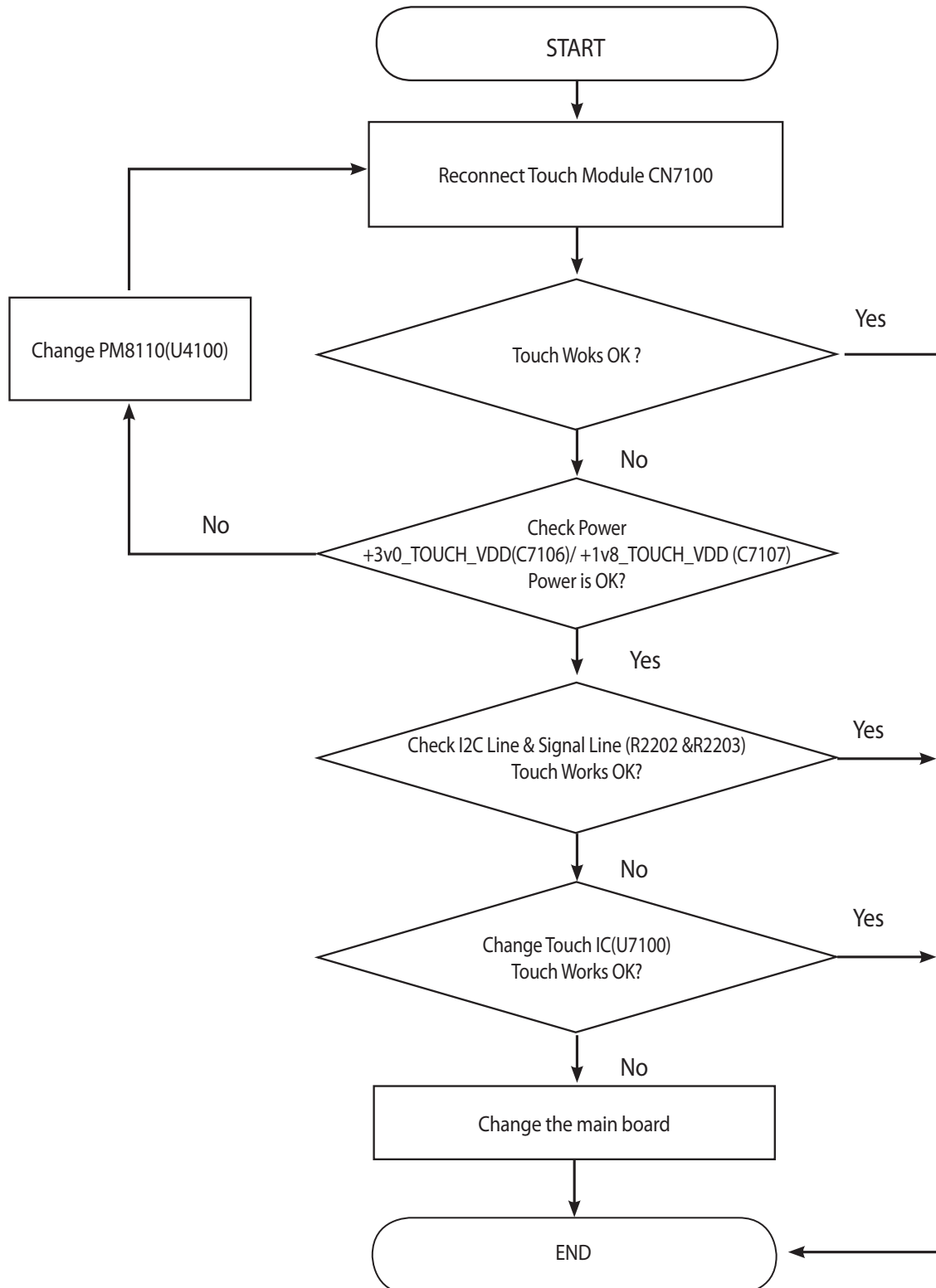
-> Control the screen's on/off operation automatically

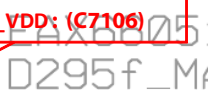


3. TROUBLE SHOOTING



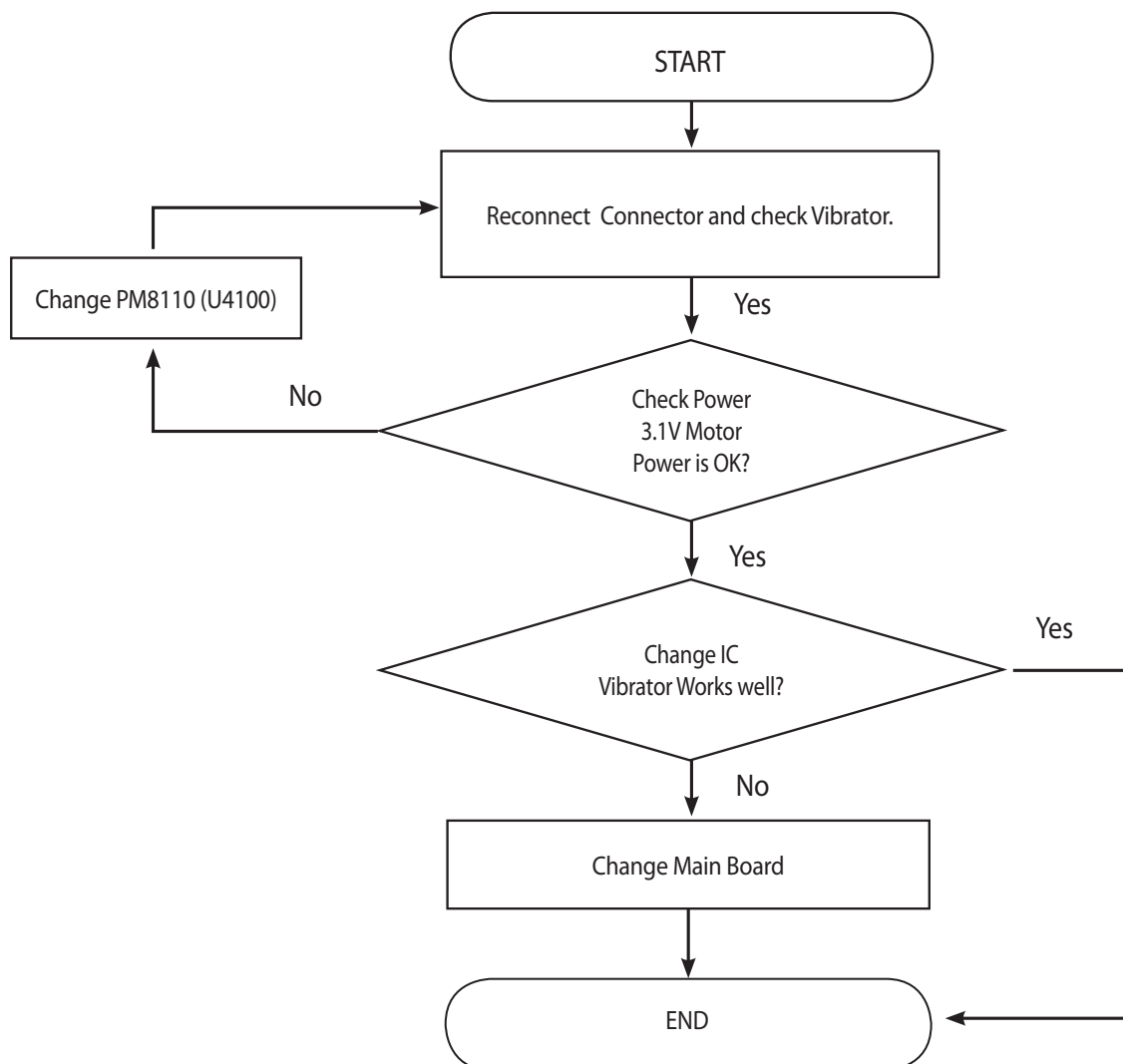
3.14. Touch Sensor Trouble Shooting



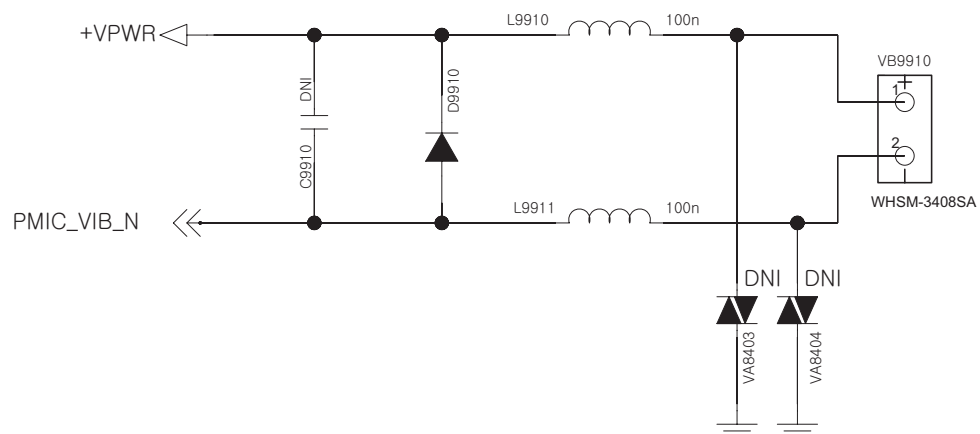
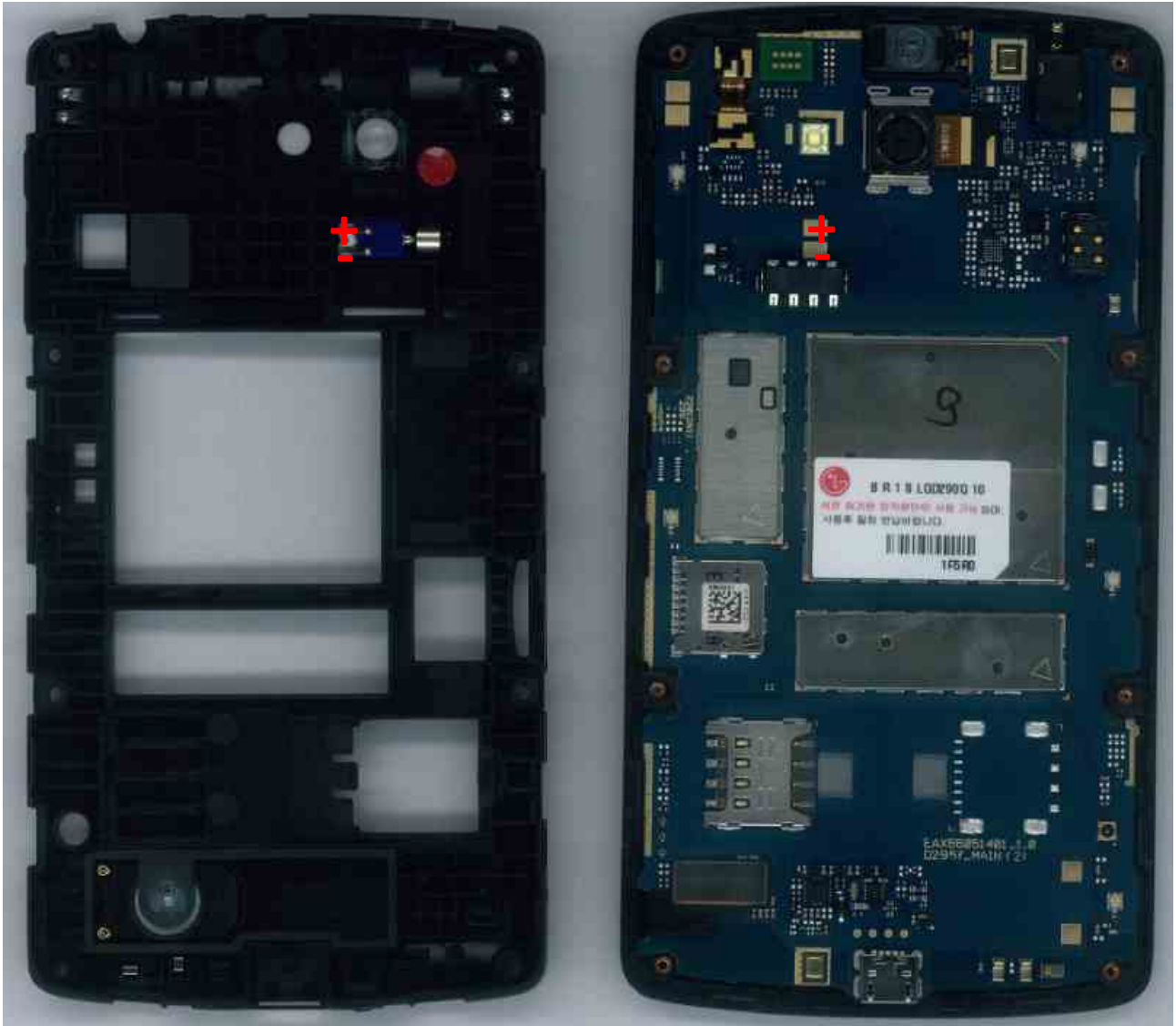


3.15. Vibrator Trouble Shooting

Vibrator signal path is PM8110(U4100) -> Vibrator. Motor Type is DC Motor



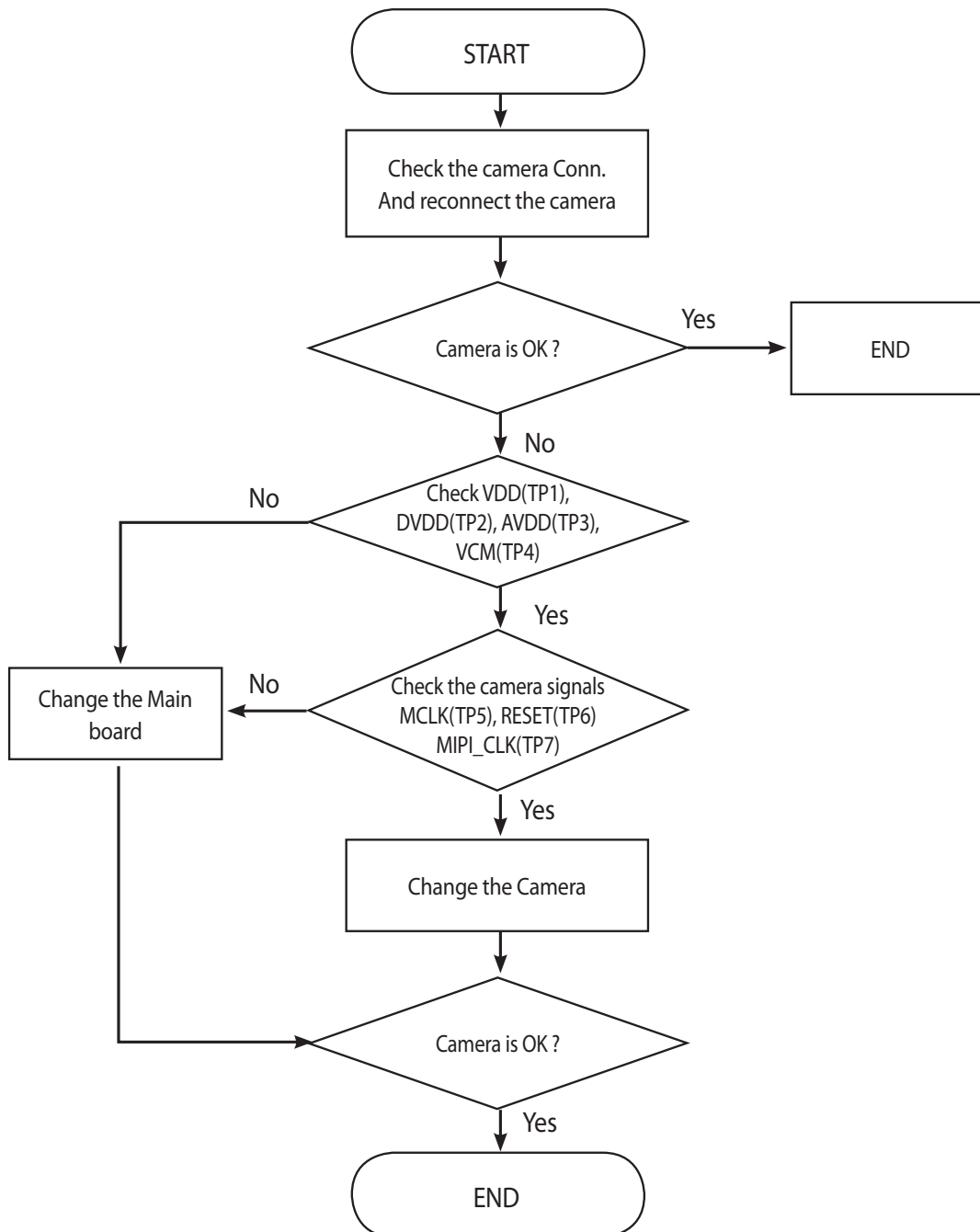
3. TROUBLE SHOOTING



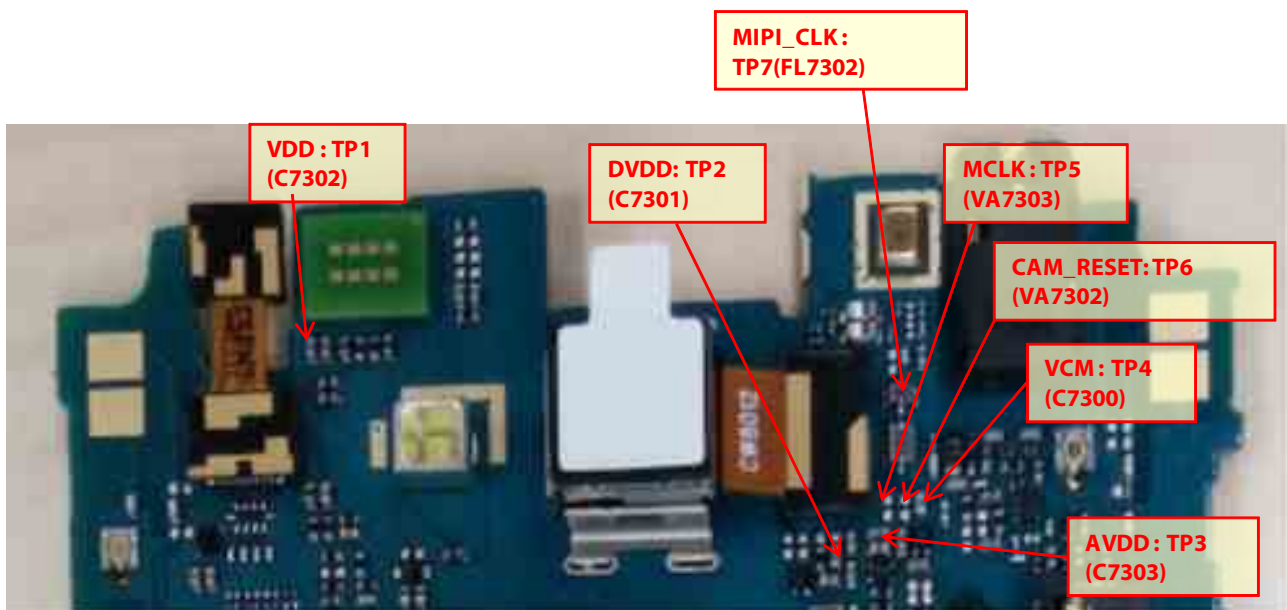
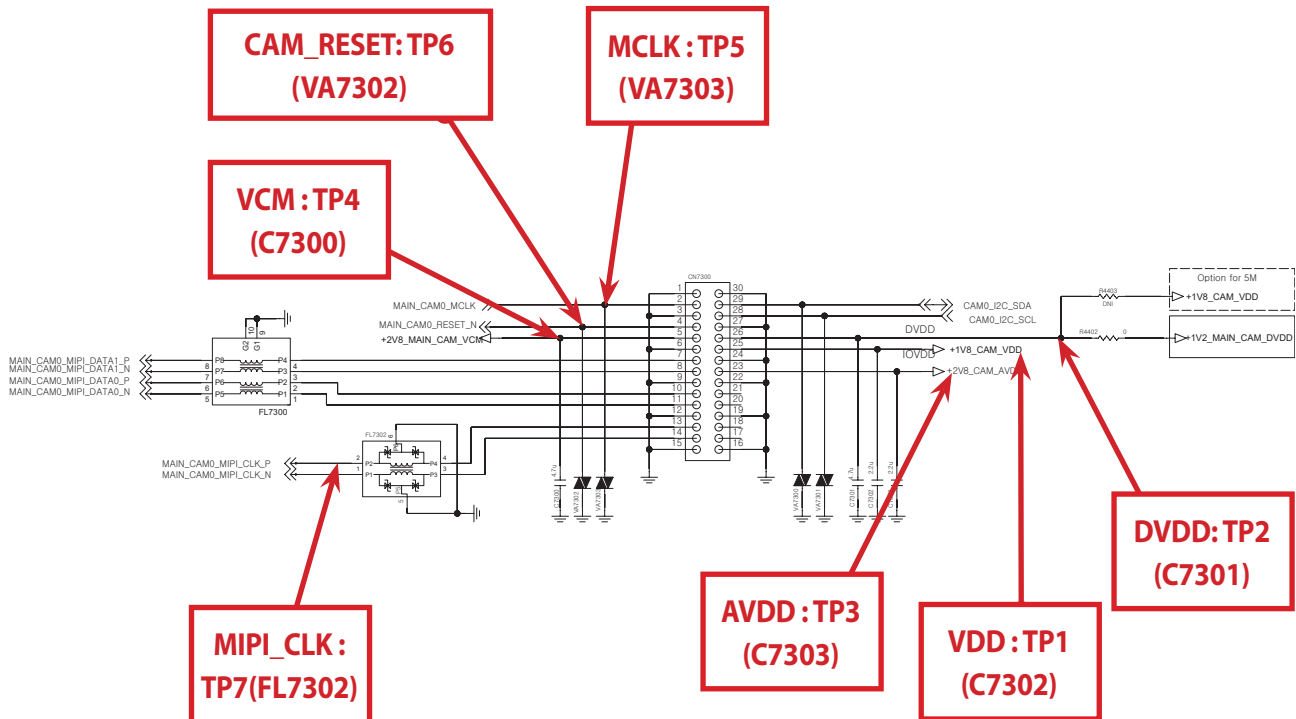
3.16 Camera TS

3.16.1 8M Main Camera TS

8M camera control signals are generated by MSM8212(U2100) and powered by voltage regulator (U4101, U4102, U4105 : LDO) , PM8110 (U4100 : PMIC).

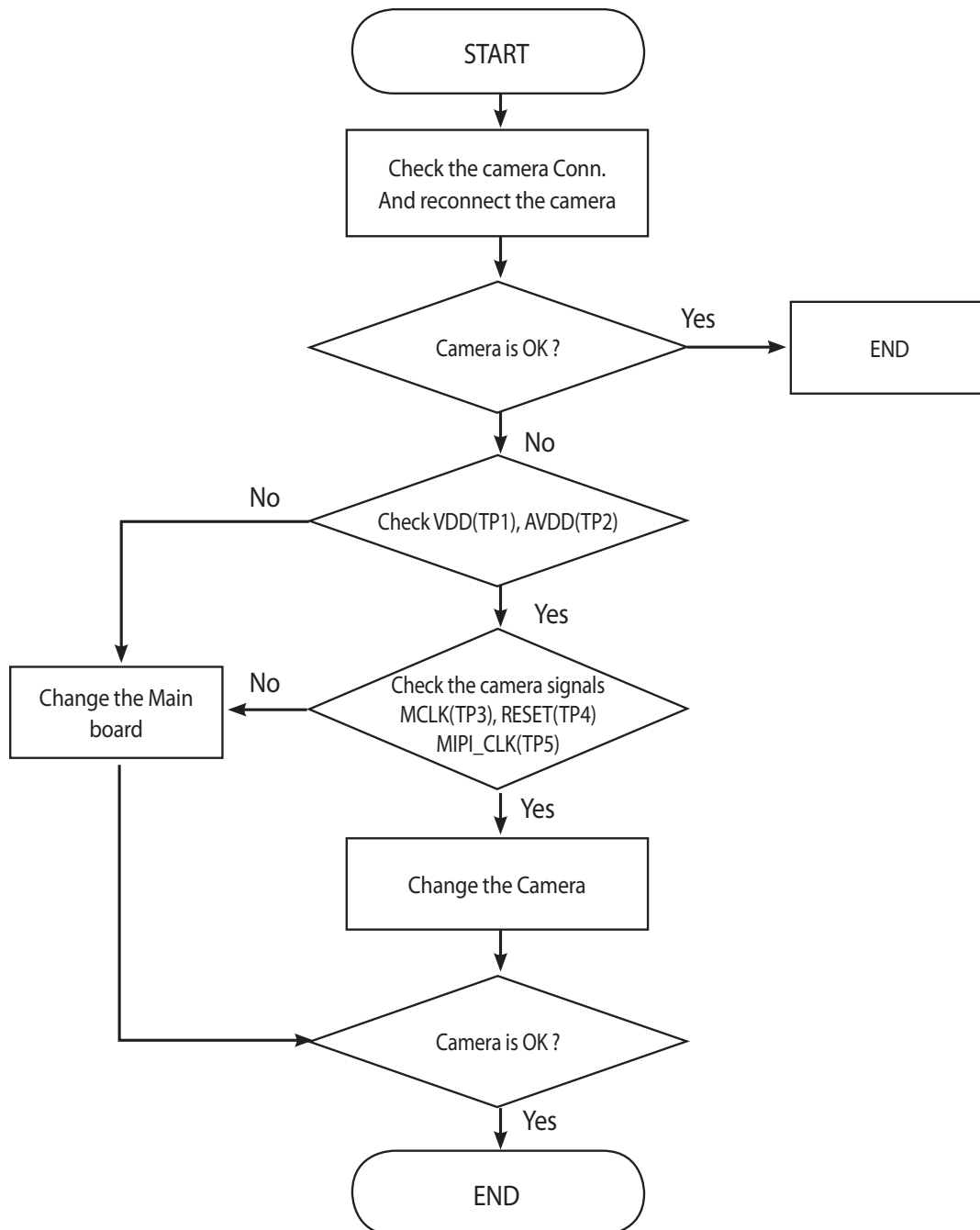


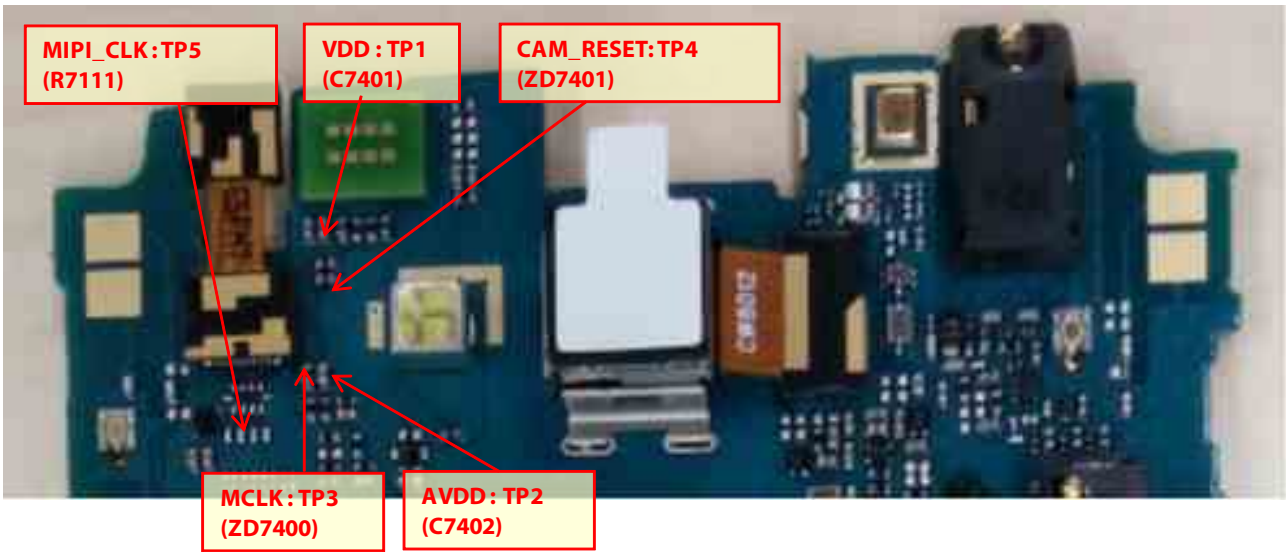
3. TROUBLE SHOOTING



3.16.2 VGA Sub Camera TS

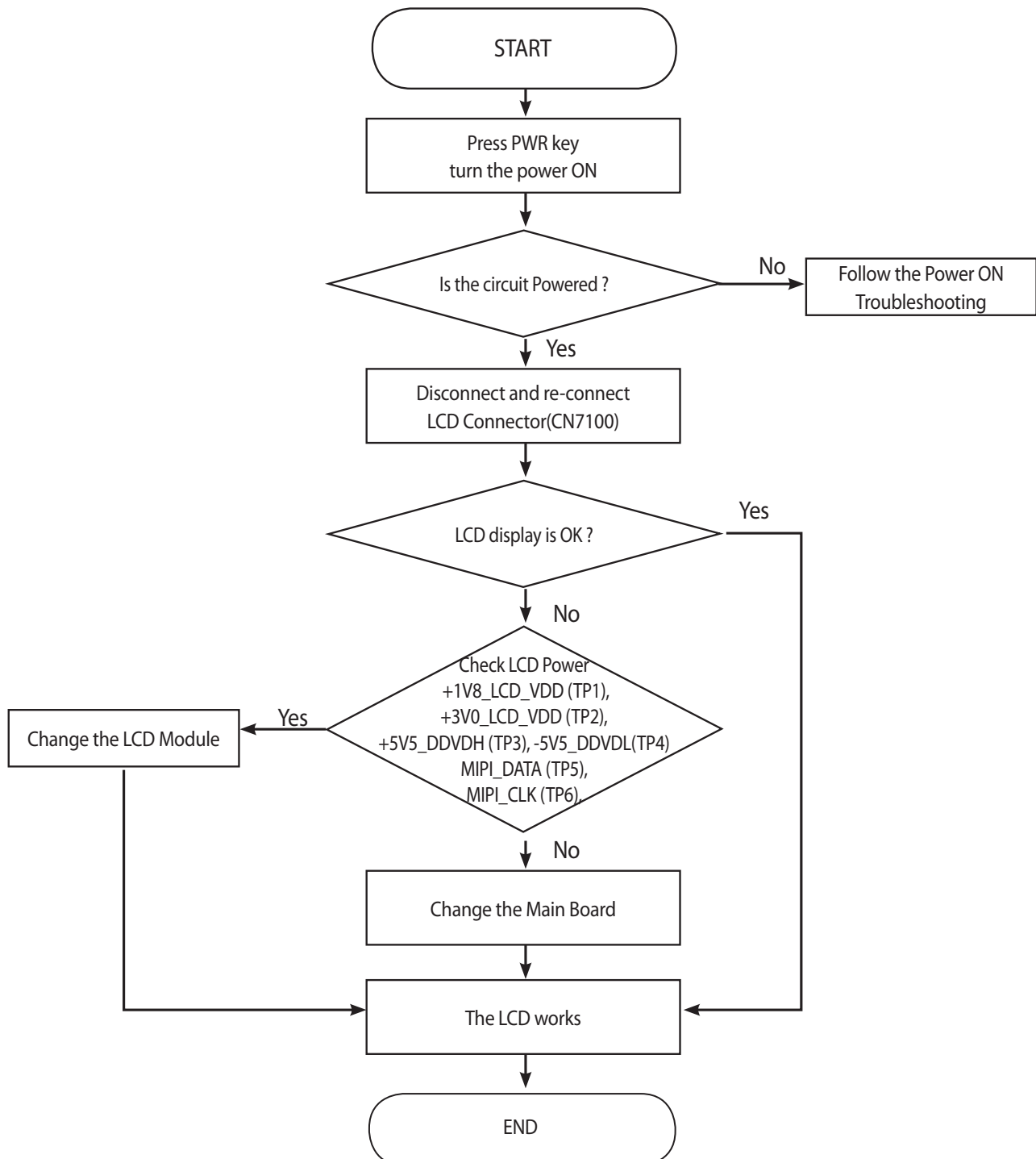
VGA camera control signals are generated by MSM8212(U2100) and powered by voltage regulator (U4102 : LDO) , PM8110 (U4100 : PMIC).

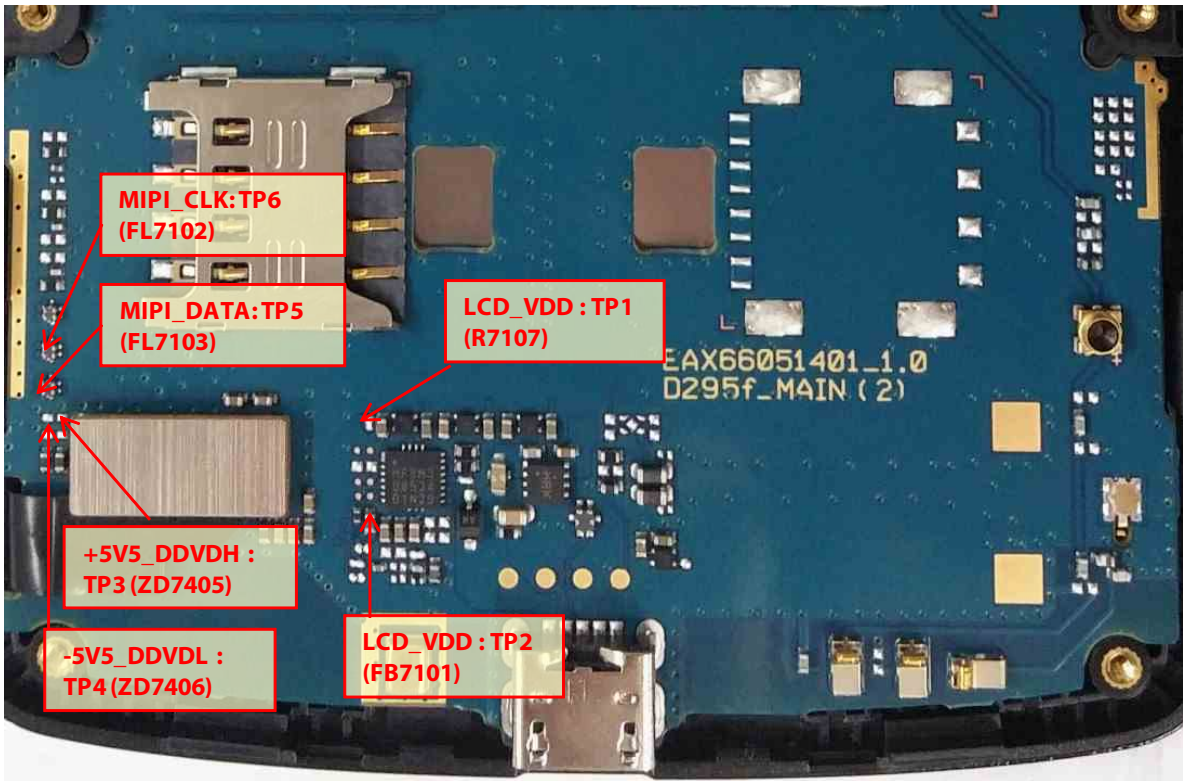




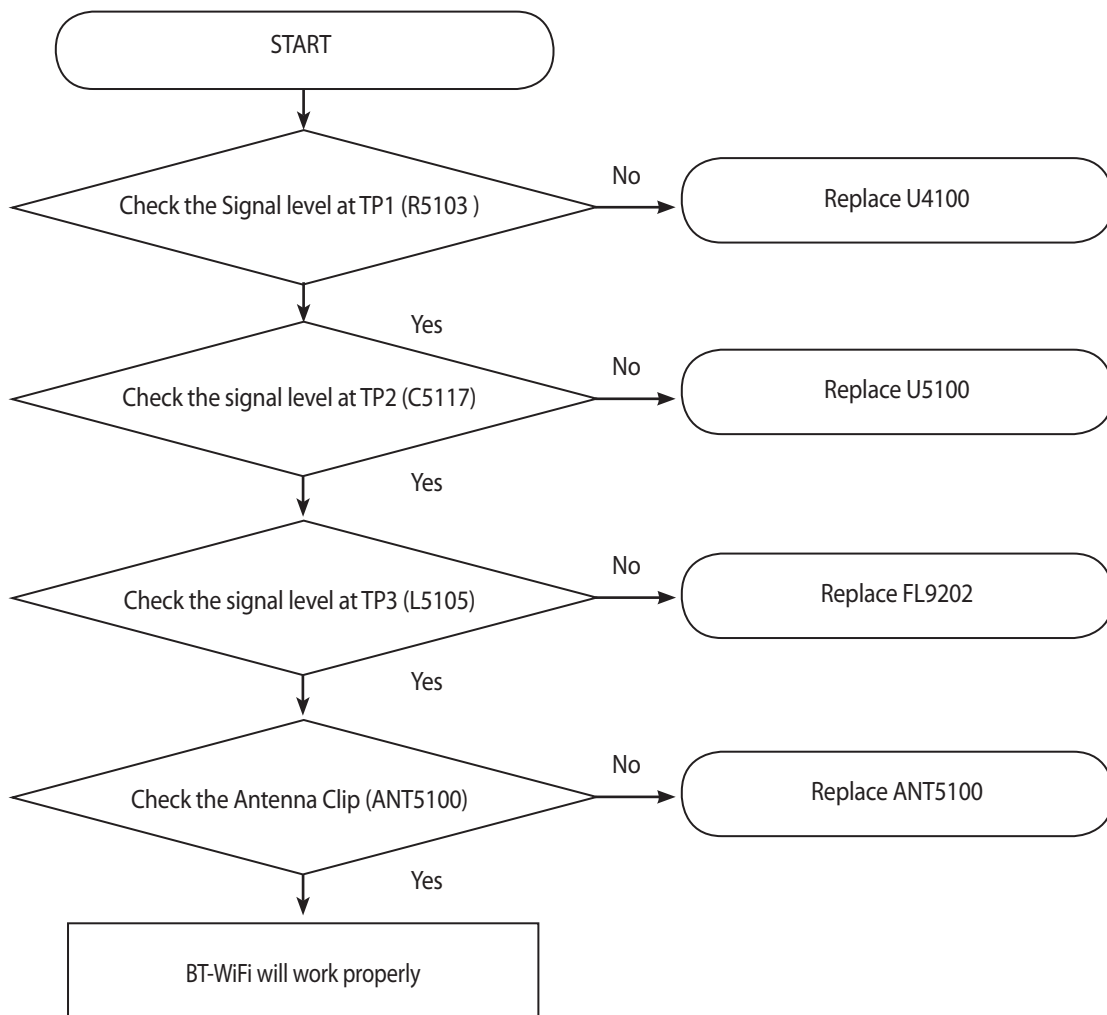
3.17. Main LCD TS

The LCD control signals are generated by MSM8212(U2100) and powered by PM8110 (U4100 : PMIC).

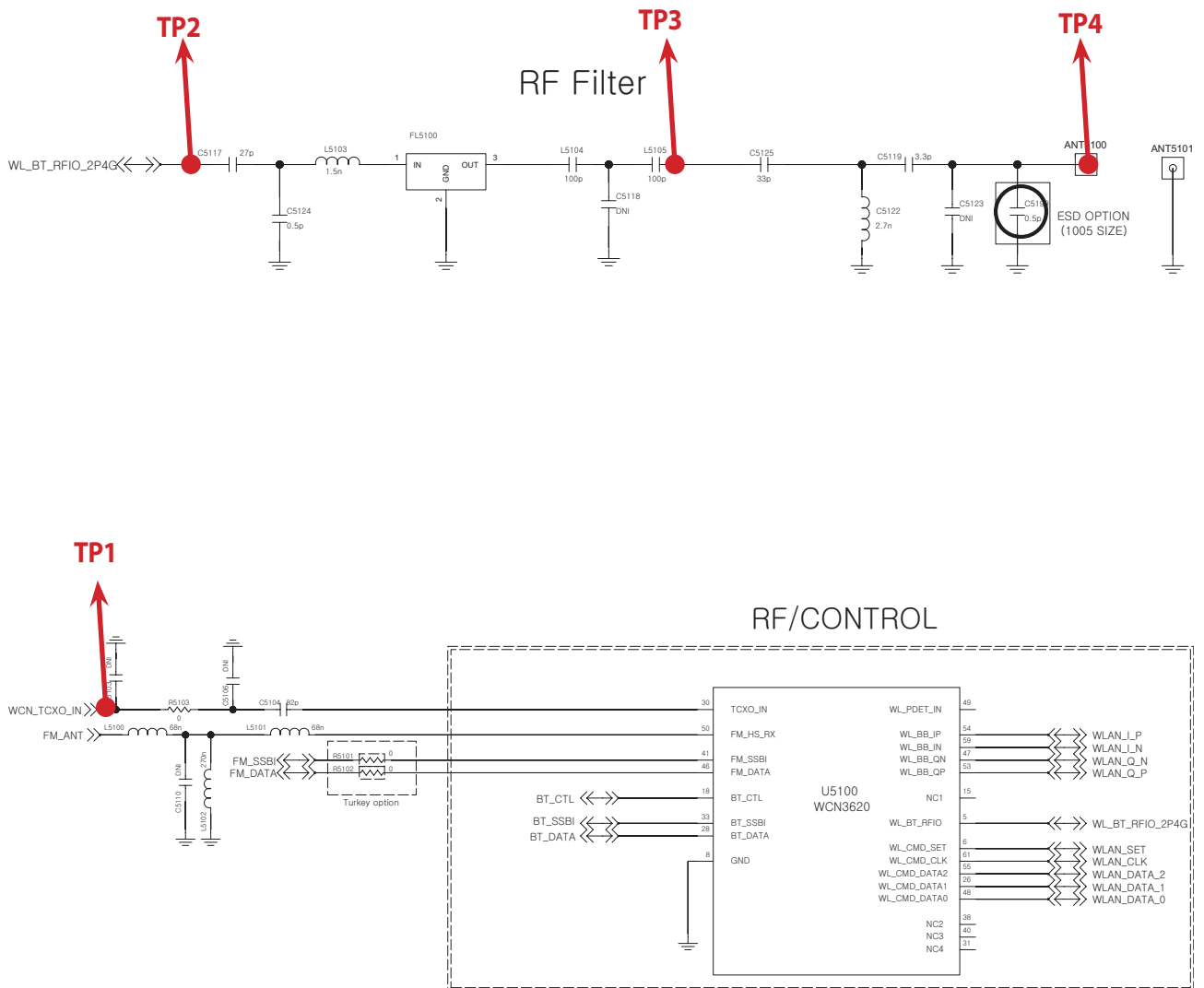


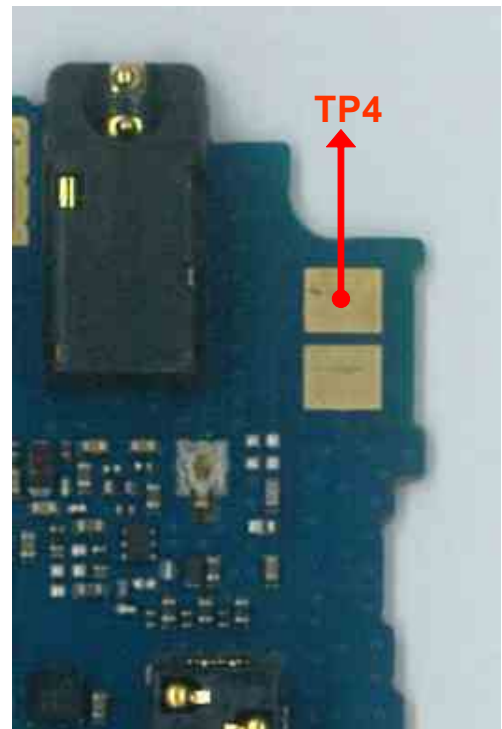
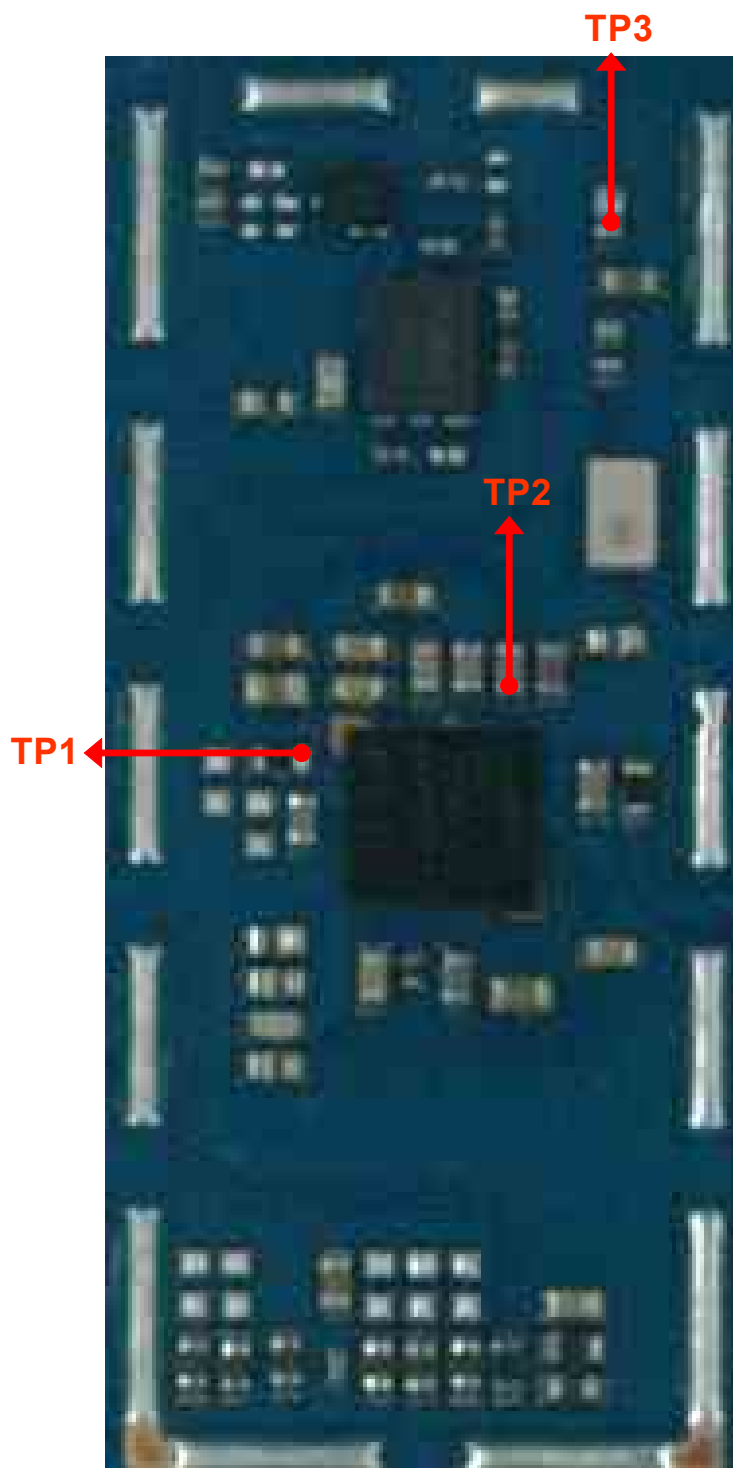


3.18. BT/Wifi TS

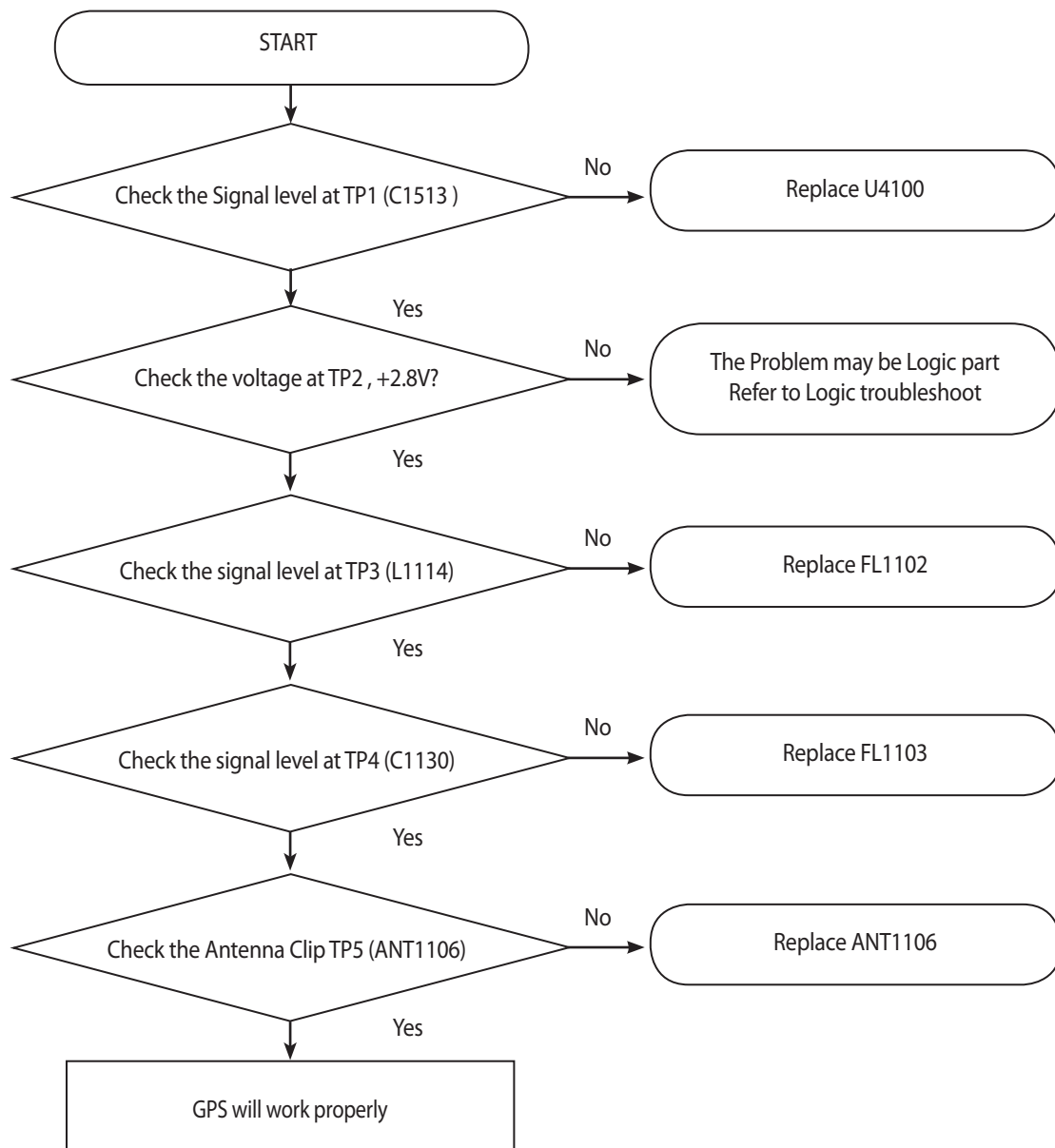


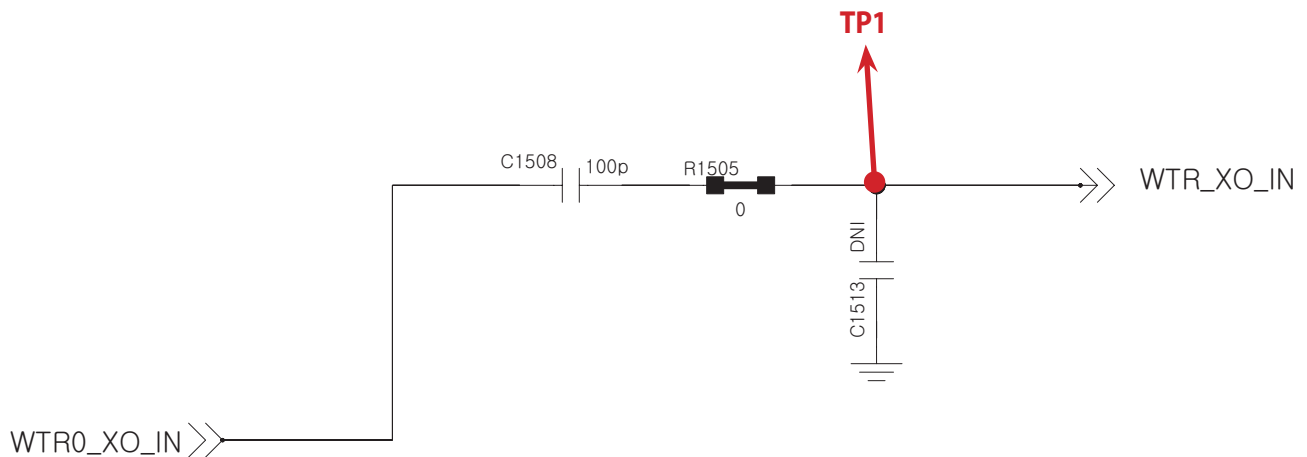
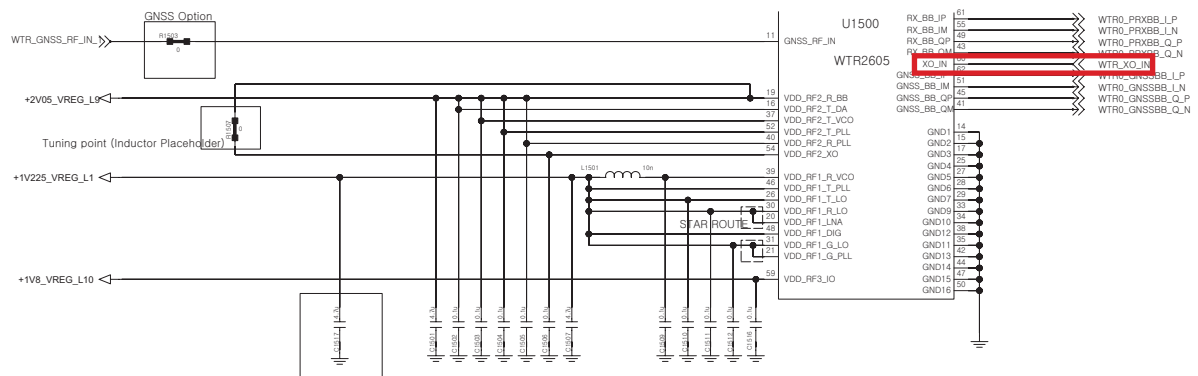
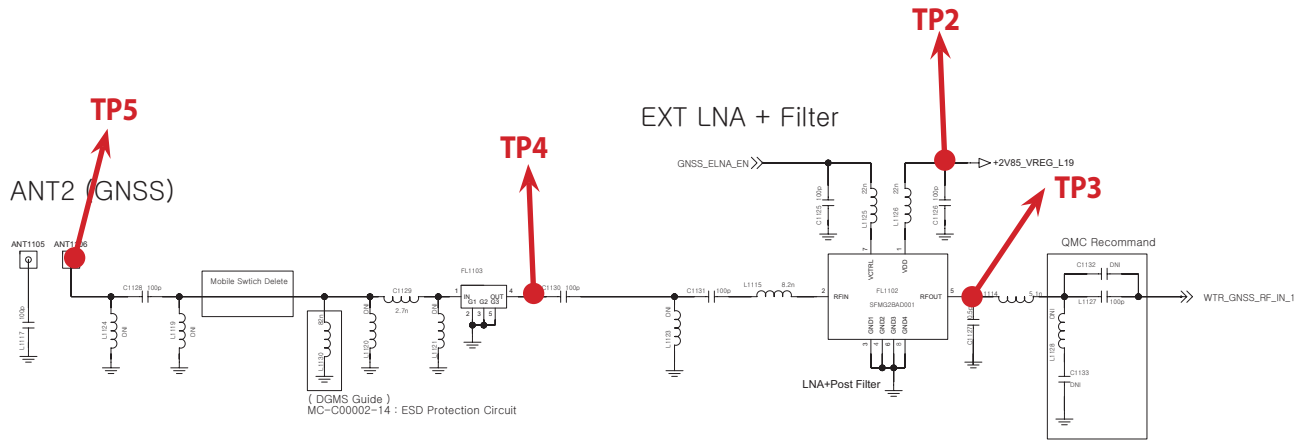
3. TROUBLE SHOOTING



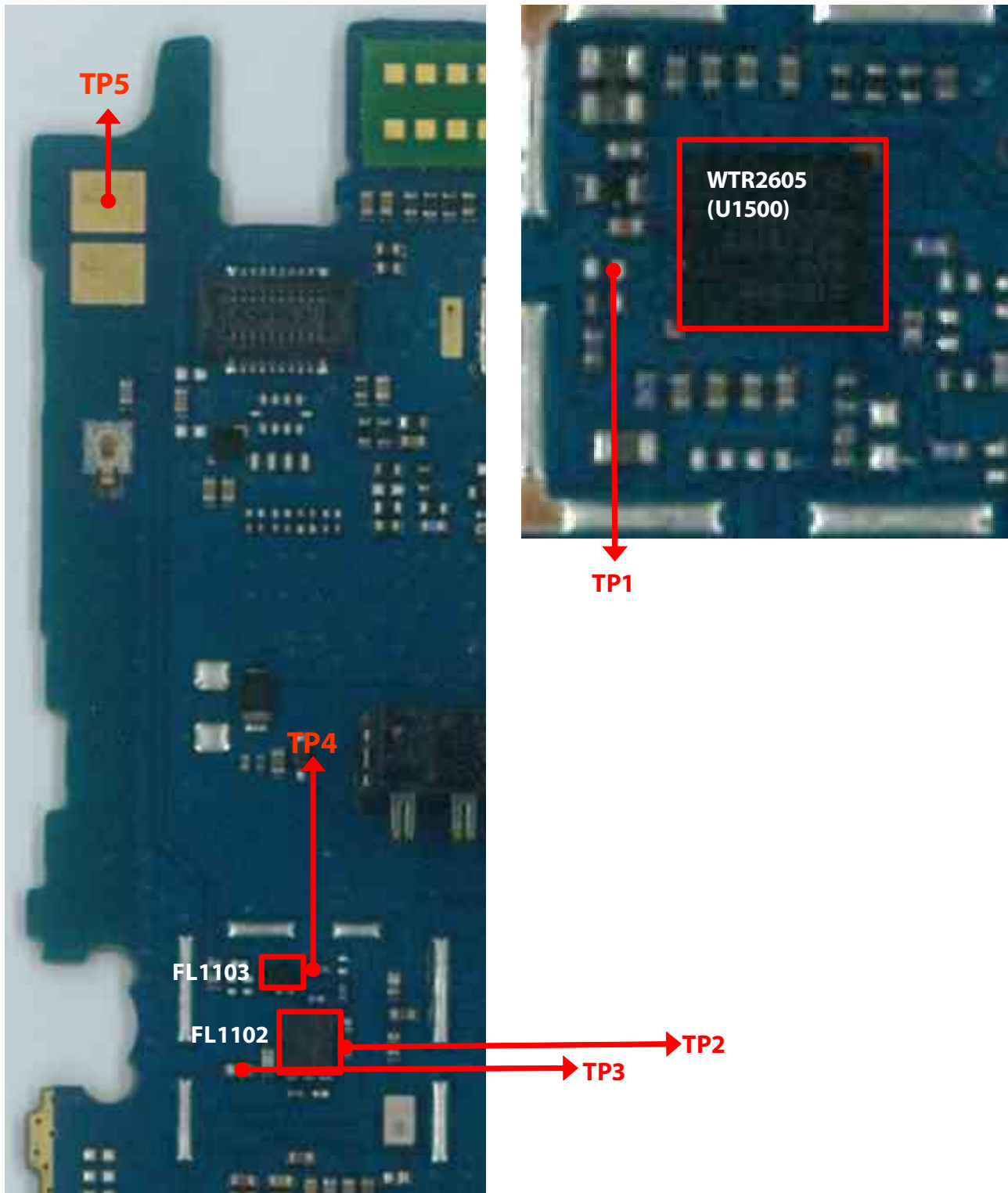


3.19. GPSTS

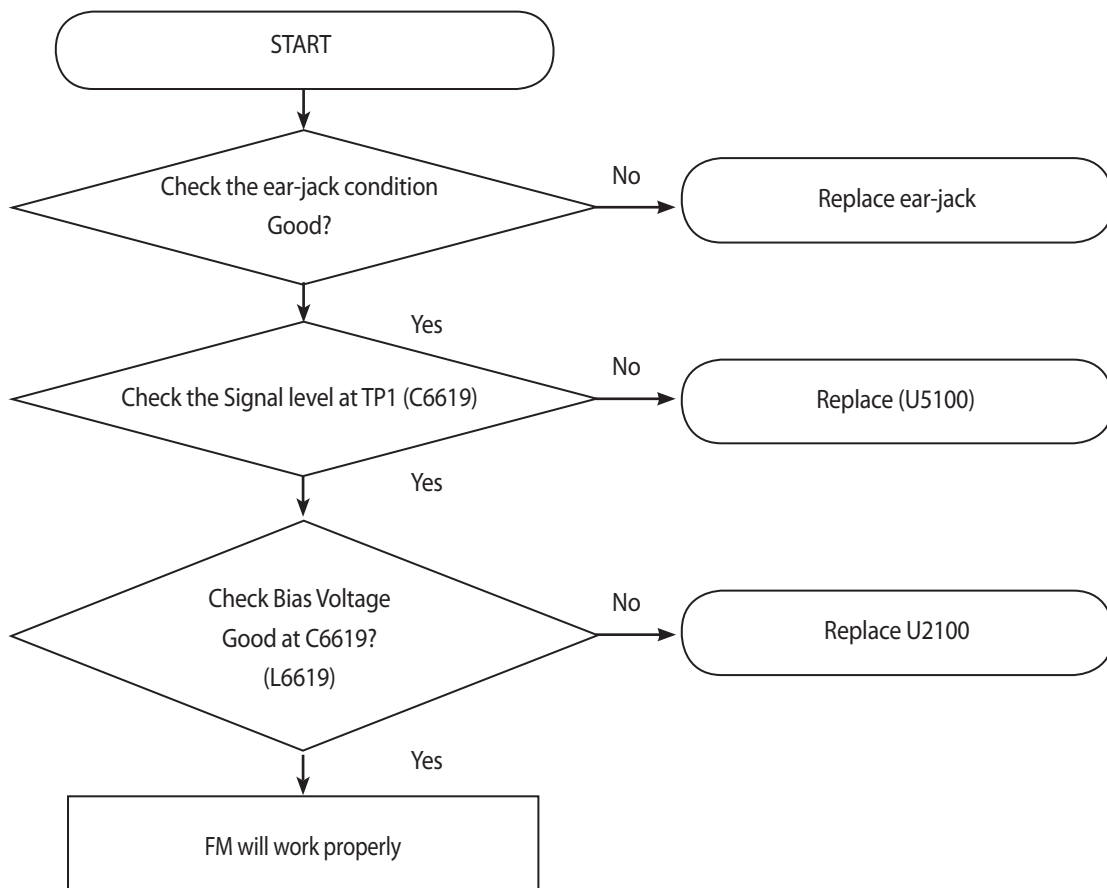




3. TROUBLE SHOOTING



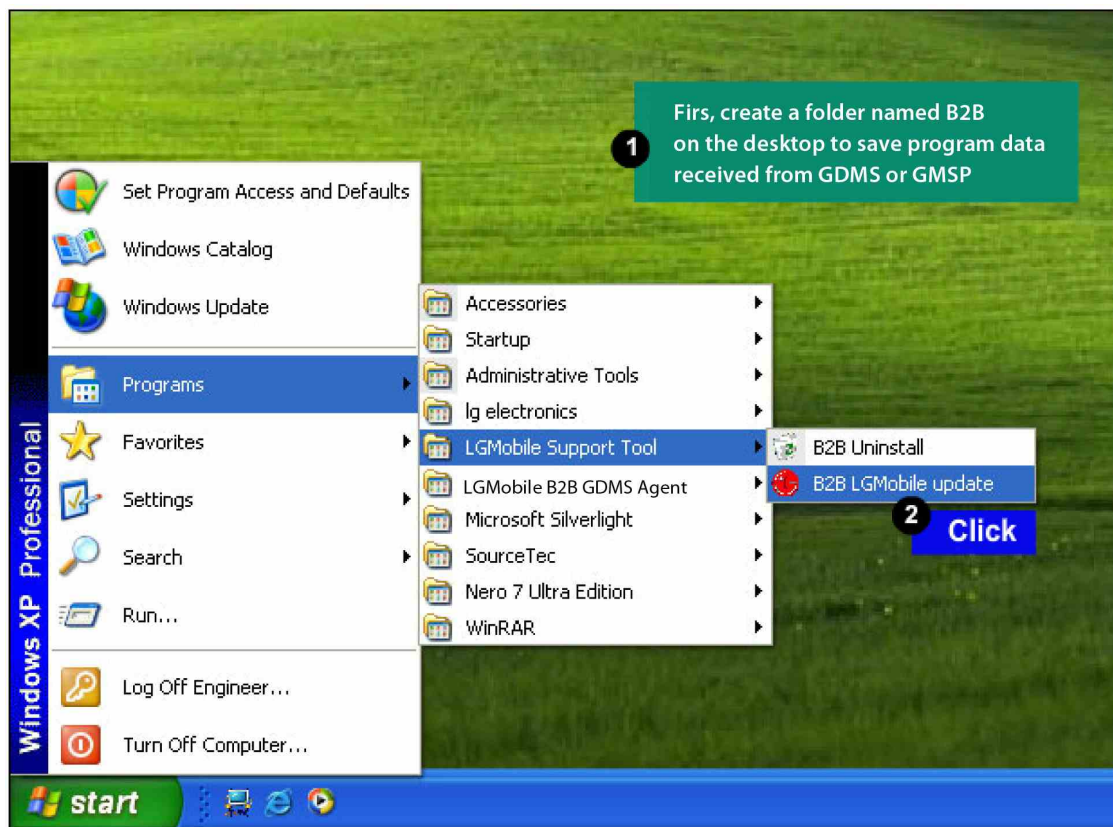
3.20. FM TS



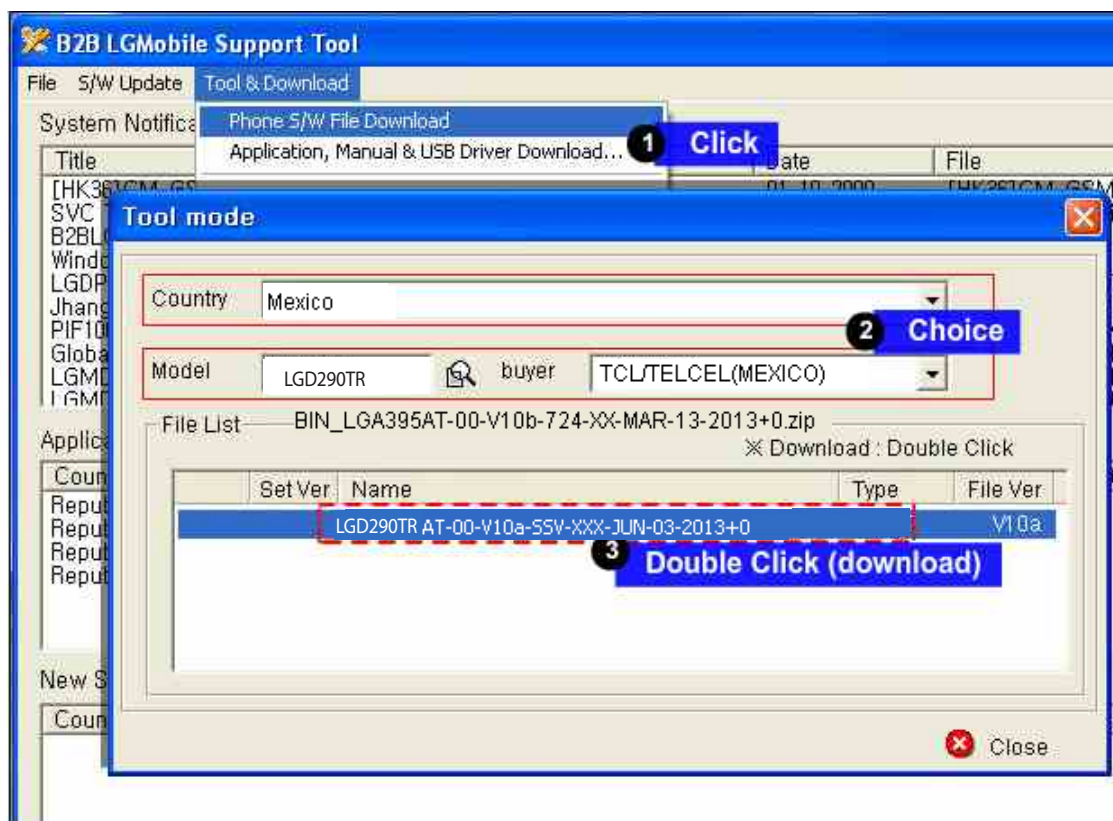
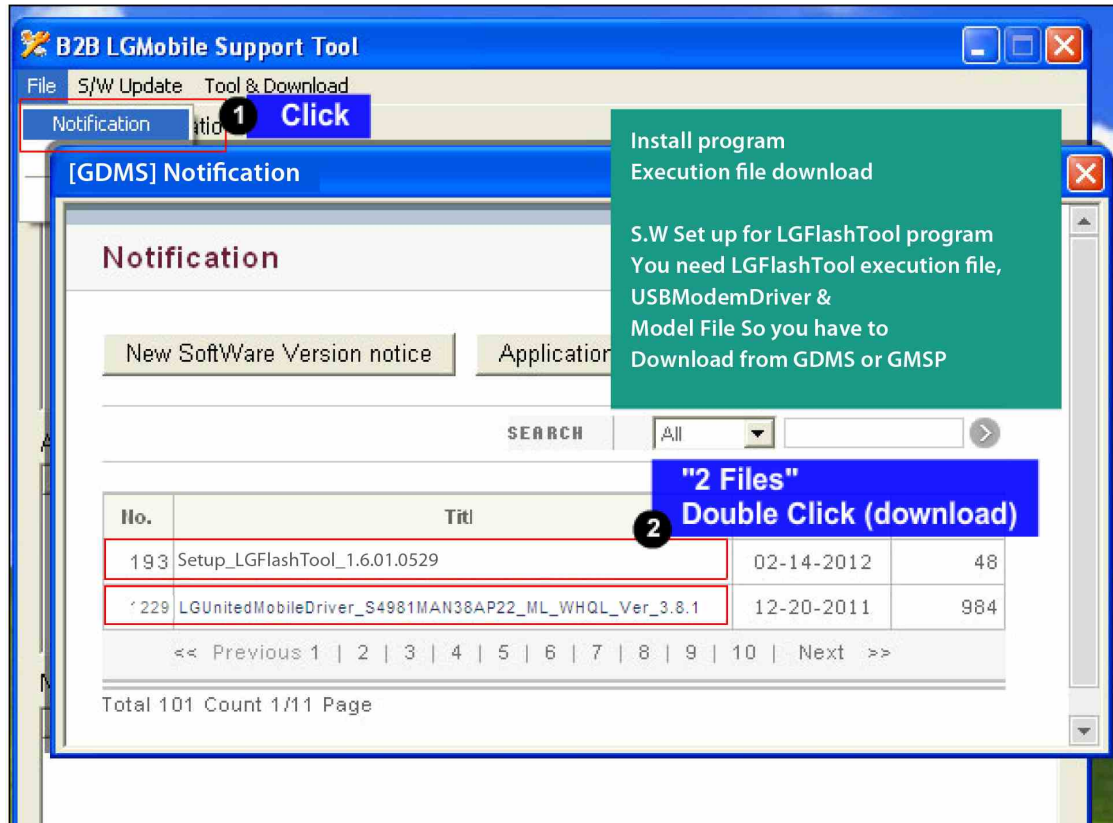


4. DOWNLOAD

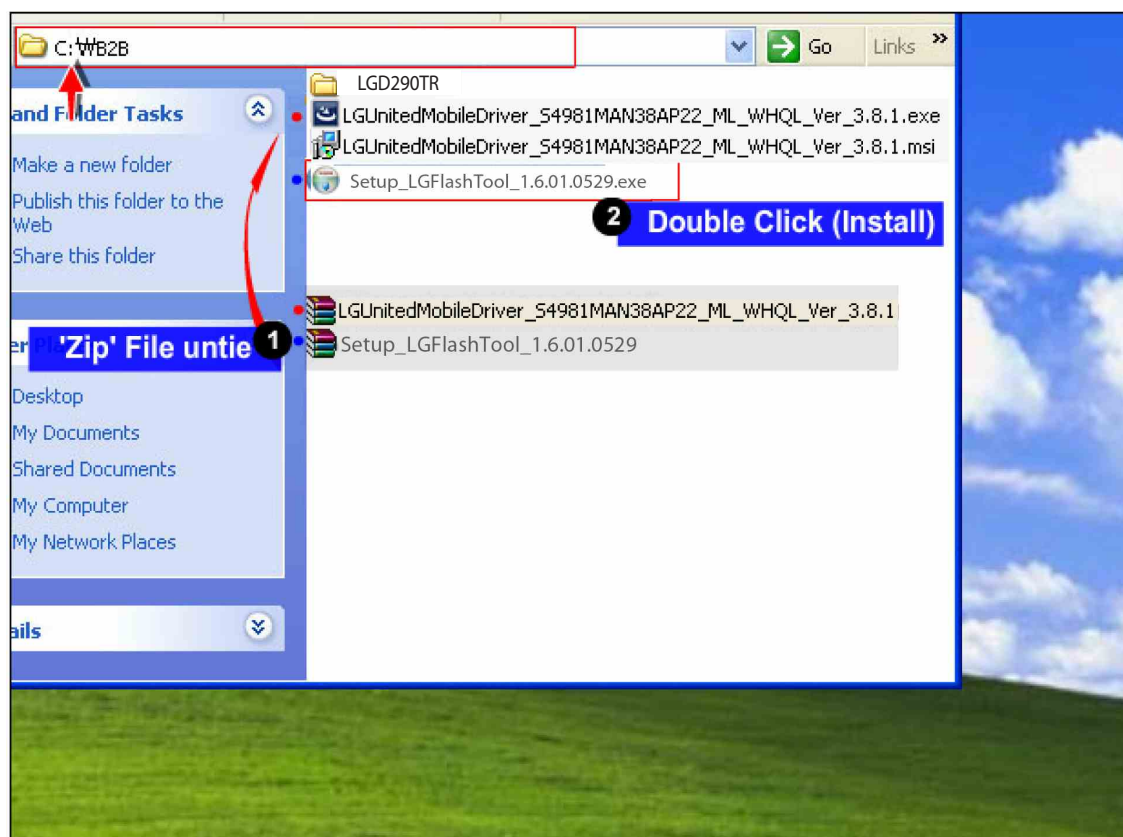
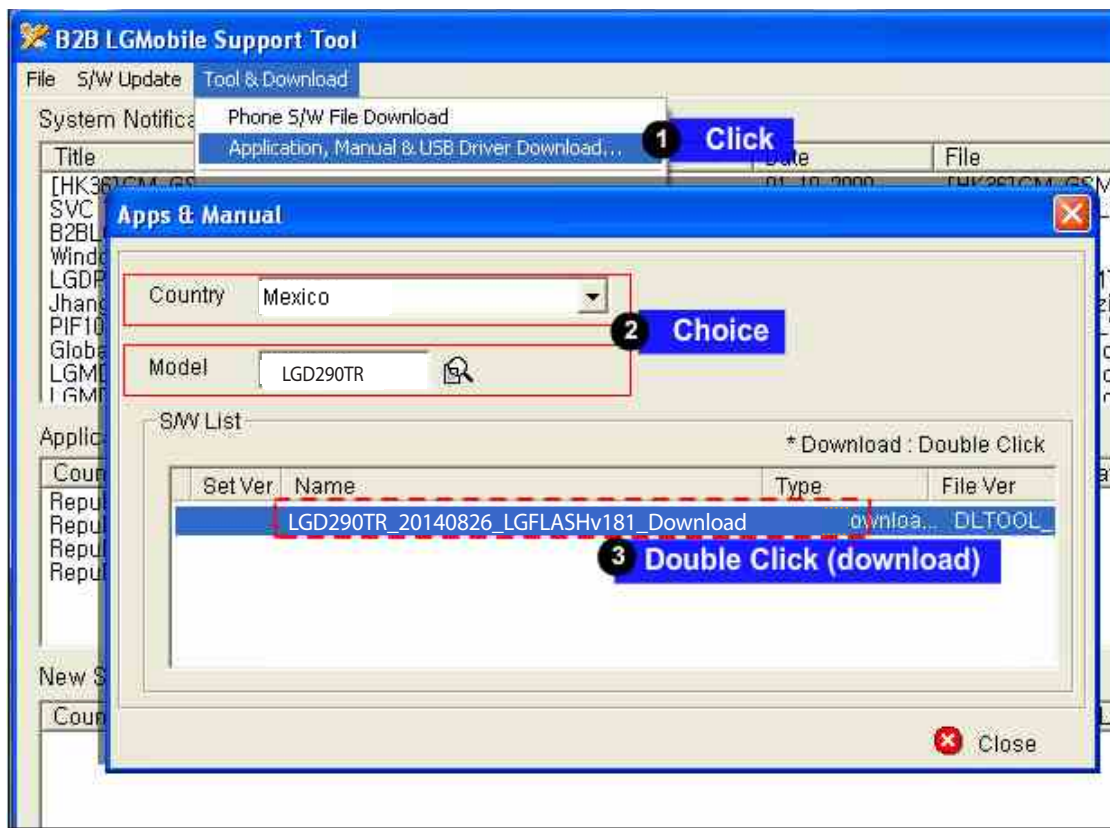
TOOL INFORMATION			
TOOL VERSION	DLL NAME	USB DRIVER	
Please Check the Version to “B2B”			
H/W			
	Name	Part No.	SW
D/L Cable			



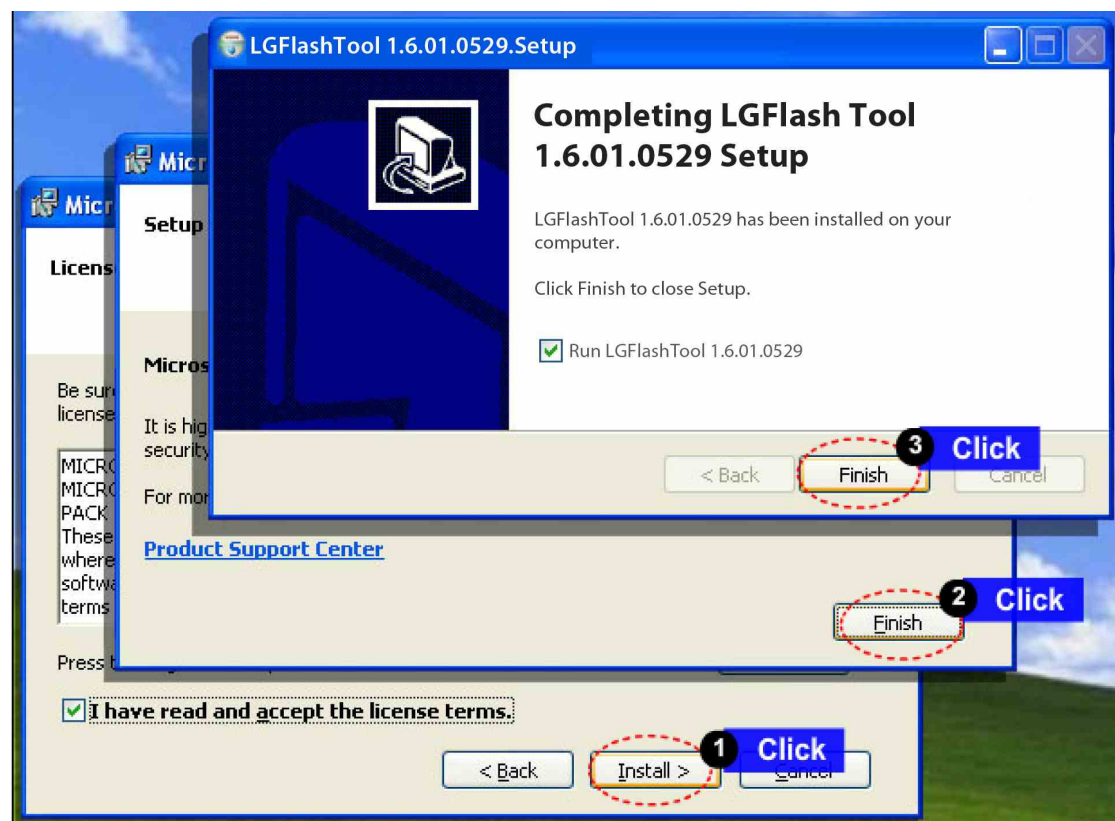
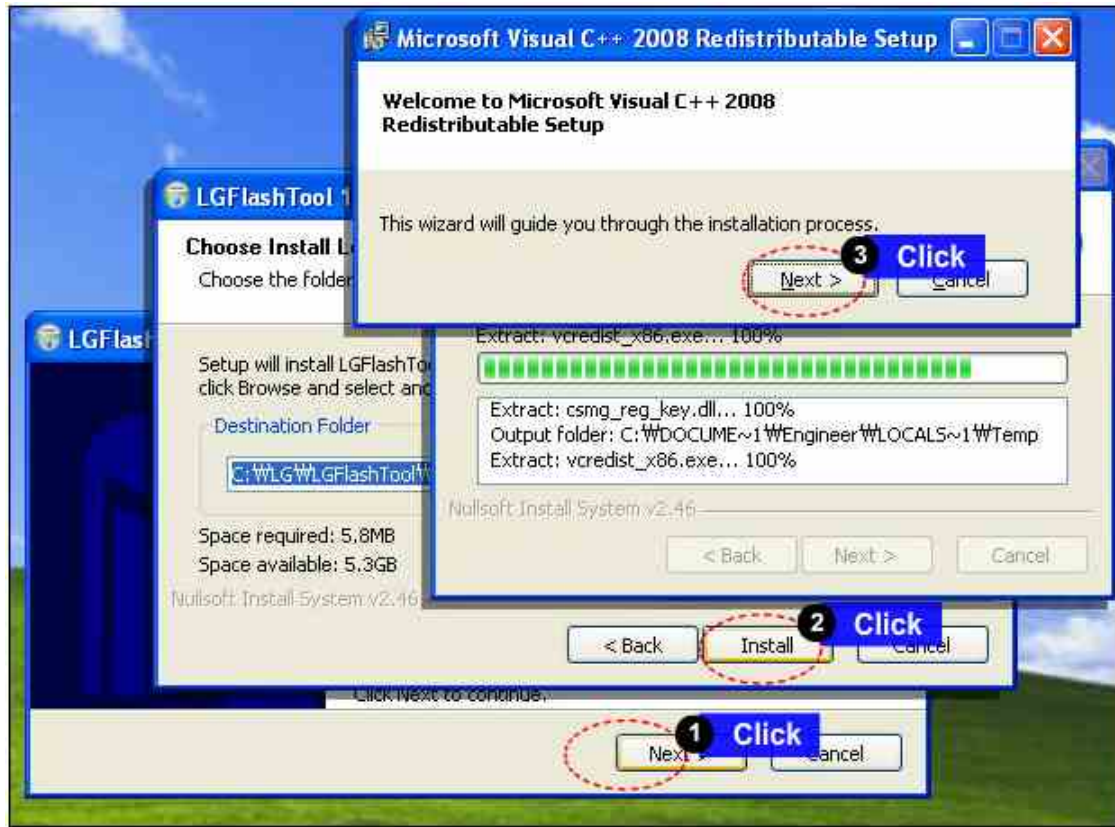
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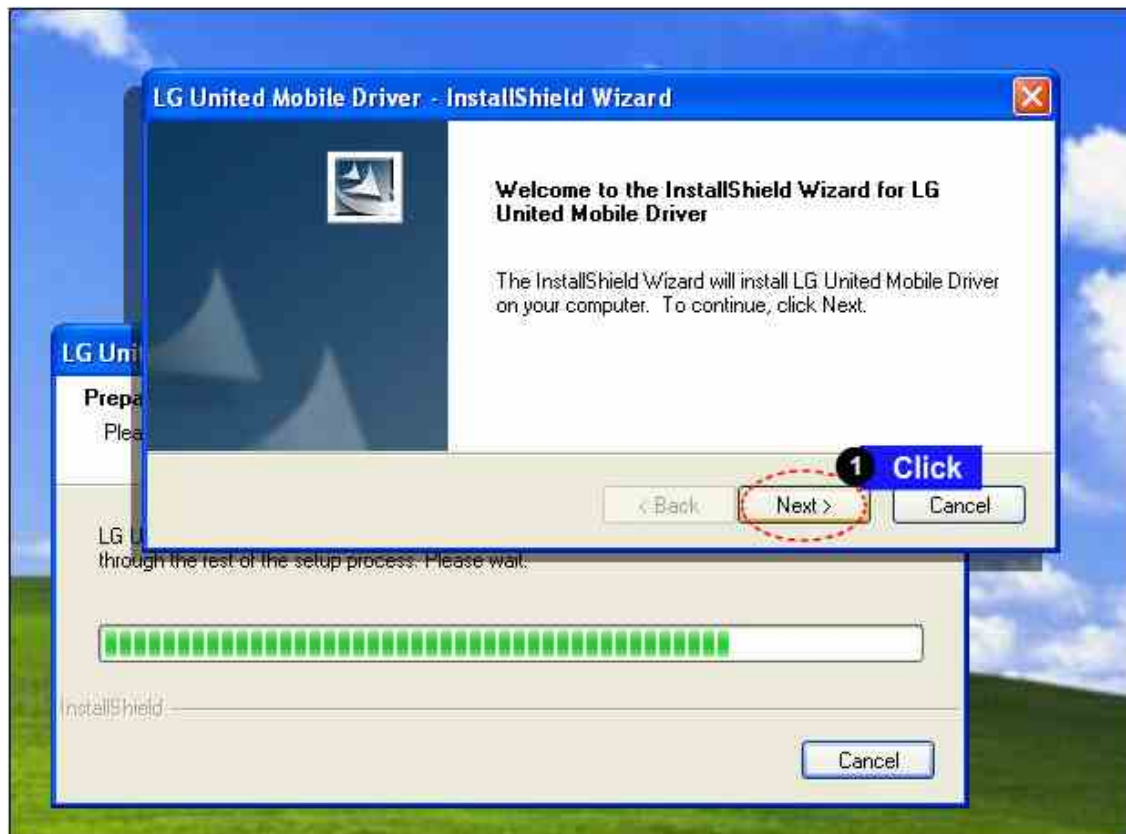
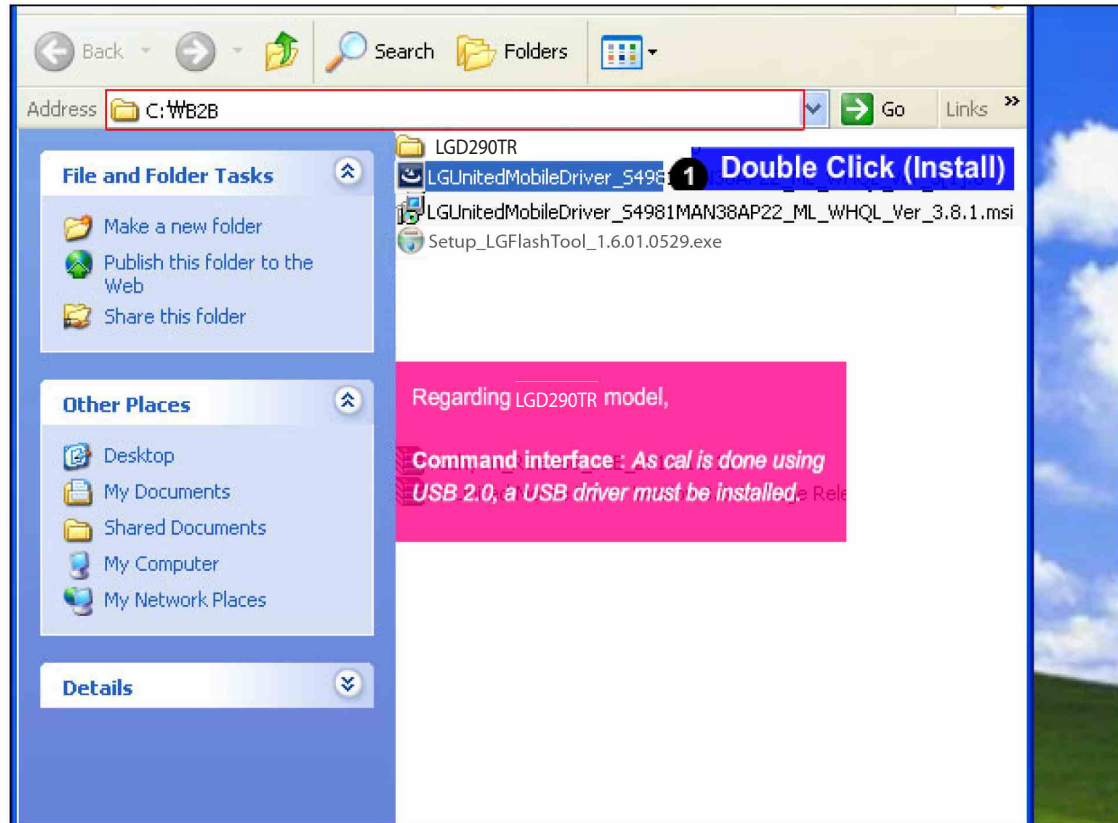
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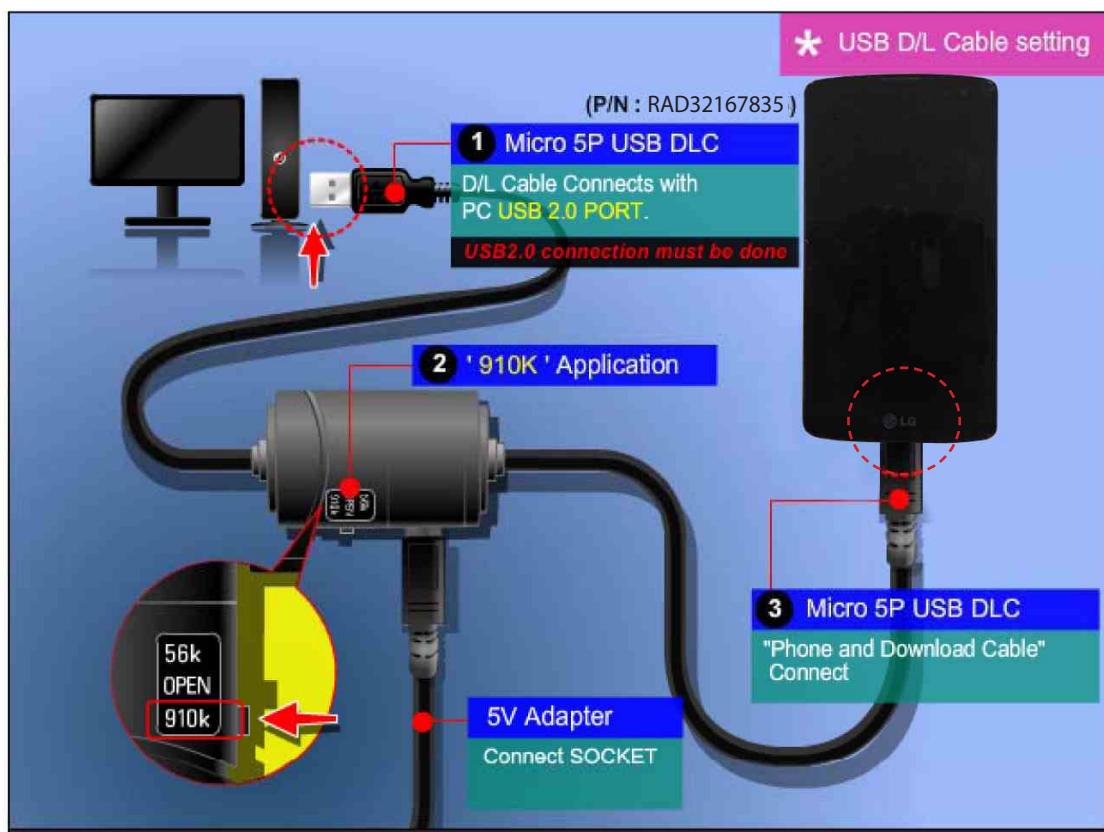
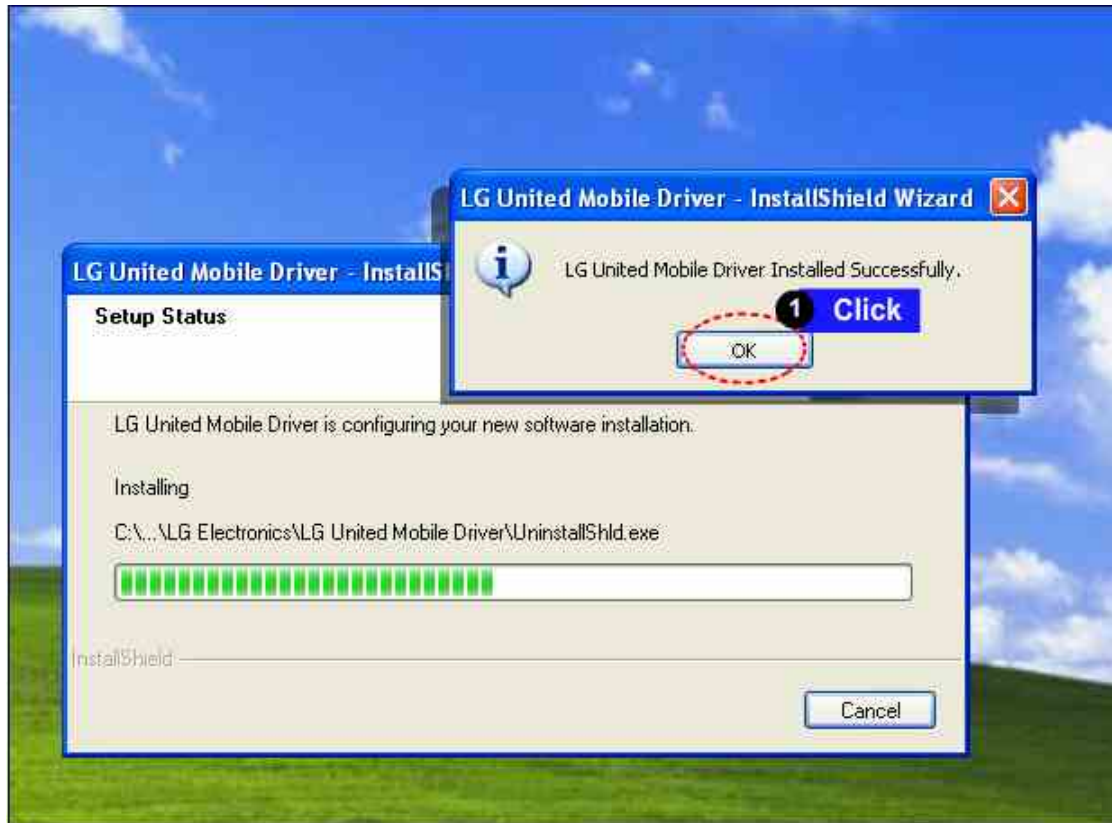
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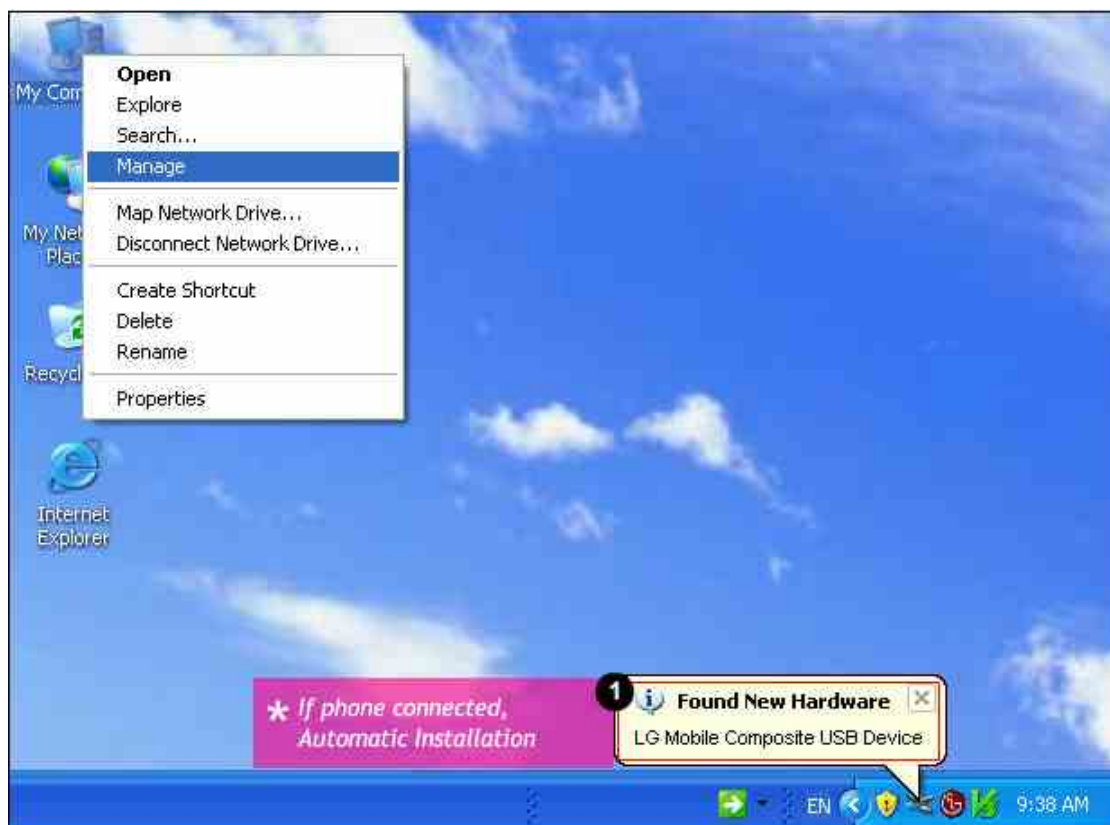
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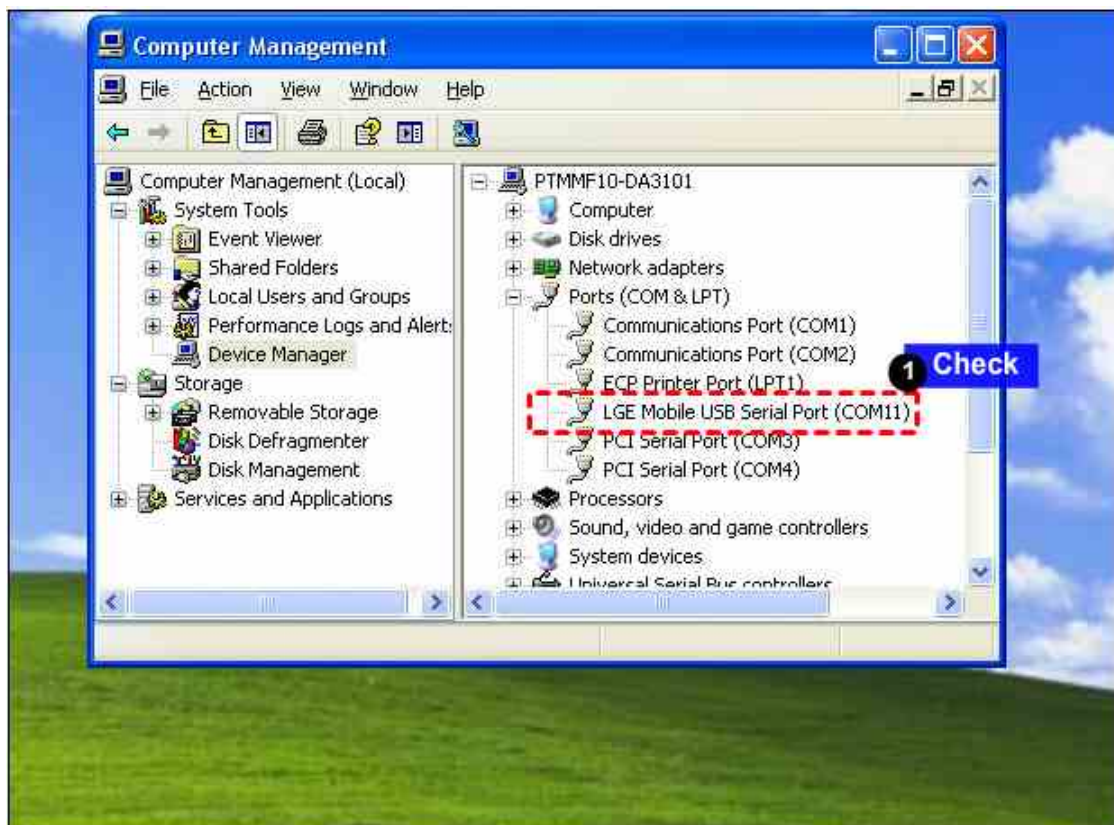
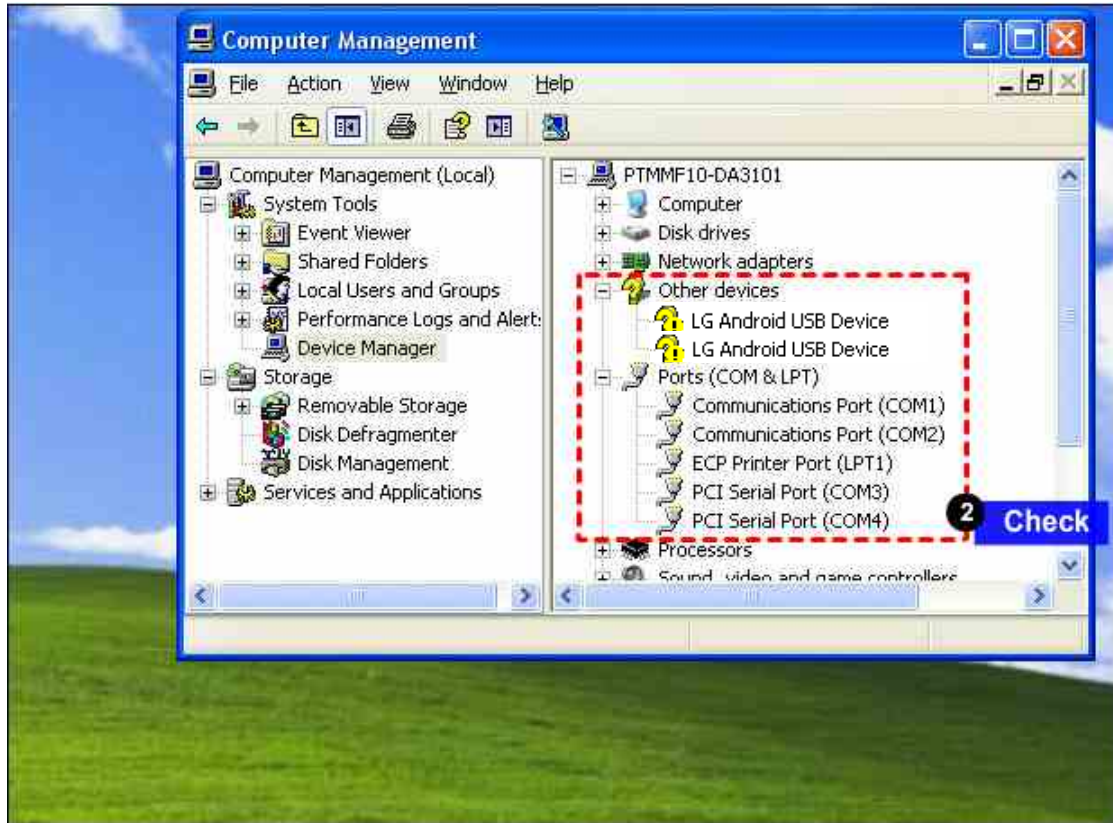


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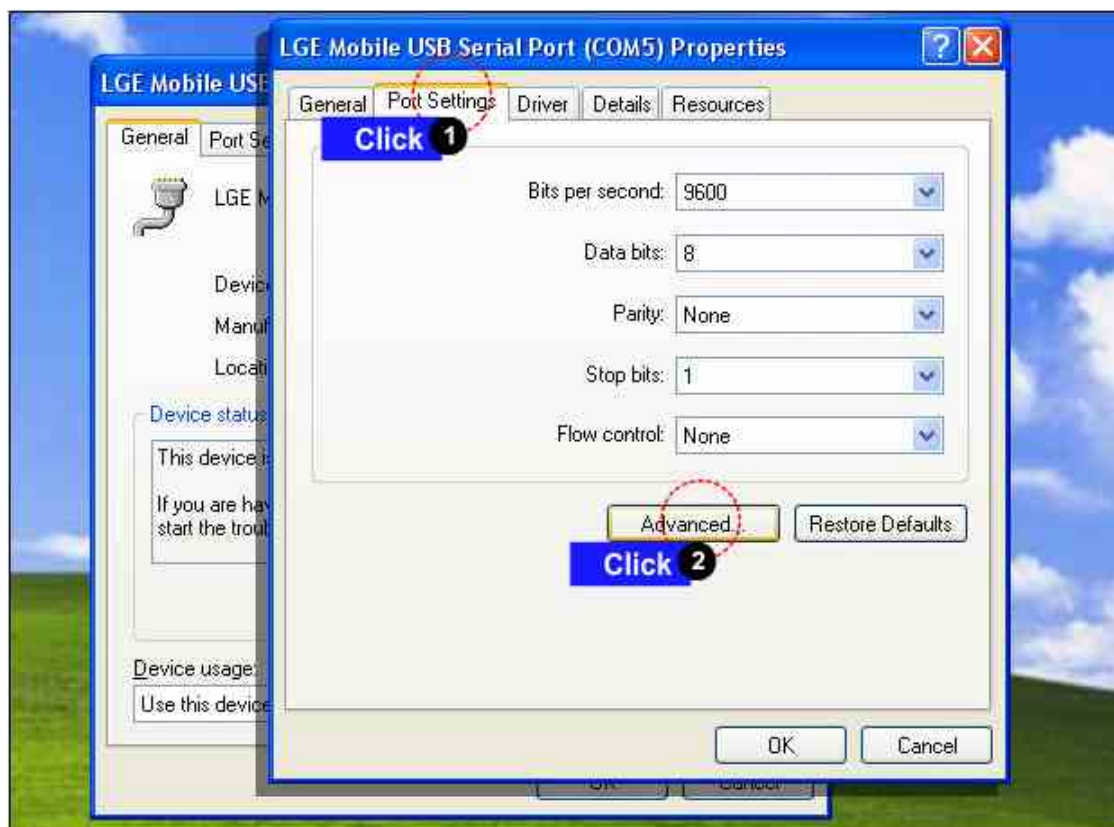
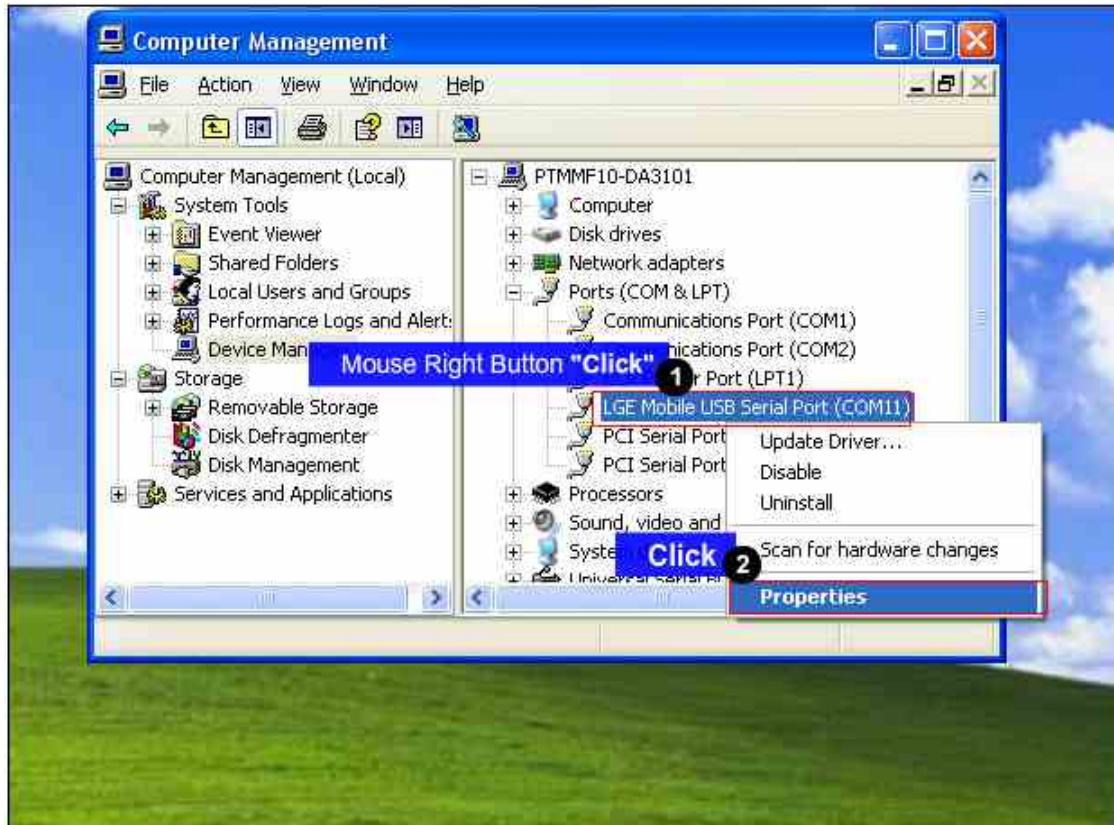


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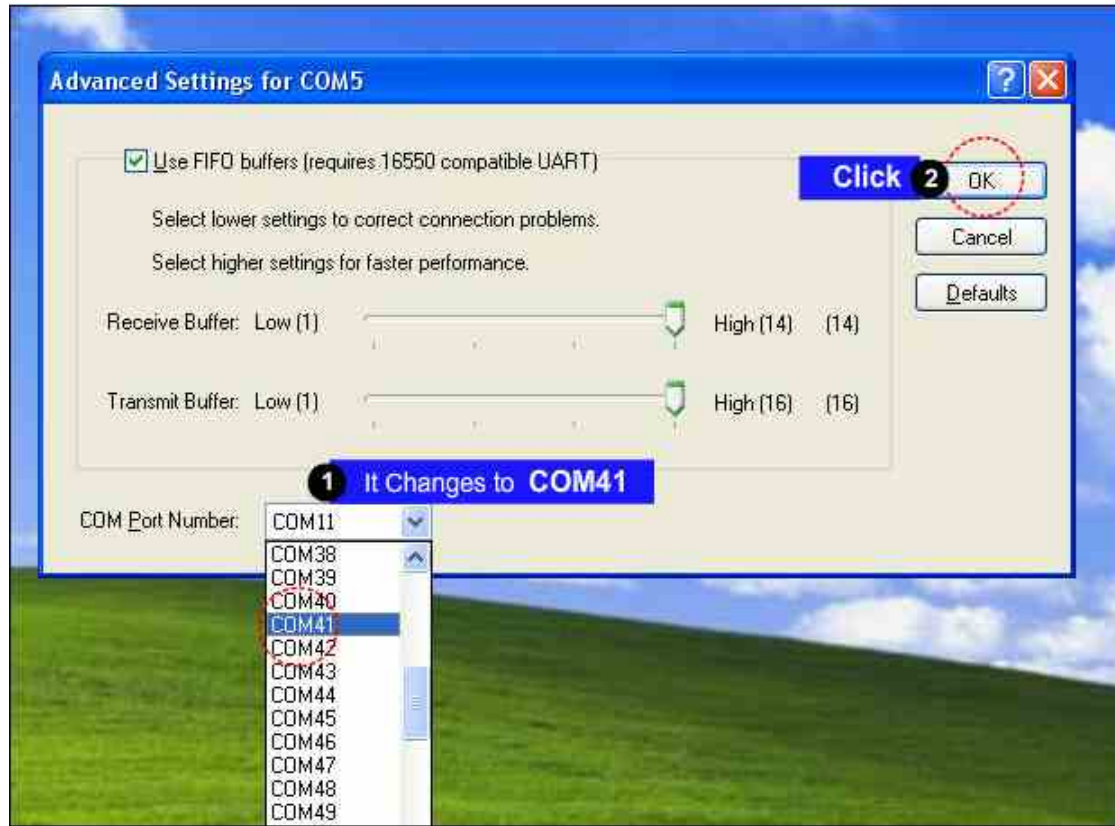


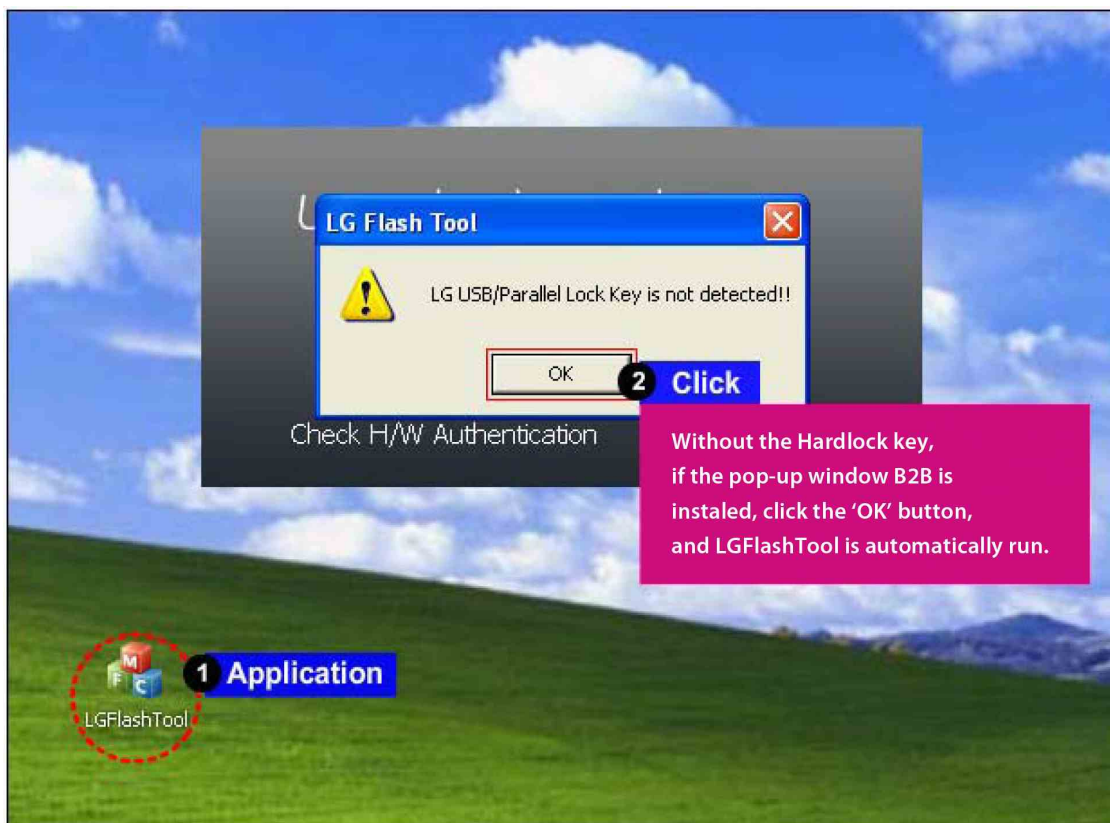
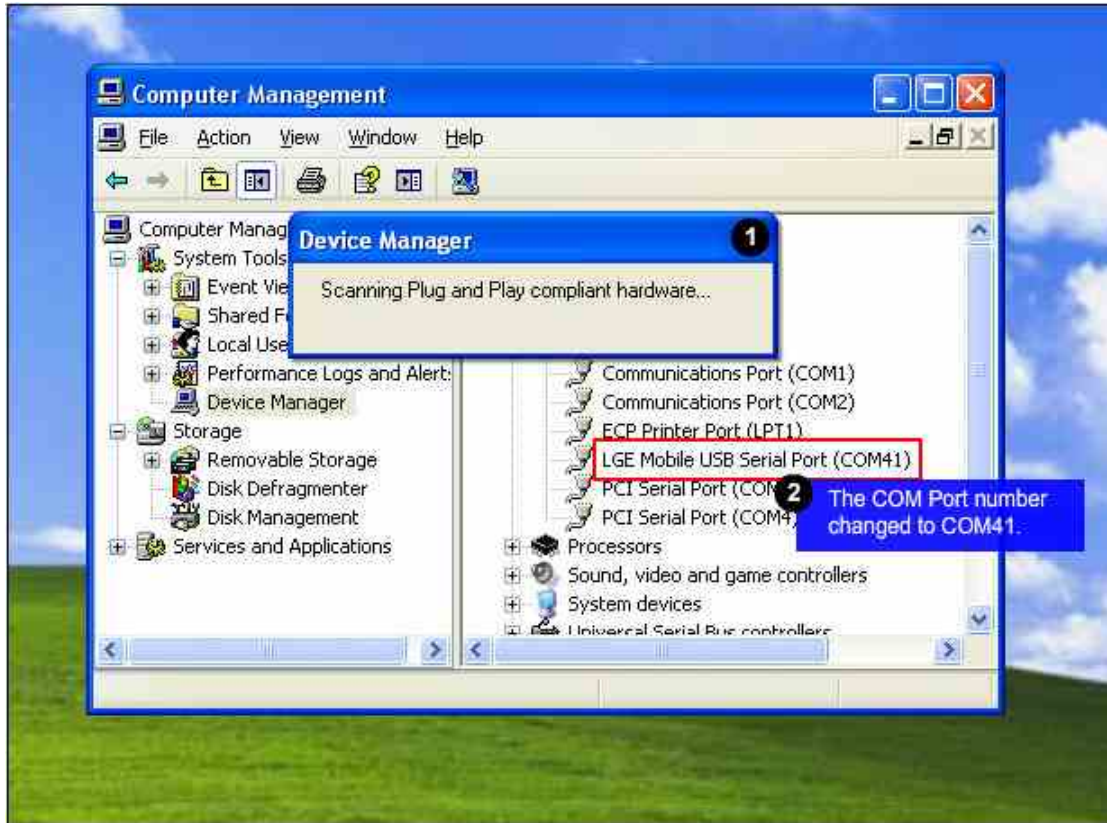


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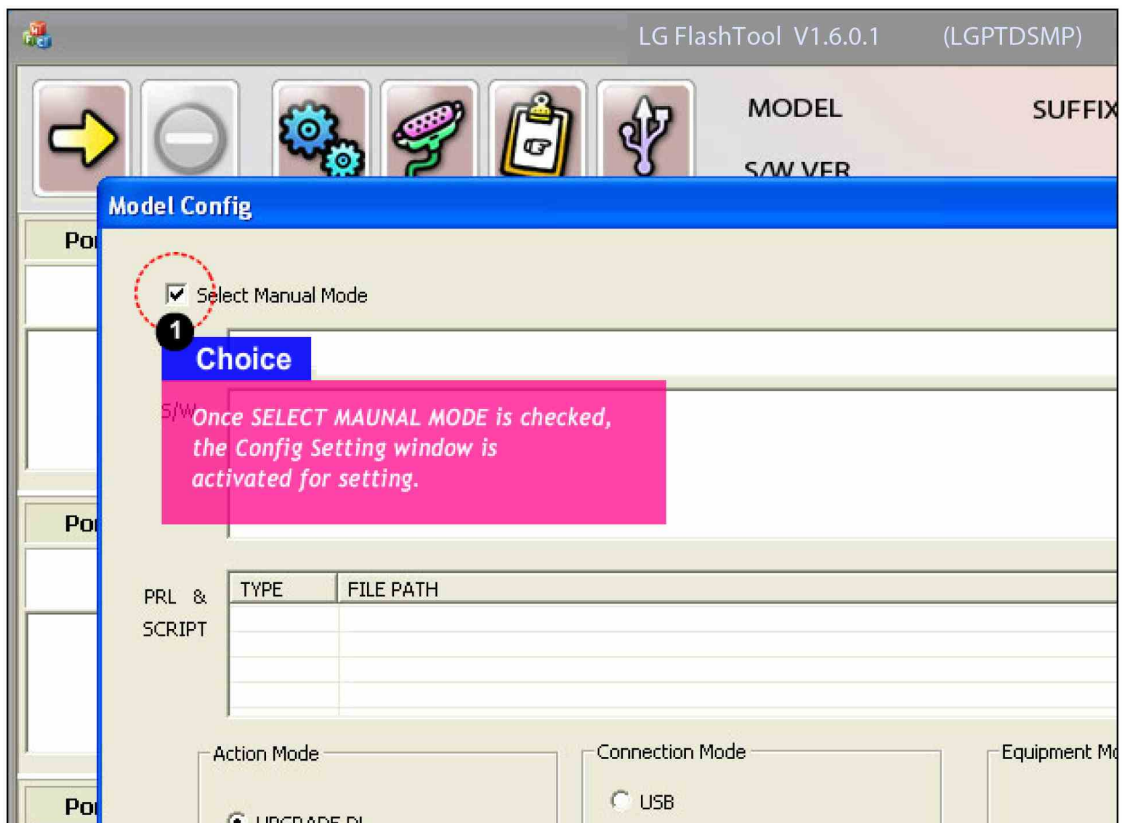
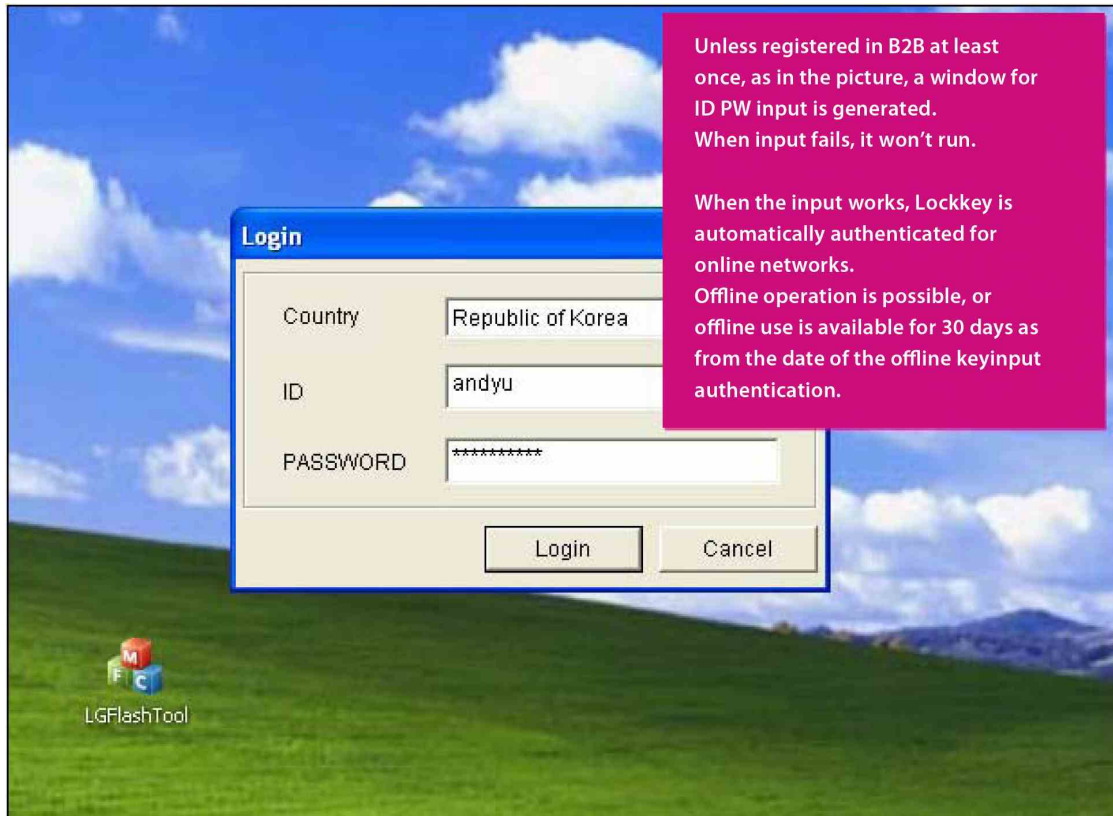


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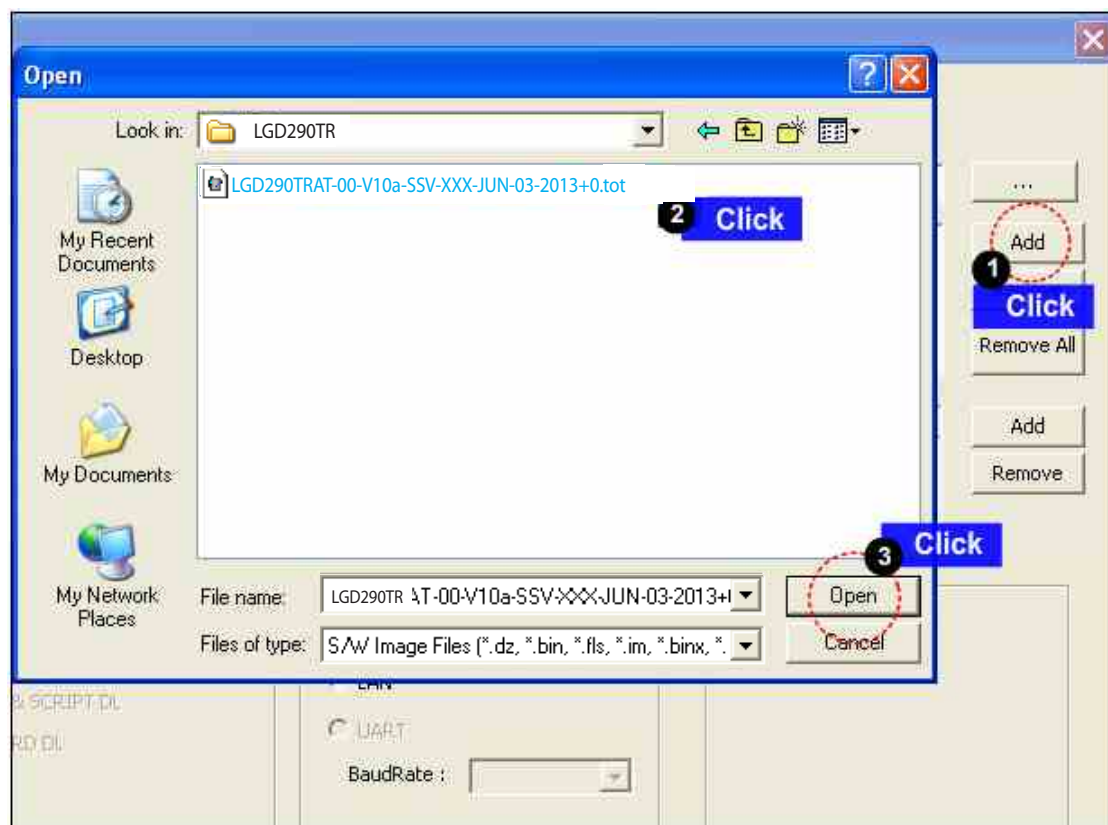
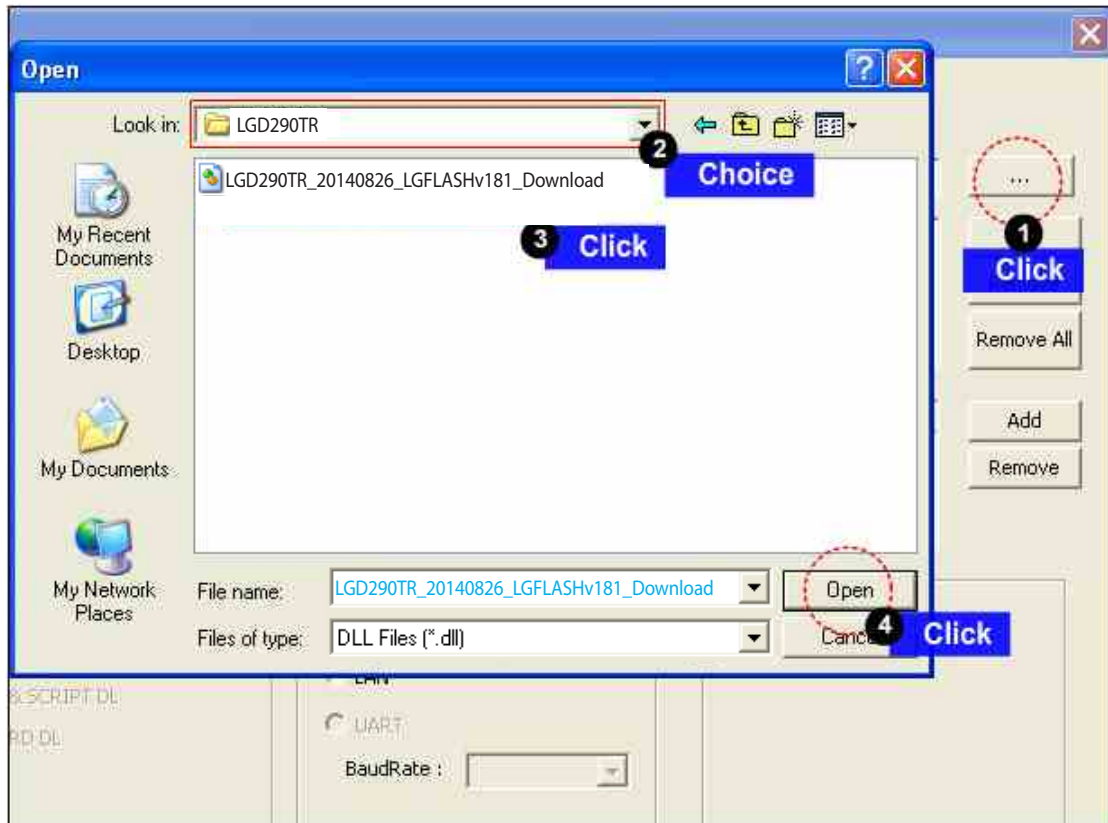




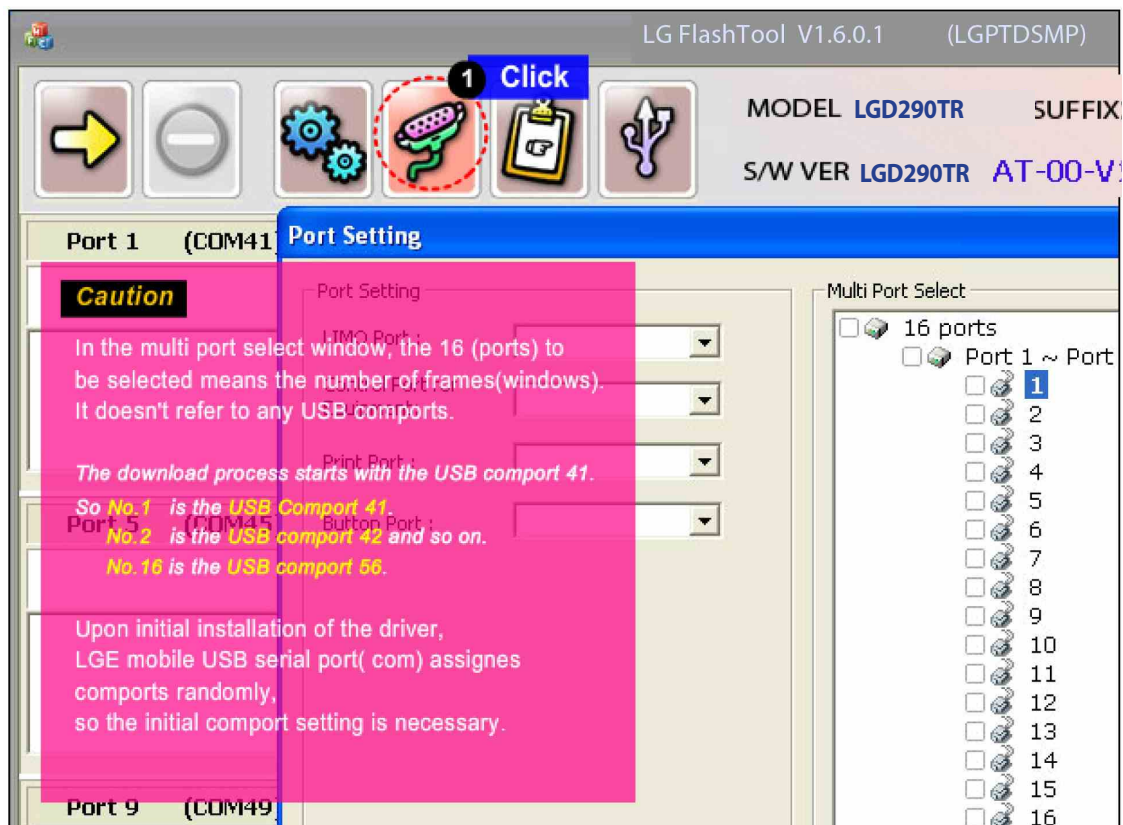
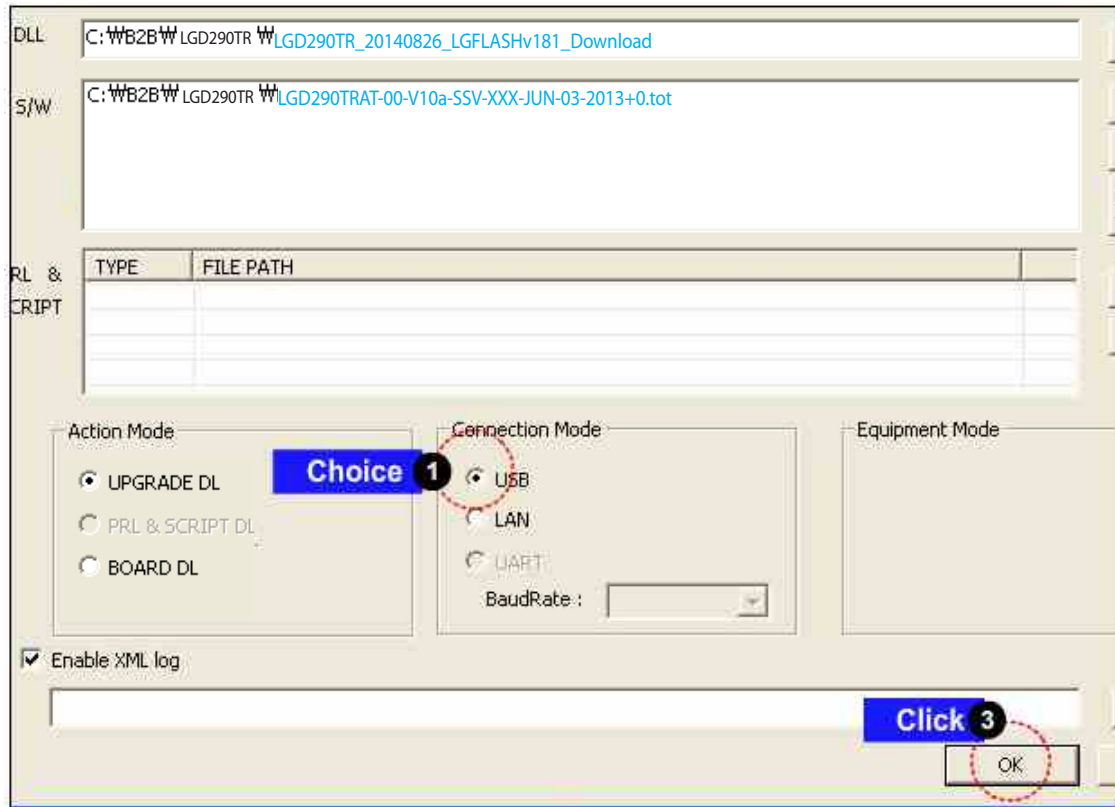
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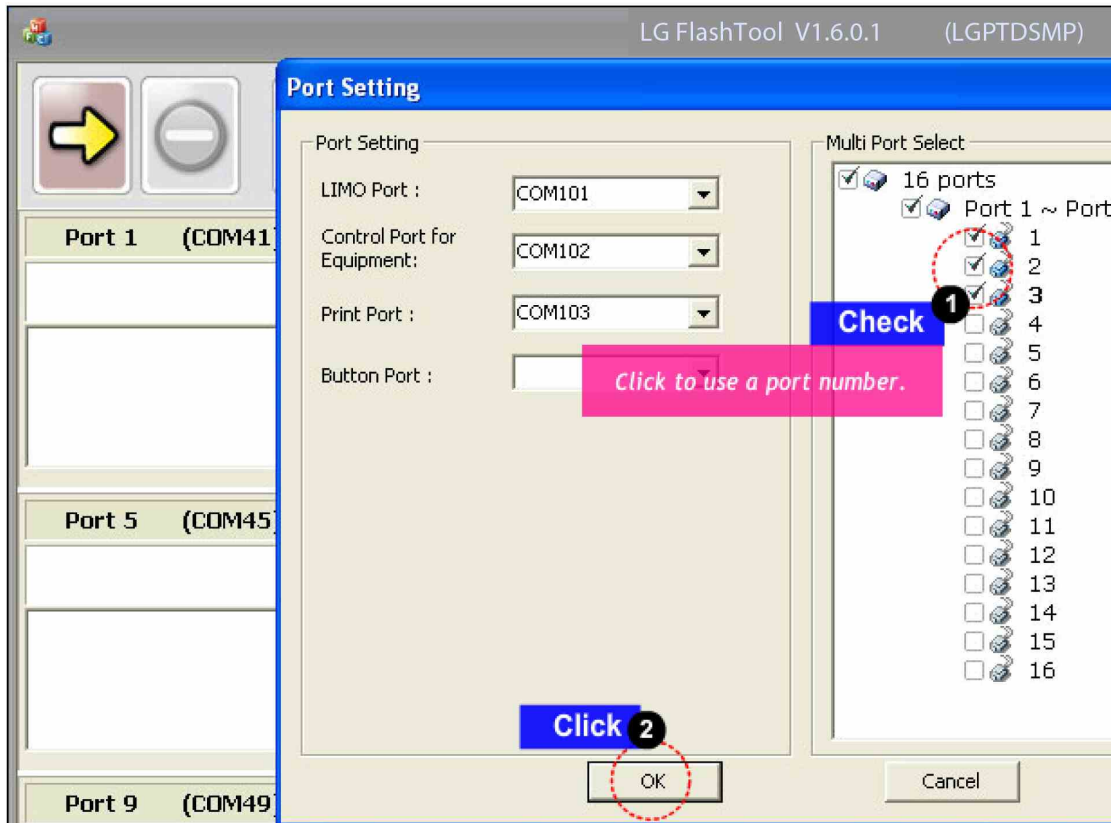
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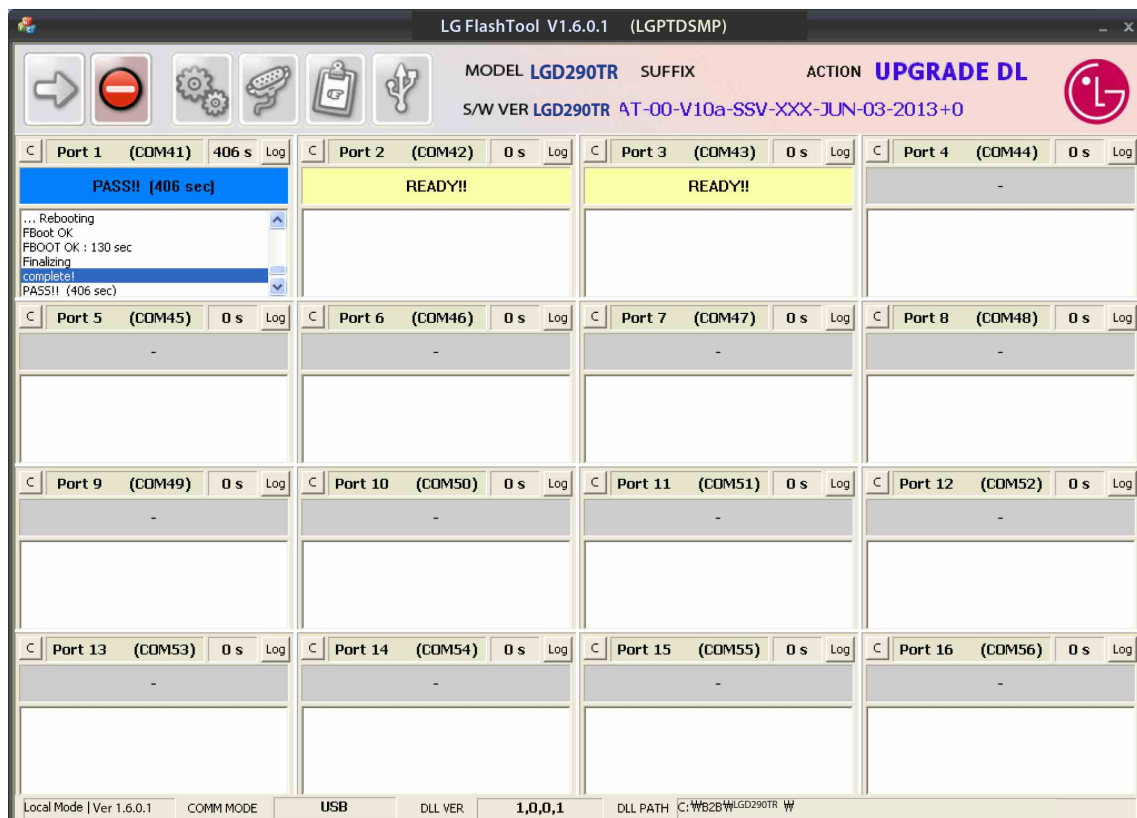
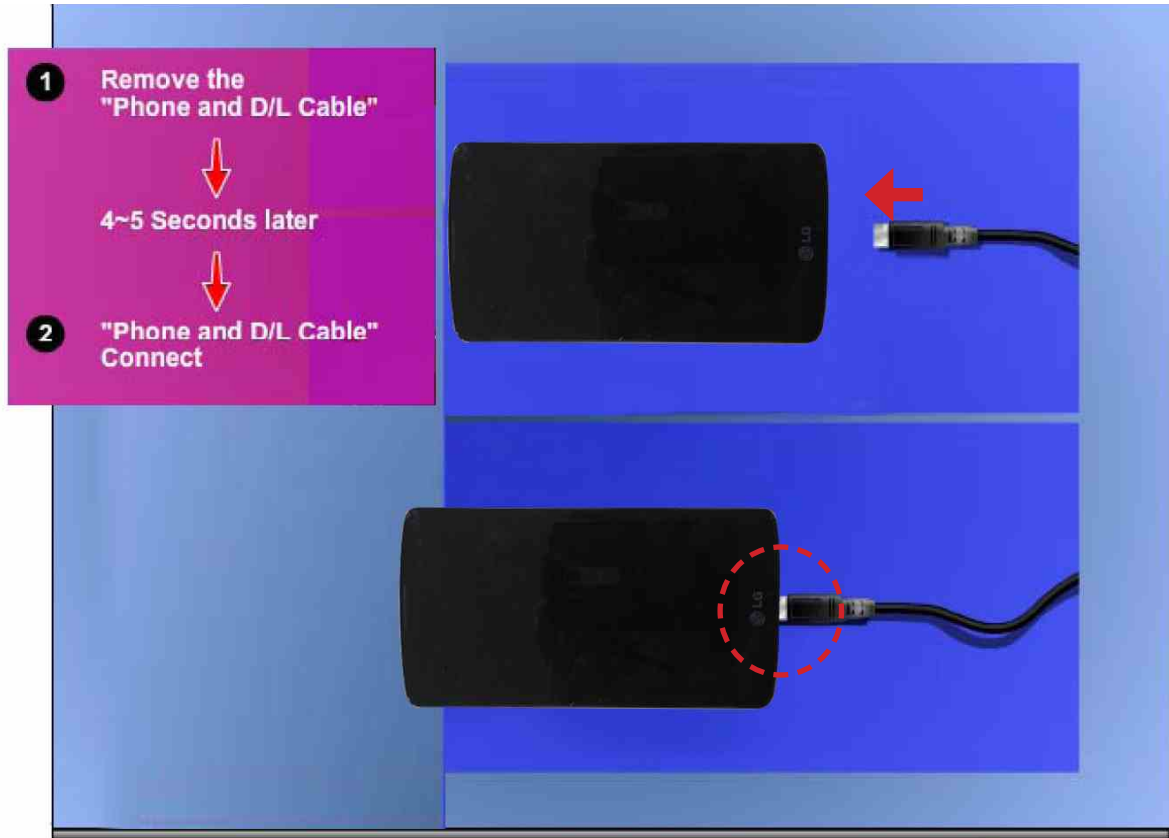
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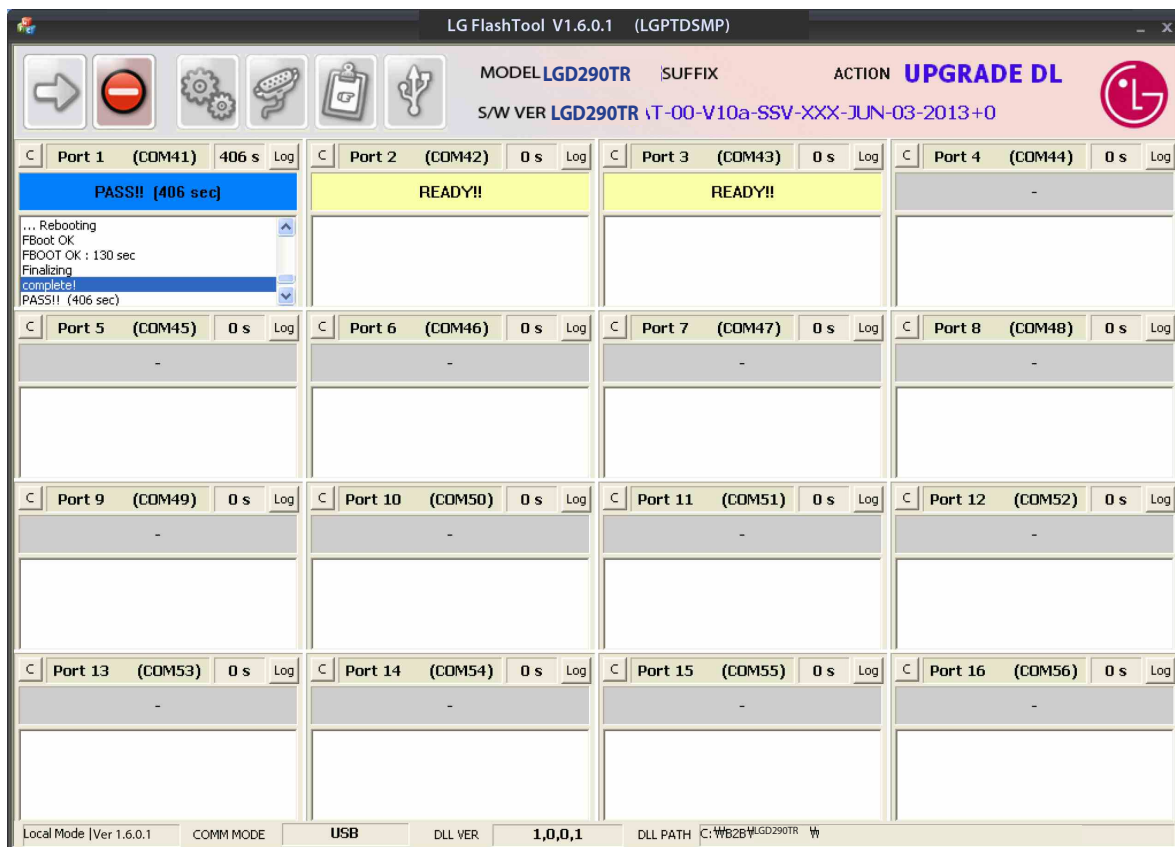
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4. DOWNLOAD

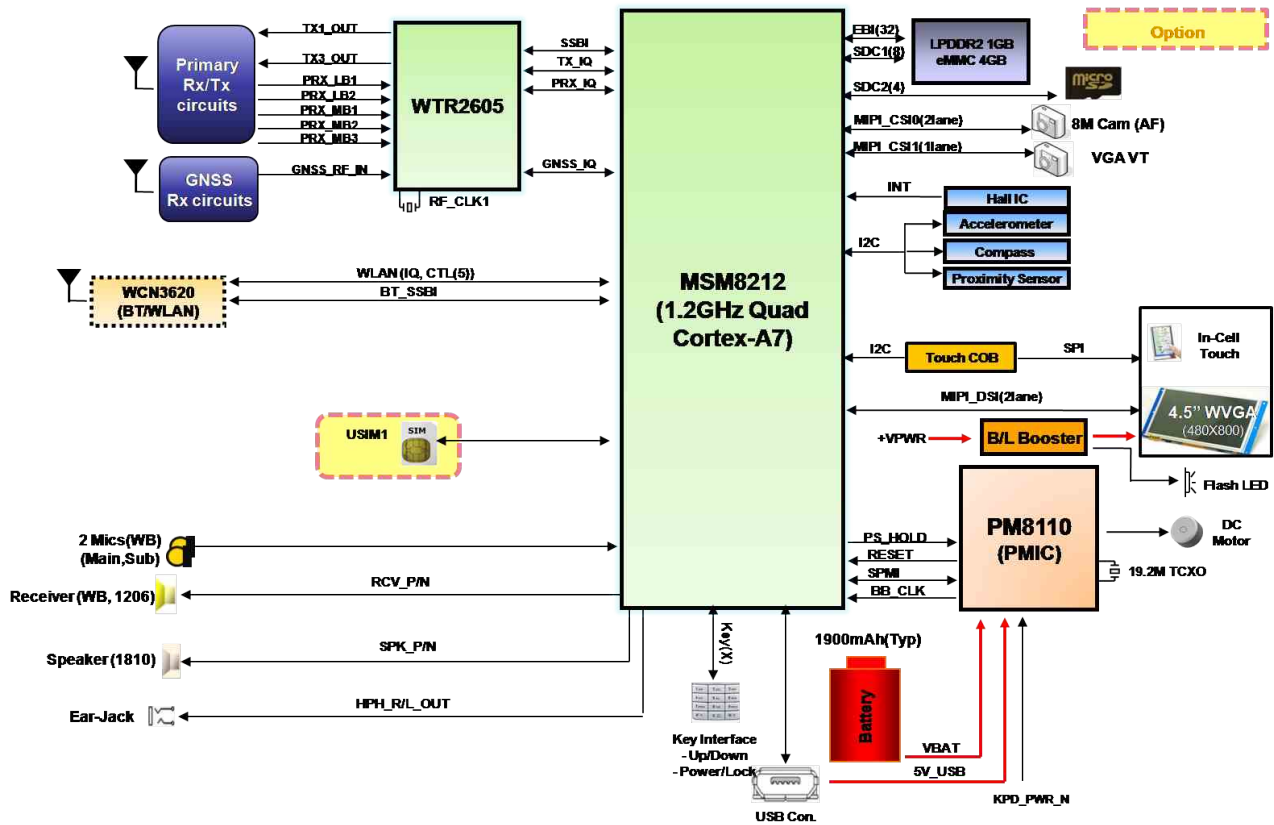


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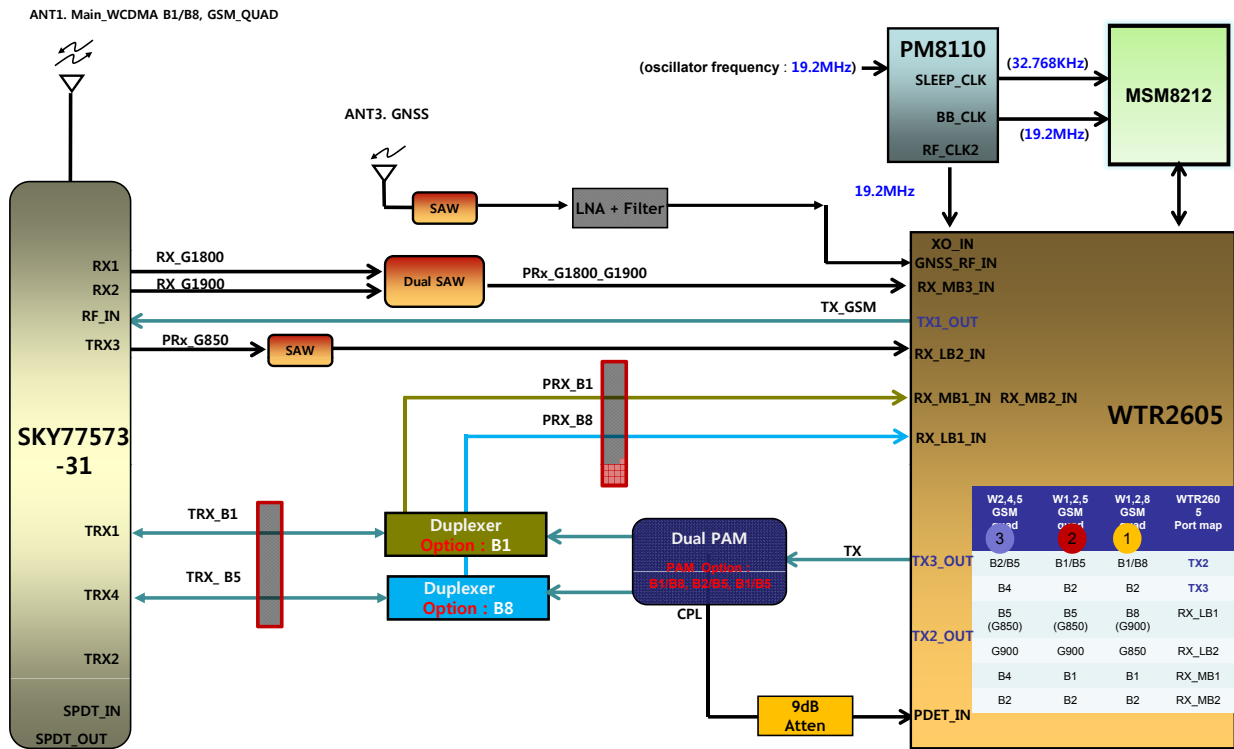
5. BLOCK DIAGRAM

[D290TR] Total Block Diagram



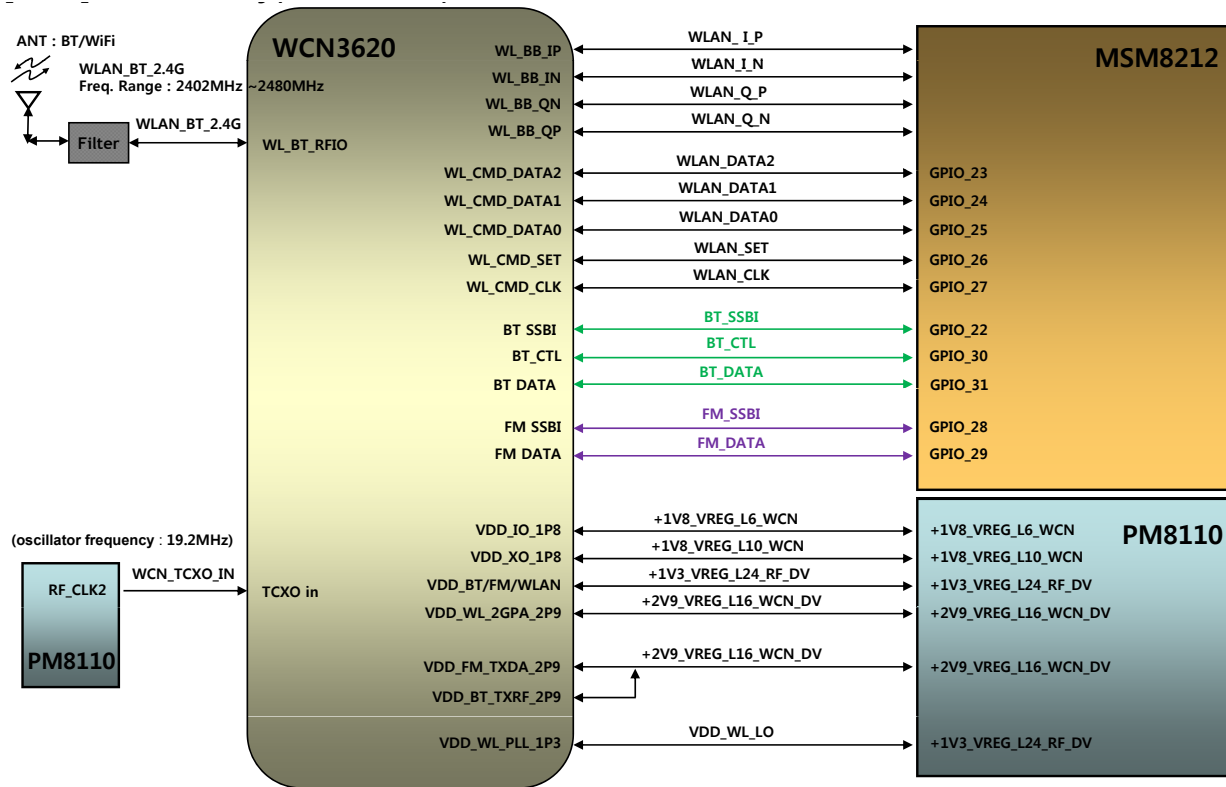
5. BLOCK DIAGRAM

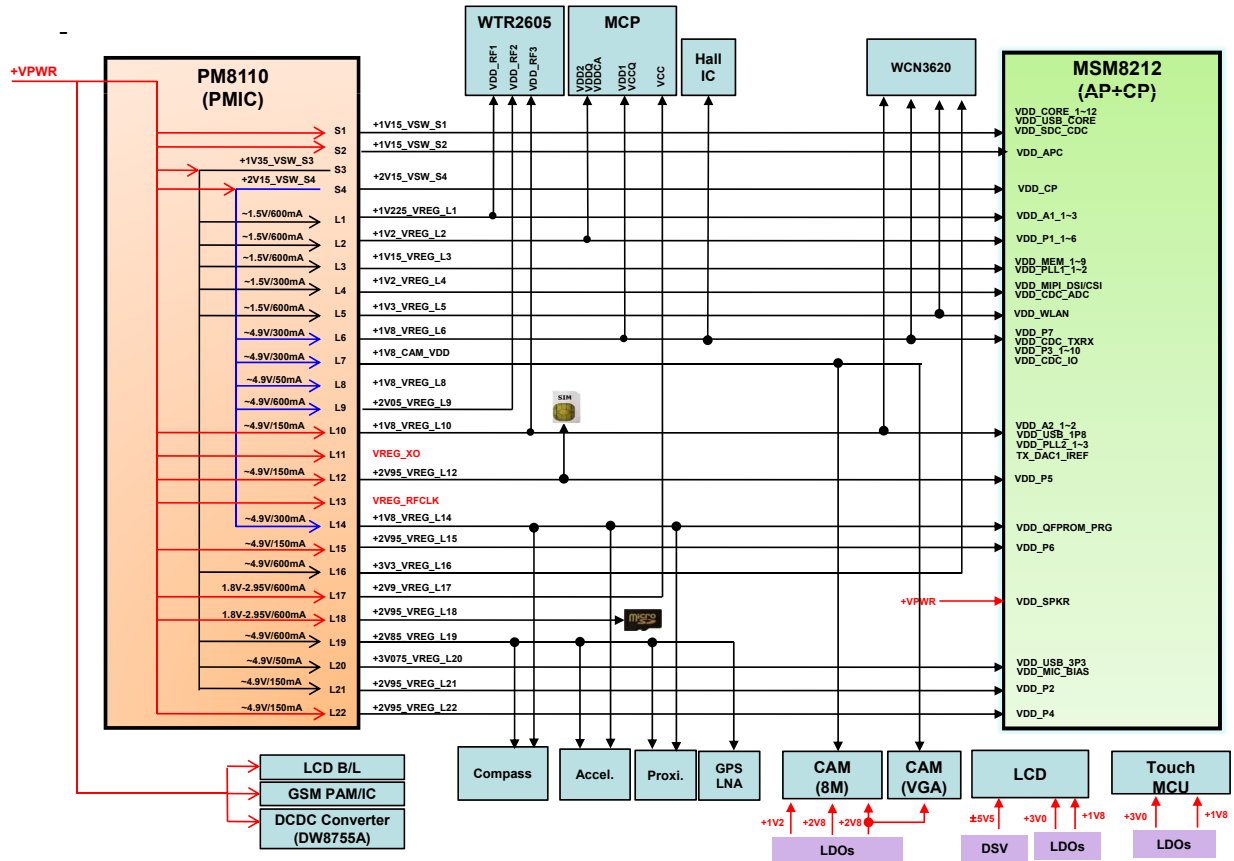
[D290TR] RF Main (3G & GSM)



5. BLOCK DIAGRAM

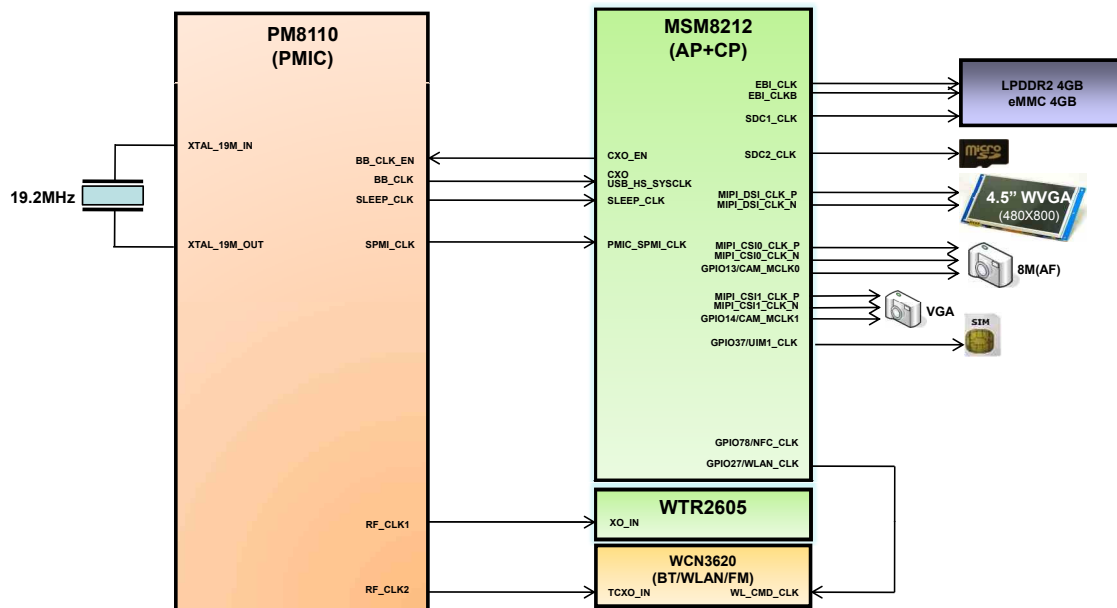
[D290TR]Connectivity(BT/WiFi/FM)





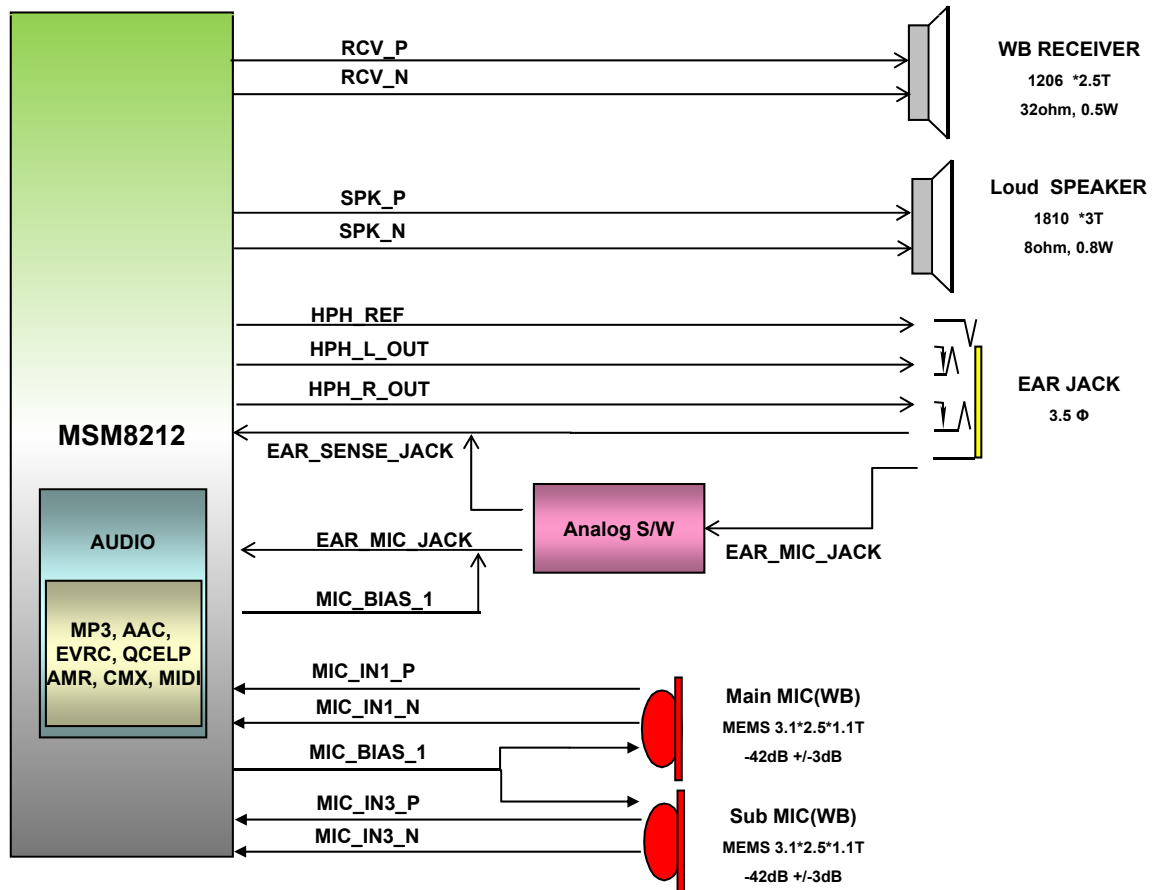
5. BLOCK DIAGRAM

[D290TR] Clock



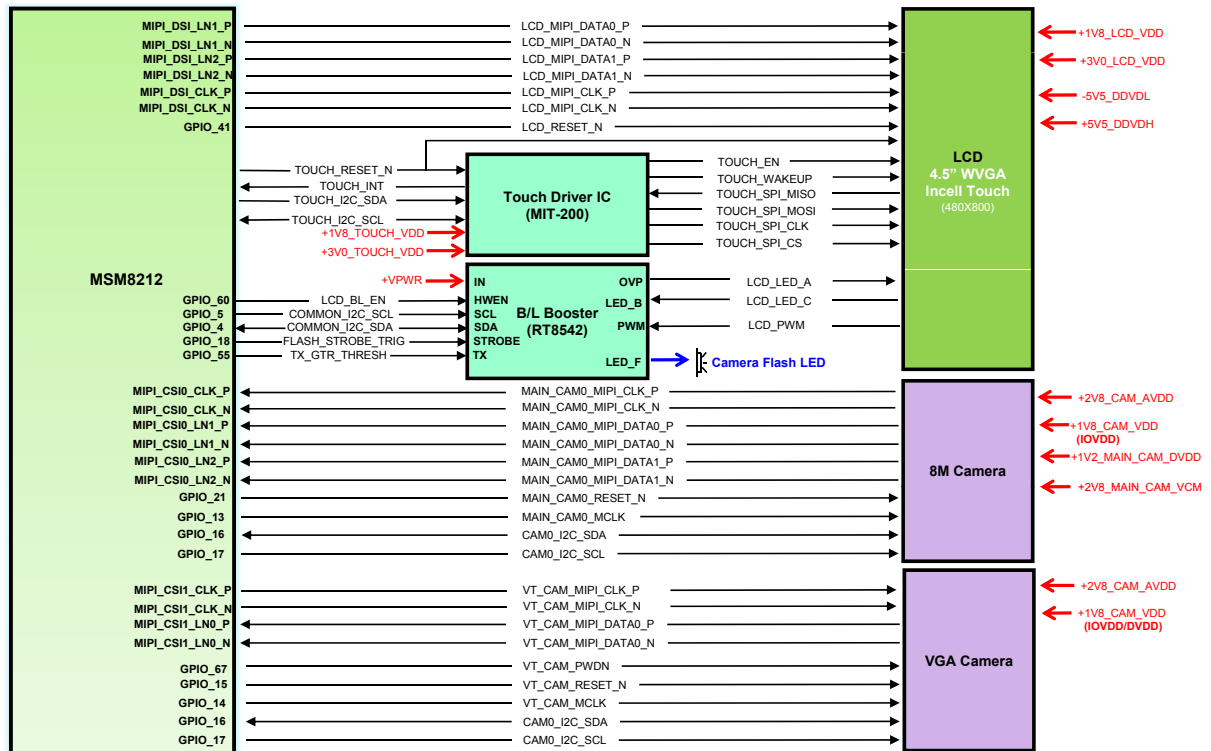
5. BLOCK DIAGRAM

[D290TR] Audio



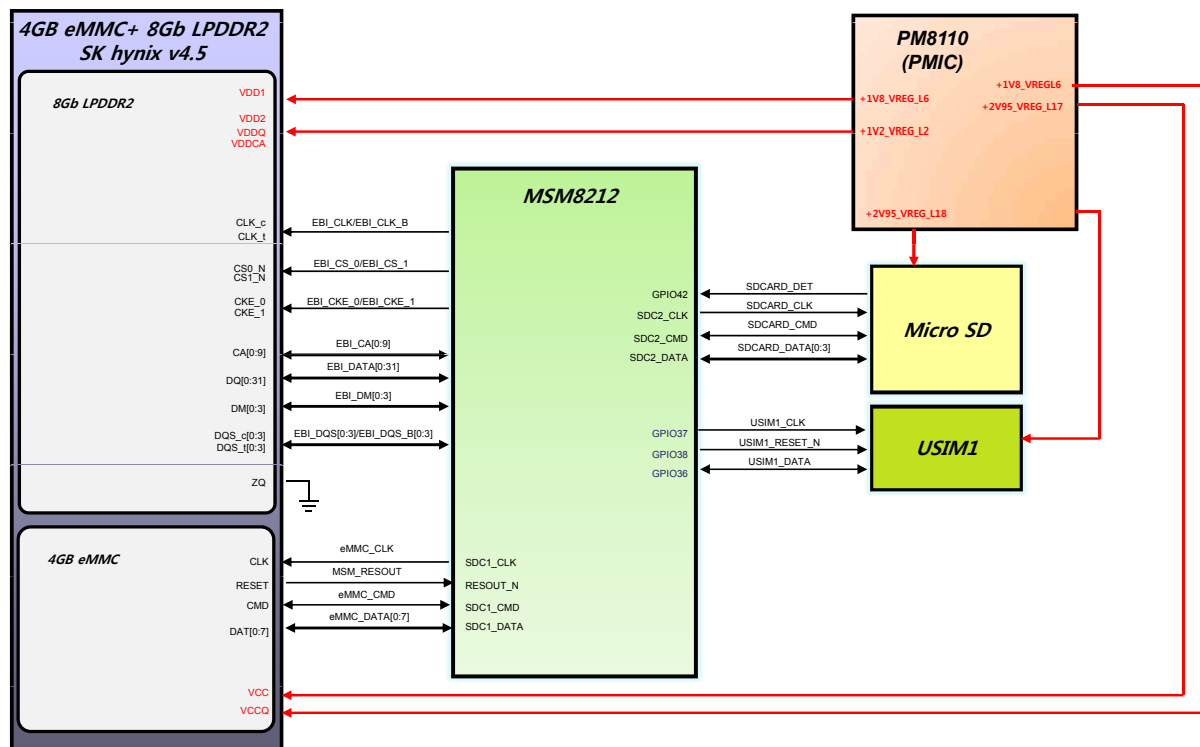
5. BLOCK DIAGRAM

[D290TR] LCD/Touch/Camera



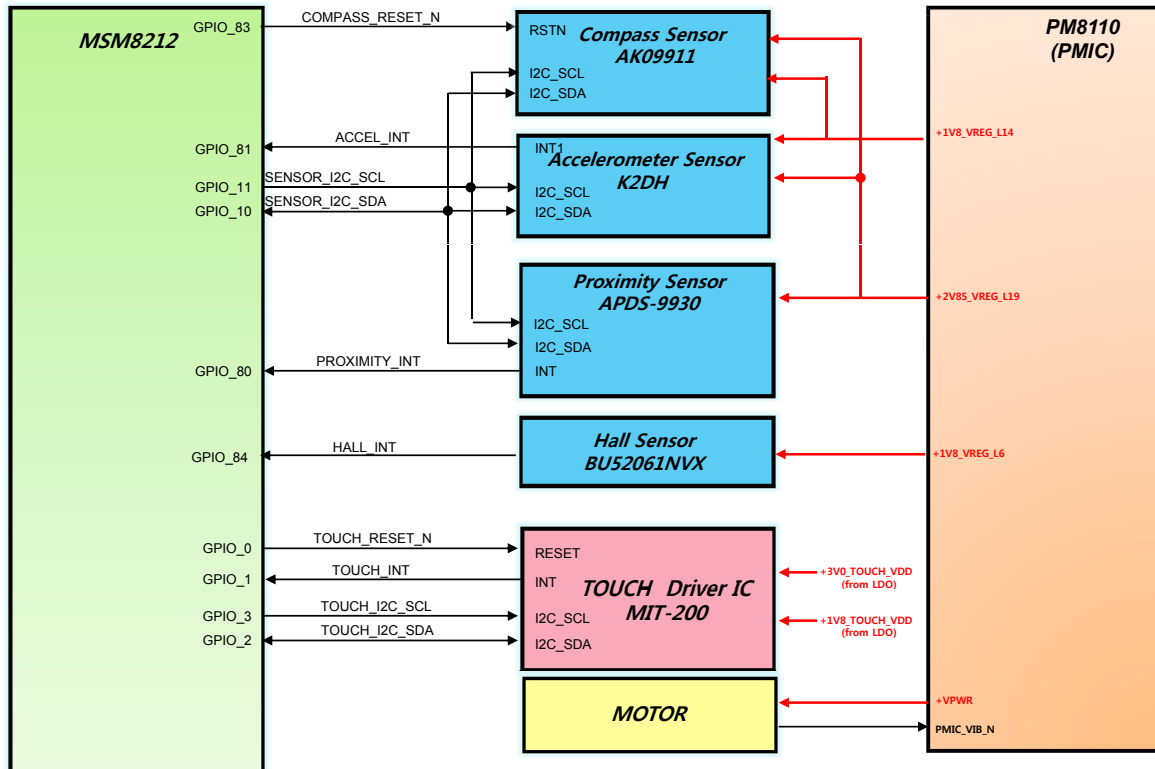
5. BLOCK DIAGRAM

[D290TR] MCP/Peripherals



5. BLOCK DIAGRAM

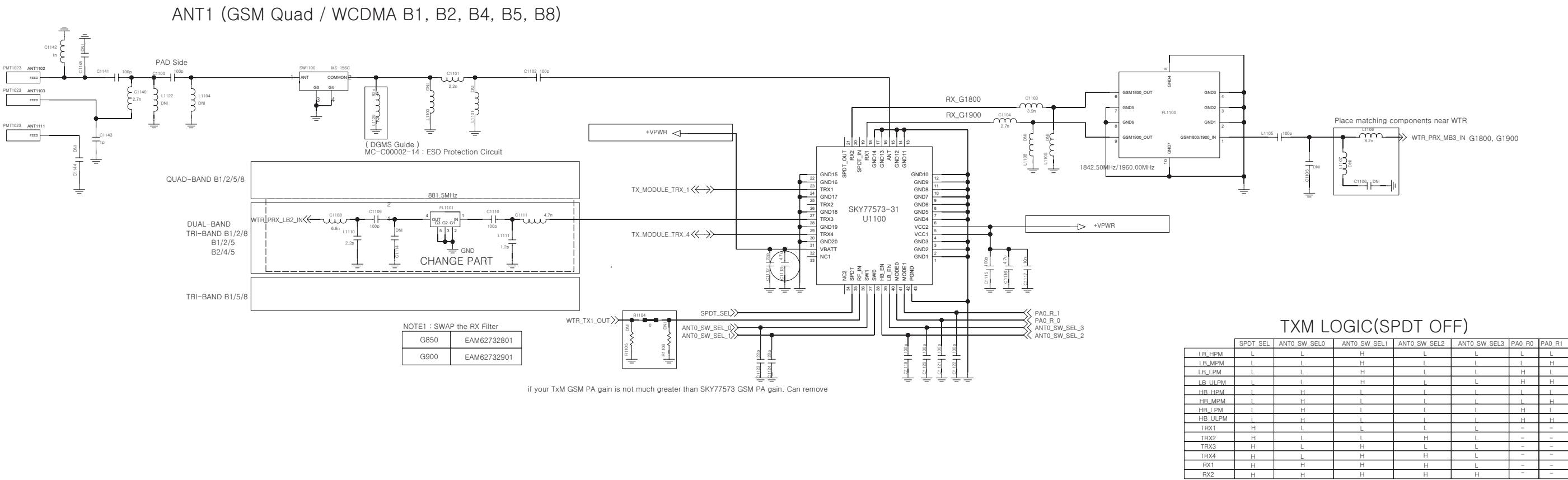
[D290TR] Sensors



6. CIRCUIT DIAGRAM

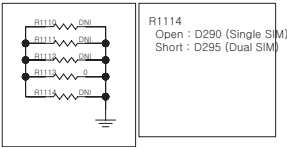
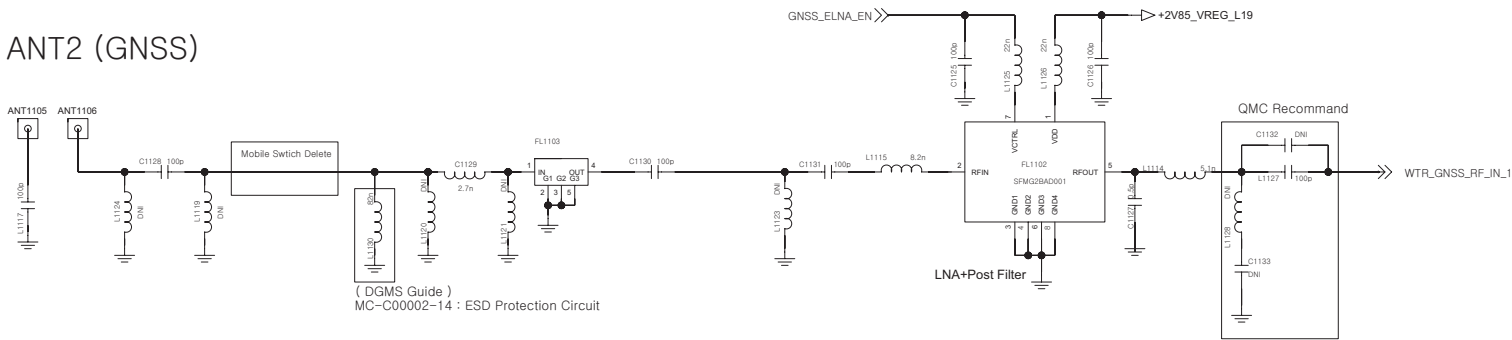
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Rev_0.6



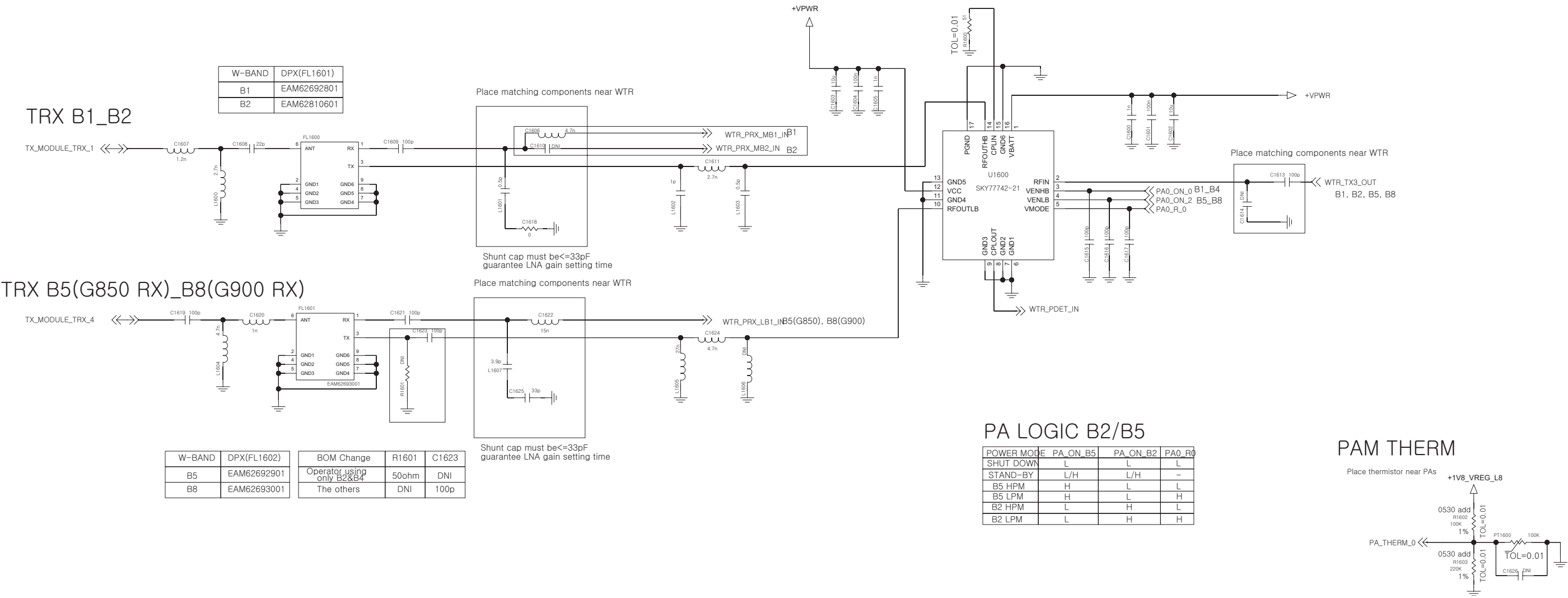
EXT LNA + Filter

ANT2 (GNSS)



< 1-1-4-5 TRX_WCDMA_B1_B2_B5_B8 DUAL >

Rev 0.4

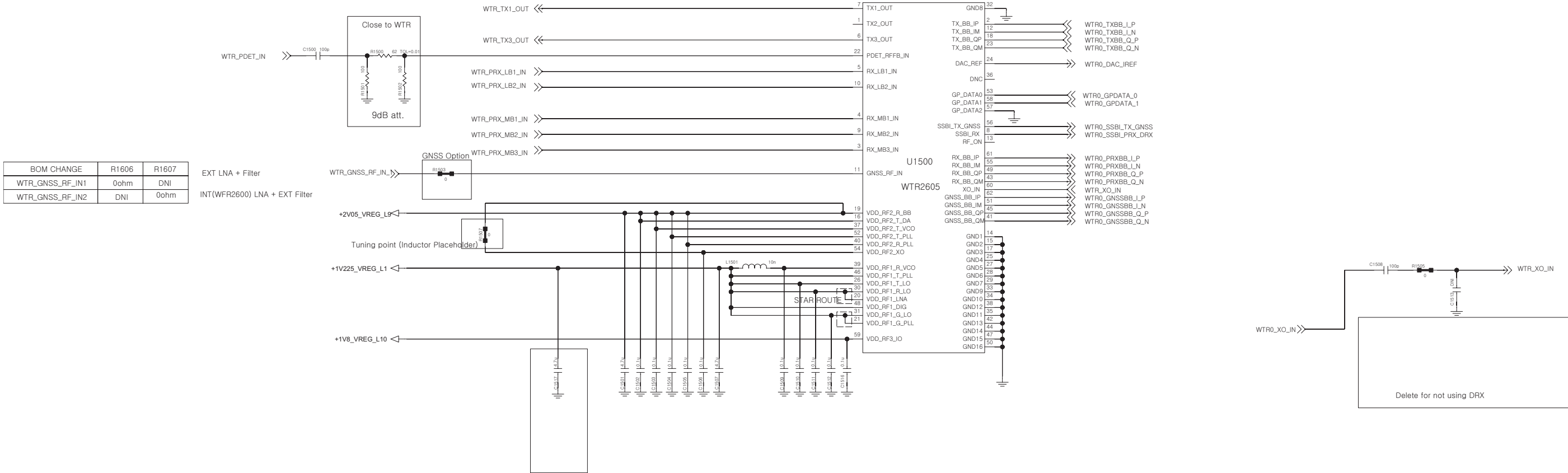


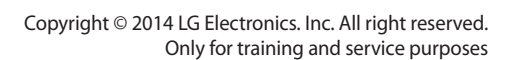
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Rev_0.6

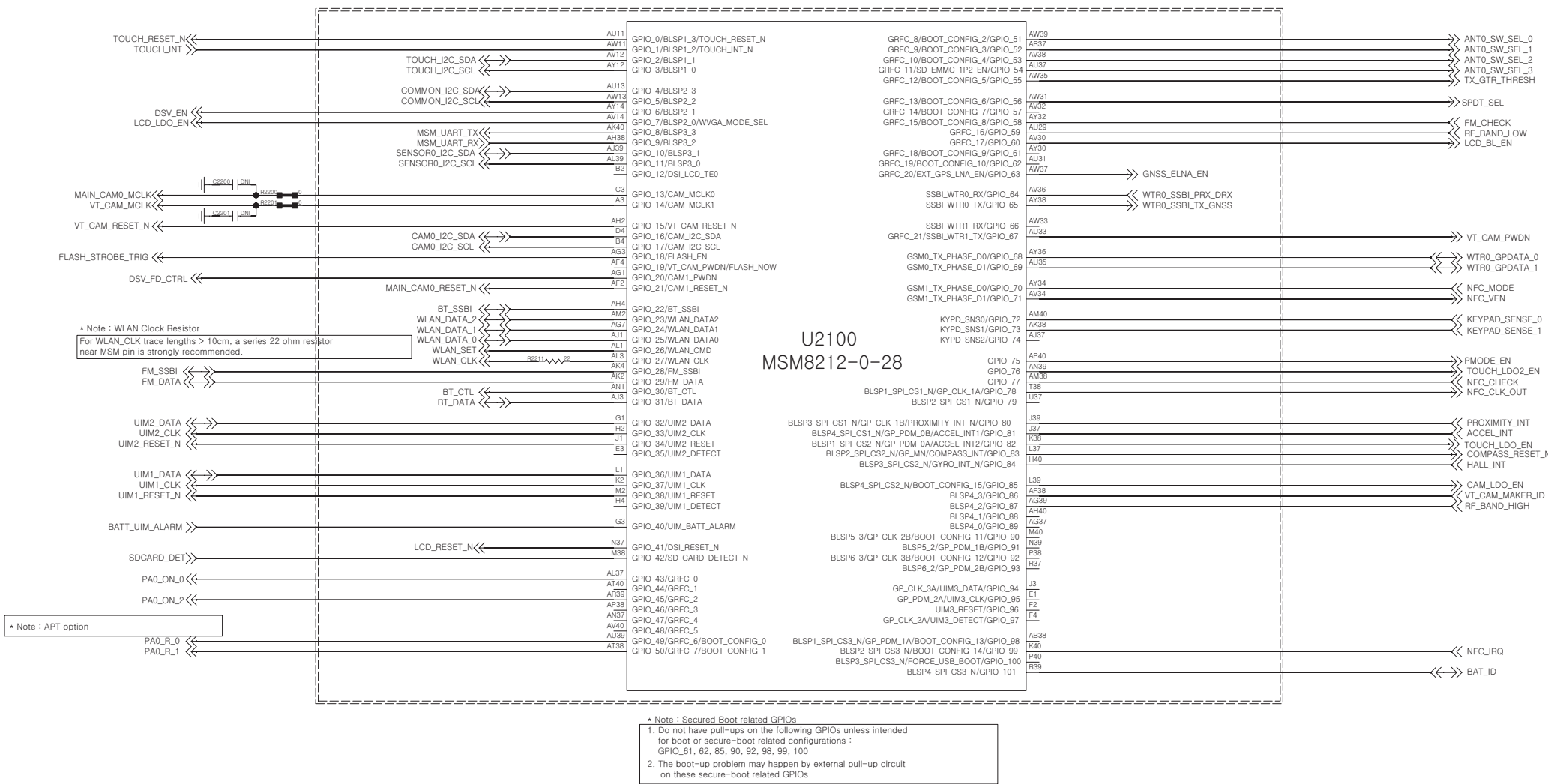
CAUTION: YOU MUST CHECK & MODIFY RF PORT AND GPIO ON YOUR SCHEMATICS.

WCDMA								CDMA			
WTR2605	B2, B4, B5	B1, B2, B5	B1, B2, B8	B1, B5, B8	B2, B5	B1, B5	B1, B8	WTR2605	VZW	SPCS	Metro PCS
TX1_OUT	GSM	GSM	GSM	GSM	GSM	GSM	GSM	TX1_OUT			
TX2_OUT	B4	B1	B1	B5				TX2_OUT	BC0(BC10)	BC0(BC10)	BC0(BC10),BC15
TX3_OUT	B2, B5	B2, B5	B2, B8	B1, B8	B2, B5	B1, B5	B1, B8	TX3_OUT	BC1	BC1	BC1
RX_LB1_IN	B5(G850)	B5(G850)	B8(G900)	B8(G900)	B5(G850)	B5(G850)	B8(G900)	RX_LB1_IN	BC0(BC10)	BC0(BC10)	BC0(BC10)
RX_LB2_IN	G900	G900	G850	B5(G850)	G900	G900	G850	RX_LB2_IN			
RX_MB1_IN	B4	B1	B1	B1			B1	RX_MB1_IN			BC15
RX_MB2_IN	B2	B2	B2		B2	B2		RX_MB2_IN	BC1	BC1	BC1
RX_MB3_IN	G1800,G1900	G1800,G1900	G1800,G1900	G1800,G1900	G1800,G1900	G1800,G1900	G1800,G1900	RX_MB3_IN			

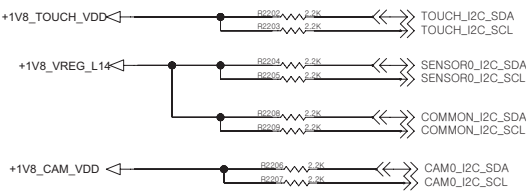




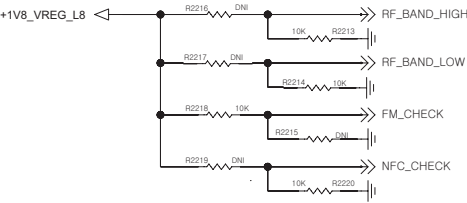
<2-1-6-1-2_MSM8212-0-28_GPIO> REV.0.3



I2C Pull-up

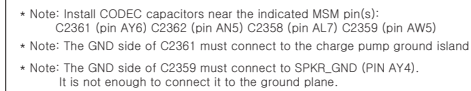


RF BAND Check

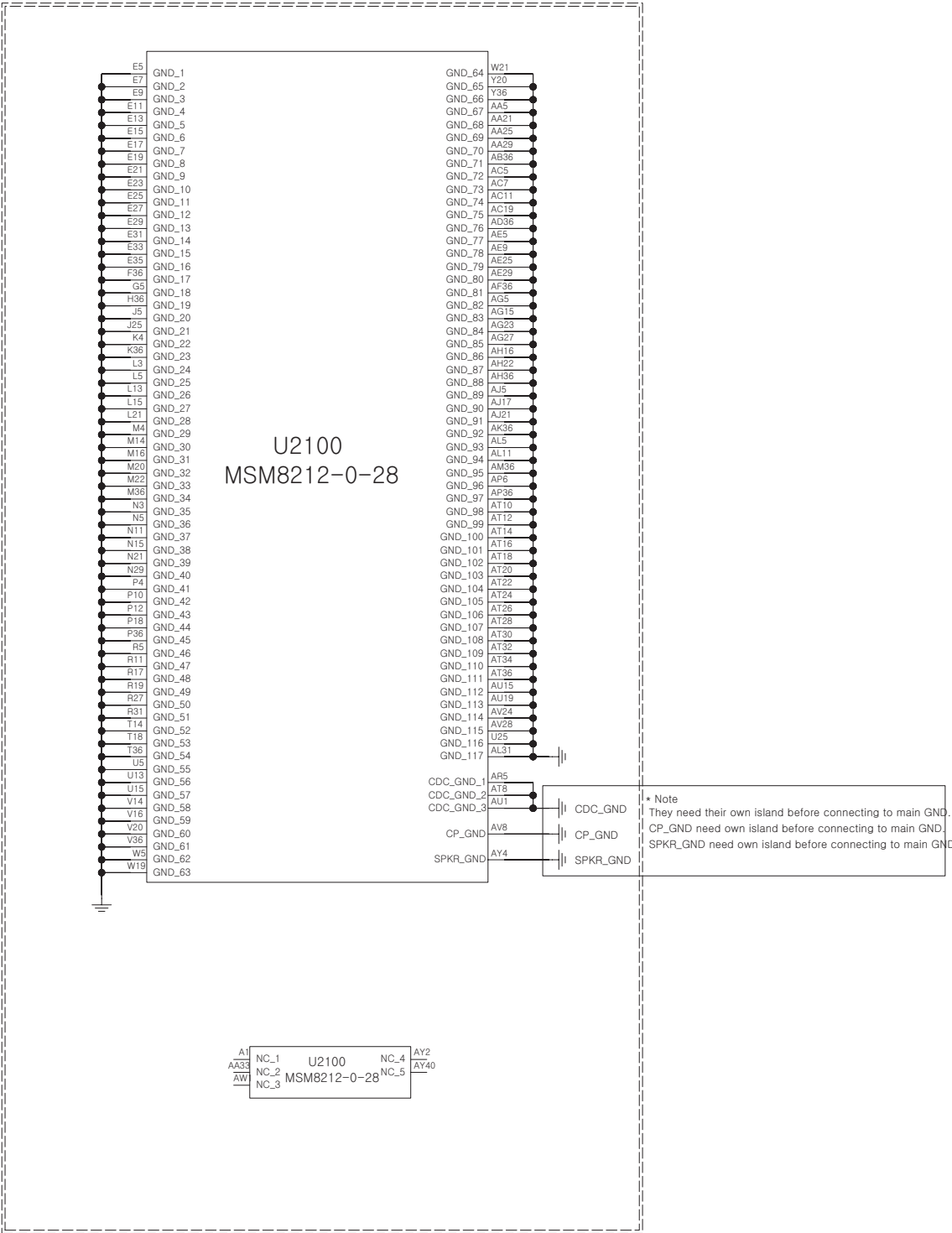


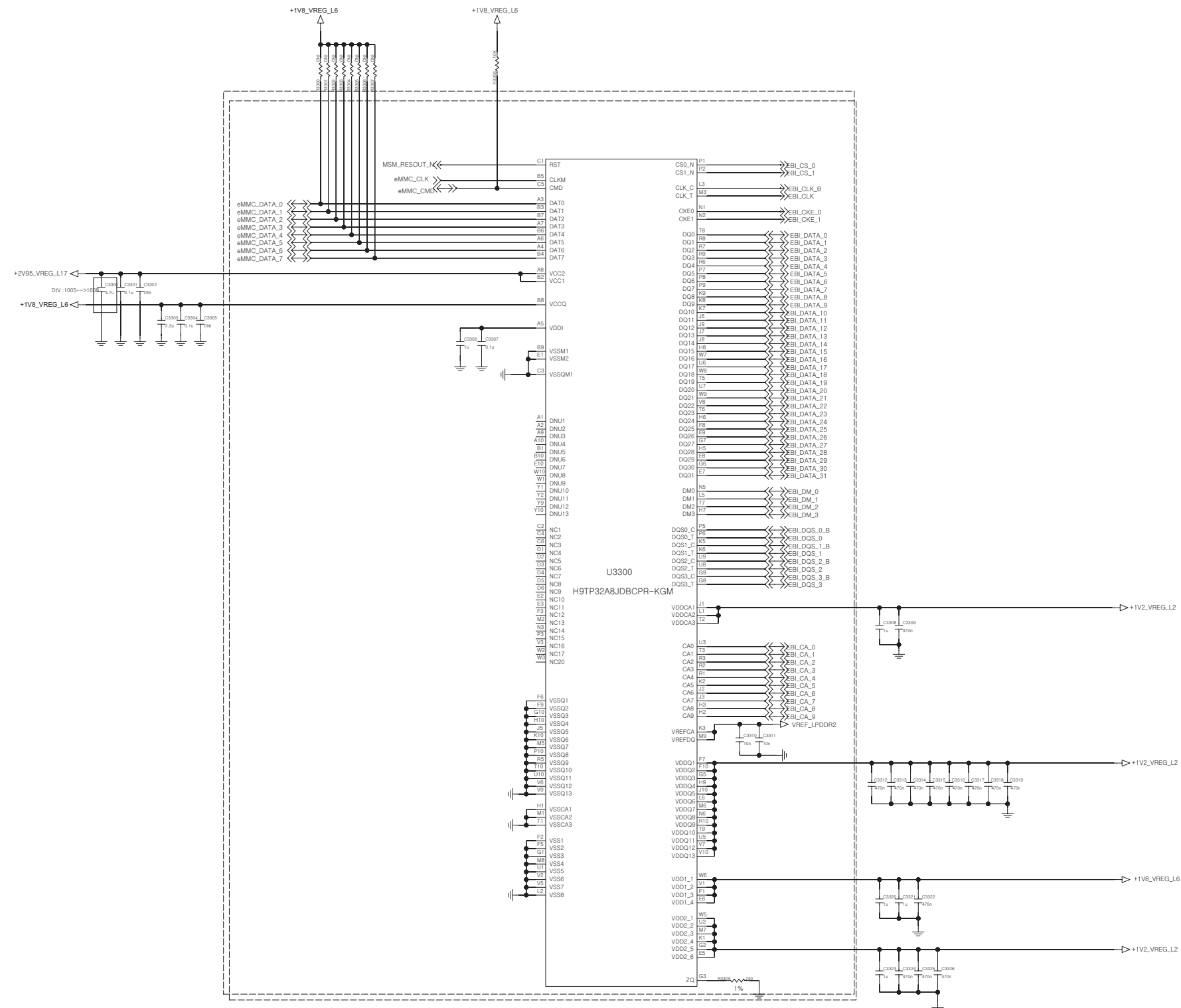
GPIO	LOW	HIGH	FM	NFC
EU_1_8	L	L	L	L
F_1_5	H	L	L	L
G_2_5	H	H	L	L
J_2_8	L	H	L	L
FM	L	L	H	L
NFC	L	L	L	H

REV.0.3



<2-1-6-1-4_MSM8212-0-28_GND/NC>
REV.0.3

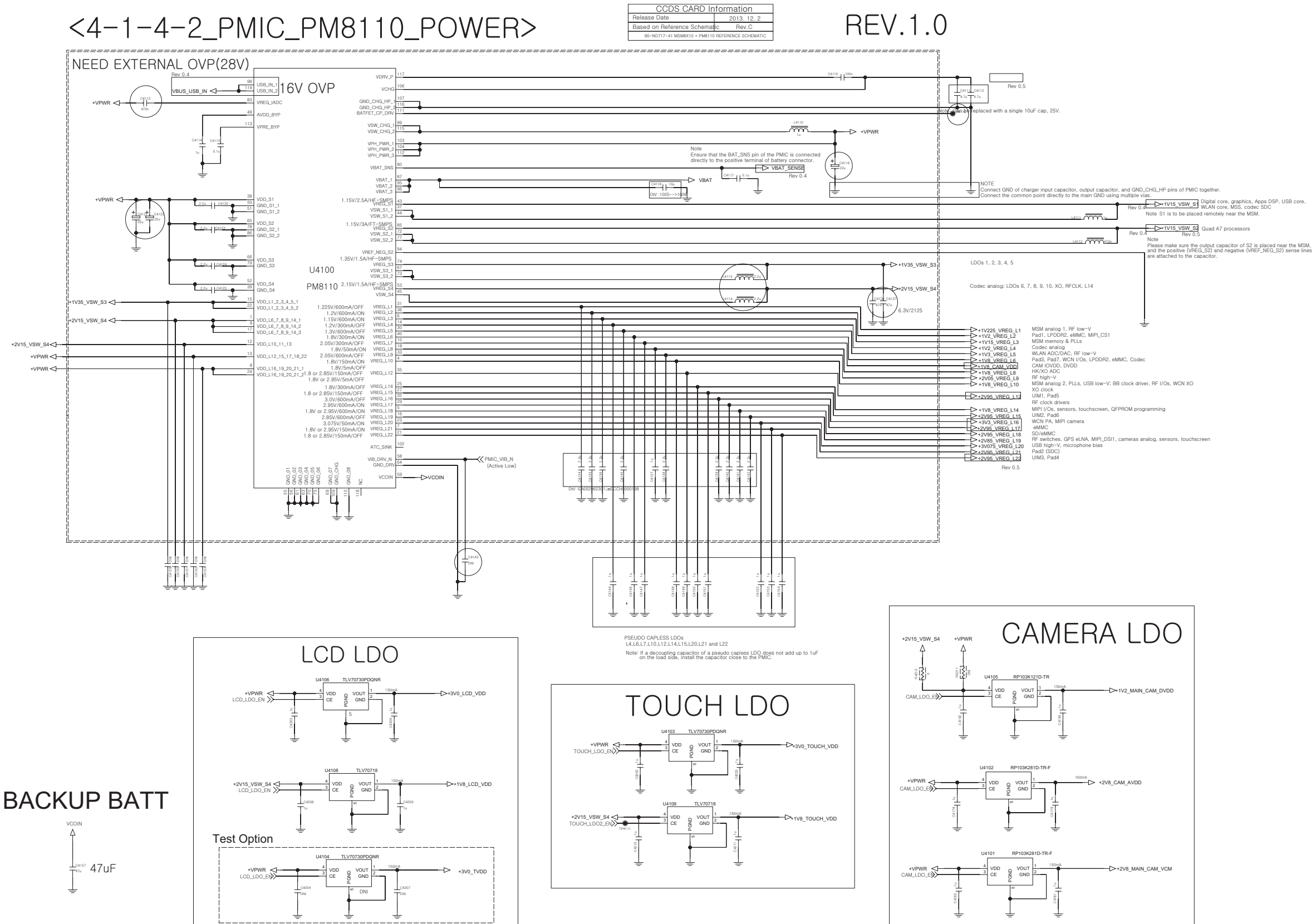




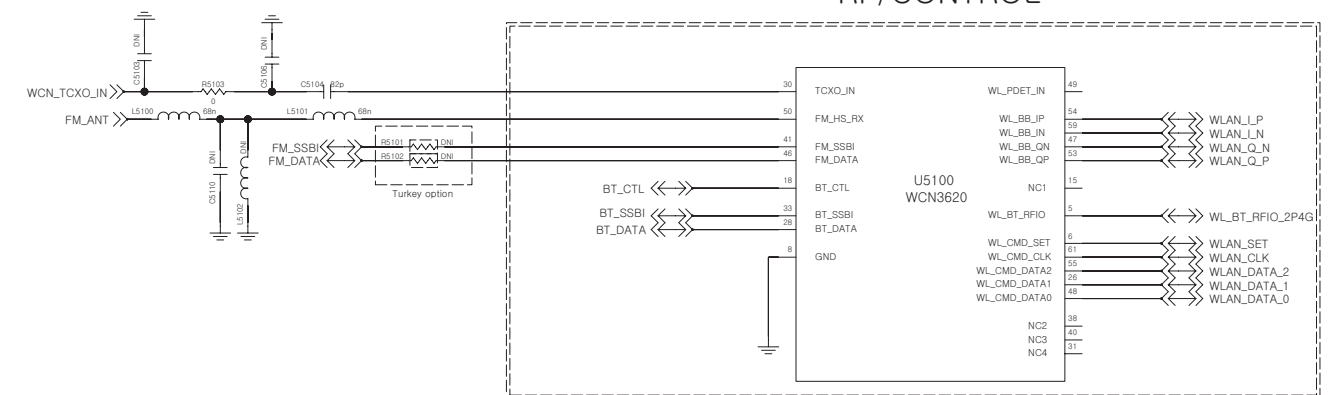
CCDS CARD Information	
Release Date	2013. 12. 2
Based on Reference Schematic	Rev.C
80-ND717-41 MSM8X10 + PM8110 REFERENCE SCHEMATIC	



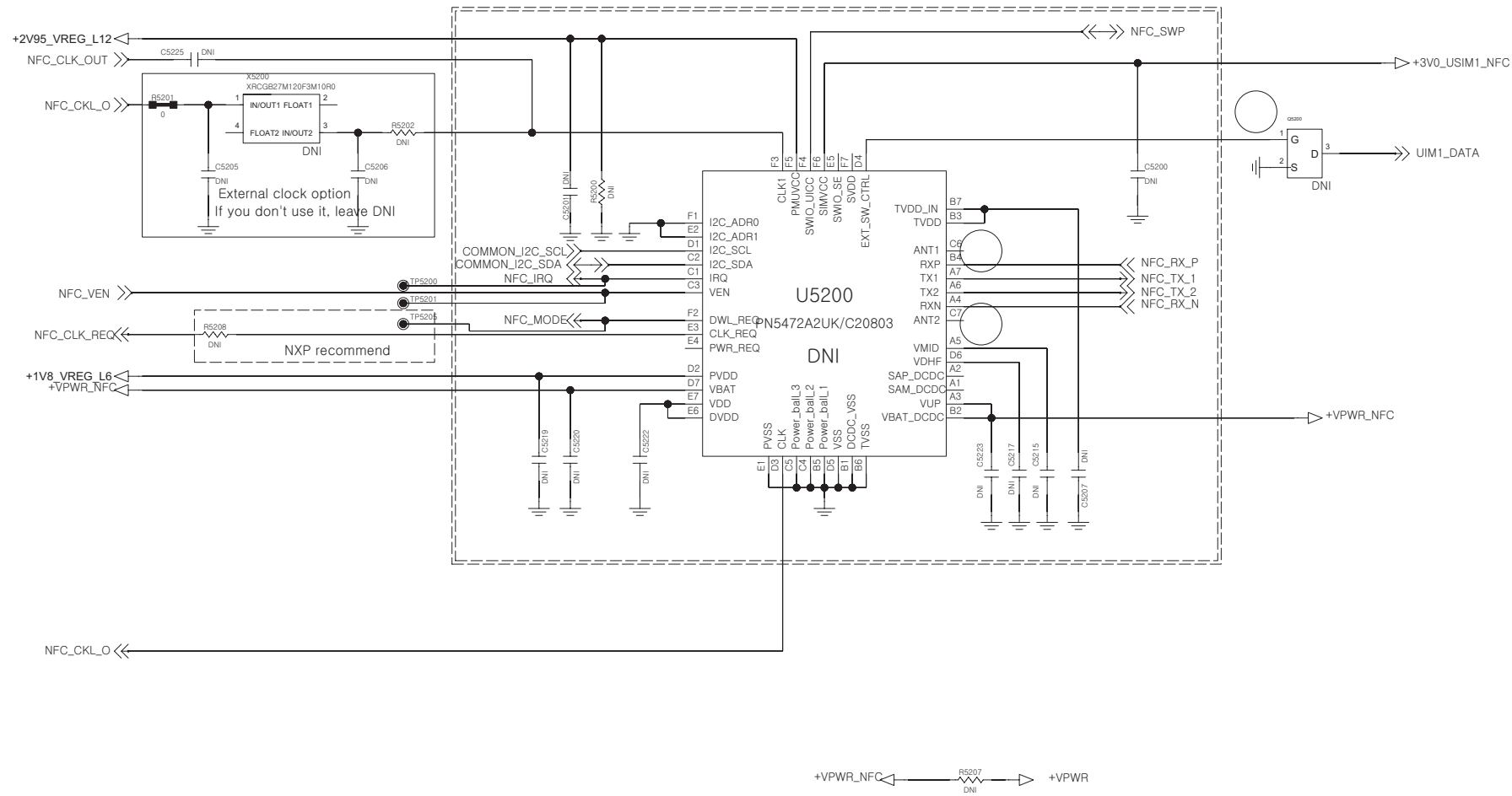
Rev.D



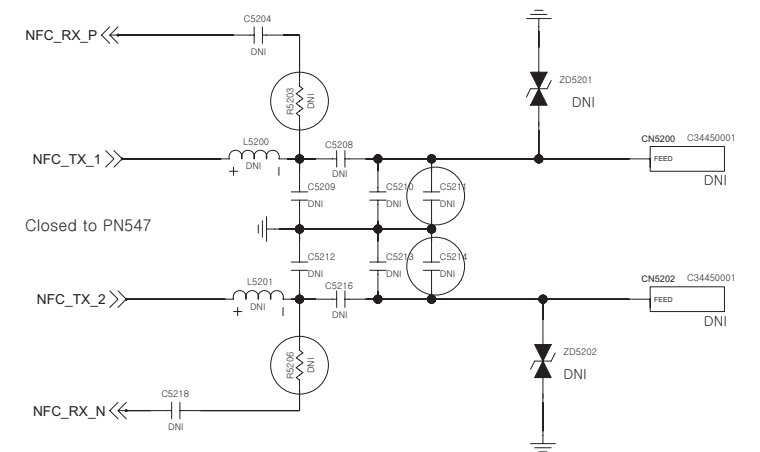
REV 1.0



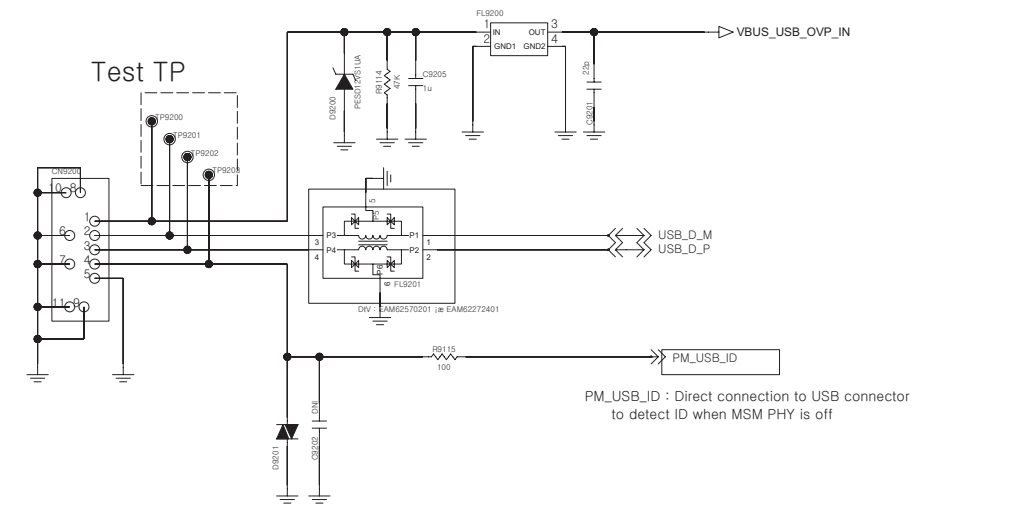
<5-2-1-4_NFC_PN547> Rev_1.1



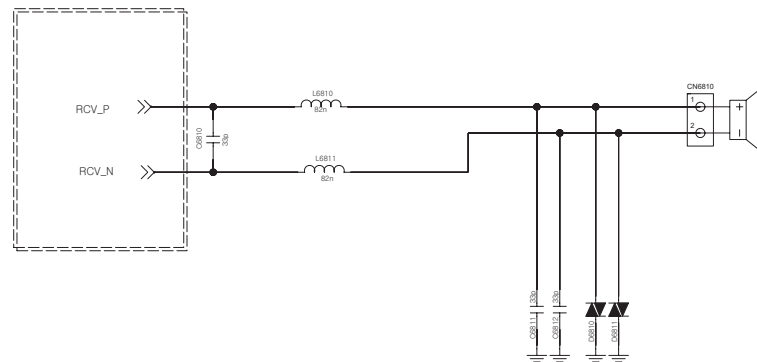
NFC Antenna



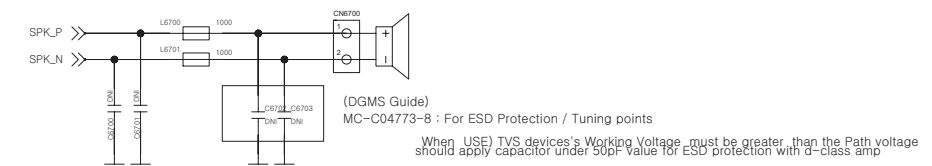
< 9-2-1_IO_USB2.0 > Rev_0.5



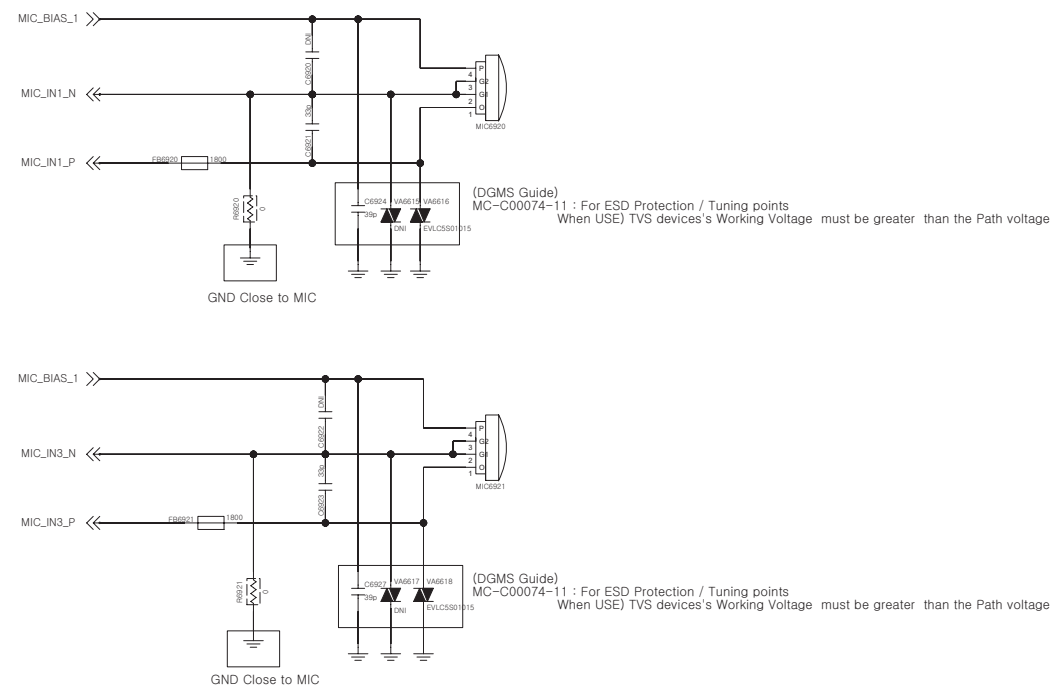
< 6-8-1_Receiver_Only > Rev_0.3



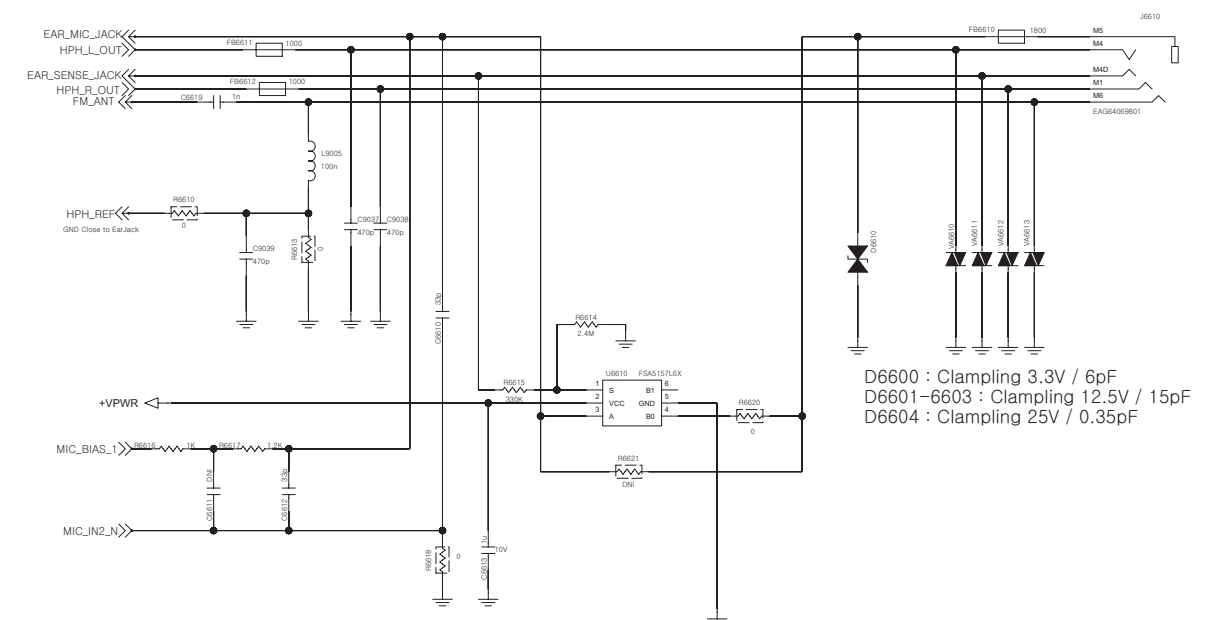
< 6-7-1_Speaker > Rev_0.4



< 6-9-2_2MIC > Rev_0.4



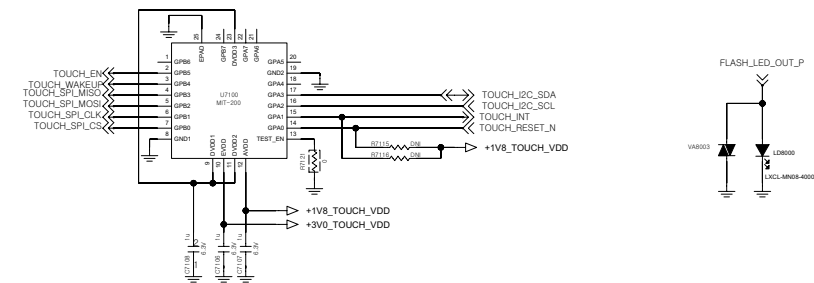
< 6-6-1_Earjack >



EN	FD	Action
LOW	LOW	Device OFF (Fast discharge active(on) after IC disable)
LOW	High	Device OFF (The status of dual outputs is floated)
High	LOW/High	Device ON (Don't care about FD Status)

[illegible]

<9-8-3_Flash_LED>



[DGMs Guide]

MC-C00081-17: 1. Add ESD Protection circuit for Camera signal lines
2. Use under 100ohm/15pF ESD protection for over 2MP Camera

Rev_0.3

VT_CAM_MIPI_CLK_I<1>
VT_CAM_MIPI_CLK_I<2>
VT_CAM_MIPI_DATA0_I<1>
VT_CAM_MIPI_DATA0_I<2>

VT_CAM_MCLK<1>
VT_CAM_MAKER_ID<1>
+1V8_VREG_L8<1>

ESD110
ESD111
ESD112
ESD113

FL7400

P1<1>
P2<1>
P3<1>
P4<1>

S<1>
S<2>
S<3>
S<4>

DIV

CN7400

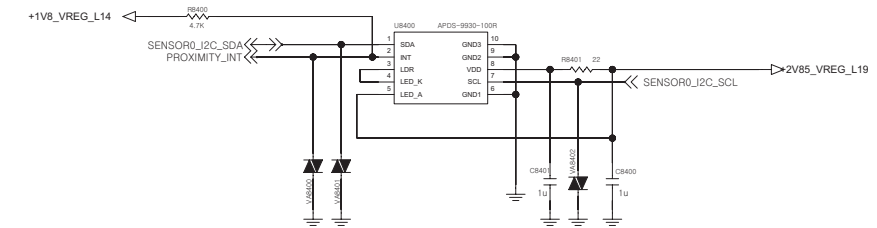
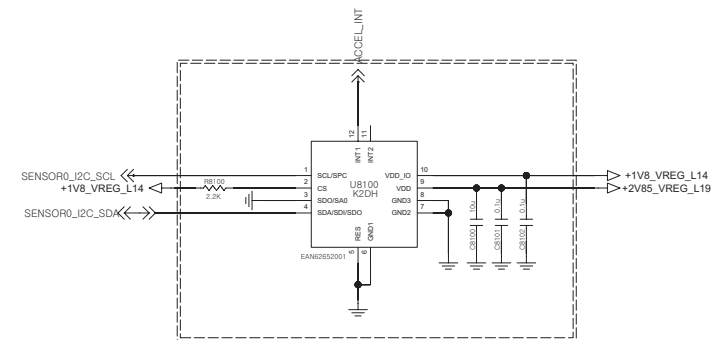
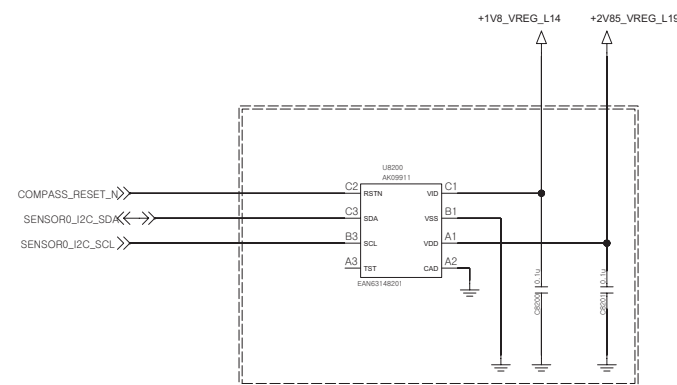
1V8_CAM_AVDD<1>
1V8_CAM_VDD<1>
1V8_CAM_VDD<2>
CAM0_I2C_SDA<1>
CAM0_I2C_SCL<1>
VT_CAM_PWDN<1>
VT_CAM_RESET_N<1>

[DGMs Guide]
MC-C01845-13 : Separate Analog Power and Digital Power

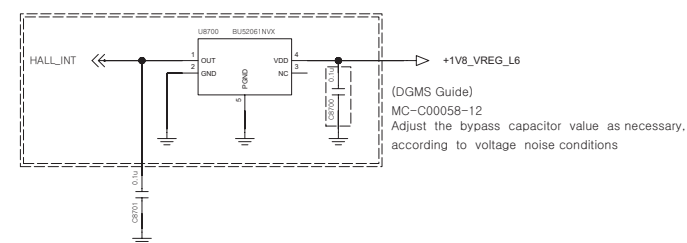
[DGMs Guide]
MC-C00060-13 : Check the pull-up resistor PAD for I2C line

[illegible]

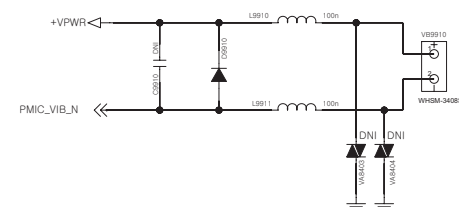
<8-2-1-3_Compass_AK09911 > <8-1-1-3_Accelerometer_K2DH > < 8-4-1-2_Proximity_APDS-9190 >
Rev_0.3 Rev_0.3 Rev_1.0



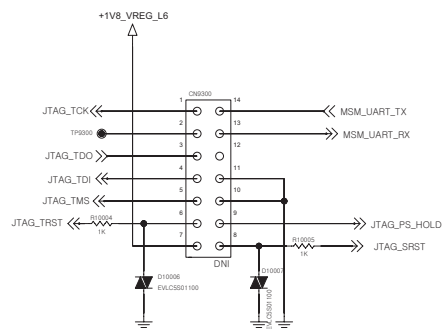
<8-7-1-6_Hall_IC_BU52033NVX>
Rev_1.0



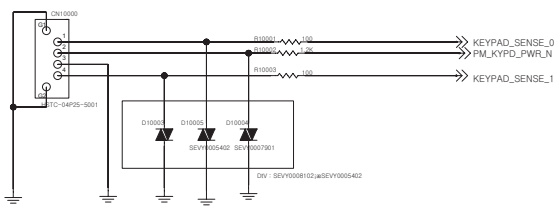
< 9-9-1-1_Motor >
Rev_0.3



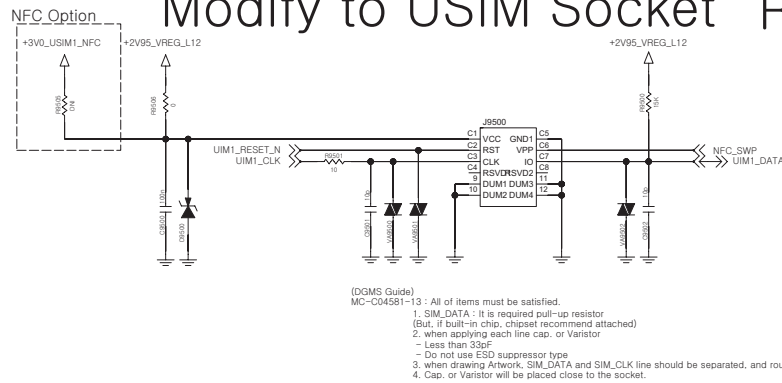
< 9-3-1_JTAG >
Rev_0.4



Back Volume Key

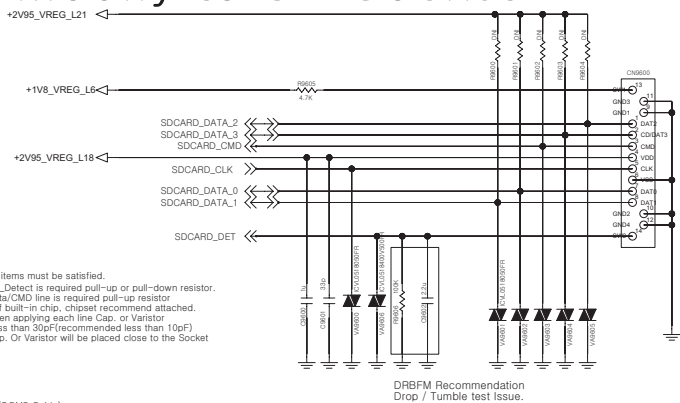


<9-5-2-2_Micro_SIM_8P_1_5T>
Modify to USIM Socket Rev_0.3



(DGMS Guide)
MC-C04581-13 : All of items must be satisfied.
1. SIM_DATA : It is required pull-up resistor
(But, if built-in chip, chipset recommend attached)
2. when applying each line cap. or Varistor
- Less than 33pF
- Do not use ESD suppressor type
3. when drawing Artwork, SIM_DATA and SIM_CLK line should be separated, and routing.
4. Cap. or Varistor will be placed close to the socket.

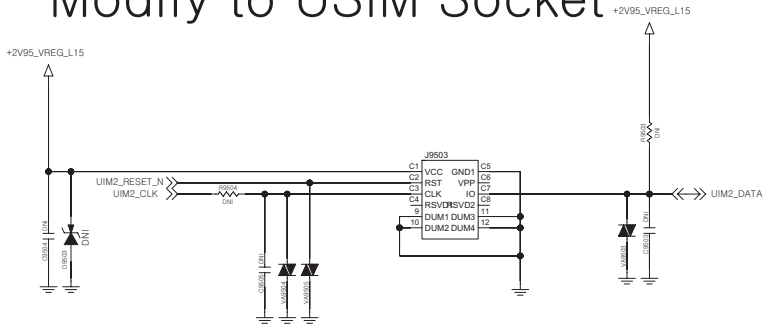
< 9-6-1_Socket_Micro_SD >
Modify to SD Socket



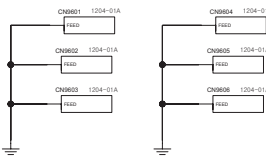
(DGMS Guide)
MC-C04582-13All of items must be satisfied.
1. SD_Detect is required pull-up or pull-down resistor.
2. Data/CMD line is required pull-up resistor.
(But, if built-in chip, chipset recommend attached).
3. when applying each line Cap. or Varistor
- Less than 30pF(recommended less than 10pF)
4. Cap. Or Varistor will be placed close to the Socket

Card Detect	SW1 SW2	SD_CARD_DET
Card Installed		1.8V
Card Not Installed		GND

<9-5-2-2_Micro_SIM_8P_1_5T>
Modify to USIM Socket

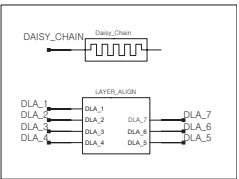


(DGMS Guide)
MC-C04581-13 : All of items must be satisfied.
1. SIM_DATA : It is required pull-up resistor
(But, if built-in chip, chipset recommend attached)
2. when applying each line cap. or Varistor
- Less than 33pF
- Do not use ESD suppressor type
3. when drawing Artwork, SIM_DATA and SIM_CLK line should be separated, and routing.
4. Cap. or Varistor will be placed close to the socket.



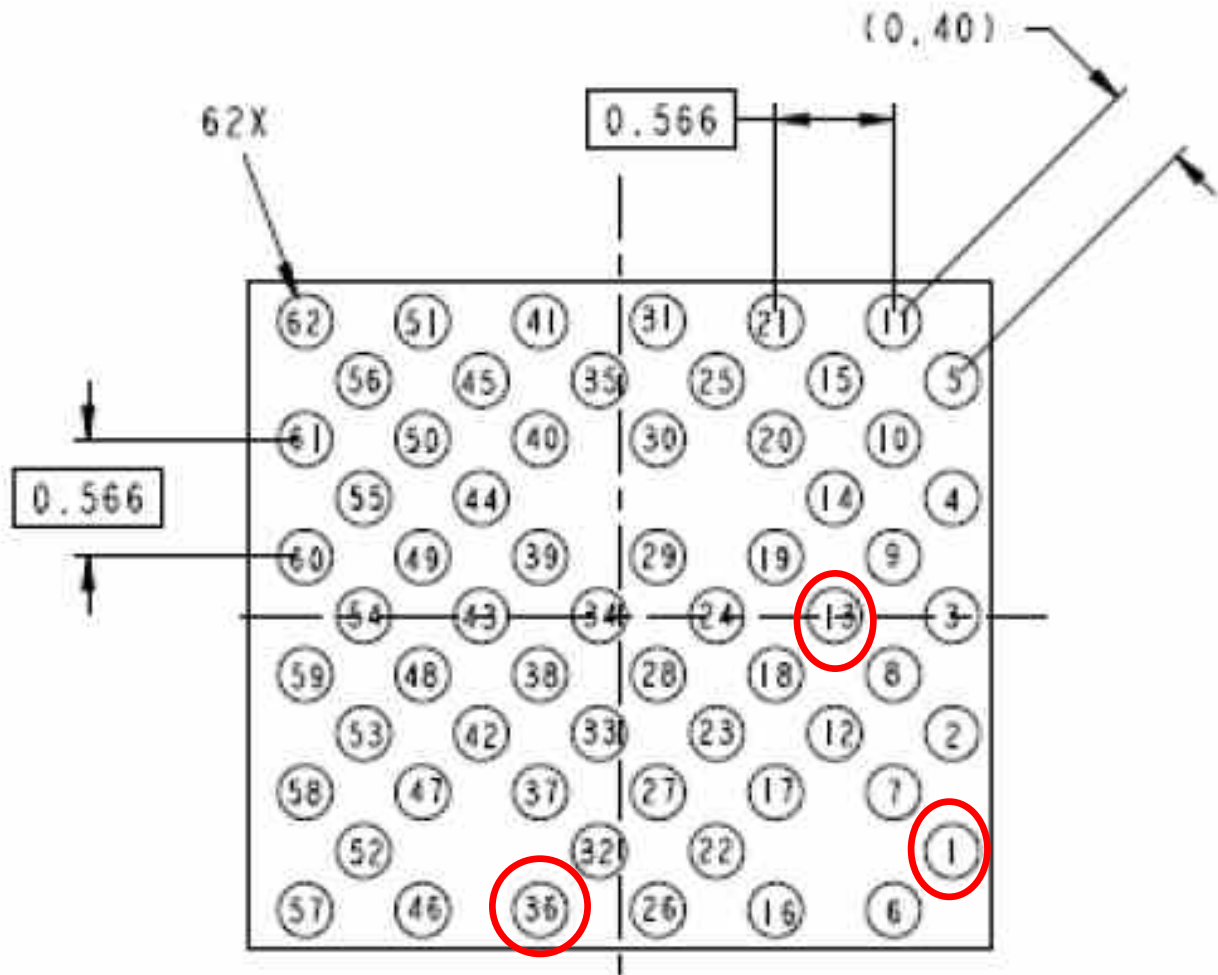
Shield CAN

CAMERA BRACKET



7. BGA PIN MAP

U1500 WTR2605



BOTTOM VIEW

- Not Used
- Used

7. BGA PIN MAP

U4100 PM8110

	1 VDD _L0_7 8 9 14		2 XTAL 19M_IN		3 VREG _RFCLK		4 VREG _L10		5 VREG _L18		6 VREG _L3		7 VREG _L21		8 VDD _L18_19 20 21
9 VDD _L0_7 8 9 14		10 VREG _L7		11 XTAL 19M_OUT		12 VDD _L10_11 13		13 VDD _L12_15 17 18 22		14 VREG _L4		15 VDD _L1_2 3 4 5		16 VREG _L19	
	17 VDD _L0_7 8 9 14		18 VREG _L1		19 VREG _XO		20 BB _CLK		21 VREG _L22		22 VDD _L1_2 3 4 5		23 VREG _L20		24 VDD _L18_19 20 21
25 VREG _L14		26 REF _BYP		27 VREG _L15		28 RF _CLK2		29 VREG _L17		30 VREG _L5		31 VREG _L1		32 VREG _L18	
	33 VREG _L9		34 GND _REF		35 VREG _L12					36 VREG _L2		37 VSW _S1		38 VDD _S1	
39 GND _S4		40 VREG _L8		41 RF _CLK1						42 GND _XO		43 VREG _S1		44 VSW _S1	
	45 VSW _S4		46 BB _CLK_EN		47 XO _THERM				48 GND _RF		49 AVDD _BYP		50 GND _S1		51 GND _S1
52 VDD _S4		53 VREG _S4		54 SLEEP _CLK				55 GND		56 GND		57 RESIN _N		58 VIB _DRV_N	
	59 VCOIN		60 MPP _04		61 GND			62 GND _XOADC		63 GND		64 GND _DRV		65 VDD _S2	
66 VDD _S3		67 VSW _S3		68 MPP _03				69 GND		70 GND		71 SPMI _DATA		72 VSW _S2	
	73 VSW _S3		74 VREG _S3					75 GND		76 SPMI _CLK		77 VSW _S2		78 GND _S2	
79 GND _S3		80 VBAT _SNS		81 GPIO _02		82 MPP _01		83 VREG _UADC		84 PS _HOLD		85 VREG _S2		86 GND _S2	
	87 VBAT		88 GPIO _04		89 GPIO _03		90 GPIO _01		91 MPP _02		92 PONL _RST_N		93 VREF _LPDDR		94 VREF _NEG_S2
95 VBAT		96 VBAT		97 BAT _THERM		98 USB _IN		99 VSW _CHG		100 HW_ID		101 KPD _PWR_N		102 ATC _SINK	
	103 VPH _PWR		104 VPH _PWR		105 BAT _ID		106 VCHG		107 GND _CHG_HP		108 VREF _BAT_THM		109 GND _CHG		110 GND
111 BATFET _CP_DRV		112 VPH _PWR		113 VPRE _BYP		114 USB _IN		115 VSW _CHG		116 GND _CHG_HP		117 VDRV_P		118 NC	

INPUT
PWR MGT

OUTPUT
PWR MGT

GEN HK

USER IF

IC IF

GPIO or
MPP

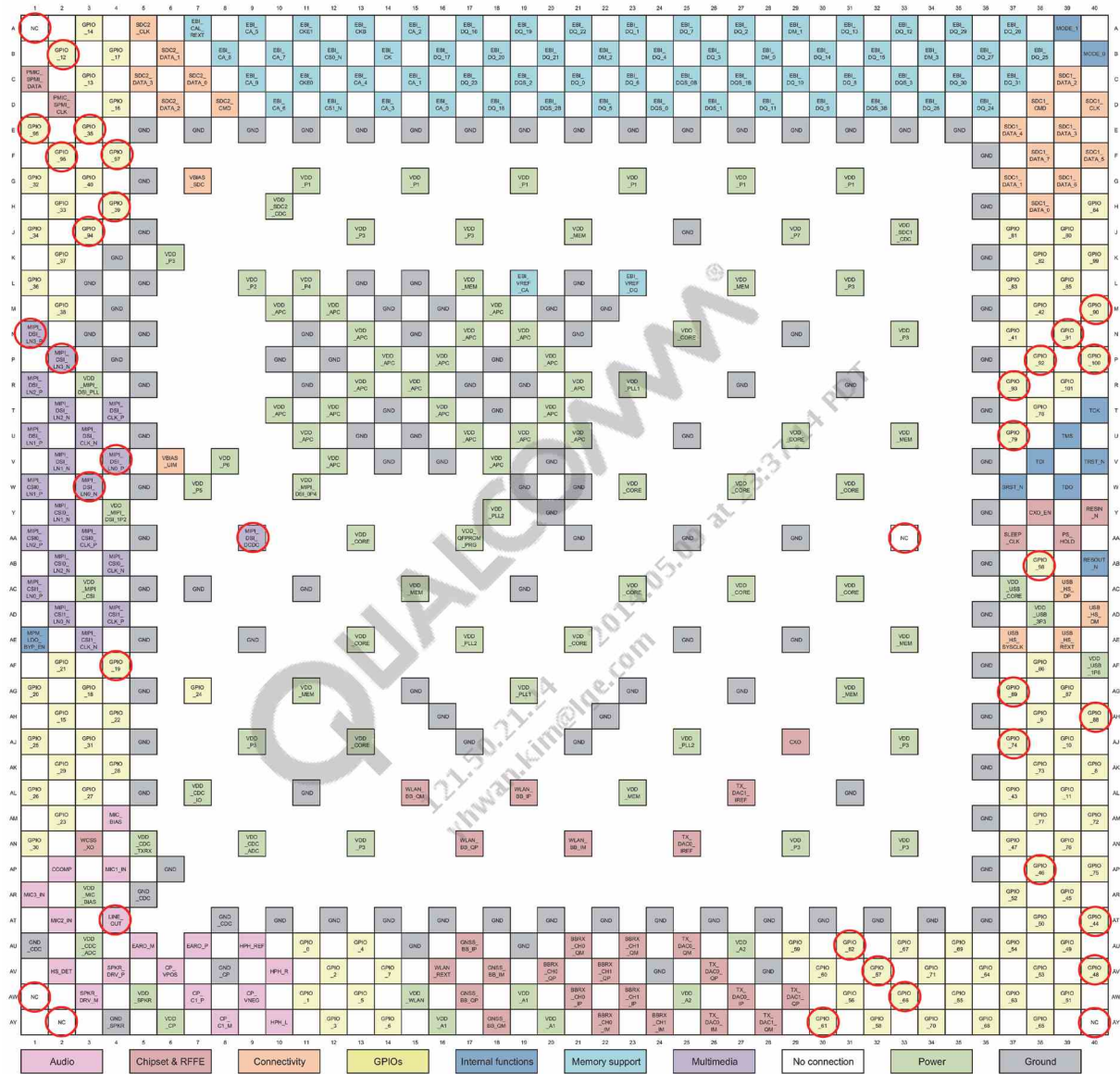
Power

Ground

○ Not Used

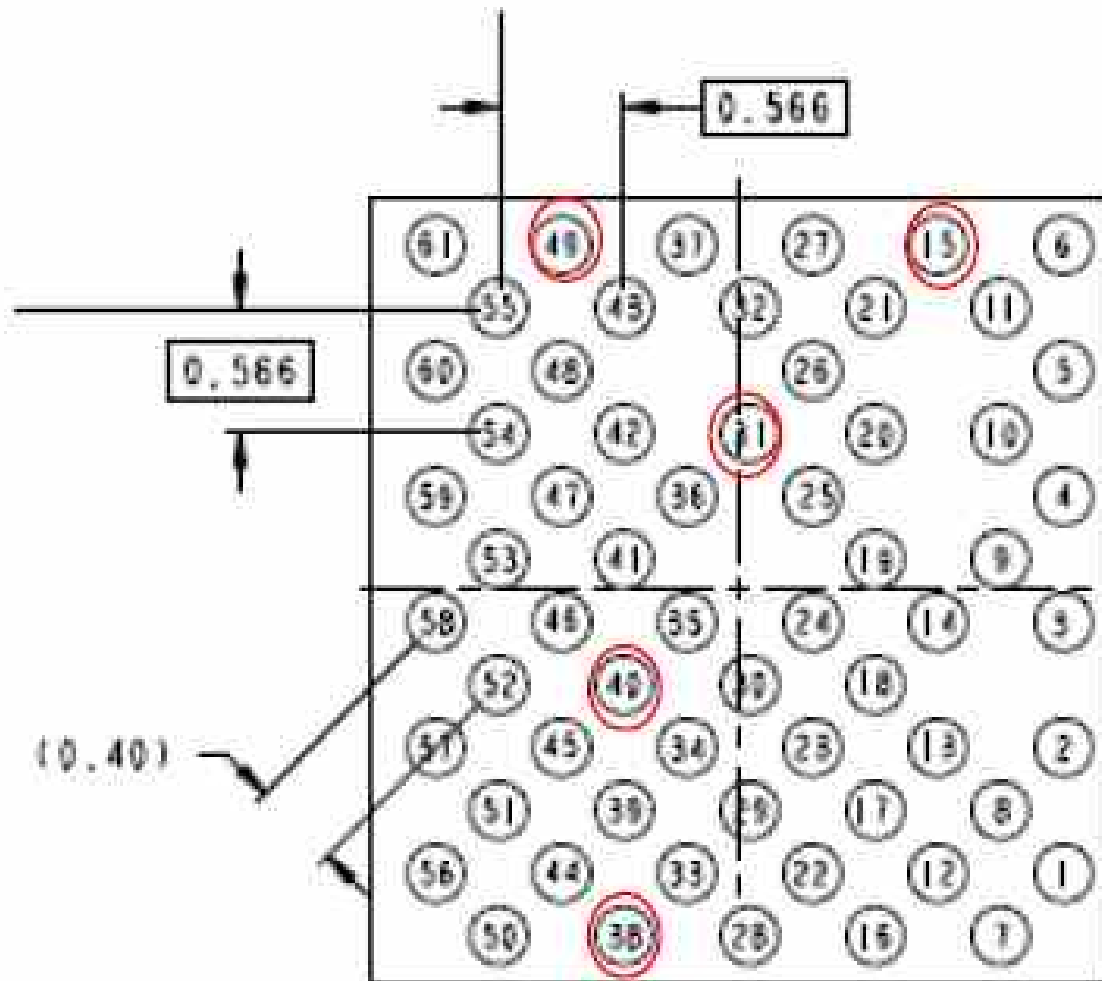
7. BGA PIN MAP

U2100 MSM8212



○ Not Used

U5100 WCN3620

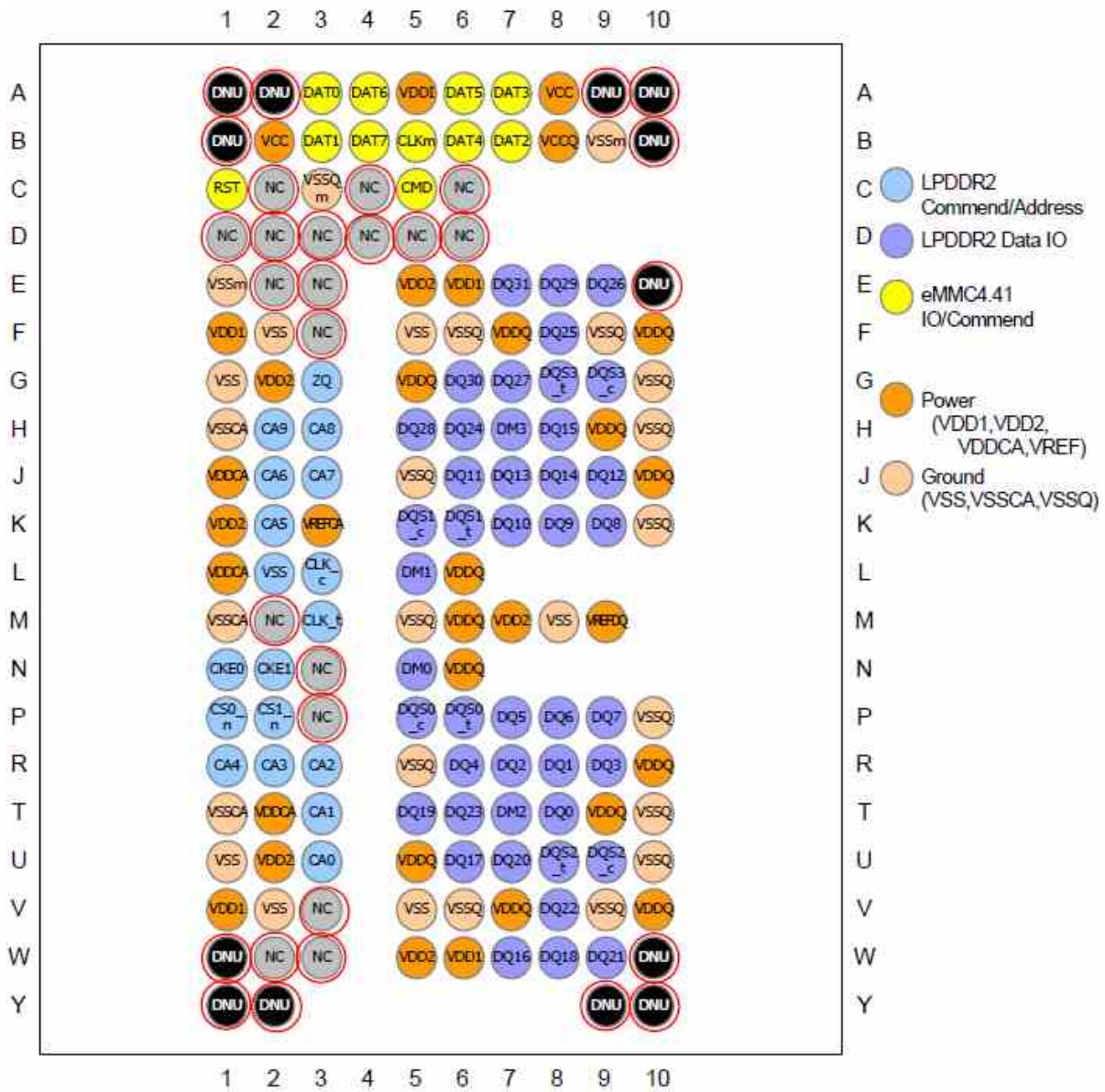


BOTTOM VIEW

- Not Used
- Used

7. BGA PIN MAP

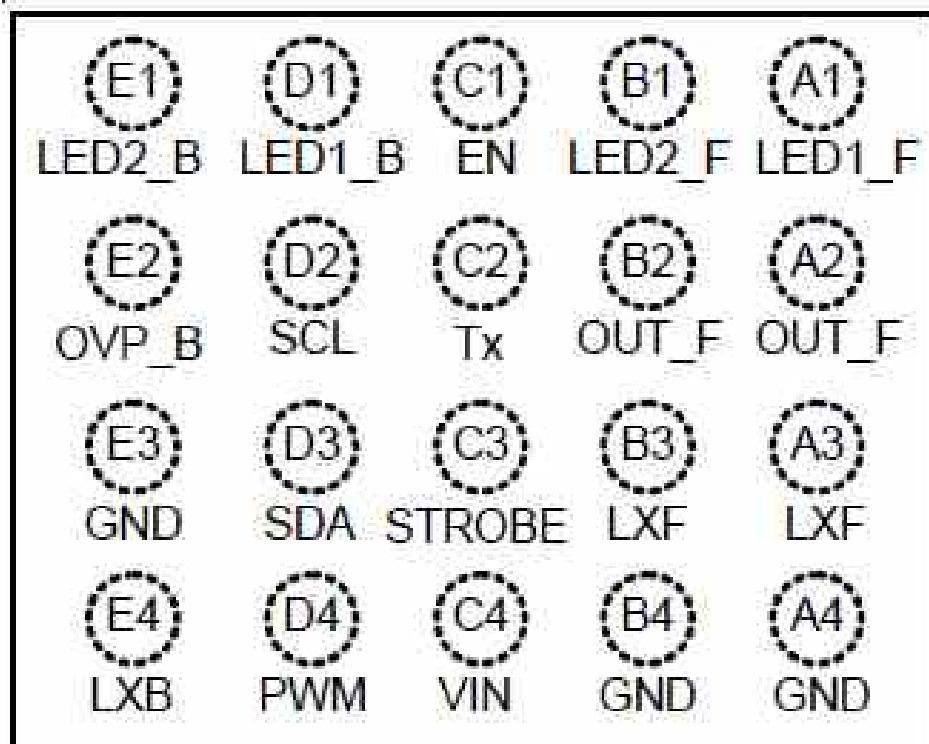
U3300 H9TP32A8JDBCPR-KGM



○ Not Used

U4400 RT8542

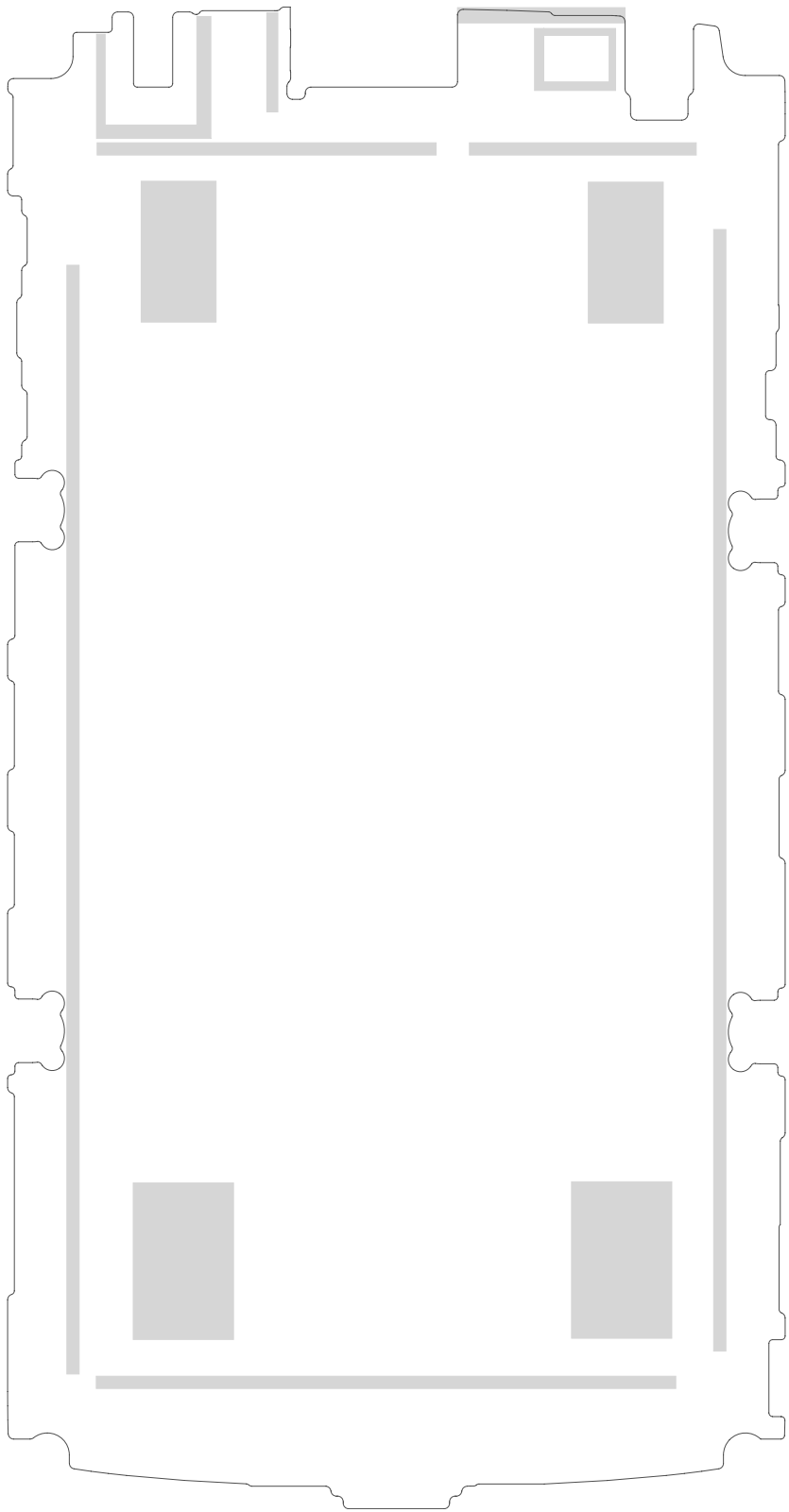
(TOP VIEW)



WL-CSP-20B 1.85x2.35 (BSC)

 Not Used

8. PCB LAYOUT

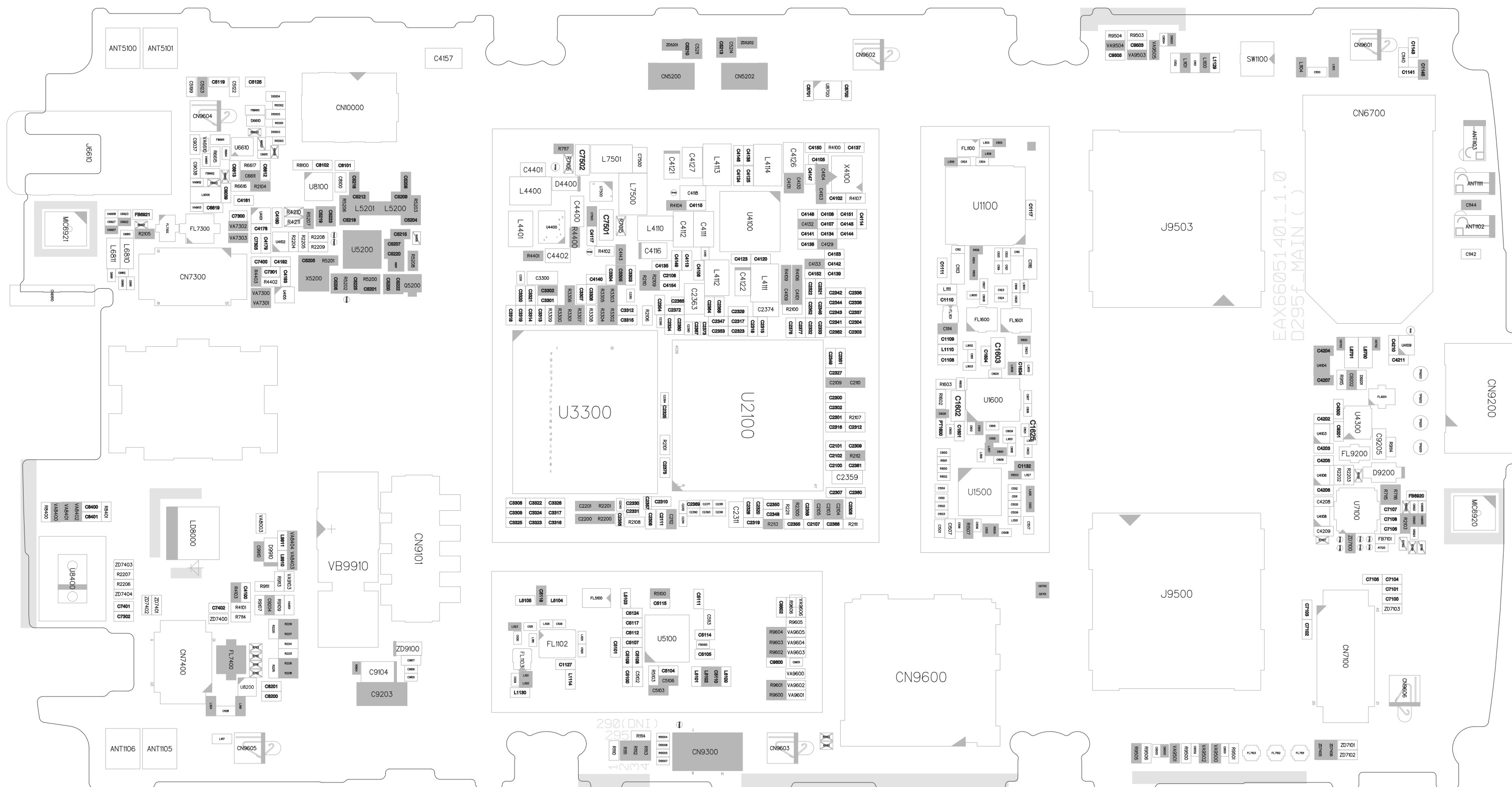


LG-D290TR_MAIN_1.0_TOP

: Search parts by part number may not work on this layout. 206p to search parts.



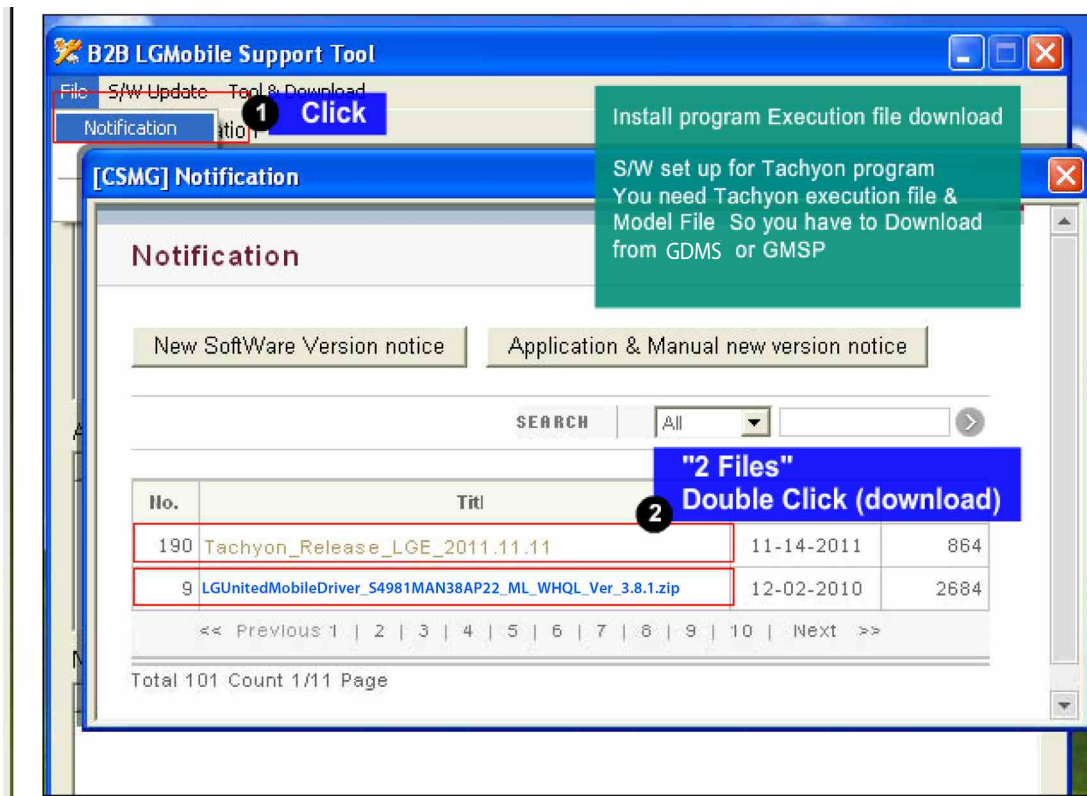
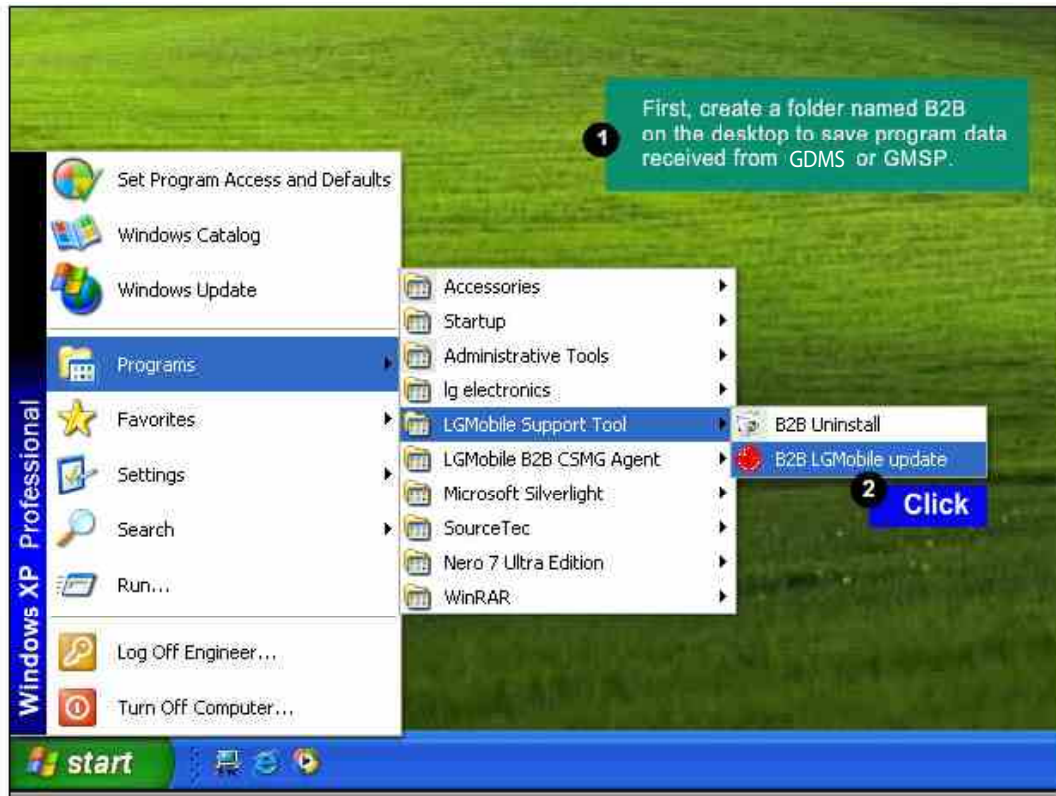
LG-D290TR_MAIN_1.0_BOT



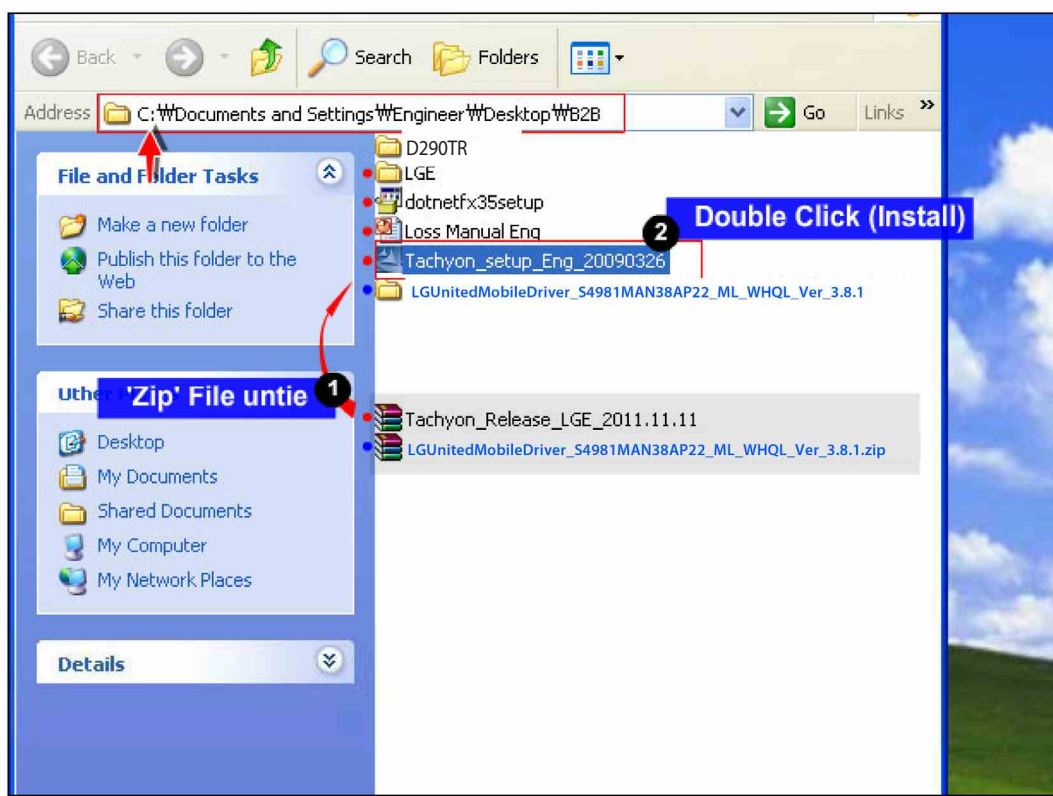
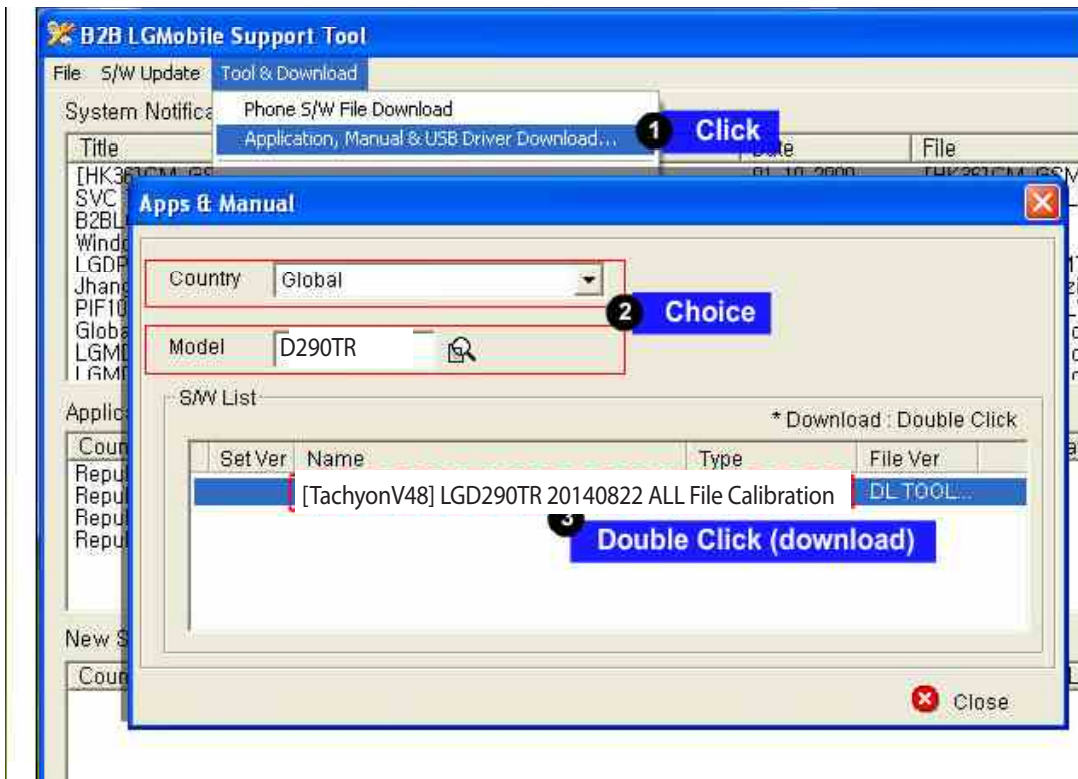
9. CALIBRATION

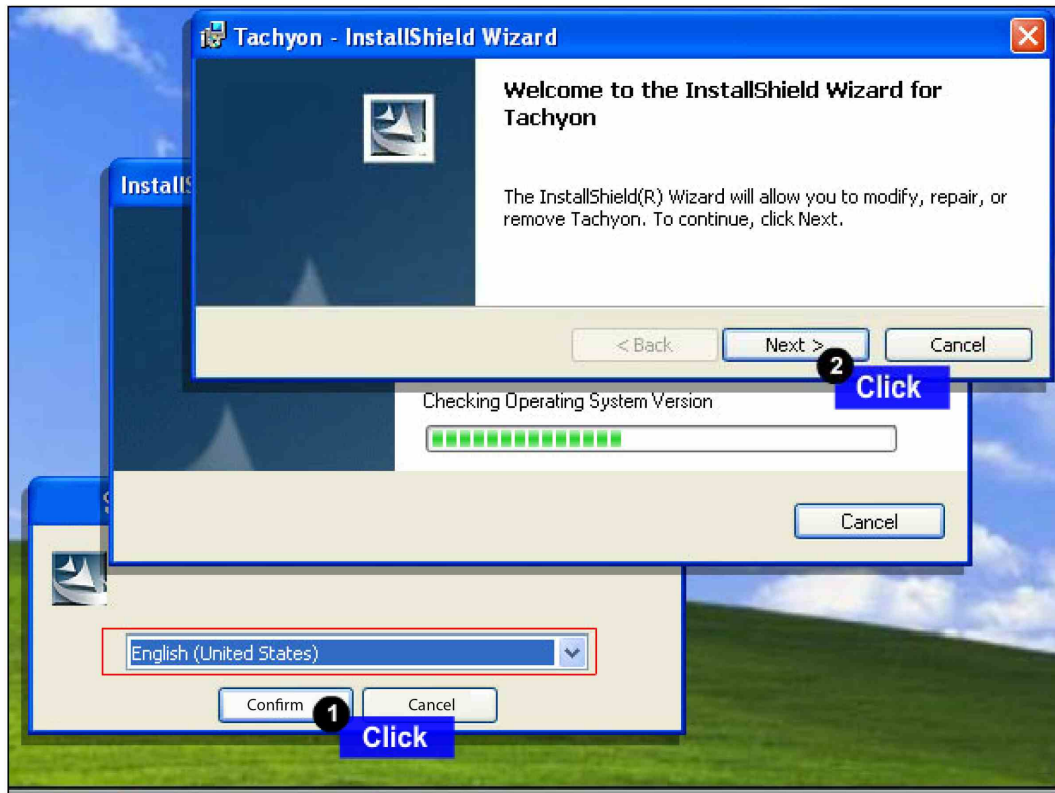
CAL INFORMATION		
S/W VERSION		
Please Check the Version to "B2B"		
H/W		
	Name	Part No.
PIF	PIF200(All Type)	BJAY0024021
USB Cable	USB Cable	RAD32247898
Power Cable	DC Power Cable	RAD32247878
I/O Cable		
RF Cable_Main		
Power Supply_PIF	Power Supply 5.3V	
Power Supply_Phone	Power Supply 5.0V	
RF Test Equipment	E5515C(8960)	
NOTICE		

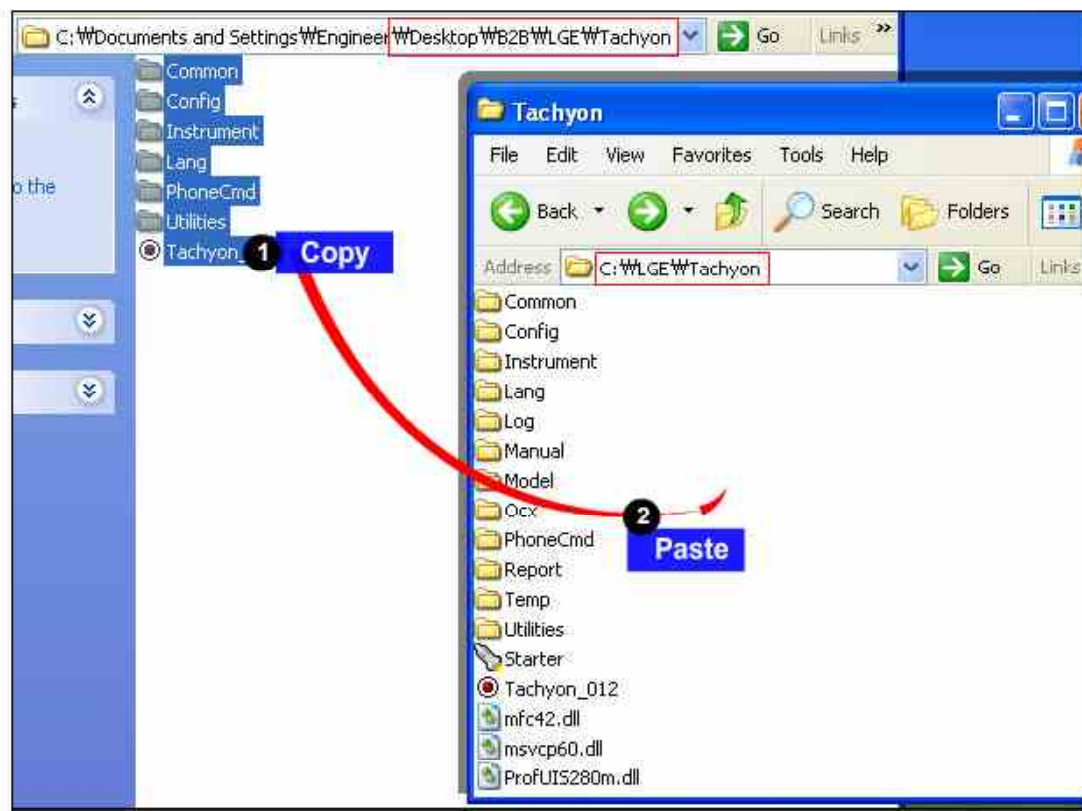
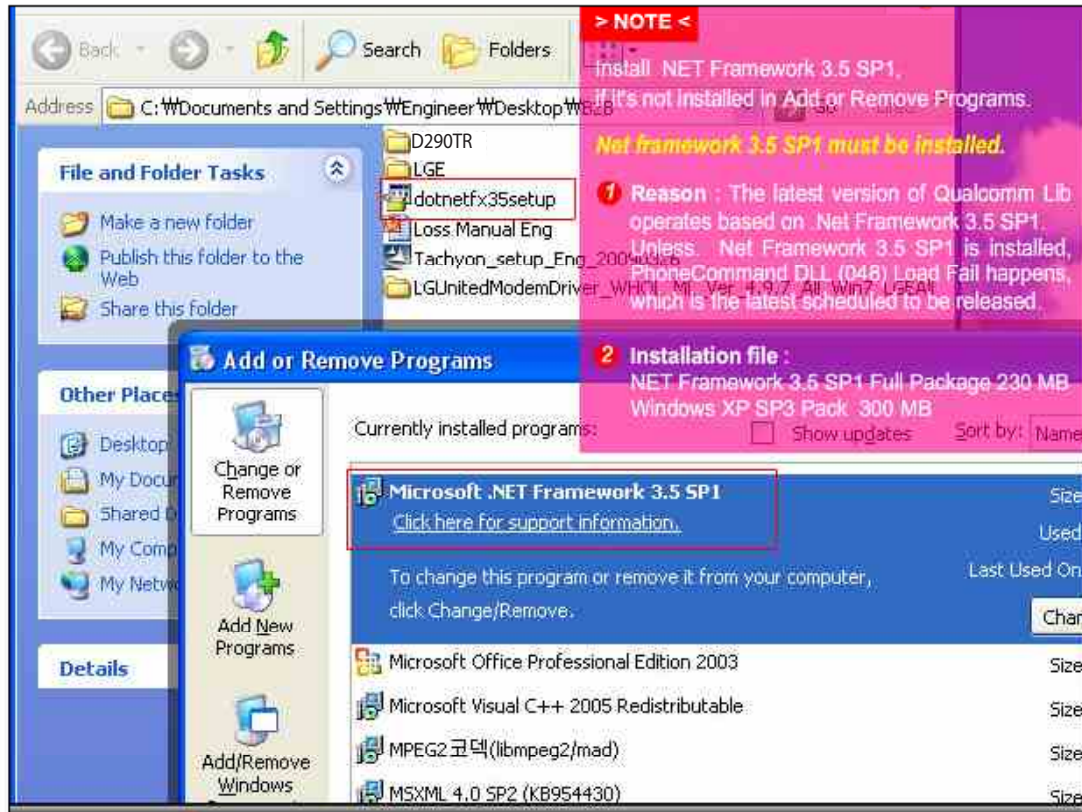
9. CALIBRATION



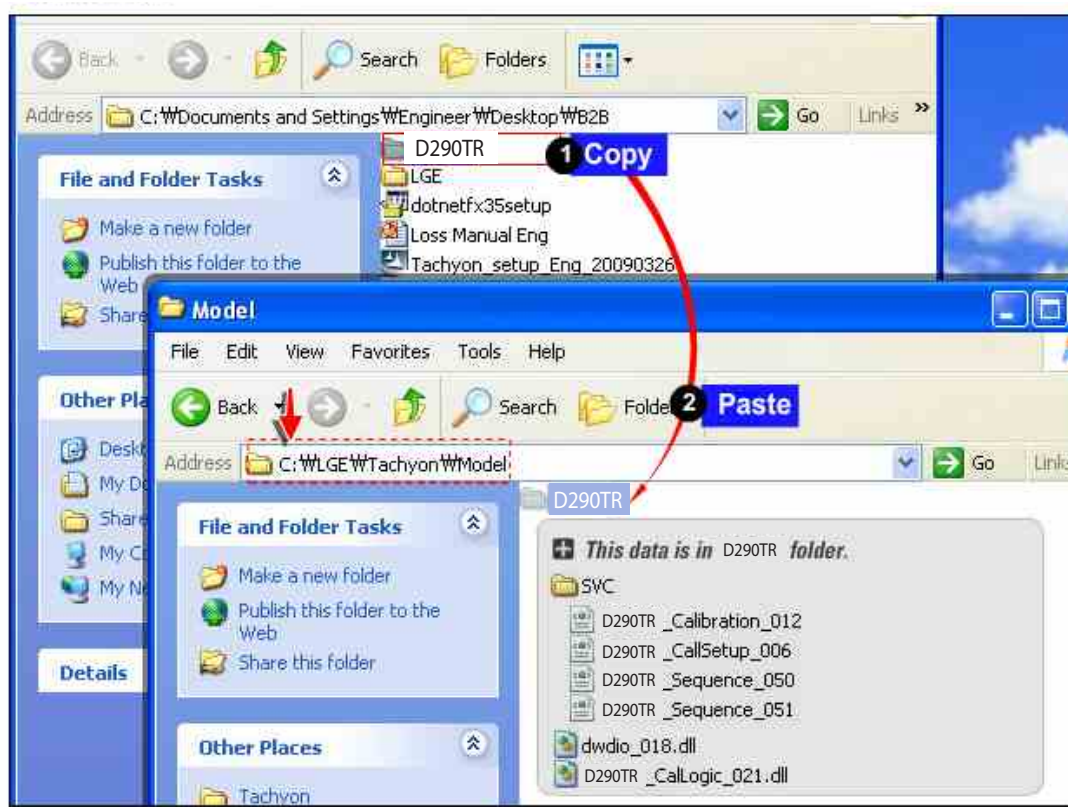
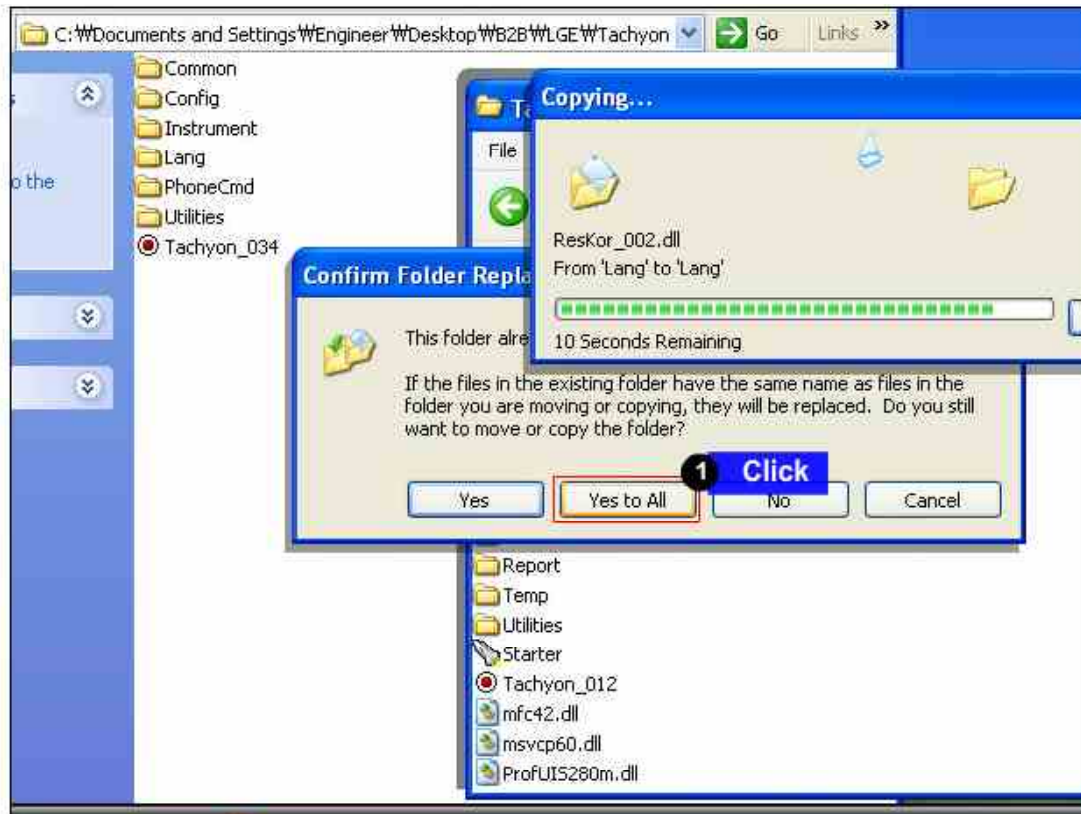
9. CALIBRATION



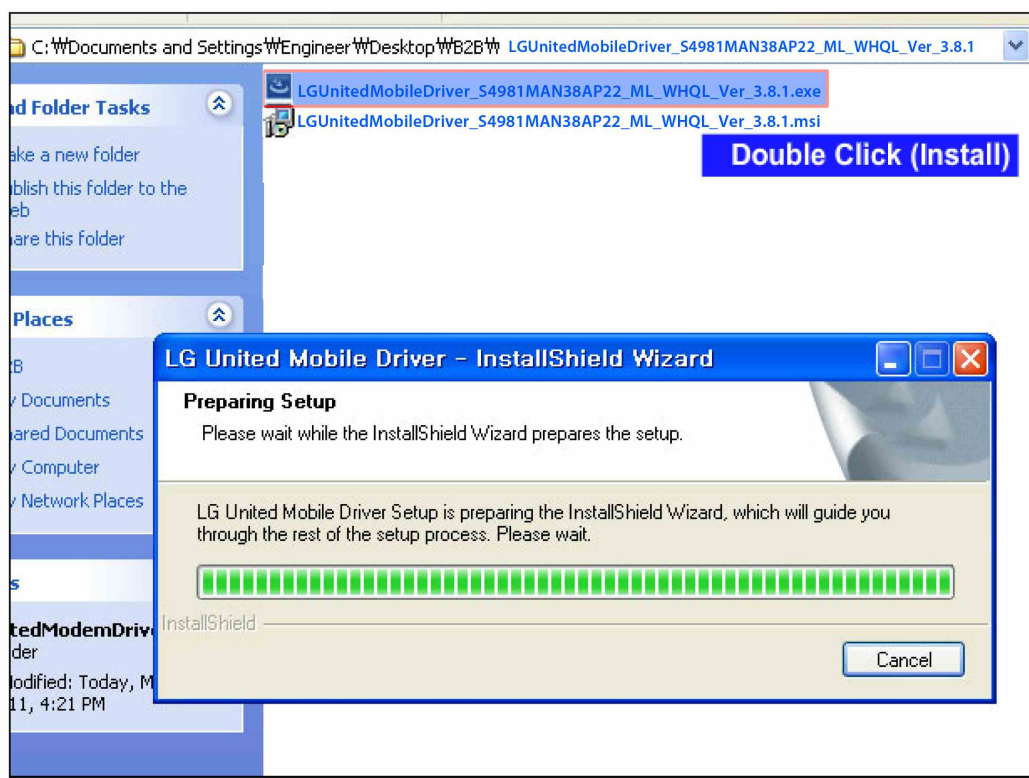
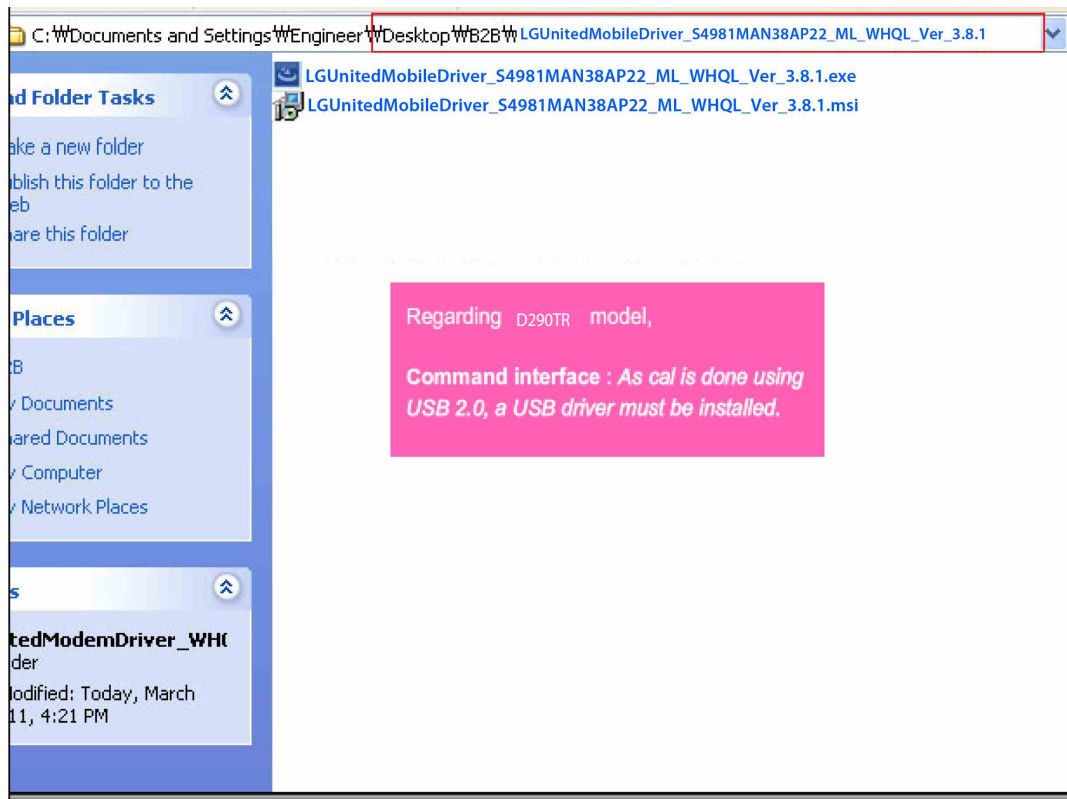




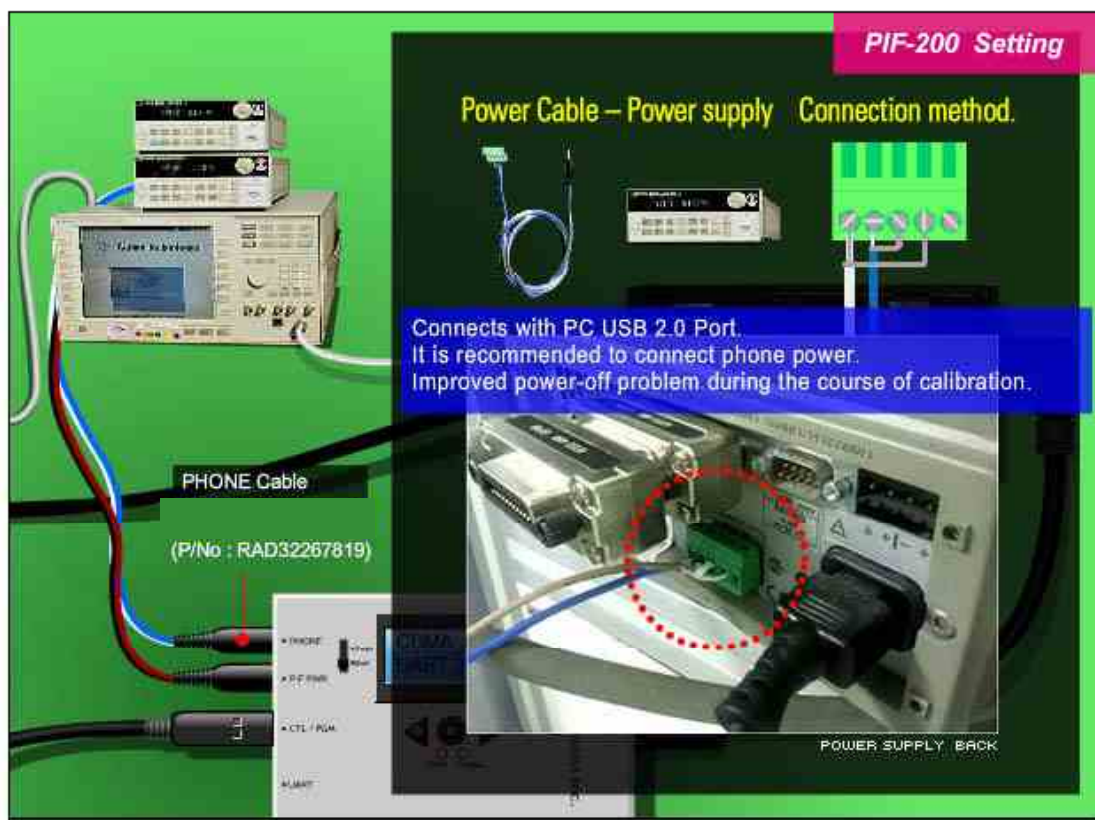
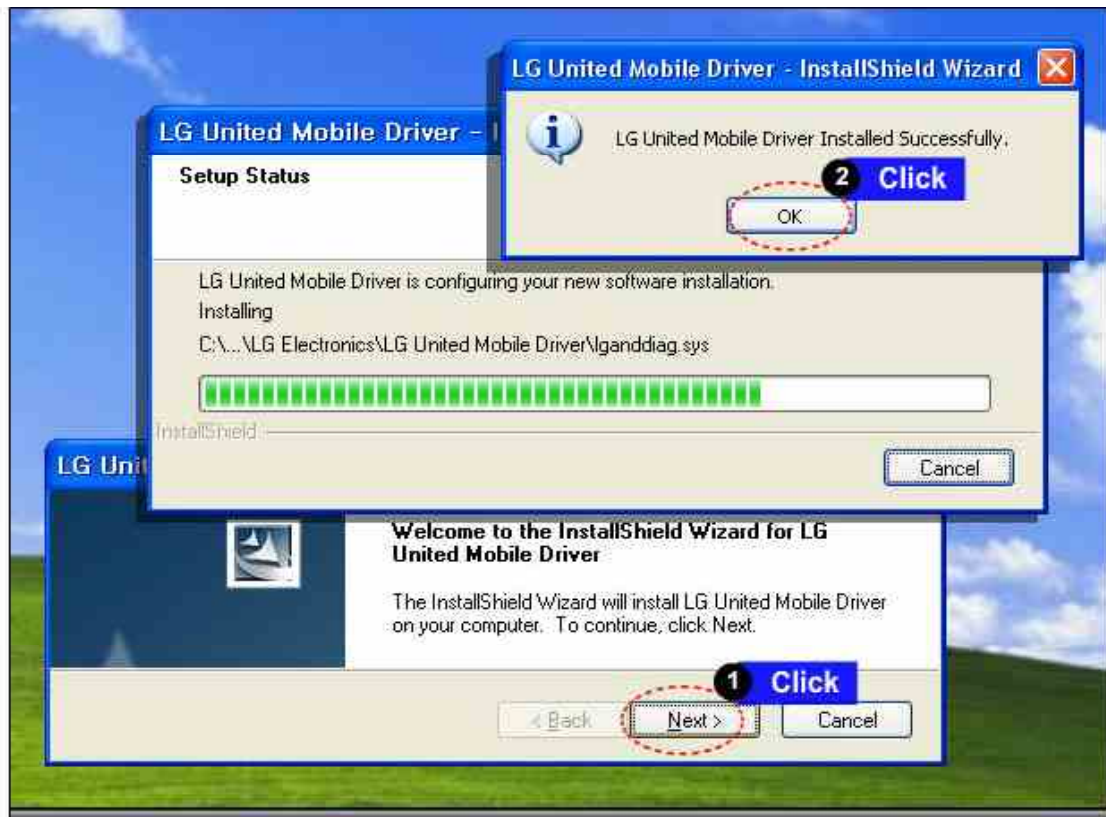
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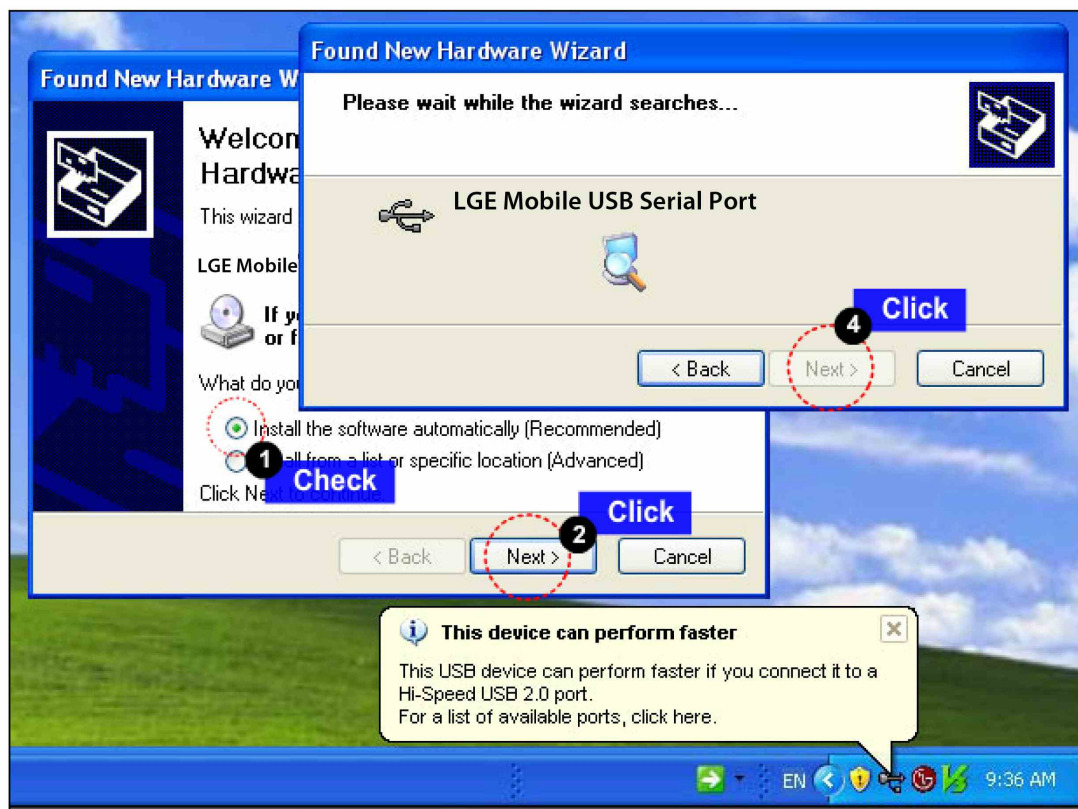
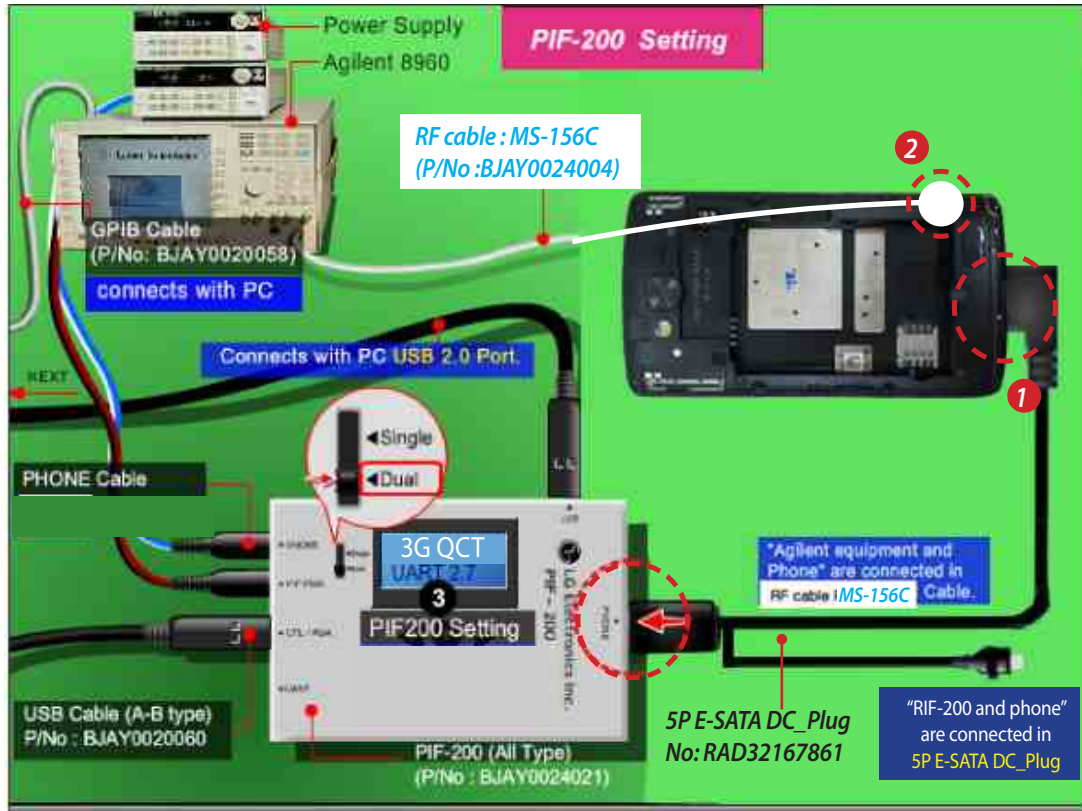


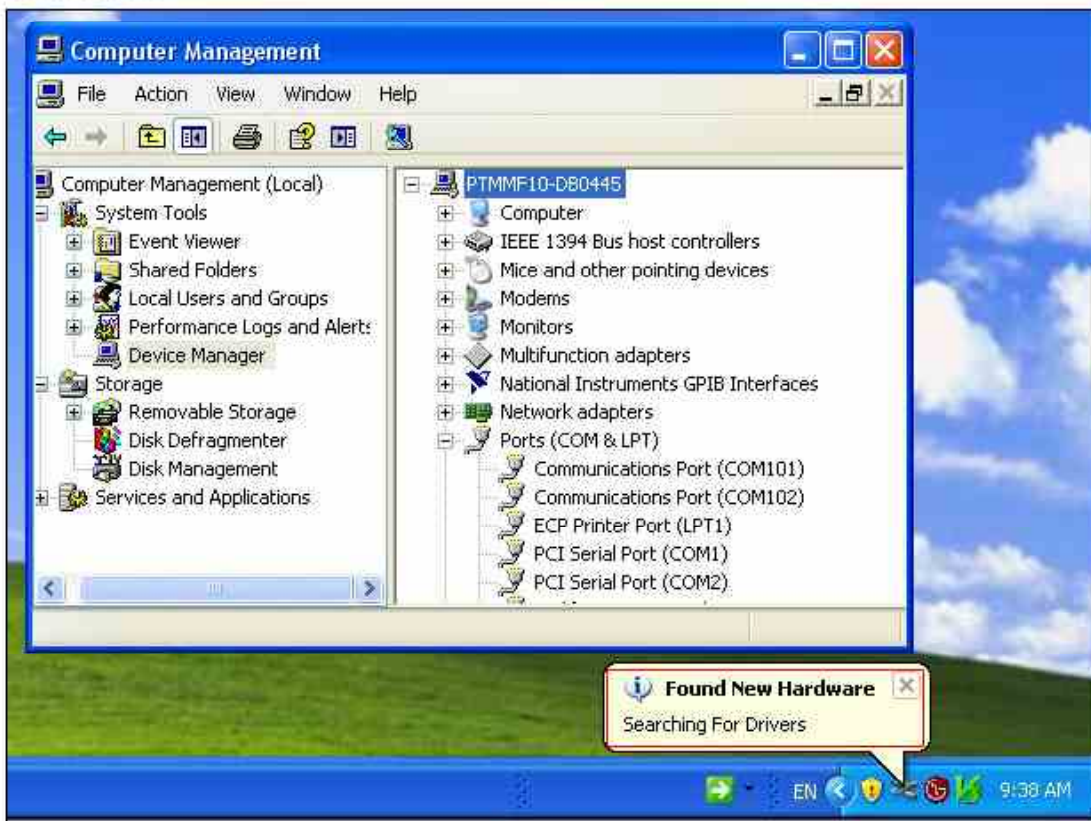
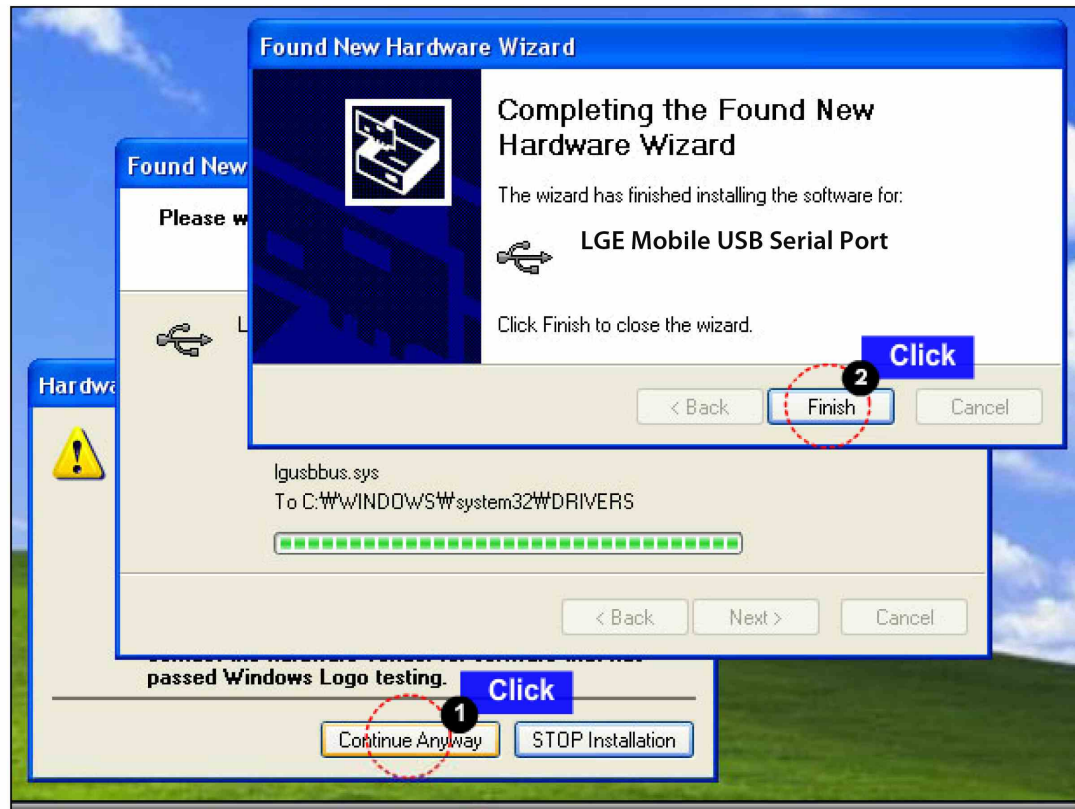
9. CALIBRATION

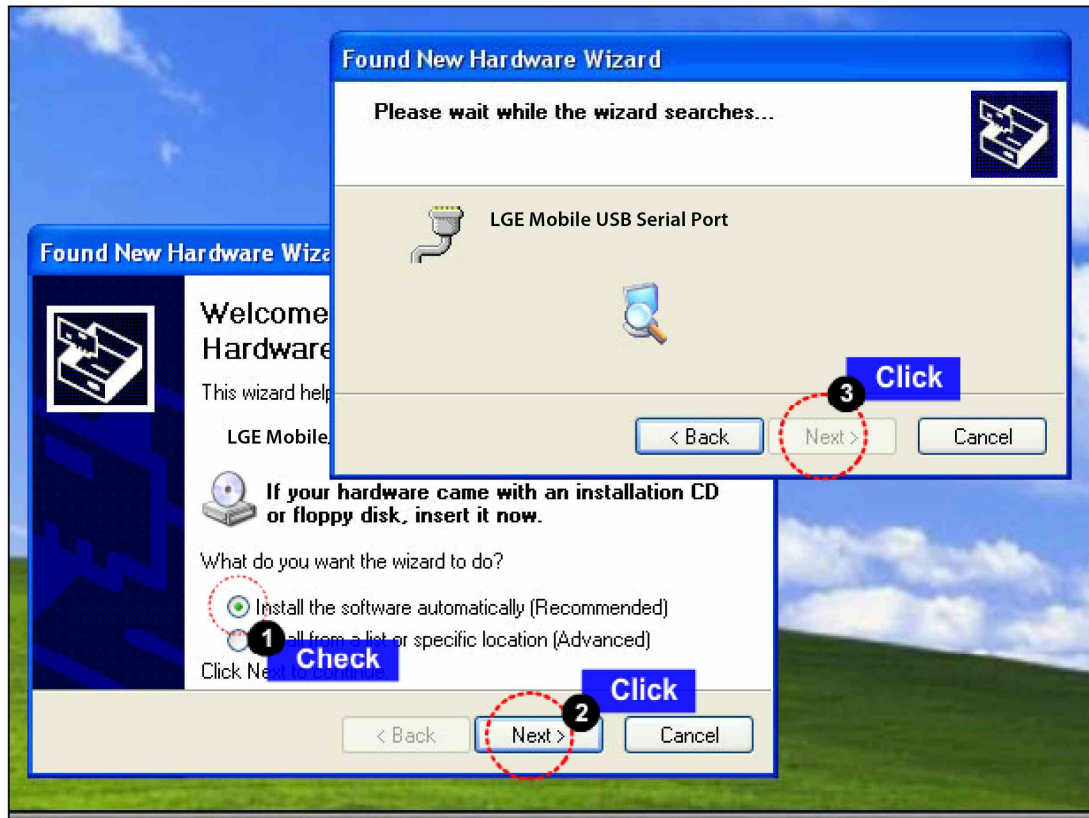


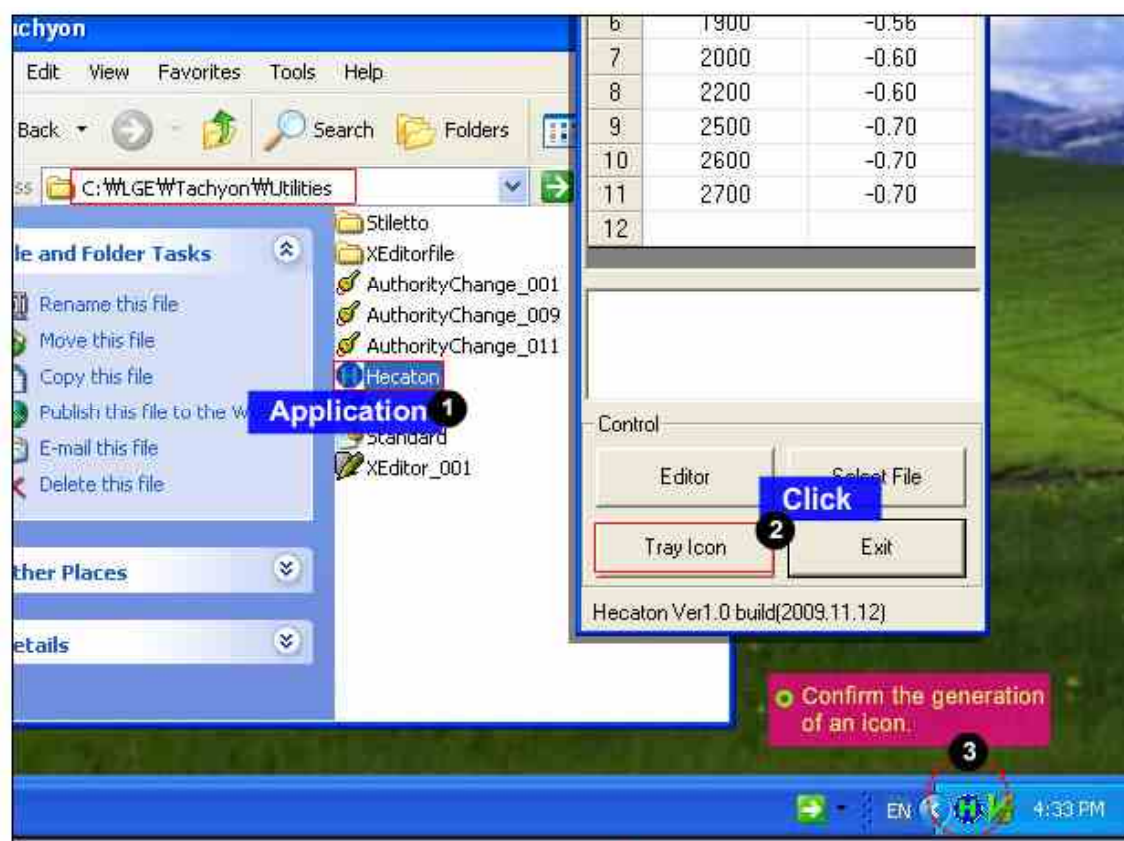
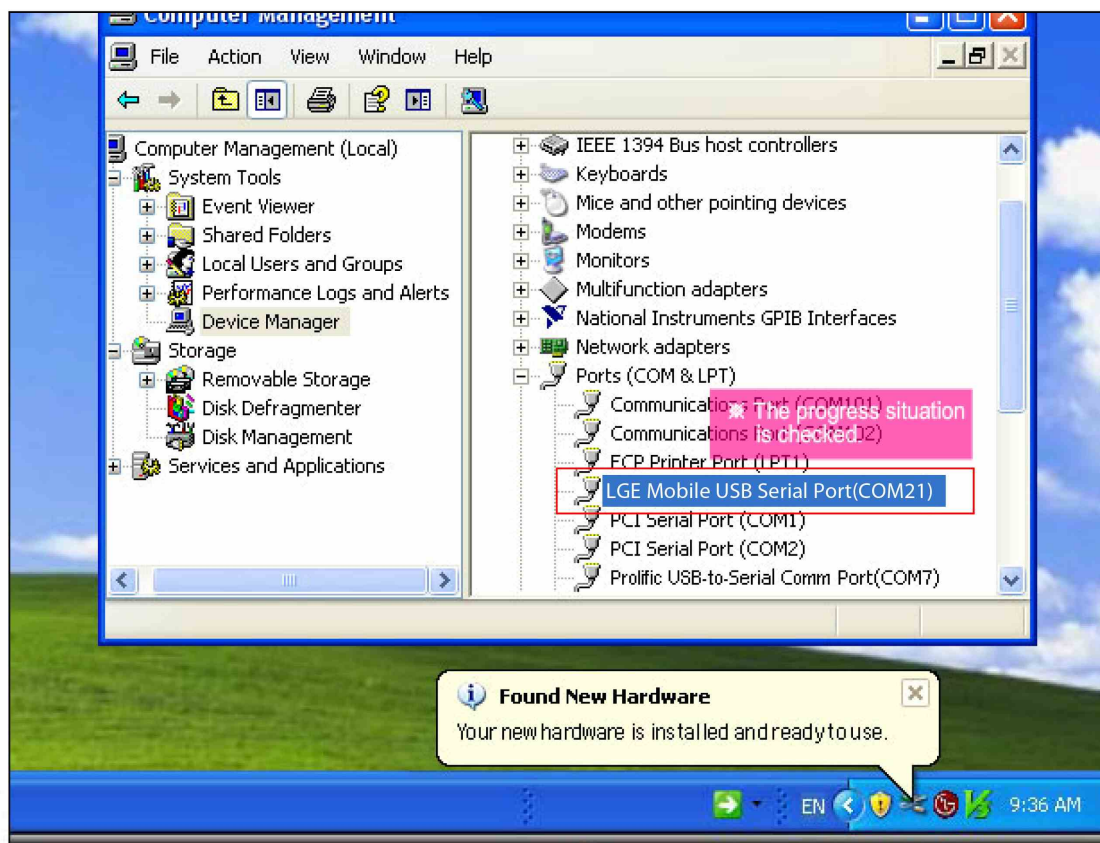
9. CALIBRATION



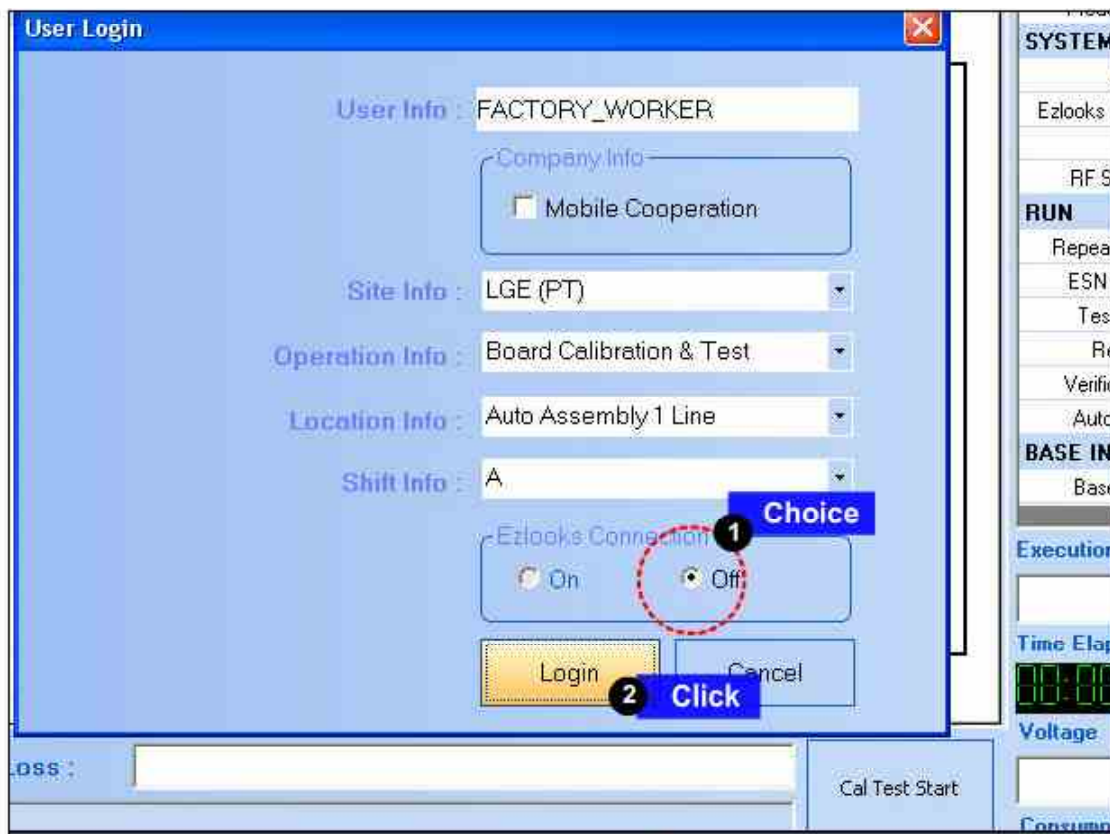
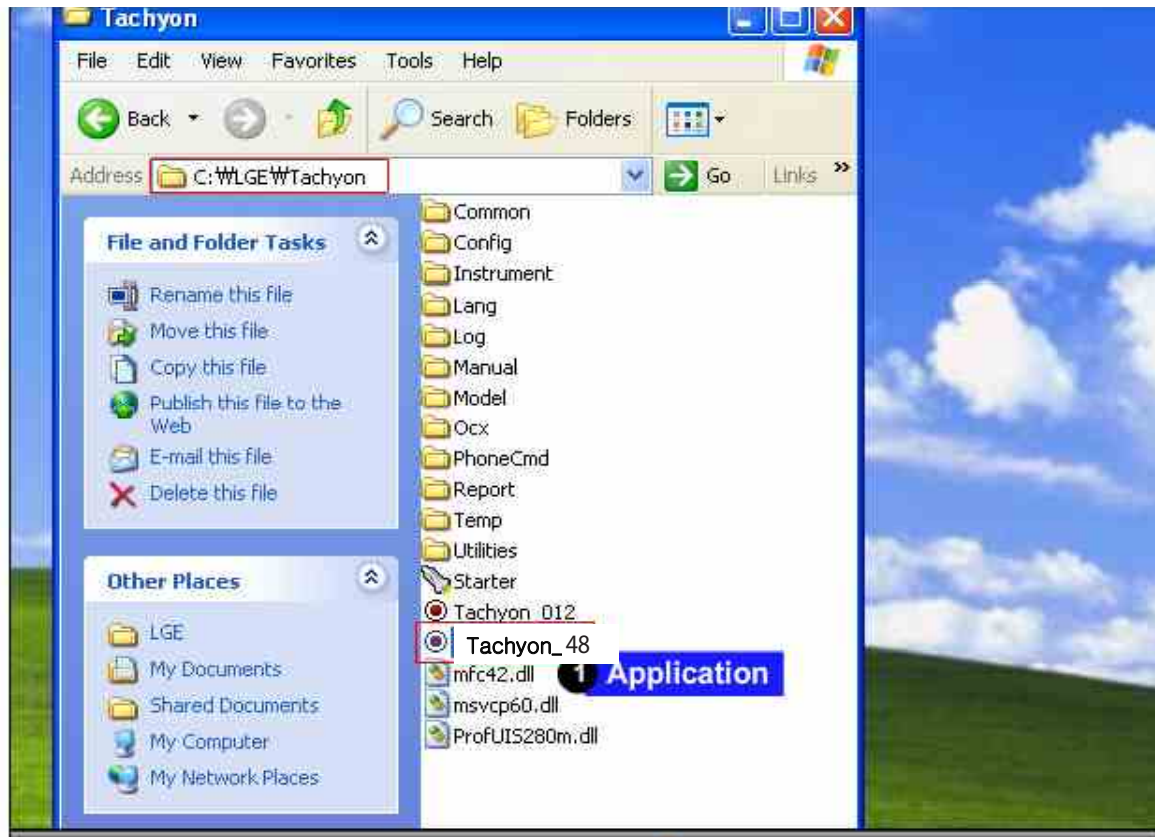


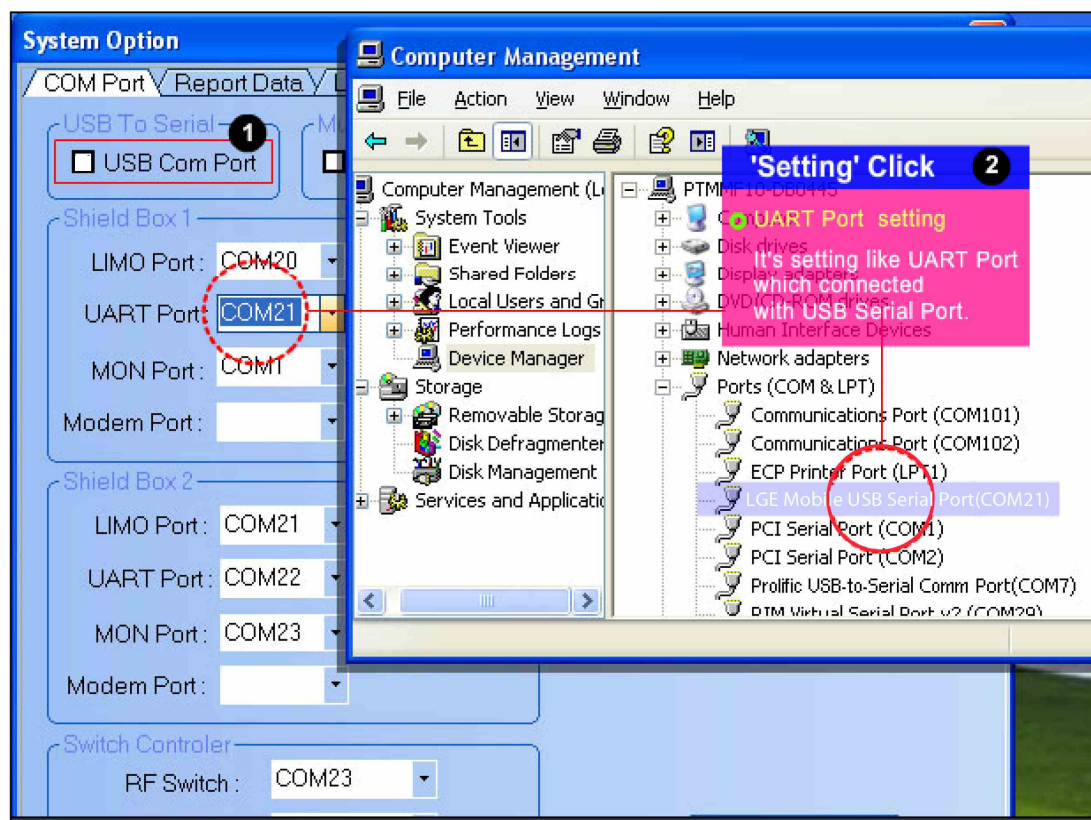
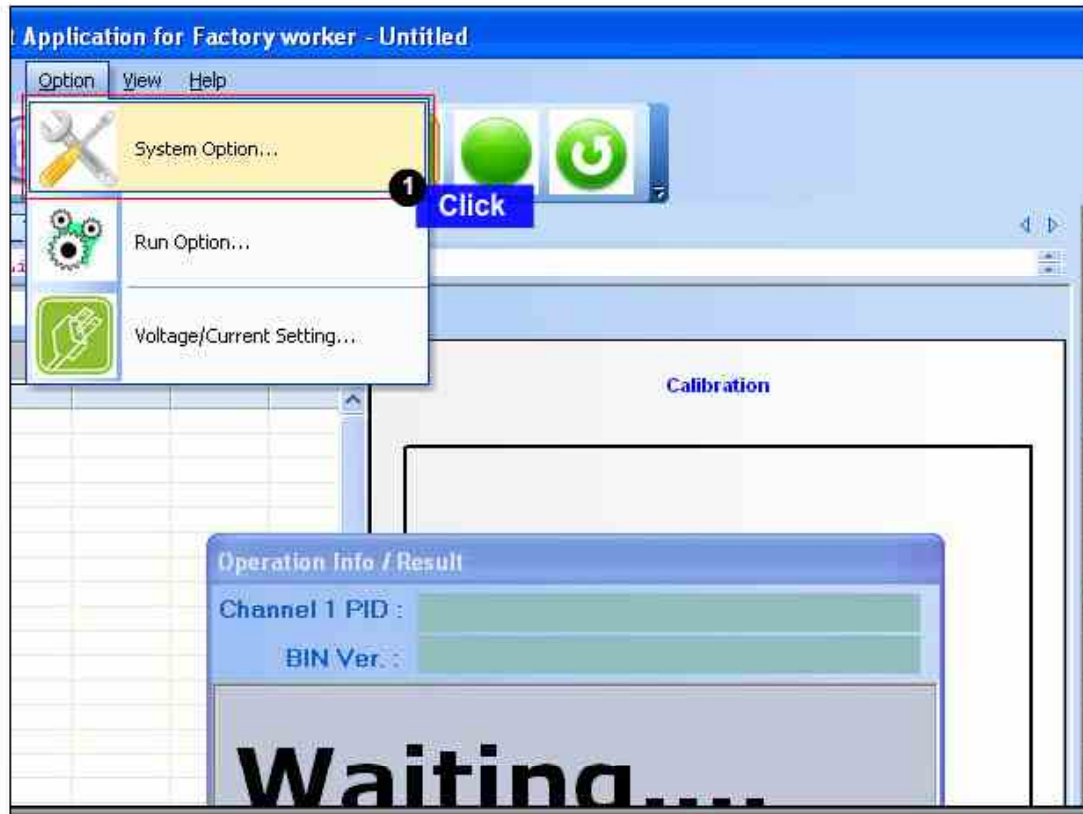




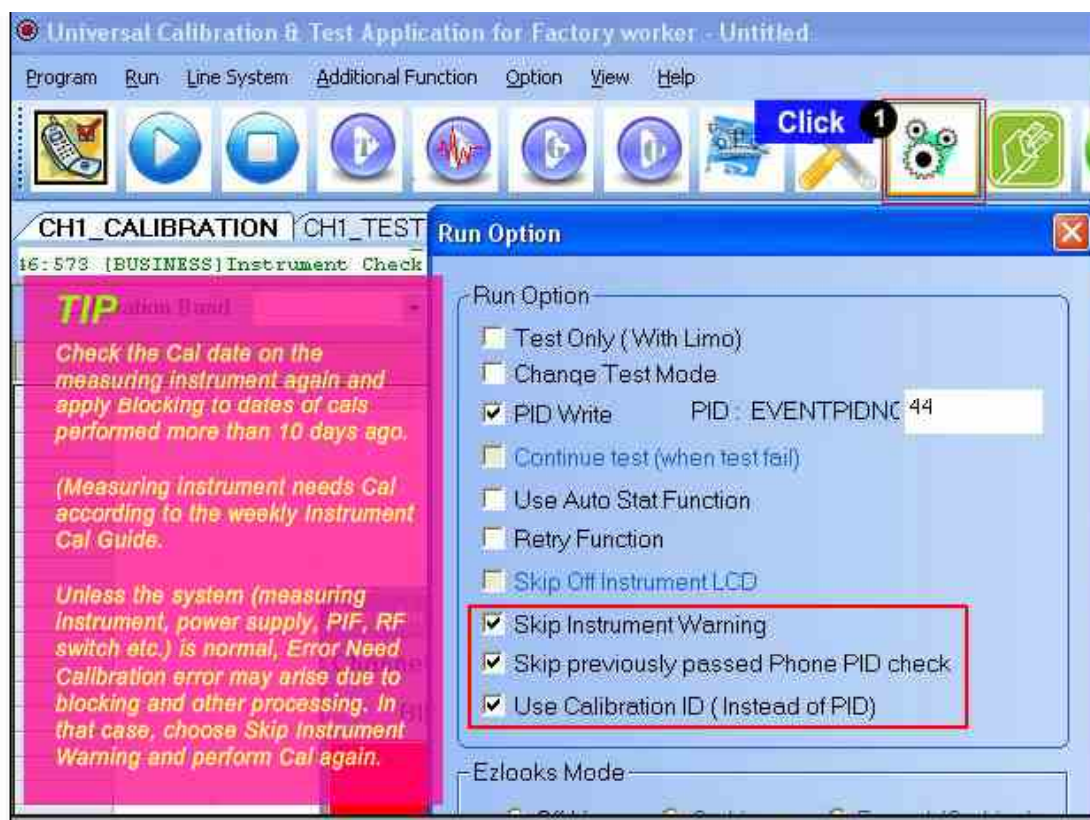
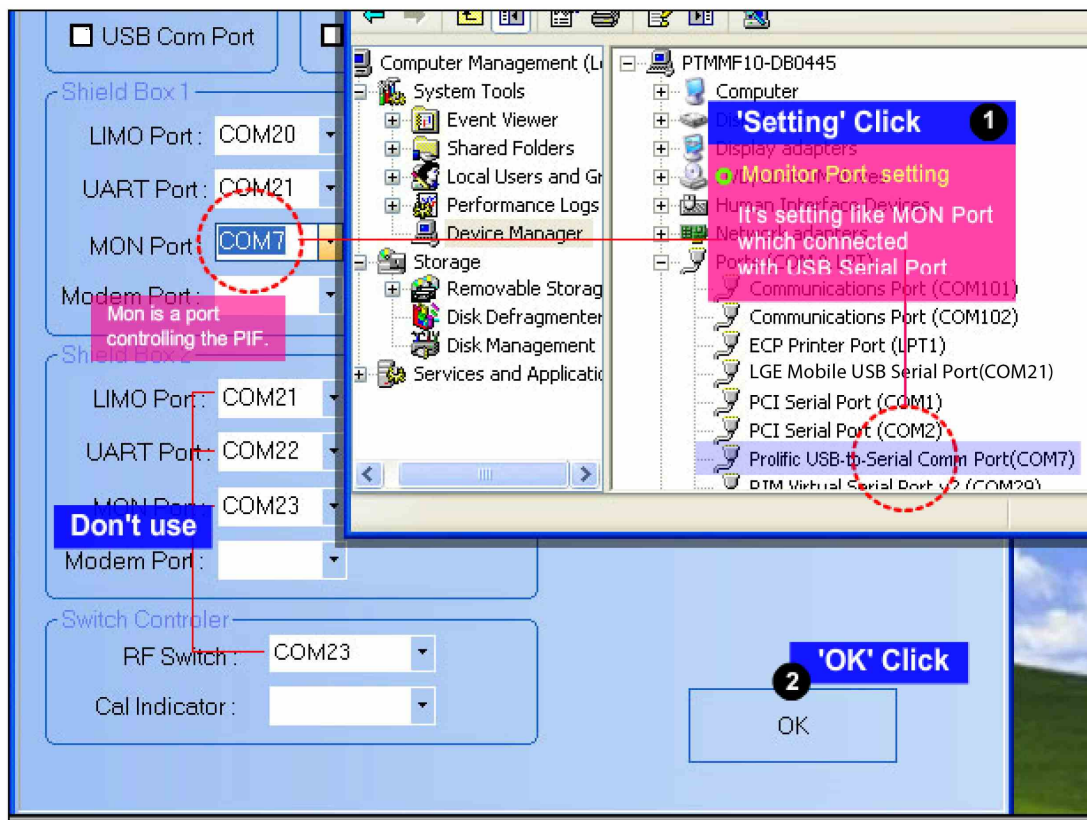


9. CALIBRATION

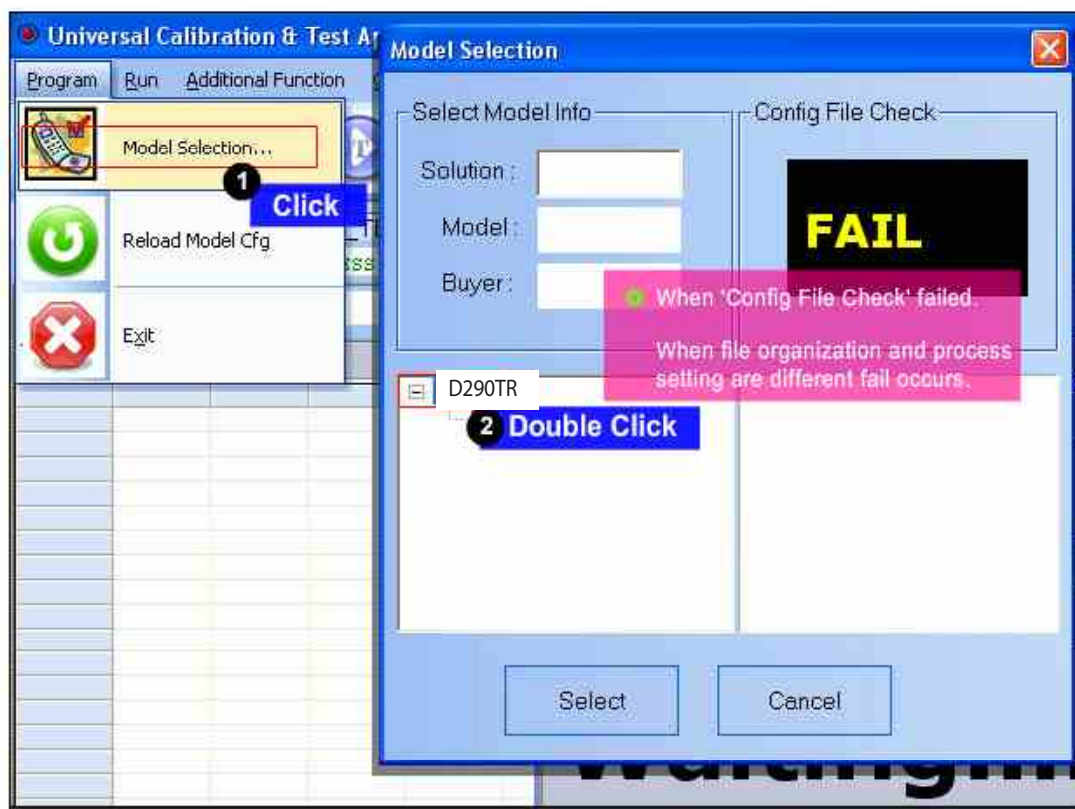
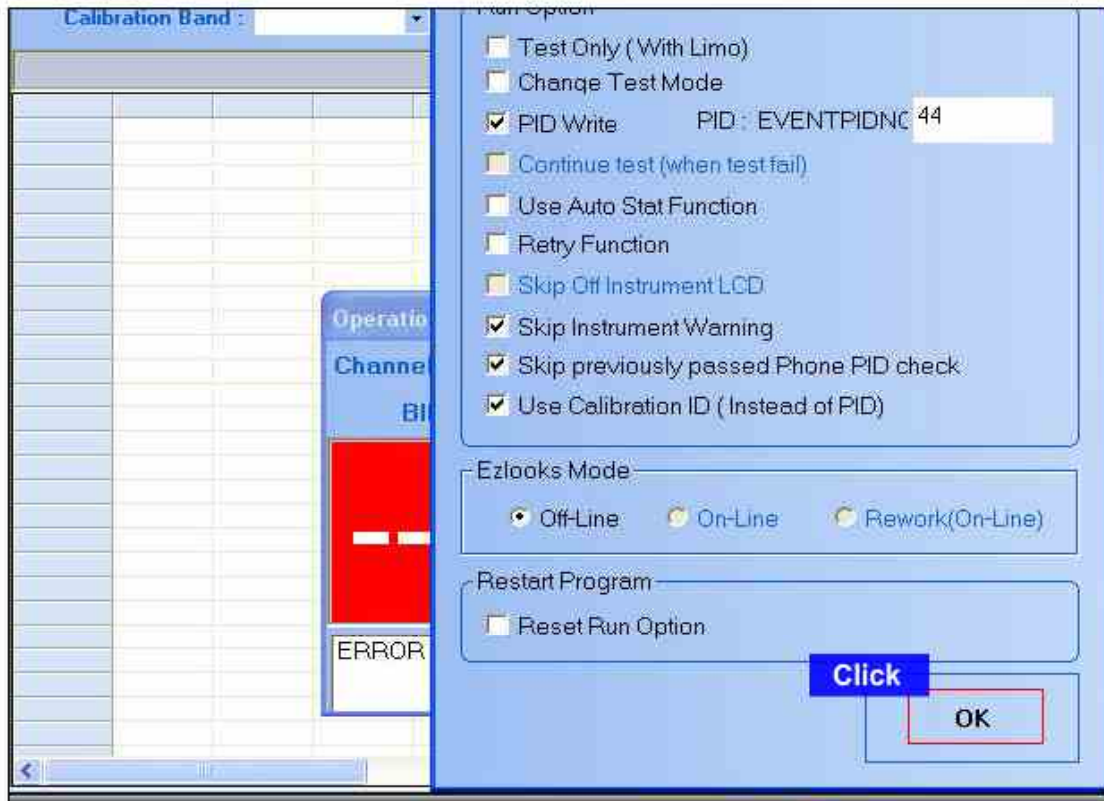




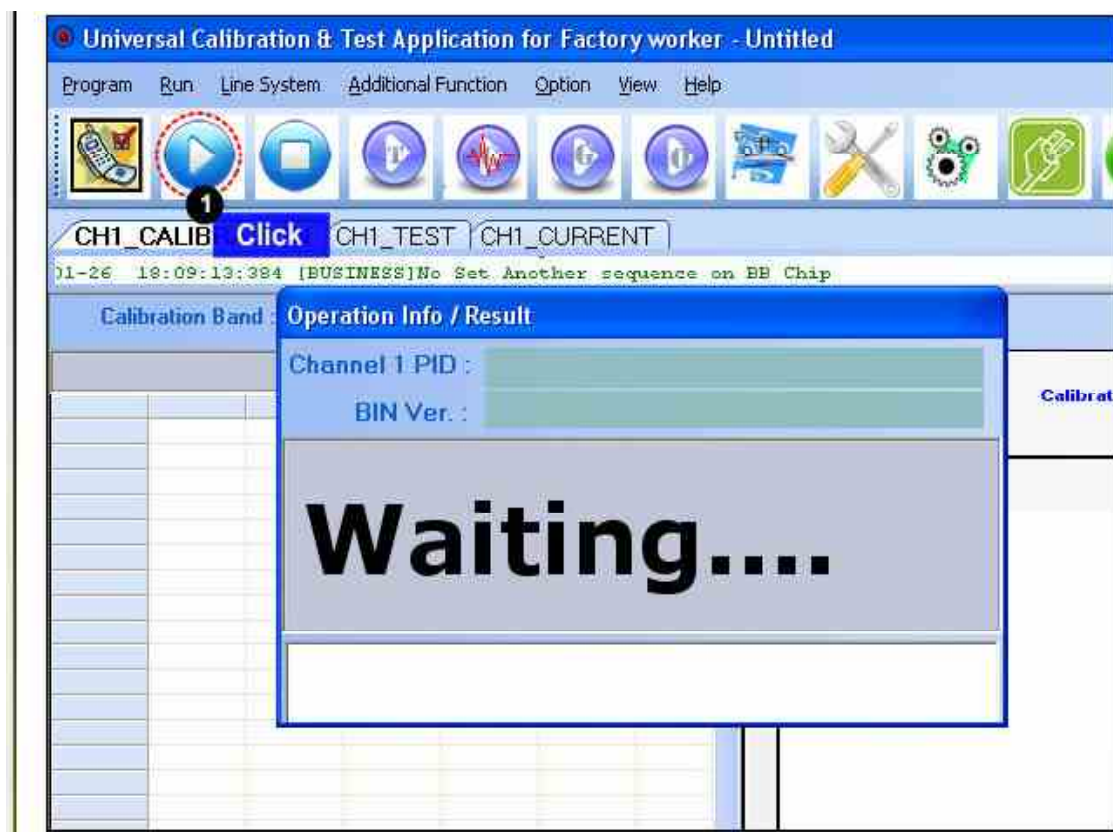
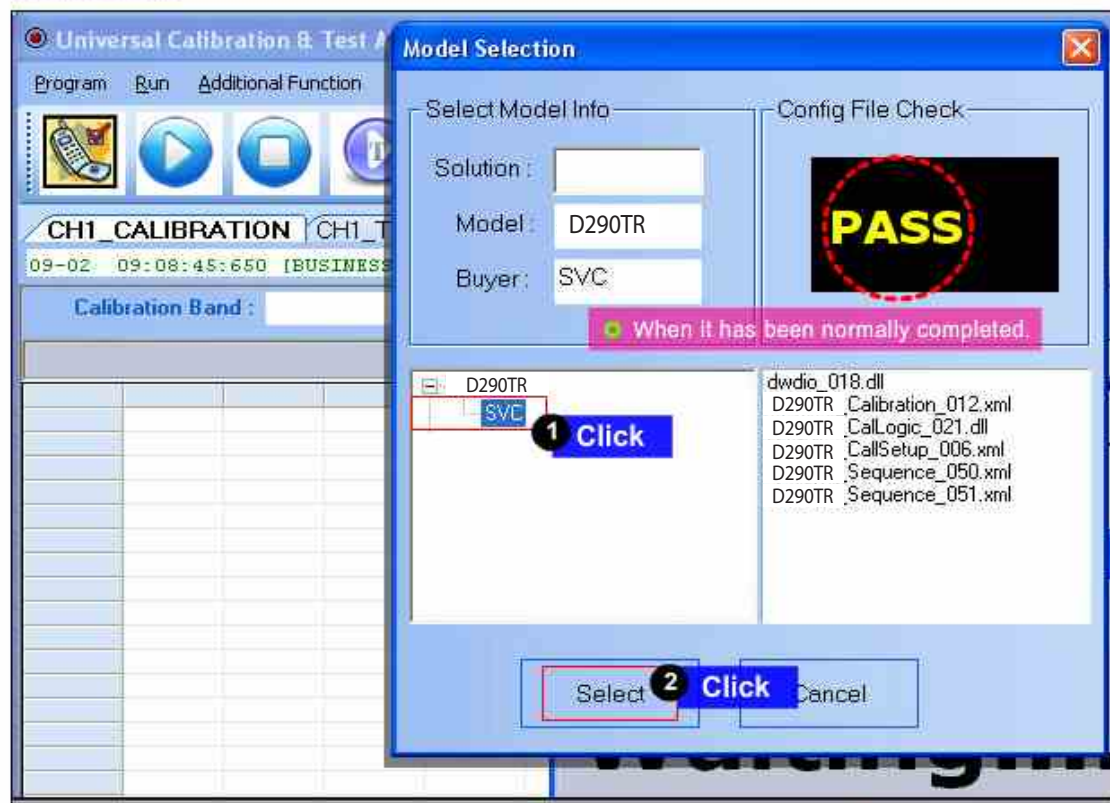
9. CALIBRATION

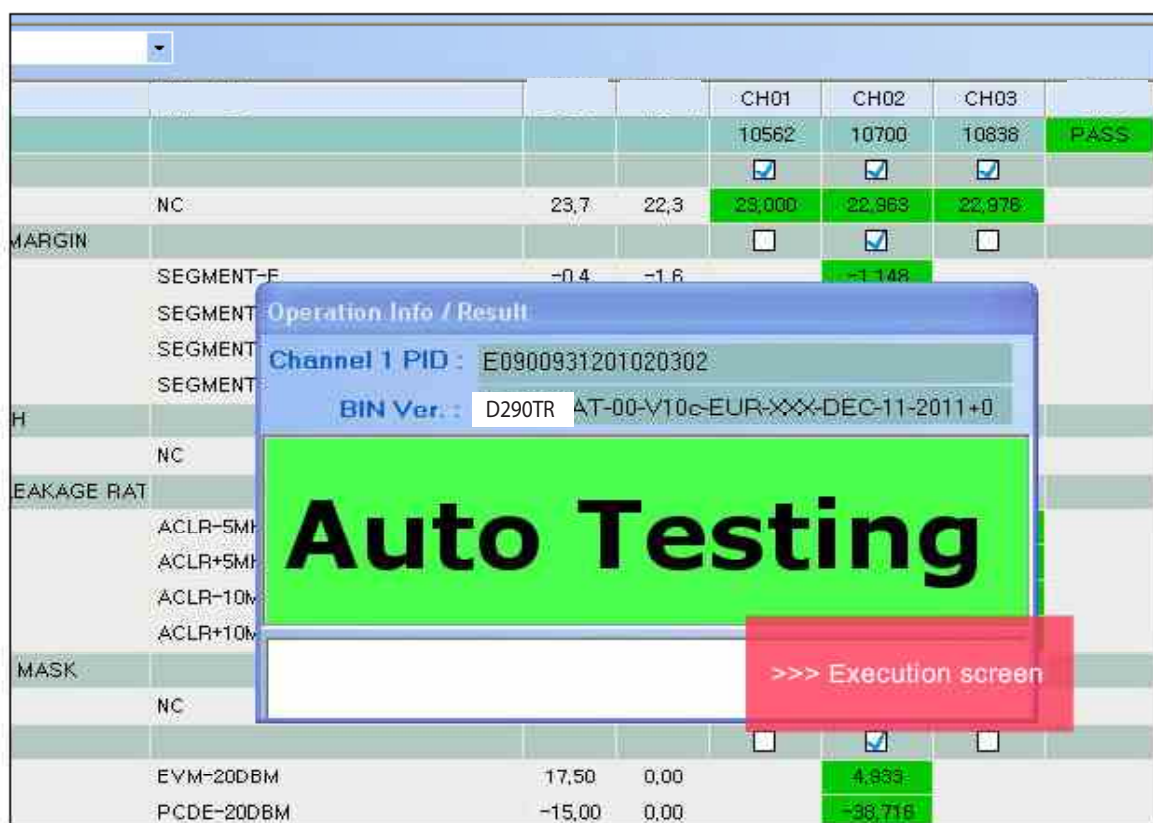
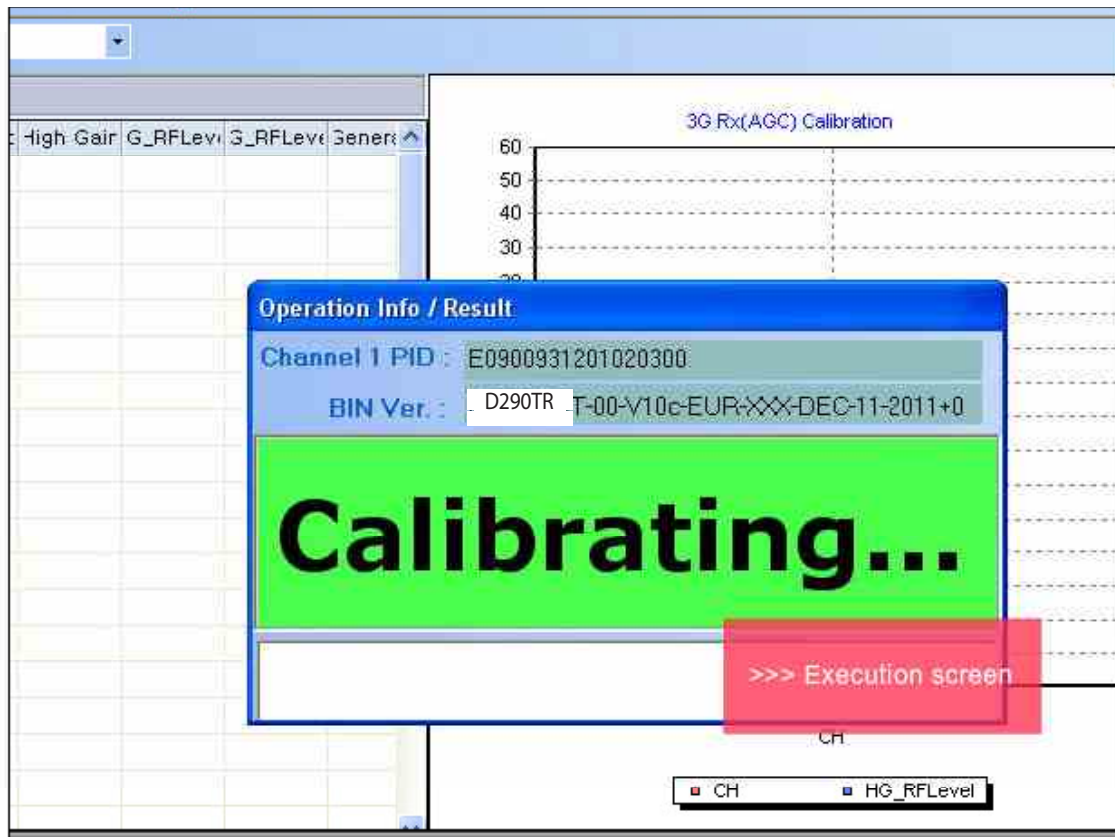


9. CALIBRATION



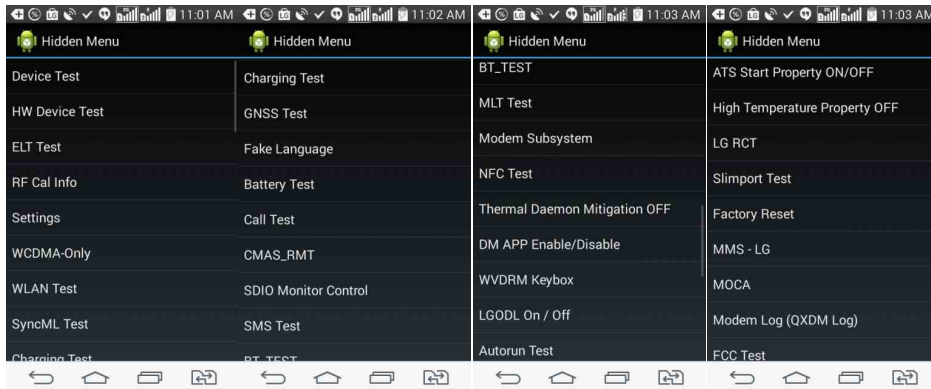
9. CALIBRATION





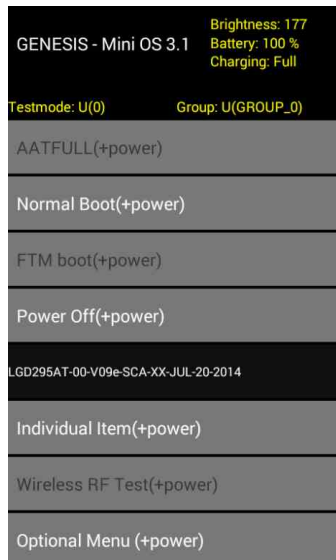
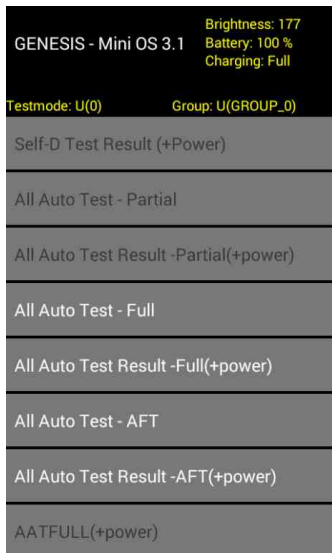
	Condition	USL	LSL	CH01	CH02	CH03	P/F
				10562	10700	10838	PASS
				☒	☒	☒	
	NC	23.7	22.3	23.065	22.963	23.013	
MARGIN				☐	☒	☐	
SEGMENT	Operation Info / Result Channel 1 PID : E0900931201020299 BIN Ver. : D290TR AT-00-V10c-EUR-XXX-DEC-11-2011+0 --- PASS ---						
SEGMENT							
SEGMENT							
SEGMENT							
TH							
	NC						
LEAKAGE RAT							
ACLR-5M							
ACLR+5M							
ACLR-10M							
ACLR+10M							
IN MASK							
	NC	0.5	-0.5				
Y							☐
EVM-20DBM		17.50	0.00		-4.720		
PCDE-20DBM		-15.00	0.00		-38.441		
'PASS' The End							
System Loss : MySystem(MS) .gms RF900 6C.grf							

10. HIDDEN MENU



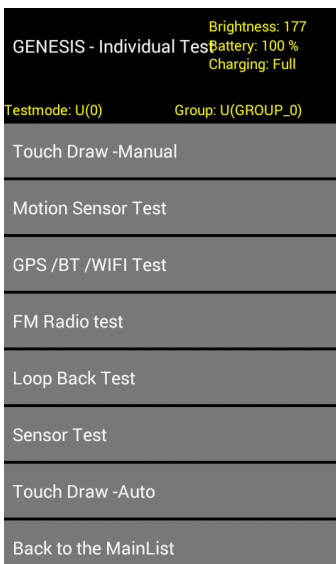
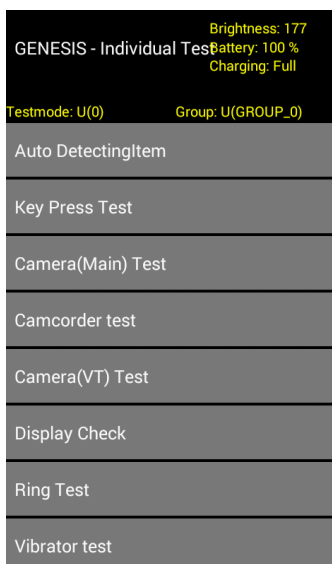
Hidden Menu Start

Start shortcut
keys:3845#*290#



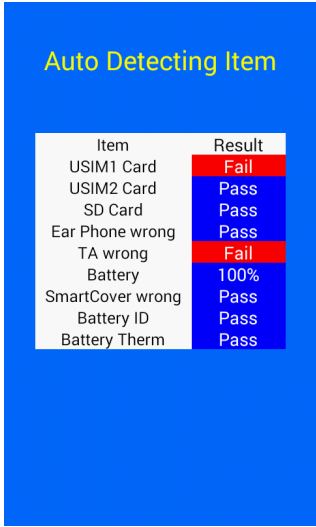
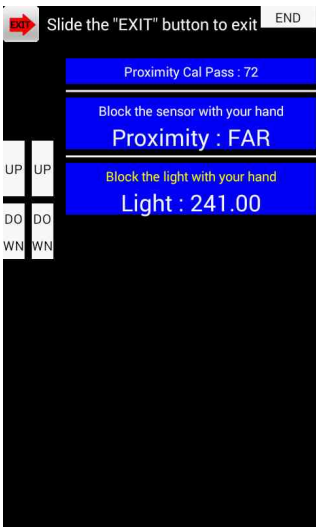
All Auto Test

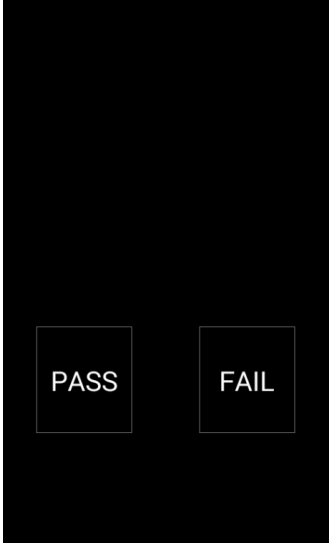
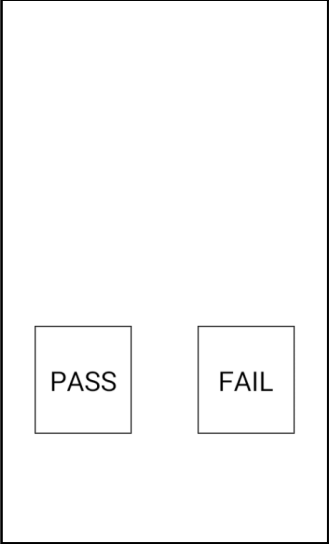
0. Self-D Test Result (+Power)
1. All Auto Test – Partial
2. All Auto Test Result– Partial(+power)
3. All Auto Test – Full
4. All Auto Test Result– Full(+power)
5. All Auto Test – AFT
6. All Auto Test Result– AFT(+power)
7. AATFULL (+power)
8. Normal Boot (+power key)
9. FTM Boot (+power key)
10. Power Off (+power key)
11. Version Name
12. Individual Item(+power)
13. Wireless RF Test(+power)
14. Optional Menu(+power)



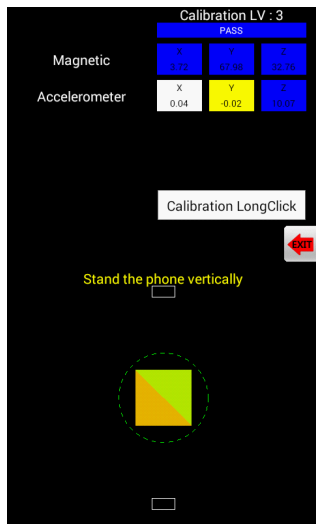
Individual Item List:

1. Auto DetectingItem
2. Key Press Test
3. Camera(Main) test
4. Camcorder Test
5. Camera(VT) test
6. LED / Display check Test
7. Ring Test
8. Vibrator Test
9. Touch Draw-Manual
10. Motion Sensor Test
11. GPS BT WIFI Test
12. NFC Test (supported in D290n only)
13. FM Radio Test
14. Loopback Test
15. Sensor Test
16. Touch Draw-Auto

 <table border="1"> <thead> <tr> <th>Item</th> <th>Result</th> </tr> </thead> <tbody> <tr> <td>USIM1 Card</td> <td>Fail</td> </tr> <tr> <td>USIM2 Card</td> <td>Pass</td> </tr> <tr> <td>SD Card</td> <td>Pass</td> </tr> <tr> <td>Ear Phone wrong</td> <td>Pass</td> </tr> <tr> <td>TA wrong</td> <td>Fail</td> </tr> <tr> <td>Battery</td> <td>100%</td> </tr> <tr> <td>SmartCover wrong</td> <td>Pass</td> </tr> <tr> <td>Battery ID</td> <td>Pass</td> </tr> <tr> <td>Battery Therm</td> <td>Pass</td> </tr> </tbody> </table>	Item	Result	USIM1 Card	Fail	USIM2 Card	Pass	SD Card	Pass	Ear Phone wrong	Pass	TA wrong	Fail	Battery	100%	SmartCover wrong	Pass	Battery ID	Pass	Battery Therm	Pass	Auto Detecting Test : USIM, SD card, Ear jack, TA, Battery • Check the SIM card • Check the SD card • Check Ear phone(jack) • Check TA • Check Battery - Ejected TA is normal - Ejected ear jack is normal * In single SIM model case, USIM2 Card item is not supported. - Single SIM model : D290n, D290g, D290TR, D290TR(D290 series)
Item	Result																				
USIM1 Card	Fail																				
USIM2 Card	Pass																				
SD Card	Pass																				
Ear Phone wrong	Pass																				
TA wrong	Fail																				
Battery	100%																				
SmartCover wrong	Pass																				
Battery ID	Pass																				
Battery Therm	Pass																				
 Slide the "EXIT" button to exit END Proximity Cal Pass : 72 Block the sensor with your hand Proximity : FAR Block the light with your hand Light : 241.00 UP UP DO DO WN WN	Key/Compass Test Up/Down key : Check Recognition Proximity Sensor : Phone contact with your fingers in the top of the sensor determine the sensor response Far Near																				
 Touch to take a picture 3264x2448	Camera Test 1 Main Camera <test step> Touch to take a picture Taking a picture Verity the picture																				

	Camera Test 2 Camcorder <test step> Taking a video Verify the video	
	Camera Test 1 VT Camera Preview	
		LED / Display check Test : LCD Black & White test (MURA 2.0 mm) • Check the Black display • Check the White display • Check MURA 2.0 mm

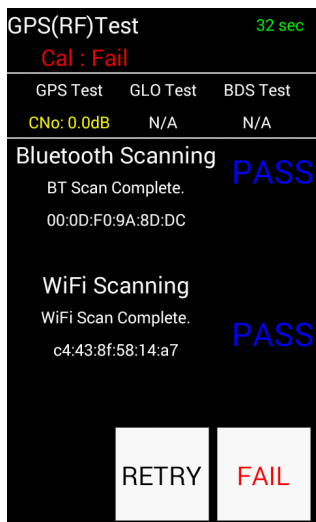
	Ring Test : Check sound • Check the sound
	Vibrator test A case-by-state vibration tests
	Touch window test Check the channel status & raw data And do touch drawing along the guide line



Accelerometer Test

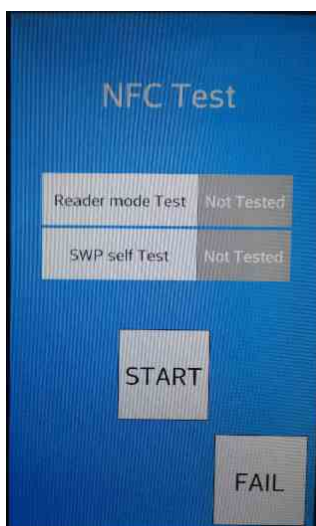
Magnetic sensor test, Accelerometer sensor test

- Check the magnetic sensor
- Check the accelerometer sensor



GPS BT WI-FI Test

- Check the GPS scanning
- Check the WIFI scanning
- Check the BT scanning

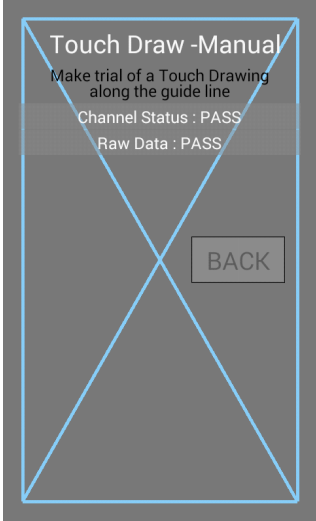
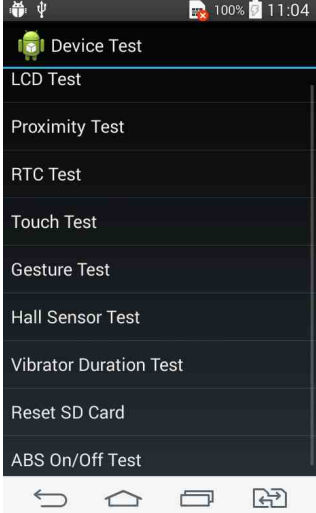


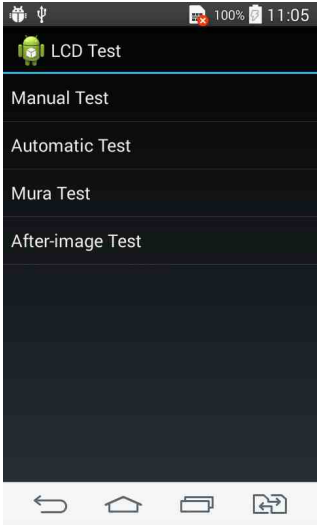
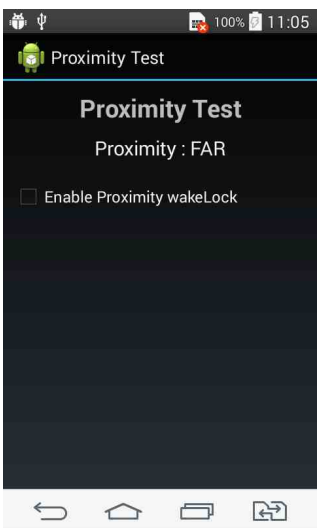
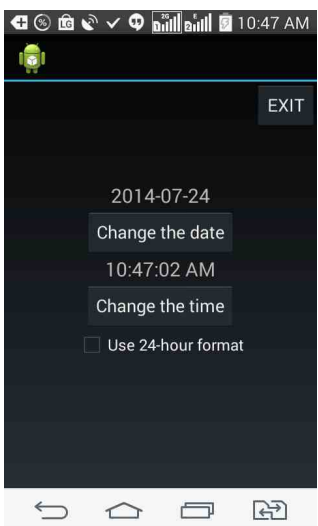
NFC Test

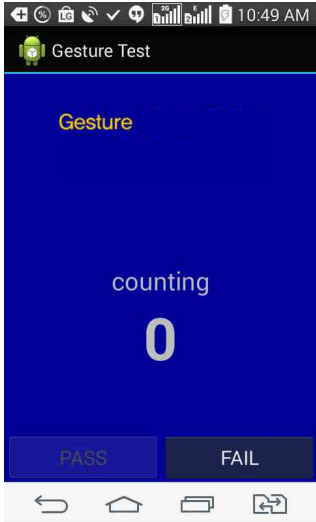
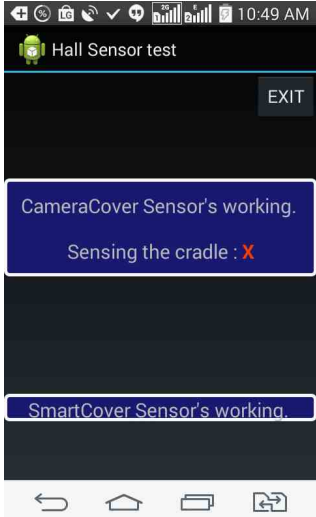
: Reader mode Test, SWP self Test

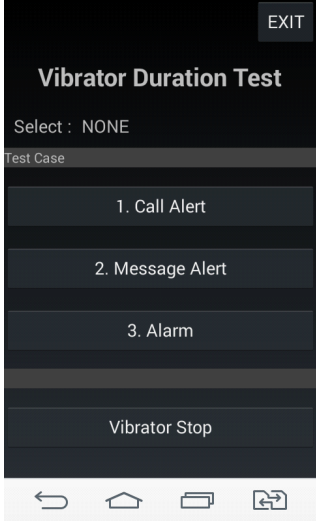
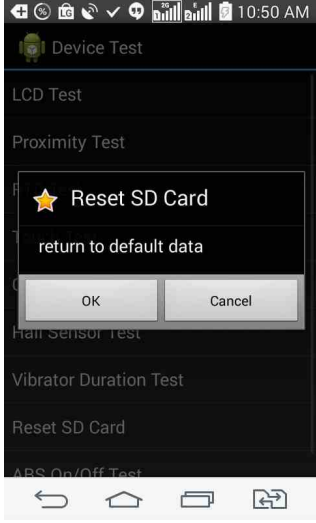
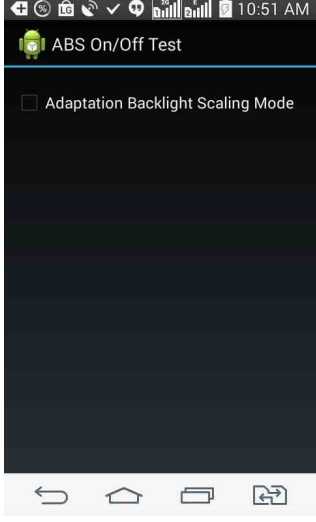
- Check the Reader with NFC tag
- Check the SWP (NFC Sim)

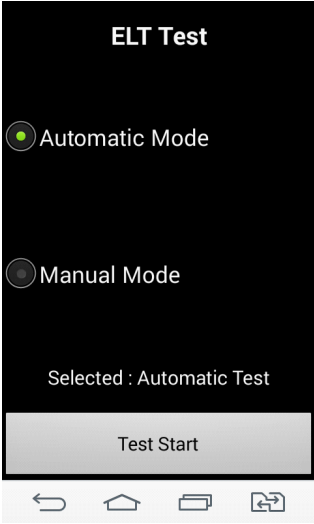
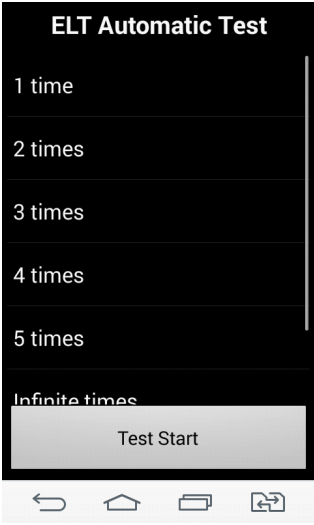
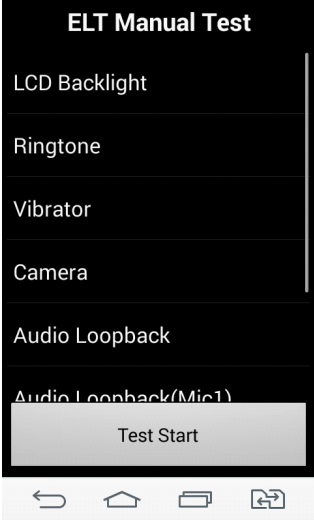
	FM Radio Test • Check the FM Radio
	Audio Loopback Test • Check the Receiver loopback • Check the speaker loopback
	Sensor Test

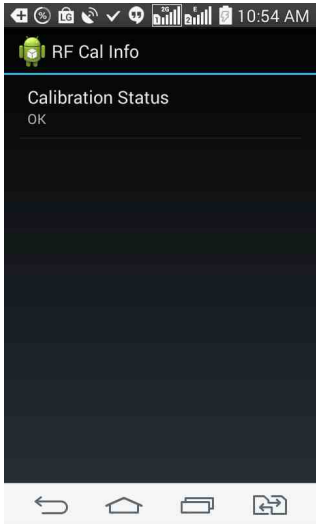
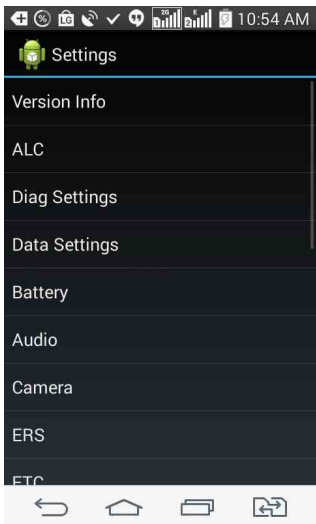
 The screenshot shows a screen titled "Touch Draw -Manual". It contains the text "Make trial of a Touch Drawing along the guide line", "Channel Status : PASS", and "Raw Data : PASS". A blue rectangular frame with a diagonal 'X' is overlaid on the screen. A "BACK" button is located in the bottom right corner of the frame.	Touch Draw -Auto
 The screenshot shows a screen titled "GENESIS - Optional Menu". It displays various status information at the top: "Brightness: 177", "Battery: 100 %", and "Charging: Full". Below this, it shows "Testmode: U(0)" and "Group: U(GROUP_0)". A list of menu items follows: "Set Proximity Limit(+power)", "AATSET/ORDER Init(+power)", "Touch Location(+power)", "Factory Reset(+power)", "MFTS(+power)", "LCD Auto test(+power)", "AAT Start ignore Self-D(+power)", and "Back to the MainList".	Optional Menu List 1. Set Proximity Limit(+power) 2. AATSET/ORDER Init (+power) 3. Touch Location (+power) 4. Factory Reset (+power) 5. MFTS (+power) 6. LCD Auto test (+power) 7. AAT Start ignore Self-D (+power)
 The screenshot shows a screen titled "Device Test". It lists various tests: "LCD Test", "Proximity Test", "RTC Test", "Touch Test", "Gesture Test", "Hall Sensor Test", "Vibrator Duration Test", "Reset SD Card", and "ABS On/Off Test". The screen has a dark background with white text. At the bottom, there are navigation icons: a back arrow, a home button, a recent apps button, and a refresh/circular arrow button.	HW Device Test 1. LCD 2. Proximity 3. RTC 4. Touch 5. Gesture 6. Hall Sensor 7. Vibrator Duration 8. Reset SD Card 9. ABS On/Off Test

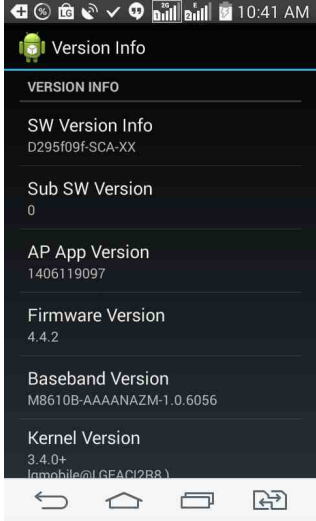
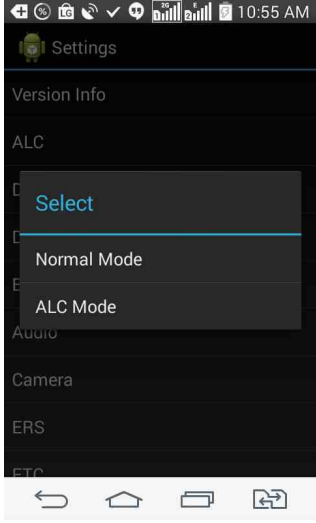
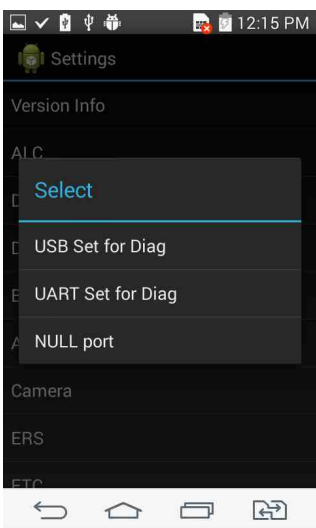
	LCD list - Manual Test - Automatic Test - Mura Test - After-image Test
	Proximity
	RTC

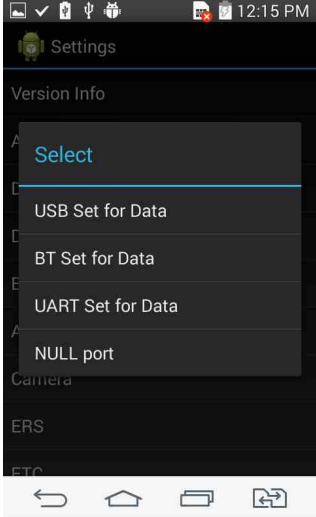
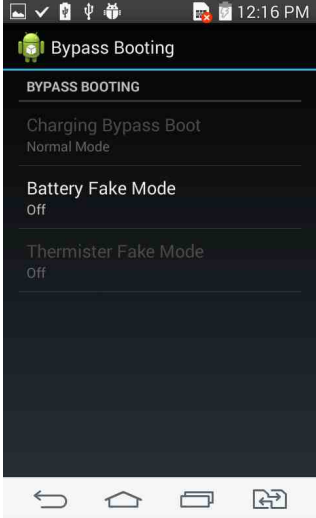
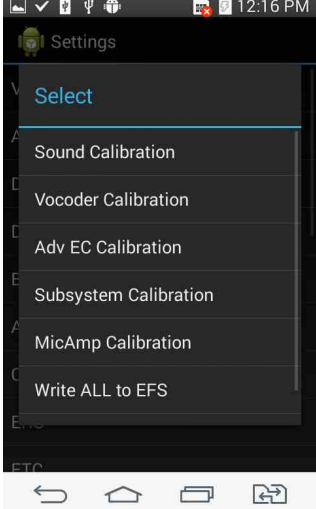
	Touch list - Multi Touch Test - Finger Paint - Dot Finger Paint - Touch Sensing Test 5*9 - key Touch Sensing Test 5*7 - Grid Touch Test 5*7 - Touch Key Test - Grid Touch Color change - Block Touch Test - Touch Firmware Version - Touch Firmware Upgrade - Check Touch Panel
	Gesture
	Hall Sensor

	Vibrator Duration
	Reset SD Card
	ABS On/Off Test

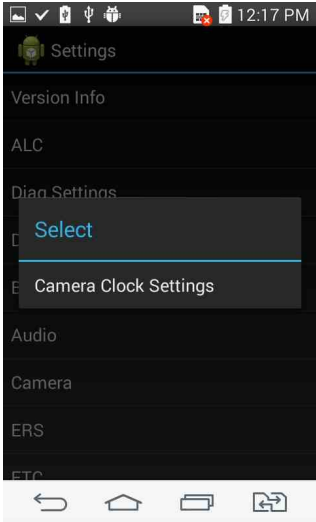
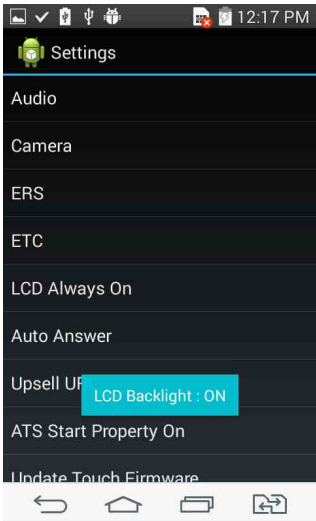
	ELT Test Automatic Mode : Test Automatically Manual Mode : Test selectivity
	ELT Test Automatic Mode : LCD Automatic on/off test -> time setting
	ELT Manual Test -LCD Backlight -Ringtone -Vibrator -Camera -Audio Loopback -Audio Loopback(Mic1) -Audio Loopback(Mic2) -Audio Loopback(Mic3) -> test on the device is working (The ability to use plant)

	RF Cal Info
	Settings -Version Info -ALC -Diag Settings -Data Settings -Battery -Audio -Camera -ERS (not implemented) -ETC -LCD Always On -Auto Answer -Upsell URL -ATS Start Property On -Update Touch Firmware (not implemented) -APN Delete (not implemented) -FastDormancy Mode -Default IMS PDN Setting (not implemented) -KnockOn On/Off Setting -DataRecovery Mode -CPJ Setting On (not implemented)

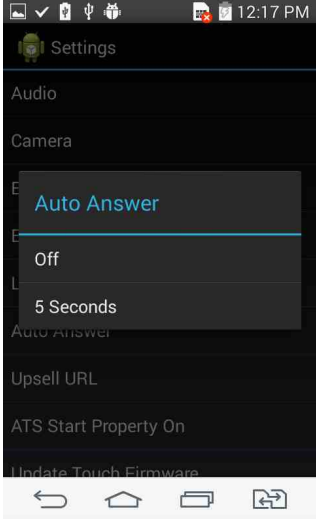
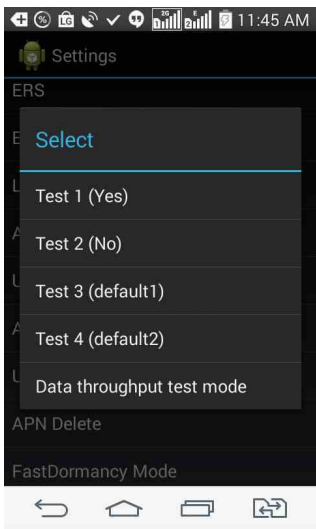
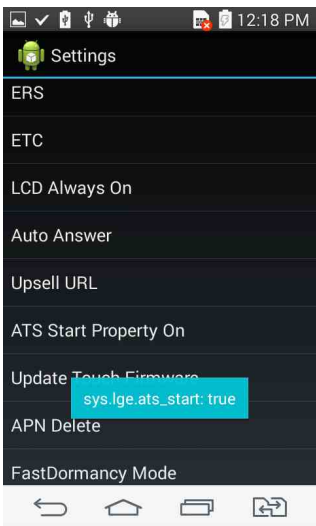
	Version Info (Classified Information representation) -SW Version Info -Sub SW Version -AP App Version -Firmware Version -Baseband Version -Kernel Version -HW Version -Factory Version -SW Original Version -Speed Bin -AVS Bin -NT code -SWFV -S/N
	ALC Normal Mode : Change to Normal Mode ALC Mode : Change to ALC Mode
	Diag Settings - USB Set for Diag - UART Set for Diag - Null port

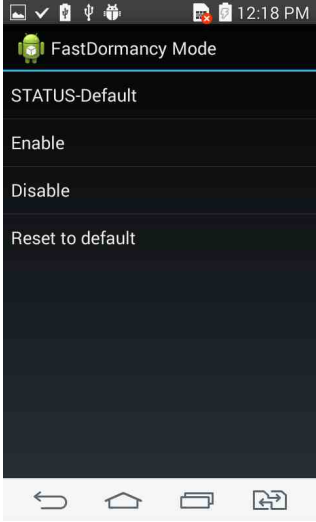
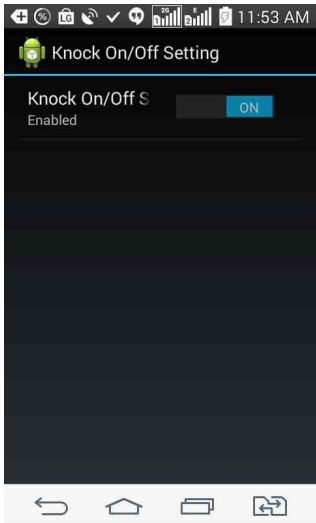
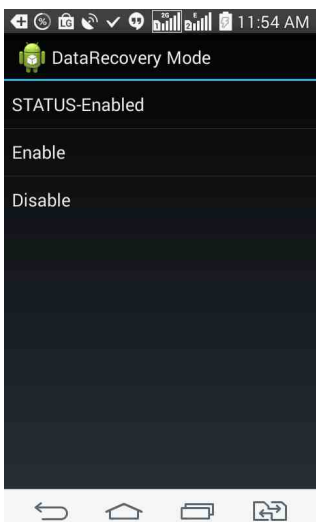
	Data Settings - USB Set for Data - BT Set for Data - UART Set for Data - Null port
	Battery -Battery Fake Mode(On/Off)
	Audio - Sound Calibration - Vocoder Calibration - Adv EC Calibration - Subsystem Calibration - MicAmp Calibration - Write All to EFS - Write All to MEM

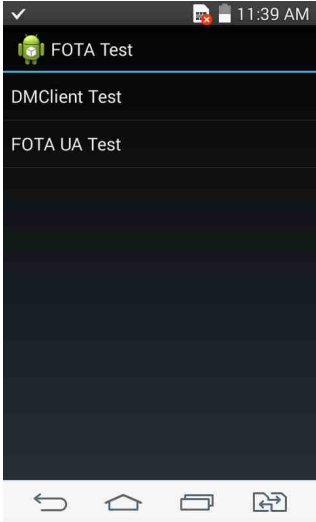
10. HIDDEN MENU

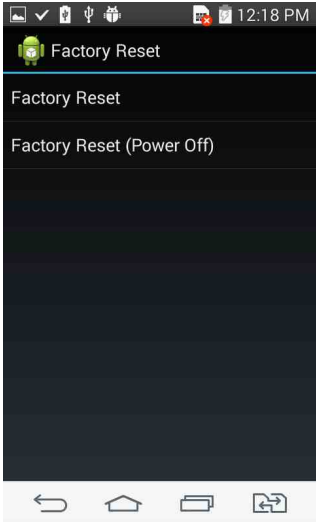
	Camera - Camera Clock Settings
	ETC ADC Info
	LCD Always On Switch to LCD Backlight ON/OFF

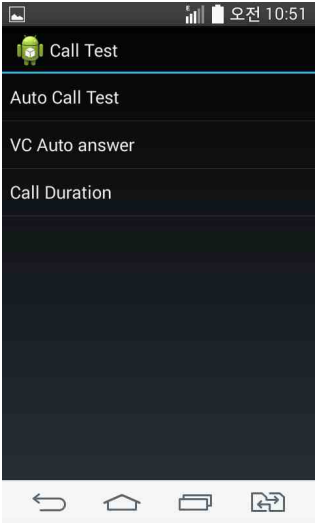
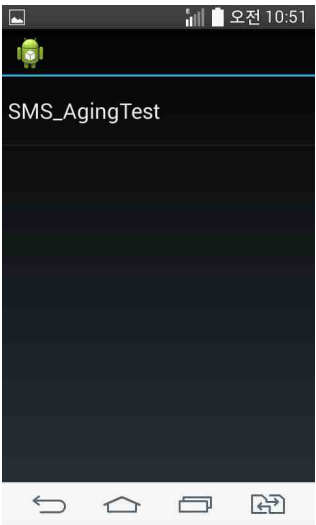
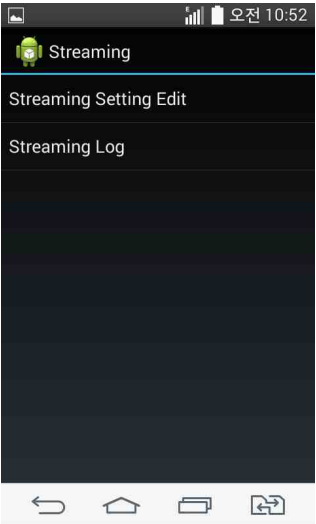
10. HIDDEN MENU

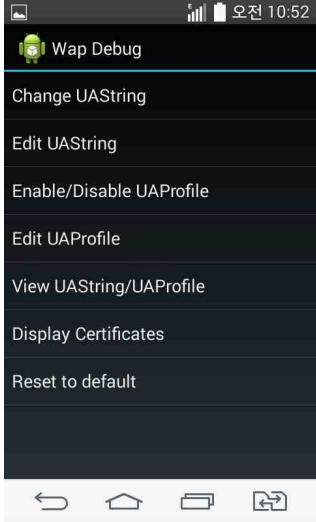
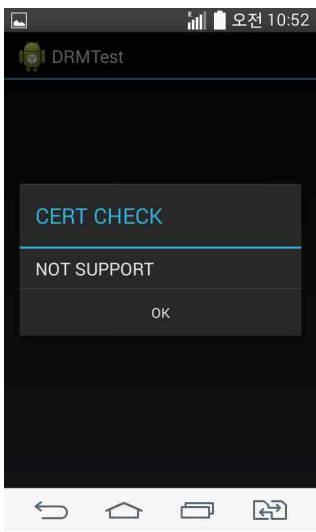
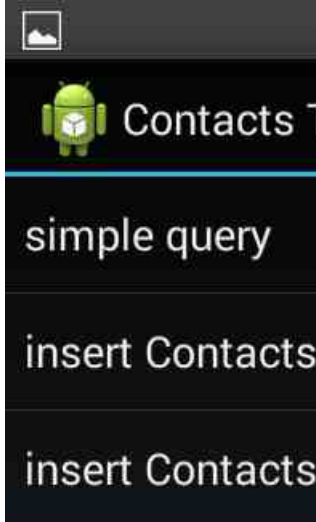
	Auto Answer
	Upsell URL - Test 1 (Yes) - Test 2 (No) - Test 3 (default1) - Test 4 (default2) - Data throughput test mode
	ATS Start Property On

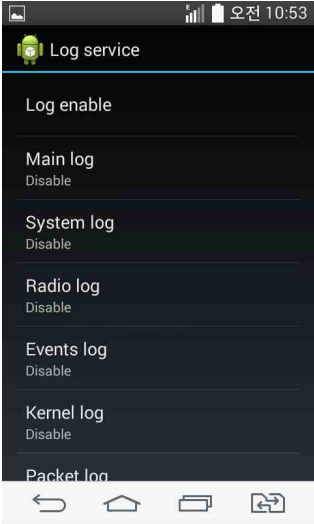
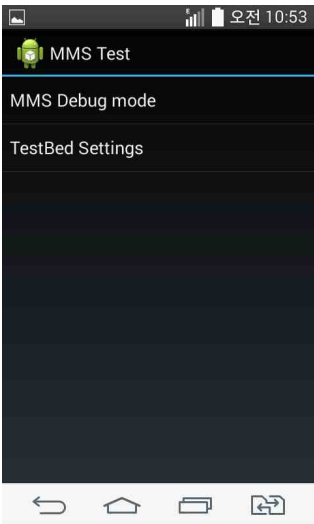
	FastDormancy
	Knock On/Off Setting
	DataRecovery Mode

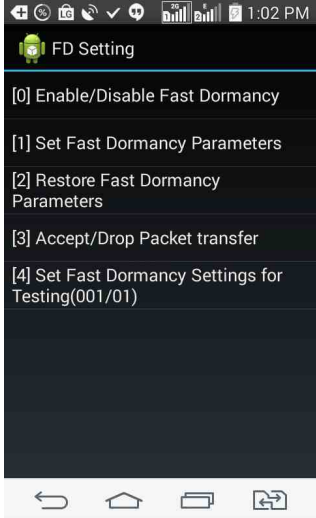
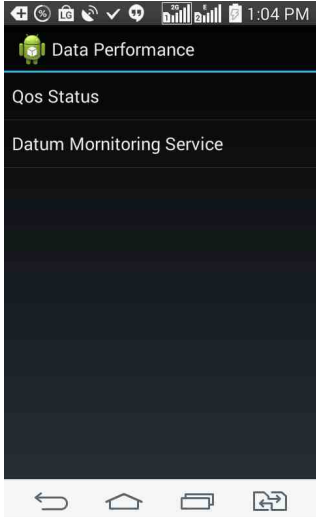
 The screenshot shows a mobile phone screen with the title 'WCDMA Only'. Below the title is a list of test options: DM/FOTA Test, Factory Reset, Modem Settings, Module Test, SIM Test, Call Test, Aging Test, Streaming, and WAP Debug. The bottom of the screen shows a standard Android navigation bar with back, home, and recent apps buttons.	WCDMA Only test list -DM/FOTA Test -Factory Reset -Modem Settings -Module Test -SIM Test(Not Implemented) -Call Test -Aging Test -Streaming -WAP Debug -DRM Test -Contacts Test -Log service -Fake Language -Mms Test -Flex test -FastDormancy Setting -Data Performance -FCC Test
 The screenshot shows a mobile phone screen with the title 'FOTA Test'. Below the title is a list of test options: DMClient Test and FOTA UA Test. The bottom of the screen shows a standard Android navigation bar with back, home, and recent apps buttons.	DM/FOTA Test: -DMClient Test -FOTA UA Test

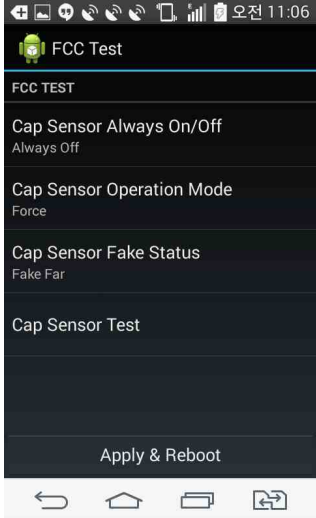
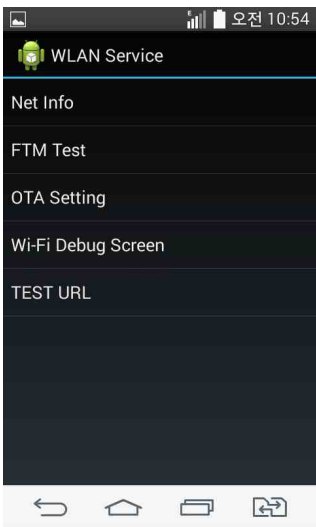
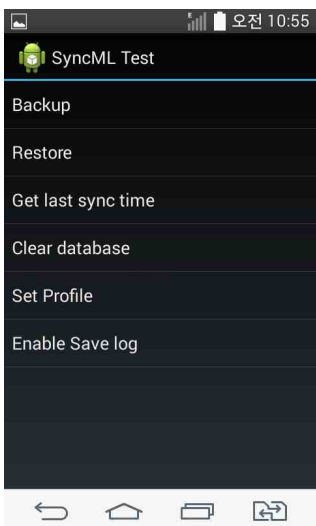
	Factory Reset: -Factory Reset :Changing the Factory -Factory Reset (Power Off) : Reset and Power off
	Modem Settings - Engineering Mode - RRC Version Setting - QCRIL Log On/Off - PDP Setting - HSDPA Category - GSM A5 Algorithm - Protocol Test - GCF Flag - Arm9 Log On/Off - Network Mode - RAT Selection
	Module Test list: -SMS Prefer Set -FOTA Test -LCD Always On -FS_TEST -Heap Free Info -Stability Test -Charging Test -ANT Setting -SMPL Counter -BT DUT Setting

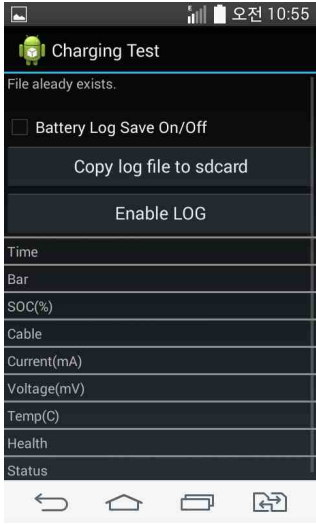
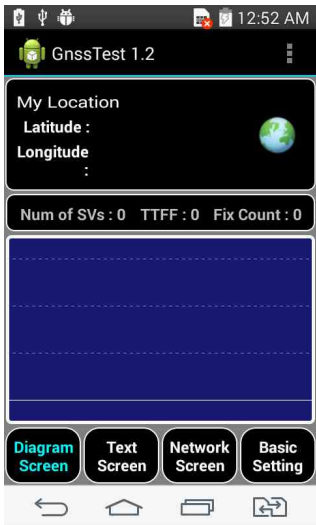
	Call Test -Auto Call Test -VC Auto answer -Call Duration
	Aging Test
	Streaming - Streaming Setting Edit ->Edit Enable or not - Streaming Log ->Log Enable or not

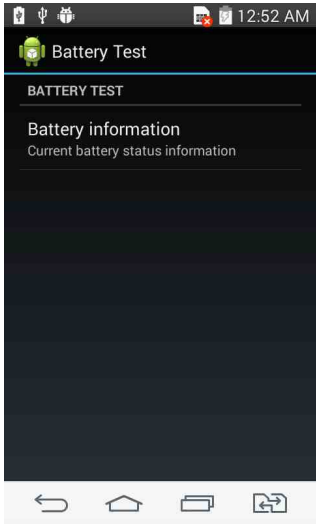
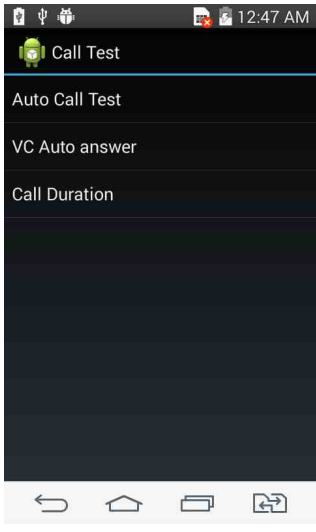
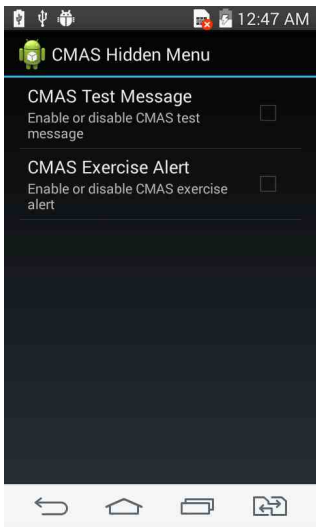
 The screenshot shows the 'Wap Debug' application interface. At the top, there's a status bar with signal strength, battery, and time (오전 10:52). Below the title bar, a list of options is displayed: 'Change UAStrng', 'Edit UAStrng', 'Enable/Disable UAProfile', 'Edit UAProfile', 'View UAStrng/UAProfile', 'Display Certificates', and 'Reset to default'. The bottom of the screen shows standard Android navigation icons.	Wap Debug -Change UAStrng -Edit UAStrng -Enable/Disable UAProfile -Edit UAProfile -View UAStrng/UAProfile -Display Certificates -Reset to default
 The screenshot shows the 'DRMTest' application. A dialog box is displayed with the text 'CERT CHECK' in blue, 'NOT SUPPORT' in white, and an 'OK' button at the bottom. The background is dark, and the status bar at the top shows the time as 오전 10:52.	DRM Test
 The screenshot shows the 'Contacts Test' application. It features a green Android robot icon and the text 'Contacts Test'. Below this, there are three menu options: 'simple query', 'insert Contacts', and 'insert Contacts'. The bottom of the screen shows standard Android navigation icons.	Contacts Test Insert specific number of contact info

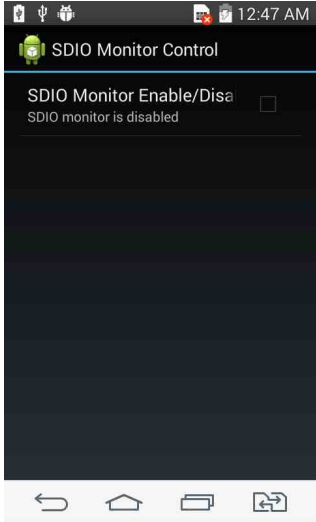
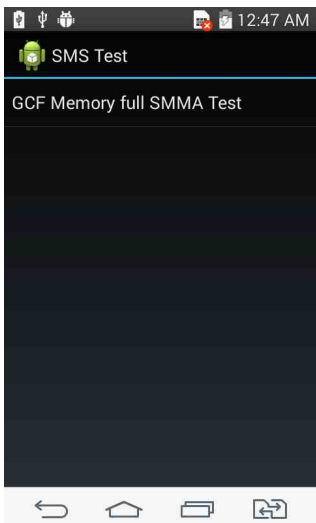
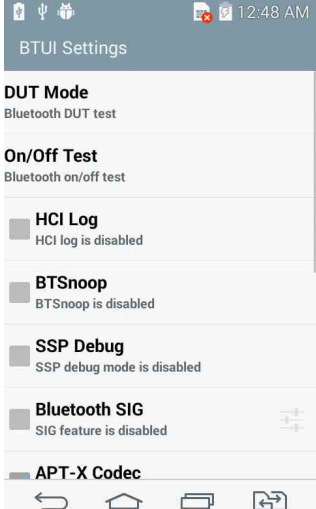
	Log service Enable of disable creating below log files. - Main log - System log - Radio log - Events log - Kernel log - Packet log
	Fake Language
	MMS test

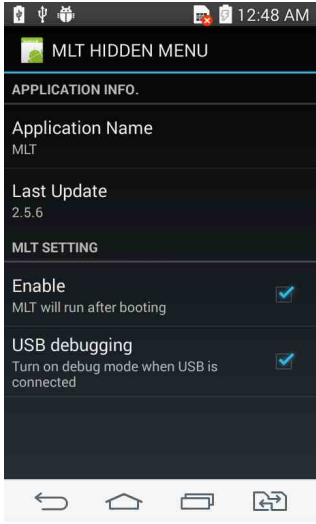
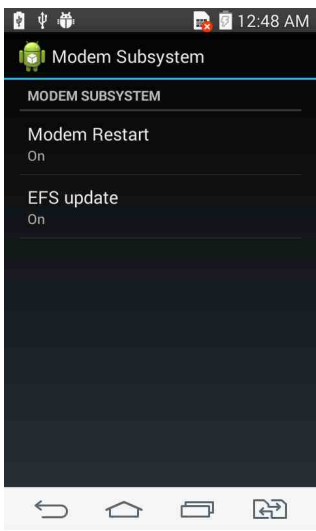
 The screenshot shows the 'Flex test' menu with the following options: FLEX MODE ON/OFF (Disabled), IMSI, MCC-MNC(ex:45000), ICCID, GID, SPN, and NTCODE-MCC(ex:45000). The status bar at the top shows the time as 10:53.	Flex Test
 The screenshot shows the 'FD Setting' menu with the following options: [0] Enable/Disable Fast Dormancy, [1] Set Fast Dormancy Parameters, [2] Restore Fast Dormancy Parameters, [3] Accept/Drop Packet transfer, and [4] Set Fast Dormancy Settings for Testing(001/01). The status bar at the top shows the time as 1:02 PM.	FastDormancy Setting
 The screenshot shows the 'Data Performance' menu with the following options: Qos Status and Datum Mornitoring Service. The status bar at the top shows the time as 1:04 PM.	Data Performance

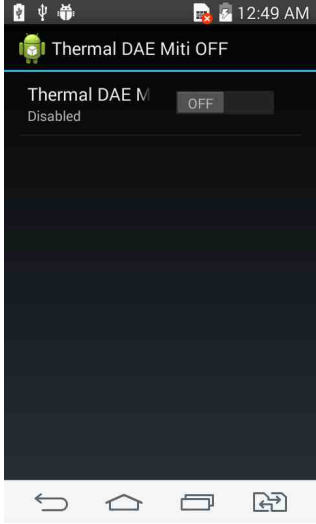
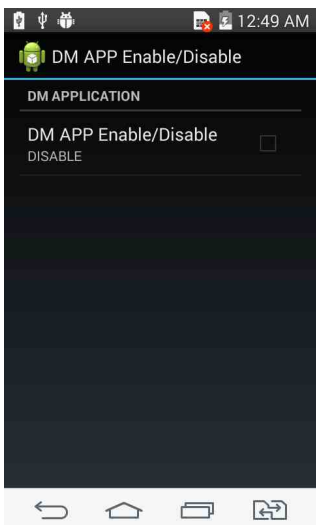
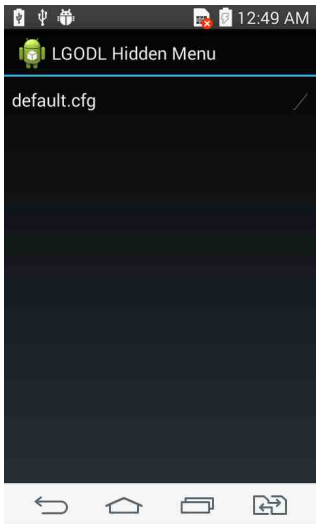
	FCC Test
	WLAN Test WLAN performance on SW
	SyncML Test

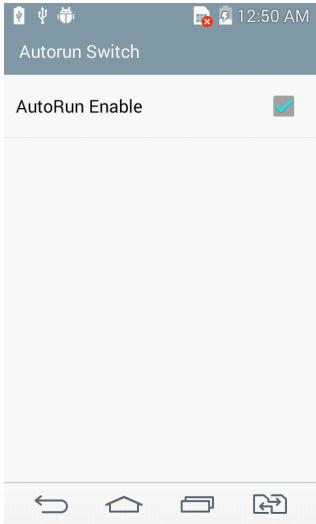
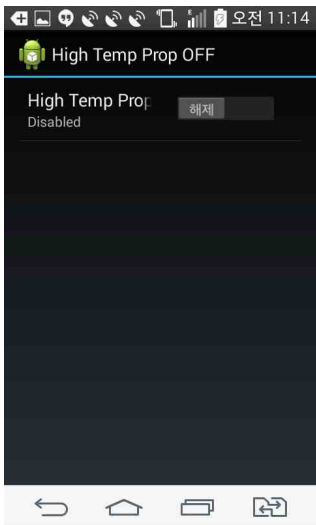
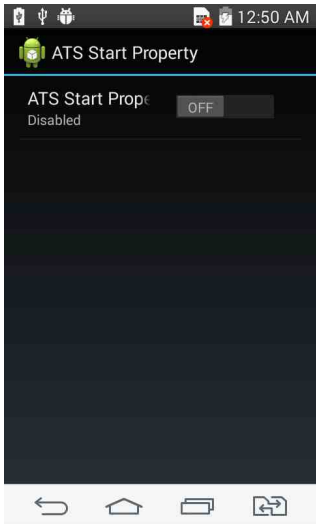
	Charging Test
	Gnss Test
	Same as WCDMA Only – Fake Language

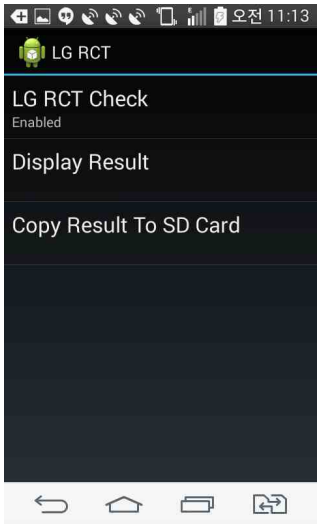
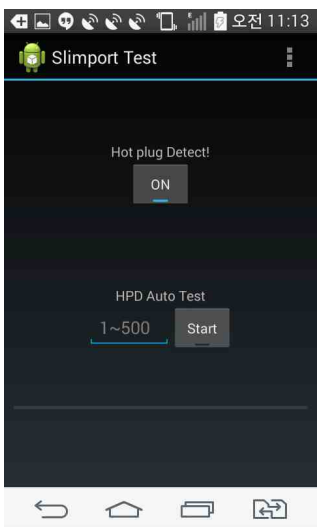
	Battery Test
	Call Test
	CMAS_RMT

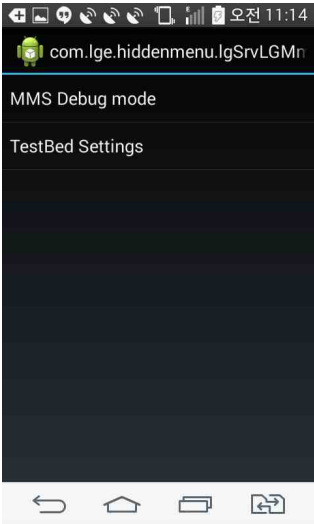
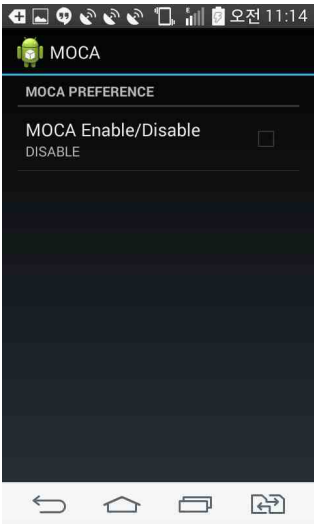
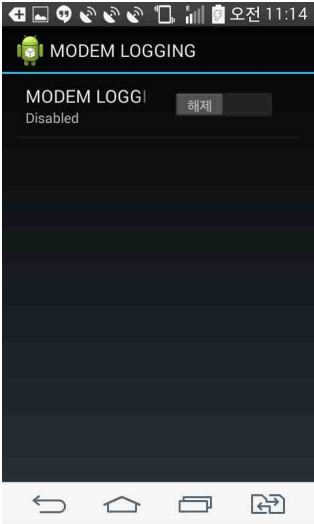
	SDIO Monitor Control
	SMS Test
	BT_TEST

	MLT Test
	Modem Subsystem
	NFC Test ** Touch 'Set Test Environment' If you want to Exit, touch Exit menu (do not touch back key) * NFC Test item is supported in D290n only.

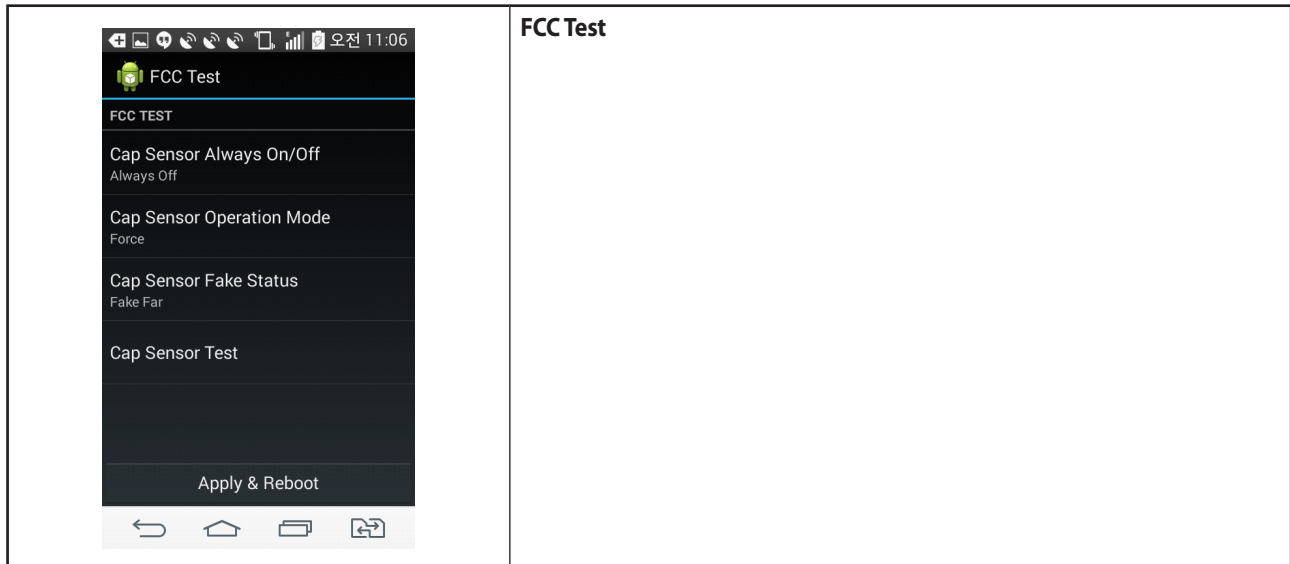
 A screenshot of an Android phone screen showing the 'Thermal DAE Mitigation' settings. The title bar says 'Thermal DAE Mitigation OFF'. Below it, 'Thermal DAE M' is followed by a toggle switch set to 'OFF'. Underneath, it says 'Disabled'. The bottom of the screen shows the standard Android navigation bar with back, home, and recent apps icons.	Thermal Daemon Mitigation OFF
 A screenshot of an Android phone screen showing the 'DM APP Enable/Disable' settings. The title bar says 'DM APP Enable/Disable'. Below it, 'DM APPLICATION' is followed by a toggle switch set to 'DISABLE'. Underneath, it says 'DISABLE'. The bottom of the screen shows the standard Android navigation bar with back, home, and recent apps icons.	DM APP Enable/Disable
 A screenshot of an Android phone screen showing the 'LGODL Hidden Menu'. The title bar says 'LGODL Hidden Menu'. Below it, 'default.cfg' is listed with a checkmark to its right. The bottom of the screen shows the standard Android navigation bar with back, home, and recent apps icons.	First Execute, default.cfg is made

	Autorun Test
	Hiht Temperature Perperty OFF
	ATS Start Property ON/OFF

	LG RCT Test
	Slimport Test
	Facroty Reset

	MMS LG
	MOCA
	Modem Log (QXDM Log)

10. HIDDEN MENU



11. DISASSEMBLE GUIDE

11.1 DISASSEMBLE GUIDE

1. Disassemble Battery Cover , Rear	
Disassemble sequence : 1 -> 2 -> 3 -> 4 	1 
2 	3 

4



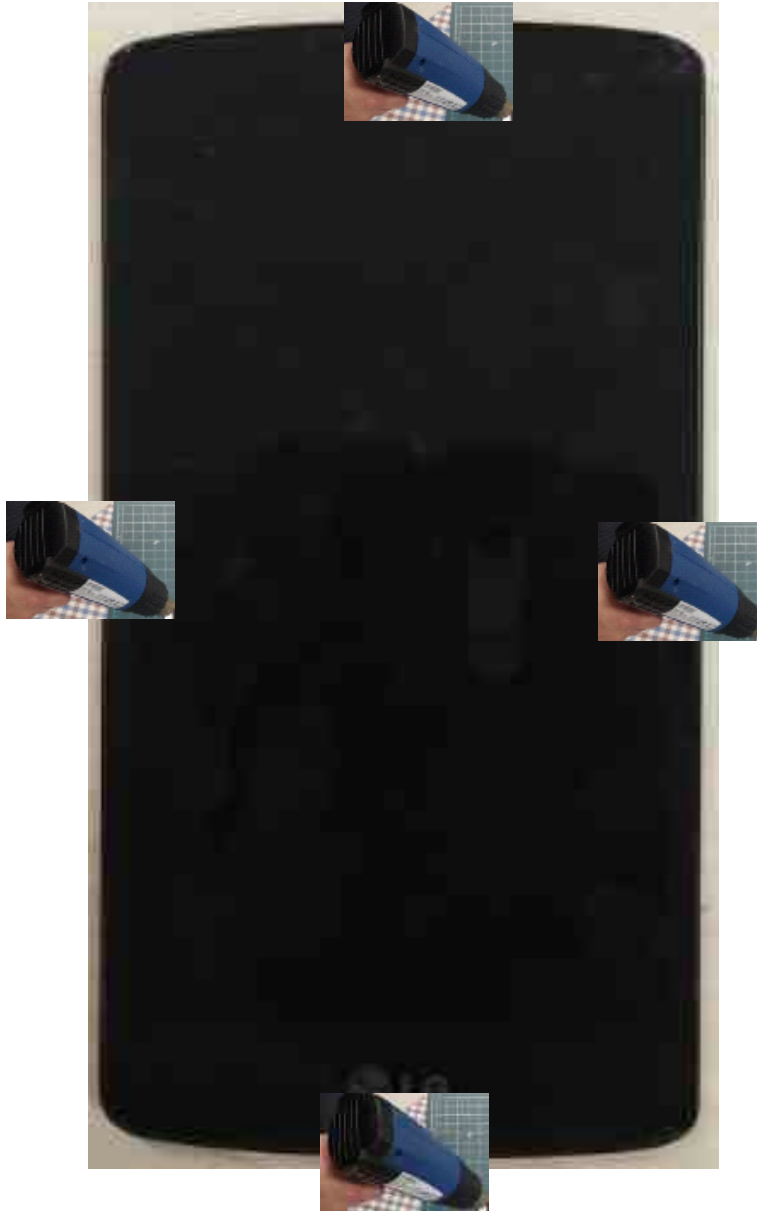
1. Remove Battery Cover
2. Remove 8 Screws from Rear
3. Remove Rear Cover

2. Disassemble PCB



1. Remove LCD Connector from PCB
2. Remove PCB

3. Disassemble LCD



1. Use the Hair drier to LCD window front for 15 seconds



2. Press the 1 holes of below images with Tool.
and disassemble the window.



3. Complete as below.

11. DISASSEMBLE GUIDE



Remove LCD from Front

11.2 SHIELD CAN REWORK GUIDE

HW Environment		
 		
BE/T 853A : Hot Plate		WHA3000P : Heat Gun
		
Soldering Iron	PINSET	H/F Solder

11. DISASSEMBLE GUIDE



NEOZOL



H/F Flux

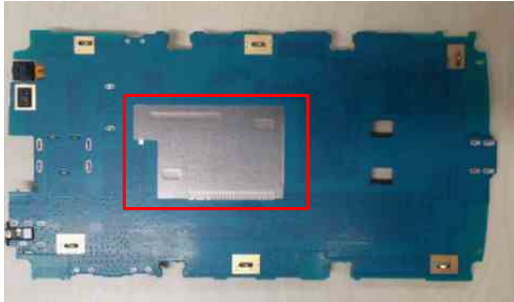


Bracket



Brush

Setting Guide

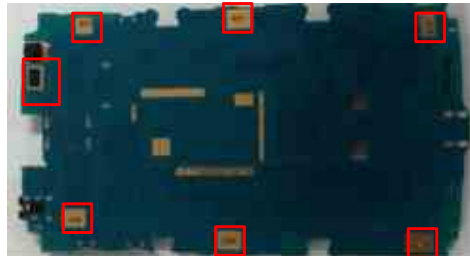


1. Remove the gasket & cameras from the PCB.



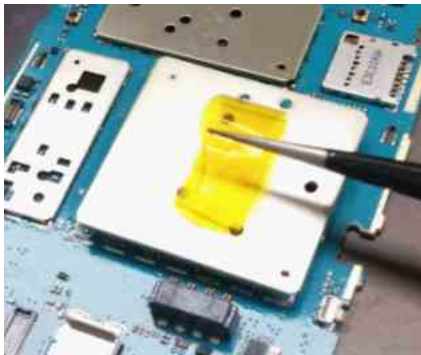
2. Set the BE/T 853A (Hot Plate) temperature to 200°C
3. Pre-Heat the PCB for 1 minute on the Board support guide as image

*Caution : C-Clip & Sensor Damage



4. Set the WHA3000P (Heat Gun) as below
TEMP AIR PREHEAT
400°C 15L OFF

Disassembly Rework Guide



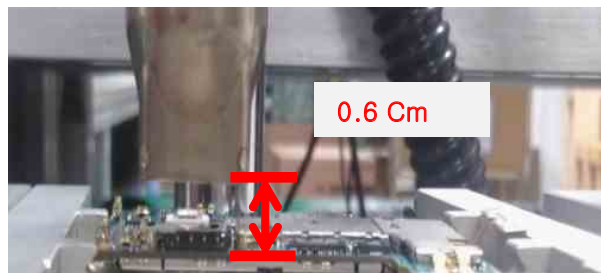
1. Remove the shield can with pin set

* Caution 1 : Lift the 'shield can' for vertical direction.
If not, that is cause of defect.



* Caution 2 : Protect the near ICs with 'bracket' from HEAT DAMAGE

* Caution 3 : Keep distance 0.6Cm PCB and Heat Gun nozzle



2. Apply additional solder on the soldering spot on PCB
3. Clean the PCB with Neozol.

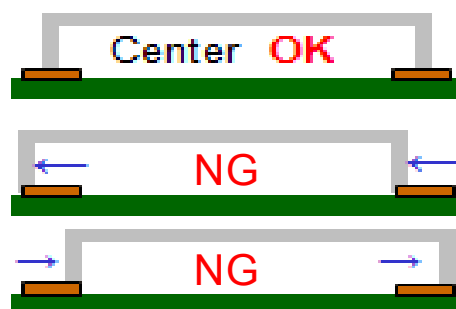
Assembly Rework Guide



1. Set the PCB as "Setting Guide"



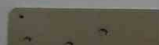
2. Assemble the "shield can" on right place of PCB.
*Caution : Shield can location



11.3 REWORK CAUTION POINTS

1. Issue : Use new part & rework short time, When change shieldcan
2. Cause : Shieldcan can be yellowish
 * It's appeared to customer for Phone Ass'y
3. Recommend rework time (200°C) : Finish rework within 2min 30sec
 -> It's base on the below S/C yellowish test on Hot Plate



Time	New	1min	2min	3min	4min	5min
1Piece Shield can						
						
						
Enlarge Image						

Not Good

12. EXPLODED VIEW & REPLACEMENT PART LIST
